



Survey Certification

7327 West Barton Road
Casper, WY 82604
(307)-472-6621 Fax (307) 472-5439

Operator	Oasis Petroleum
Well Name & No.	Lewis Federal 5300 11-31 3B
API #	33-053-06548
County & State	McKenzie County, ND
SDI Job #	OP.05419
Rig	Nabors B21
Survey Date	22-Jan-2019

I, Seth M. Burstad, having personal knowledge of all the facts, hereby certify that the attached directional survey run from a measured depth of 0 feet to a measured depth of 2743 feet is true and correct as determined from all available records.

Seth M. Burstad

Signature

22-Jan-2019

Date

Seth M. Burstad

Rockies Region Well Planner
Scientific Drilling - Rocky Mountain District



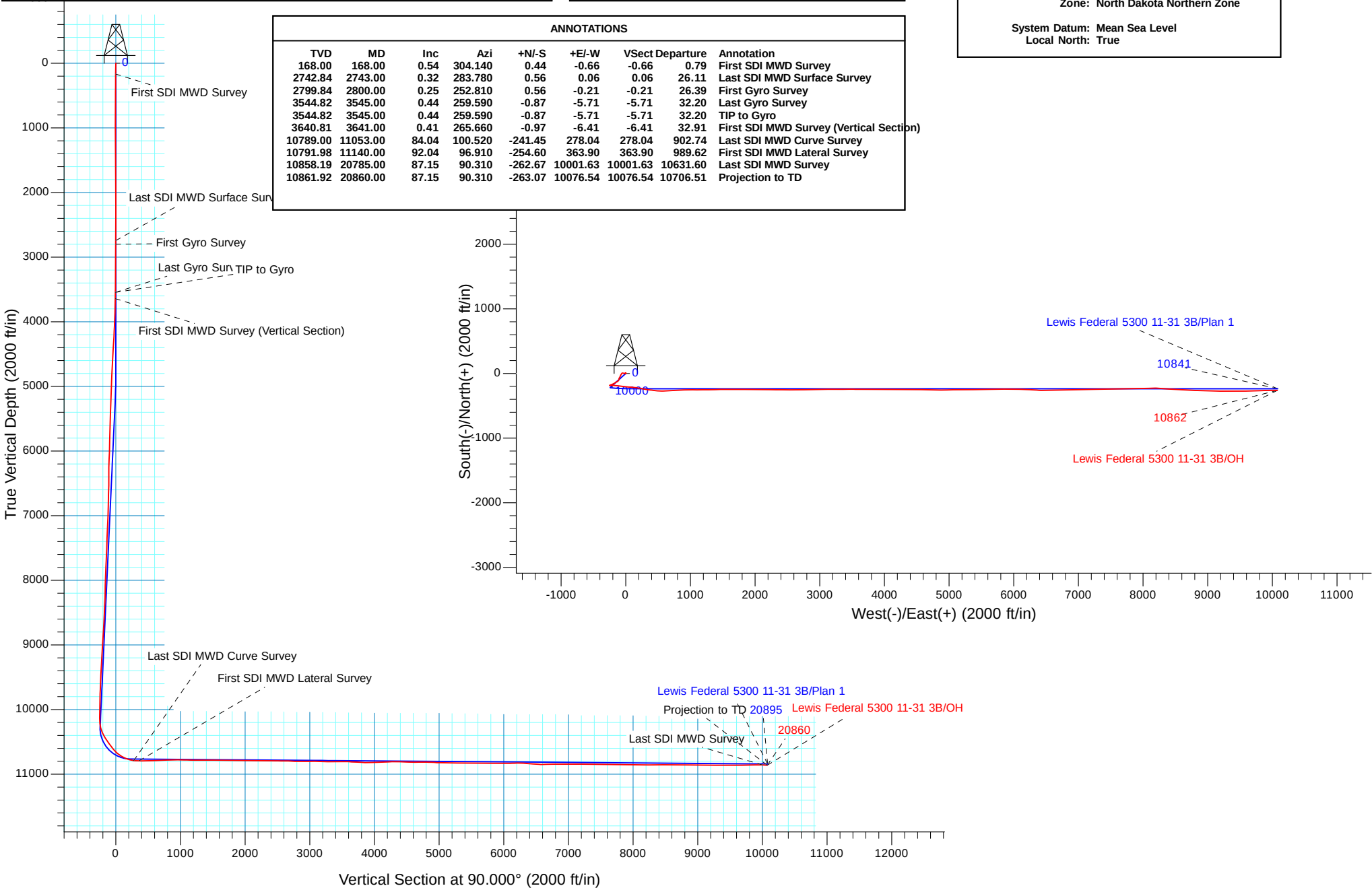
Project: McKenzie County, ND
Site: Lewis Federal
Well: Lewis Federal 5300 11-31 3B
Wellbore: OH
Design: OH



WELL DETAILS: Lewis Federal 5300 11-31 3B			
GL 2110' & KB 25' @ 2135.00ft (Nabors B21)	2110.00		
Northing	Easting	Latitude	Longitude
393129.76	1209541.15	48° 2' 8.980 N	103° 36' 11.110 W

Design: OH (Lewis Federal 5300 11-31 3B/OH)
Created By: Seth Burstad

PROJECT DETAILS: McKenzie County, ND
Geodetic System: US State Plane 1983
Datum: North American Datum 1983
Ellipsoid: GRS 1980
Zone: North Dakota Northern Zone
System Datum: Mean Sea Level
Local North: True





Oasis Petroleum

McKenzie County, ND

Lewis Federal

Lewis Federal 5300 11-31 3B

OH

Design: OH

Standard Survey Report

22 January, 2019



Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Lewis Federal 5300 11-31 3B
Project:	McKenzie County, ND	TVD Reference:	GL 2110' & KB 25' @ 2135.00ft (Nabors B21)
Site:	Lewis Federal	MD Reference:	GL 2110' & KB 25' @ 2135.00ft (Nabors B21)
Well:	Lewis Federal 5300 11-31 3B	North Reference:	True
Wellbore:	OH	Survey Calculation Method:	Minimum Curvature
Design:	OH	Database:	Casper District

Project	McKenzie County, ND		
Map System:	US State Plane 1983	System Datum:	Mean Sea Level
Geo Datum:	North American Datum 1983		
Map Zone:	North Dakota Northern Zone		

Site	Lewis Federal, Site Center: Lewis Federal 5300 11-31 2B		
Site Position:		Northing:	393,162.02 usft
From:	Lat/Long	Easting:	1,209,545.85 usft
Position Uncertainty:	0.00 ft	Slot Radius:	13.200 in
		Latitude:	48° 2' 9.300 N
		Longitude:	103° 36' 11.060 W
		Grid Convergence:	-2.31 °

Well	Lewis Federal 5300 11-31 3B, 1083' FNL 262' FWL Sec 31 T153N R100W		
Well Position	+N/-S	0.00 ft	Northing: 393,129.75 usft
	+E/-W	0.00 ft	Easting: 1,209,541.15 usft
Position Uncertainty	0.00 ft	Wellhead Elevation:	0.00 ft
		Latitude:	48° 2' 8.980 N
		Longitude:	103° 36' 11.110 W
		Ground Level:	2,110.00 ft

Wellbore	OH				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	HDGM	11/5/2018	7.88	72.70	55,778

Design	OH				
Audit Notes:					
Version:	1.0	Phase:	ACTUAL	Tie On Depth:	0.00
Vertical Section:	Depth From (TVD) (ft)	+N/-S (ft)	+E/-W (ft)	Direction (°)	
	0.00	0.00	0.00	90.000	

Survey Program	Date	1/22/2019			
From (ft)	To (ft)	Survey (Wellbore)	Tool Name	Description	
168.00	2,743.00	Survey #1 - Surface (OH)	MWD+HDGM	OWSG MWD + HDGM	
2,800.00	3,545.00	Survey #2 - Surface (GYRO) (OH)	GYRO-NS	Gyrocompass Gyro	
3,641.00	11,053.00	Survey #3 - Vertical / Curve (OH)	MWD+HDGM	OWSG MWD + HDGM	
11,140.00	20,785.00	Survey #4 - Lateral (OH)	MWD+HDGM	OWSG MWD + HDGM	

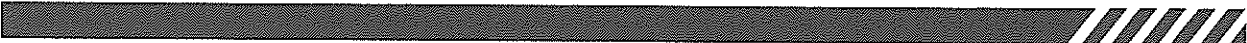
Survey										
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)	
0.00	0.00	0.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
168.00	0.54	304.140	168.00	0.44	-0.66	-0.66	0.32	0.32	0.00	
First SDI MWD Survey										
261.00	0.29	4.550	261.00	0.92	-1.00	-1.00	0.51	-0.27	64.96	
355.00	0.61	316.070	354.99	1.52	-1.33	-1.33	0.50	0.34	-51.57	
447.00	0.80	325.270	446.99	2.40	-2.03	-2.03	0.24	0.21	10.00	
537.00	0.85	323.540	536.98	3.46	-2.79	-2.79	0.06	0.06	-1.92	
626.00	0.24	338.370	625.97	4.16	-3.25	-3.25	0.70	-0.69	16.66	

Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Lewis Federal 5300 11-31 3B
Project:	McKenzie County, ND	TVD Reference:	GL 2110' & KB 25' @ 2135.00ft (Nabors B21)
Site:	Lewis Federal	MD Reference:	GL 2110' & KB 25' @ 2135.00ft (Nabors B21)
Well:	Lewis Federal 5300 11-31 3B	North Reference:	True
Wellbore:	OH	Survey Calculation Method:	Minimum Curvature
Design:	OH	Database:	Casper District

Survey									
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
725.00	0.62	348.620	724.97	4.88	-3.43	-3.43	0.39	0.38	10.35
815.00	0.79	346.470	814.96	5.96	-3.67	-3.67	0.19	0.19	-2.39
905.00	0.81	323.760	904.95	7.08	-4.19	-4.19	0.35	0.02	-25.23
996.00	0.96	344.770	995.94	8.33	-4.77	-4.77	0.39	0.16	23.09
1,086.00	0.84	322.190	1,085.93	9.58	-5.38	-5.38	0.41	-0.13	-25.09
1,176.00	0.07	75.770	1,175.93	10.11	-5.73	-5.73	0.97	-0.86	126.20
1,266.00	0.67	154.080	1,265.93	9.65	-5.44	-5.44	0.73	0.67	87.01
1,357.00	0.70	123.650	1,356.92	8.87	-4.75	-4.75	0.40	0.03	-33.44
1,447.00	0.98	127.920	1,446.91	8.09	-3.68	-3.68	0.32	0.31	4.74
1,539.00	0.99	134.820	1,538.90	7.05	-2.50	-2.50	0.13	0.01	7.50
1,629.00	0.87	147.210	1,628.89	5.92	-1.58	-1.58	0.26	-0.13	13.77
1,719.00	1.33	137.890	1,718.87	4.57	-0.51	-0.51	0.55	0.51	-10.36
1,810.00	0.75	144.370	1,809.85	3.31	0.55	0.55	0.65	-0.64	7.12
1,903.00	0.24	97.420	1,902.85	2.79	1.09	1.09	0.66	-0.55	-50.48
1,996.00	0.20	174.020	1,995.85	2.60	1.31	1.31	0.30	-0.04	82.37
2,089.00	0.26	120.440	2,088.85	2.33	1.50	1.50	0.23	0.06	-57.61
2,183.00	0.34	205.730	2,182.85	1.97	1.57	1.57	0.44	0.09	90.73
2,276.00	0.30	175.010	2,275.85	1.48	1.47	1.47	0.19	-0.04	-33.03
2,370.00	0.39	219.140	2,369.84	0.99	1.29	1.29	0.29	0.10	46.95
2,463.00	0.14	209.480	2,462.84	0.64	1.03	1.03	0.27	-0.27	-10.39
2,557.00	0.03	250.320	2,556.84	0.53	0.95	0.95	0.13	-0.12	43.45
2,650.00	0.38	267.070	2,649.84	0.51	0.62	0.62	0.38	0.38	18.01
2,743.00	0.32	283.780	2,742.84	0.56	0.06	0.06	0.13	-0.06	17.97
Last SDI MWD Surface Survey									

Design Annotations				
Measured Depth (ft)	Vertical Depth (ft)	Local Coordinates		Comment
		+N/-S (ft)	+E/-W (ft)	
168.00	168.00	0.44	-0.66	First SDI MWD Survey
2,743.00	2,742.84	0.56	0.06	Last SDI MWD Surface Survey
2,800.00	2,799.84	0.56	-0.21	First Gyro Survey
3,545.00	3,544.82	-0.87	-5.71	Last Gyro Survey
11,140.00	10,791.98	-254.60	363.90	First SDI MWD Lateral Survey

Checked By: _____	Approved By: _____	Date: _____
-------------------	--------------------	-------------



Directional Survey Certification

Operator: OASIS PETROLEUM NORTH AMERICA LLC.

Well Name: LEWIS FEDERAL 5300 11-31 3B

API Number: 33-053-06548-00-00

County & State: McKenzie County, North Dakota

Job Number: RM0519GDM387

Run Date: 1/6/2019

Surveyor: Clifton Miech

I, Robert Steele, having personal knowledge of all of the facts, hereby certify that the attached directional survey run from the measured depth of 2743.00 feet to a measured depth of 3545.00 feet is true and correct as determined from all available records.

X 

Robert Steele
Survey Specialist XI Gyrodata, Inc.

A Gyrodata Directional Survey

FINAL DEFINITIVE COPY

for

OASIS PETROLEUM NORTH AMERICA LLC

Lease: LEWIS FEDERAL Well: 5300 11-31 3B, 13 3/8" casing
Location: Nabors B21, McKenzie County, ND

Job Number: RM0119GMSW111

Run Date: 06 Jan 2019

Report Generated: 18 Nov 2019 04:14

Surveyor: Clifton Miech, Zachary Lodge, Austin Maxwell, Brian Enloe

Calculation Method: MINIMUM CURVATURE

Survey Latitude: 48.035810 deg. N

Longitude: 103.602608 deg. W

Azimuth Correction:

Gyro: Bearings are Relative to True North

Depth Reference: Rotary Table

Air Gap (RKB to Ground / RKB to MSL): 25.00

Ground Elevation (Ground to MSL / MSL to Mudline): 2118.00

Vertical Section Calculated from Well Head Location

Closure Calculated from Well Head Location

Horizontal Coordinates Calculated from Well Head Location

A Gyrodata Directional Survey

OASIS PETROLEUM NORTH AMERICA LLC

Lease: LEWIS FEDERAL Well: 5300 11-31 3B, 13 3/8" casing

Location: Nabors B21, Mackenzie County, ND

Job Number: RM0119GMSW111

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. AZIMUTH feet deg.	HORIZONTAL COORDINATES feet
---------------------------	--------------	-----------------	--	---------------------------	---------------------------------------	-----------------------------------

2743 TO 3545 FT RATE GYROSCOPIC MULTISHOT RUN INSIDE 13 3/8" CASING
ALL MEASURED DEPTHS AND COORDINATES REFERENCED TO NABORS B21 WITH AN R.K.B. OF 25 FT
TIE-IN DATA PROVIDED BY PREVIOUS SCIENTIFIC DRILLING SURVEY DATED JAN.6, 2019.

2650.00	0.38	267.07	0.01	2649.84	0.8 53.3	0.47 N 0.63 E
2743.00	0.32	283.78	0.13	2742.84	0.5 7.7	0.52 N 0.07 E
2800.00	0.25	252.81	0.29	2799.84	0.6 338.5	0.52 N 0.20 W
2900.00	0.48	279.36	0.28	2899.84	1.0 302.3	0.52 N 0.83 W
3000.00	0.67	258.42	0.28	2999.83	1.9 284.6	0.47 N 1.81 W
3100.00	0.44	255.96	0.23	3099.83	2.8 275.4	0.26 N 2.76 W
3200.00	0.42	237.60	0.14	3199.82	3.4 269.5	0.03 S 3.44 W
3300.00	0.42	253.99	0.12	3299.82	4.1 265.5	0.33 S 4.10 W
3400.00	0.33	248.64	0.10	3399.82	4.8 263.6	0.53 S 4.72 W
3500.00	0.47	245.50	0.14	3499.82	5.4 261.4	0.81 S 5.36 W
3545.00	0.44	259.59	0.26	3544.81	5.8 260.9	0.91 S 5.70 W

Final Station Closure: Distance: 5.77 ft Az: 260.88 deg.



Survey Certification

7327 West Barton Road
Casper, WY 82604
(307)-472-6621 Fax (307) 472-5439

Operator	Oasis Petroleum
Well Name & No.	Lewis Federal 5300 11-31 3B
API #	33-053-06548
County & State	McKenzie County, ND
SDI Job #	OP.05419
Rig	Nabors B21
Survey Date	22-Jan-2019

I, Seth M. Burstad, having personal knowledge of all the facts, hereby certify that the attached directional survey run from a measured depth of 3545 feet to a measured depth of 20860 feet is true and correct as determined from all available records.

Seth M. Burstad

Signature

22-Jan-2019

Date

Seth M. Burstad

Rockies Region Well Planner
Scientific Drilling - Rocky Mountain District



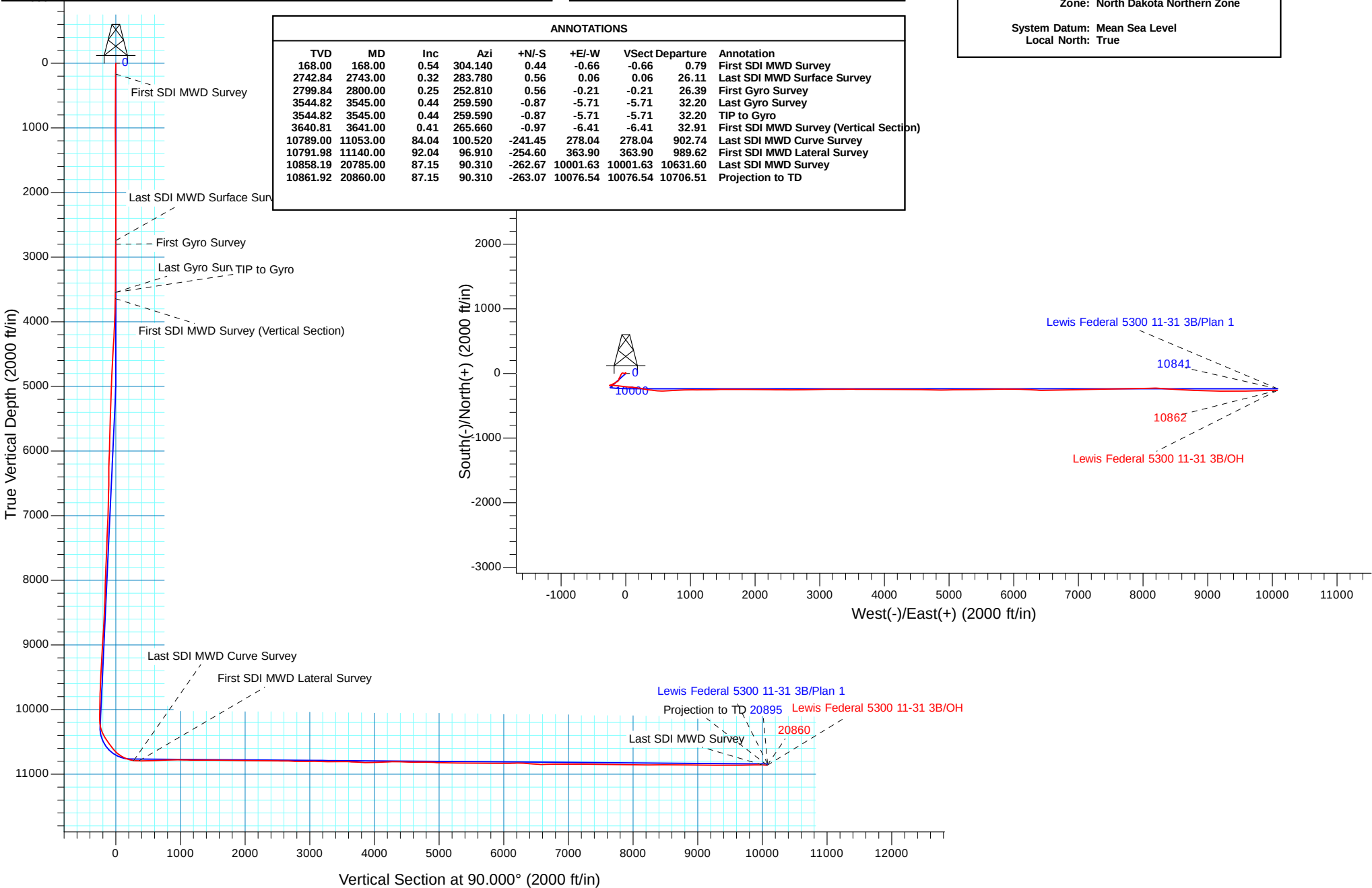
Project: McKenzie County, ND
Site: Lewis Federal
Well: Lewis Federal 5300 11-31 3B
Wellbore: OH
Design: OH



WELL DETAILS: Lewis Federal 5300 11-31 3B			
GL 2110' & KB 25' @ 2135.00ft (Nabors B21)	2110.00		
Northing	Easting	Latitude	Longitude
393129.76	1209541.15	48° 2' 8.980 N	103° 36' 11.110 W

Design: OH (Lewis Federal 5300 11-31 3B/OH)
Created By: Seth Burstad

PROJECT DETAILS: McKenzie County, ND
Geodetic System: US State Plane 1983
Datum: North American Datum 1983
Ellipsoid: GRS 1980
Zone: North Dakota Northern Zone
System Datum: Mean Sea Level
Local North: True





Oasis Petroleum

McKenzie County, ND

Lewis Federal

Lewis Federal 5300 11-31 3B

OH

Design: OH

Standard Survey Report

22 January, 2019



Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Lewis Federal 5300 11-31 3B
Project:	McKenzie County, ND	TVD Reference:	GL 2110' & KB 25' @ 2135.00ft (Nabors B21)
Site:	Lewis Federal	MD Reference:	GL 2110' & KB 25' @ 2135.00ft (Nabors B21)
Well:	Lewis Federal 5300 11-31 3B	North Reference:	True
Wellbore:	OH	Survey Calculation Method:	Minimum Curvature
Design:	OH	Database:	Casper District

Project	McKenzie County, ND		
Map System:	US State Plane 1983	System Datum:	Mean Sea Level
Geo Datum:	North American Datum 1983		
Map Zone:	North Dakota Northern Zone		

Site	Lewis Federal, Site Center: Lewis Federal 5300 11-31 2B		
Site Position:		Northing:	393,162.02 usft
From:	Lat/Long	Easting:	1,209,545.85 usft
Position Uncertainty:	0.00 ft	Slot Radius:	13.200 in
		Latitude:	48° 2' 9.300 N
		Longitude:	103° 36' 11.060 W
		Grid Convergence:	-2.31 °

Well	Lewis Federal 5300 11-31 3B, 1083' FNL 262' FWL Sec 31 T153N R100W		
Well Position	+N/-S	0.00 ft	Northing: 393,129.75 usft
	+E/-W	0.00 ft	Easting: 1,209,541.15 usft
Position Uncertainty	0.00 ft	Wellhead Elevation:	0.00 ft
		Latitude:	48° 2' 8.980 N
		Longitude:	103° 36' 11.110 W
		Ground Level:	2,110.00 ft

Wellbore	OH				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	HDGM	11/5/2018	7.88	72.70	55,778

Design	OH				
Audit Notes:					
Version:	1.0	Phase:	ACTUAL	Tie On Depth:	0.00
Vertical Section:	Depth From (TVD) (ft)	+N/-S (ft)	+E/-W (ft)	Direction (°)	
	0.00	0.00	0.00	90.000	

Survey Program	Date	1/22/2019			
From (ft)	To (ft)	Survey (Wellbore)	Tool Name	Description	
168.00	2,743.00	Survey #1 - Surface (OH)	MWD+HDGM	OWSG MWD + HDGM	
2,800.00	3,545.00	Survey #2 - Surface (GYRO) (OH)	GYRO-NS	Gyrocompass Gyro	
3,641.00	11,053.00	Survey #3 - Vertical / Curve (OH)	MWD+HDGM	OWSG MWD + HDGM	
11,140.00	20,860.00	Survey #4 - Lateral (OH)	MWD+HDGM	OWSG MWD + HDGM	

Survey										
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)	
3,545.00	0.44	259.590	3,544.82	-0.87	-5.71	-5.71	0.26	-0.07	31.31	
Last Gyro Survey - TIP to Gyro										
3,641.00	0.41	265.660	3,640.81	-0.97	-6.41	-6.41	0.06	-0.03	6.32	
First SDI MWD Survey (Vertical Section)										
3,672.00	0.35	227.310	3,671.81	-1.04	-6.59	-6.59	0.83	-0.19	-123.71	
3,766.00	1.41	265.050	3,765.80	-1.33	-7.96	-7.96	1.23	1.13	40.15	
3,859.00	2.07	263.160	3,858.76	-1.63	-10.77	-10.77	0.71	0.71	-2.03	

Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Lewis Federal 5300 11-31 3B
Project:	McKenzie County, ND	TVD Reference:	GL 2110' & KB 25' @ 2135.00ft (Nabors B21)
Site:	Lewis Federal	MD Reference:	GL 2110' & KB 25' @ 2135.00ft (Nabors B21)
Well:	Lewis Federal 5300 11-31 3B	North Reference:	True
Wellbore:	OH	Survey Calculation Method:	Minimum Curvature
Design:	OH	Database:	Casper District

Survey									
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
3,953.00	2.65	271.610	3,952.68	-1.77	-14.62	-14.62	0.72	0.62	8.99
4,046.00	2.50	287.080	4,045.59	-1.12	-18.71	-18.71	0.76	-0.16	16.63
4,140.00	2.33	287.920	4,139.50	0.07	-22.49	-22.49	0.18	-0.18	0.89
4,233.00	2.79	274.450	4,232.41	0.83	-26.54	-26.54	0.81	0.49	-14.48
4,326.00	3.79	266.050	4,325.26	0.79	-31.87	-31.87	1.19	1.08	-9.03
4,420.00	3.52	280.150	4,419.07	1.09	-37.81	-37.81	1.00	-0.29	15.00
4,513.00	3.32	296.150	4,511.90	2.78	-43.03	-43.03	1.04	-0.22	17.20
4,608.00	3.47	289.230	4,606.74	4.94	-48.22	-48.22	0.46	0.16	-7.28
4,703.00	3.99	255.250	4,701.55	5.04	-54.13	-54.13	2.35	0.55	-35.77
4,798.00	4.11	230.690	4,796.32	2.04	-59.96	-59.96	1.82	0.13	-25.85
4,893.00	3.76	212.640	4,891.10	-2.74	-64.28	-64.28	1.35	-0.37	-19.00
4,988.00	4.17	206.210	4,985.87	-8.46	-67.48	-67.48	0.64	0.43	-6.77
5,083.00	3.49	210.520	5,080.66	-14.05	-70.47	-70.47	0.78	-0.72	4.54
5,178.00	3.11	213.800	5,175.50	-18.68	-73.38	-73.38	0.45	-0.40	3.45
5,272.00	3.68	210.860	5,269.33	-23.39	-76.34	-76.34	0.63	0.61	-3.13
5,367.00	4.27	209.400	5,364.11	-29.09	-79.64	-79.64	0.63	0.62	-1.54
5,462.00	3.68	203.480	5,458.88	-34.96	-82.59	-82.59	0.76	-0.62	-6.23
5,557.00	3.35	202.510	5,553.70	-40.32	-84.87	-84.87	0.35	-0.35	-1.02
5,652.00	3.92	199.740	5,648.51	-45.95	-87.03	-87.03	0.63	0.60	-2.92
5,746.00	3.49	206.210	5,742.31	-51.54	-89.38	-89.38	0.64	-0.46	6.88
5,841.00	3.24	212.420	5,837.15	-56.40	-92.10	-92.10	0.46	-0.26	6.54
5,935.00	3.53	209.590	5,930.98	-61.16	-94.95	-94.95	0.36	0.31	-3.01
6,021.00	4.32	212.130	6,016.78	-66.20	-97.98	-97.98	0.94	0.92	2.95
6,116.00	4.88	210.580	6,111.47	-72.71	-101.94	-101.94	0.60	0.59	-1.63
6,210.00	3.85	199.950	6,205.20	-79.12	-105.05	-105.05	1.39	-1.10	-11.31
6,303.00	3.13	185.940	6,298.03	-84.58	-106.38	-106.38	1.19	-0.77	-15.06
6,396.00	3.29	184.170	6,390.88	-89.77	-106.83	-106.83	0.20	0.17	-1.90
6,490.00	3.02	191.330	6,484.74	-94.88	-107.52	-107.52	0.51	-0.29	7.62
6,583.00	2.96	193.640	6,577.62	-99.62	-108.56	-108.56	0.14	-0.06	2.48
6,677.00	2.41	211.570	6,671.51	-103.66	-110.17	-110.17	1.06	-0.59	19.07
6,770.00	2.82	216.370	6,764.42	-107.17	-112.55	-112.55	0.50	0.44	5.16
6,863.00	2.73	212.460	6,857.31	-110.88	-115.10	-115.10	0.23	-0.10	-4.20
6,957.00	3.13	218.710	6,951.18	-114.77	-117.90	-117.90	0.54	0.43	6.65
7,050.00	3.18	213.880	7,044.04	-118.90	-120.93	-120.93	0.29	0.05	-5.19
7,143.00	3.19	223.590	7,136.90	-122.91	-124.15	-124.15	0.58	0.01	10.44
7,237.00	2.70	248.740	7,230.78	-125.61	-128.02	-128.02	1.46	-0.52	26.76
7,330.00	2.71	249.530	7,323.68	-127.17	-132.12	-132.12	0.04	0.01	0.85
7,424.00	2.57	246.150	7,417.58	-128.80	-136.13	-136.13	0.22	-0.15	-3.60
7,517.00	1.97	255.090	7,510.50	-130.05	-139.58	-139.58	0.75	-0.65	9.61
7,611.00	2.10	249.720	7,604.44	-131.07	-142.76	-142.76	0.25	0.14	-5.71
7,704.00	3.18	242.570	7,697.34	-132.85	-146.64	-146.64	1.21	1.16	-7.69
7,798.00	2.91	244.250	7,791.21	-135.08	-151.11	-151.11	0.30	-0.29	1.79
7,891.00	2.74	239.410	7,884.10	-137.24	-155.15	-155.15	0.31	-0.18	-5.20

Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Lewis Federal 5300 11-31 3B
Project:	McKenzie County, ND	TVD Reference:	GL 2110' & KB 25' @ 2135.00ft (Nabors B21)
Site:	Lewis Federal	MD Reference:	GL 2110' & KB 25' @ 2135.00ft (Nabors B21)
Well:	Lewis Federal 5300 11-31 3B	North Reference:	True
Wellbore:	OH	Survey Calculation Method:	Minimum Curvature
Design:	OH	Database:	Casper District

Survey									
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
7,985.00	1.96	235.940	7,978.02	-139.28	-158.41	-158.41	0.84	-0.83	-3.69
8,078.00	1.52	230.920	8,070.98	-140.95	-160.69	-160.69	0.50	-0.47	-5.40
8,171.00	1.44	235.240	8,163.95	-142.40	-162.61	-162.61	0.15	-0.09	4.65
8,265.00	2.04	229.530	8,257.90	-144.16	-164.85	-164.85	0.66	0.64	-6.07
8,358.00	2.01	225.210	8,350.84	-146.38	-167.27	-167.27	0.17	-0.03	-4.65
8,451.00	2.57	234.410	8,443.77	-148.74	-170.12	-170.12	0.72	0.60	9.89
8,544.00	3.68	242.800	8,536.63	-151.32	-174.47	-174.47	1.29	1.19	9.02
8,638.00	4.30	246.640	8,630.40	-154.10	-180.39	-180.39	0.72	0.66	4.09
8,731.00	4.39	247.400	8,723.13	-156.85	-186.87	-186.87	0.11	0.10	0.82
8,824.00	4.07	244.470	8,815.88	-159.64	-193.14	-193.14	0.42	-0.34	-3.15
8,918.00	3.95	241.700	8,909.65	-162.61	-199.00	-199.00	0.24	-0.13	-2.95
9,011.00	3.82	243.100	9,002.44	-165.53	-204.58	-204.58	0.17	-0.14	1.51
9,105.00	3.49	246.340	9,096.25	-168.09	-210.00	-210.00	0.41	-0.35	3.45
9,198.00	3.01	248.910	9,189.10	-170.11	-214.87	-214.87	0.54	-0.52	2.76
9,292.00	3.59	247.440	9,282.94	-172.13	-219.89	-219.89	0.62	0.62	-1.56
9,386.00	3.30	248.320	9,376.77	-174.25	-225.12	-225.12	0.31	-0.31	0.94
9,481.00	2.90	252.100	9,471.63	-176.00	-229.95	-229.95	0.47	-0.42	3.98
9,575.00	2.31	249.480	9,565.53	-177.40	-233.98	-233.98	0.64	-0.63	-2.79
9,670.00	2.13	251.570	9,660.46	-178.63	-237.45	-237.45	0.21	-0.19	2.20
9,765.00	2.30	232.600	9,755.39	-180.34	-240.64	-240.64	0.79	0.18	-19.97
9,860.00	2.46	219.190	9,850.31	-183.08	-243.44	-243.44	0.61	0.17	-14.12
9,954.00	1.65	225.410	9,944.25	-185.60	-245.68	-245.68	0.89	-0.86	6.62
10,049.00	0.90	230.900	10,039.23	-187.03	-247.23	-247.23	0.80	-0.79	5.78
10,143.00	0.45	331.380	10,133.22	-187.17	-247.98	-247.98	1.15	-0.48	106.89
10,206.00	2.74	83.570	10,196.20	-186.78	-246.61	-246.61	4.67	3.63	178.08
10,237.00	6.77	90.400	10,227.09	-186.71	-244.04	-244.04	13.10	13.00	22.03
10,268.00	11.63	90.870	10,257.68	-186.77	-239.09	-239.09	15.68	15.68	1.52
10,299.00	15.88	90.190	10,287.78	-186.83	-231.72	-231.72	13.72	13.71	-2.19
10,331.00	19.88	89.680	10,318.23	-186.82	-221.90	-221.90	12.51	12.50	-1.59
10,362.00	24.28	89.890	10,346.95	-186.78	-210.25	-210.25	14.20	14.19	0.68
10,394.00	27.62	91.430	10,375.72	-186.95	-196.25	-196.25	10.65	10.44	4.81
10,425.00	27.59	91.400	10,403.19	-187.30	-181.89	-181.89	0.11	-0.10	-0.10
10,457.00	30.13	93.170	10,431.21	-187.93	-166.46	-166.46	8.37	7.94	5.53
10,488.00	32.93	96.170	10,457.64	-189.26	-150.31	-150.31	10.35	9.03	9.68
10,520.00	32.84	96.430	10,484.51	-191.17	-133.04	-133.04	0.52	-0.28	0.81
10,552.00	35.04	95.640	10,511.06	-193.05	-115.27	-115.27	7.01	6.88	-2.47
10,583.00	37.03	97.850	10,536.12	-195.20	-97.17	-97.17	7.67	6.42	7.13
10,615.00	36.54	97.650	10,561.75	-197.78	-78.18	-78.18	1.58	-1.53	-0.63
10,646.00	38.79	96.620	10,586.29	-200.13	-59.38	-59.38	7.54	7.26	-3.32
10,678.00	41.36	97.130	10,610.77	-202.60	-38.93	-38.93	8.10	8.03	1.59
10,709.00	41.92	96.370	10,633.94	-205.02	-18.48	-18.48	2.43	1.81	-2.45
10,741.00	45.61	95.140	10,657.05	-207.23	3.54	3.54	11.83	11.53	-3.84
10,772.00	49.29	94.940	10,678.01	-209.23	26.29	26.29	11.88	11.87	-0.65
10,803.00	53.11	94.530	10,697.43	-211.22	50.36	50.36	12.37	12.32	-1.32

Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Lewis Federal 5300 11-31 3B
Project:	McKenzie County, ND	TVD Reference:	GL 2110' & KB 25' @ 2135.00ft (Nabors B21)
Site:	Lewis Federal	MD Reference:	GL 2110' & KB 25' @ 2135.00ft (Nabors B21)
Well:	Lewis Federal 5300 11-31 3B	North Reference:	True
Wellbore:	OH	Survey Calculation Method:	Minimum Curvature
Design:	OH	Database:	Casper District

Survey									
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
10,834.00	57.69	93.170	10,715.03	-212.93	75.81	75.81	15.21	14.77	-4.39
10,865.00	61.64	94.970	10,730.68	-214.84	102.49	102.49	13.69	12.74	5.81
10,896.00	65.17	95.680	10,744.56	-217.41	130.09	130.09	11.57	11.39	2.29
10,928.00	68.76	98.020	10,757.08	-220.93	159.32	159.32	13.08	11.22	7.31
10,959.00	71.32	99.850	10,767.66	-225.46	188.10	188.10	9.95	8.26	5.90
10,990.00	74.45	100.000	10,776.78	-230.56	217.28	217.28	10.11	10.10	0.48
11,022.00	78.44	100.050	10,784.28	-235.98	247.90	247.90	12.47	12.47	0.16
11,053.00	84.04	100.520	10,789.00	-241.45	278.04	278.04	18.13	18.06	1.52
Last SDI MWD Curve Survey									
11,140.00	92.04	96.910	10,791.98	-254.60	363.90	363.90	10.09	9.20	-4.15
First SDI MWD Lateral Survey									
11,171.00	90.90	98.880	10,791.18	-258.86	394.59	394.59	7.34	-3.68	6.35
11,202.00	90.64	97.420	10,790.76	-263.25	425.28	425.28	4.78	-0.84	-4.71
11,233.00	90.00	96.170	10,790.59	-266.92	456.06	456.06	4.53	-2.06	-4.03
11,294.00	89.53	91.980	10,790.84	-271.25	516.89	516.89	6.91	-0.77	-6.87
11,325.00	89.46	91.590	10,791.11	-272.22	547.87	547.87	1.28	-0.23	-1.26
11,356.00	89.93	87.910	10,791.28	-272.09	578.87	578.87	11.97	1.52	-11.87
11,387.00	92.04	87.640	10,790.75	-270.88	609.84	609.84	6.86	6.81	-0.87
11,418.00	92.28	87.550	10,789.58	-269.58	640.79	640.79	0.83	0.77	-0.29
11,510.00	92.84	86.340	10,785.47	-264.68	732.56	732.56	1.45	0.61	-1.32
11,602.00	93.72	86.030	10,780.20	-258.57	824.21	824.21	1.01	0.96	-0.34
11,633.00	91.61	88.550	10,778.76	-257.11	855.14	855.14	10.60	-6.81	8.13
11,664.00	91.68	87.640	10,777.87	-256.08	886.11	886.11	2.94	0.23	-2.94
11,695.00	90.37	89.510	10,777.32	-255.31	917.09	917.09	7.36	-4.23	6.03
11,787.00	89.30	89.220	10,777.58	-254.29	1,009.08	1,009.08	1.21	-1.16	-0.32
11,878.00	87.72	89.730	10,779.95	-253.46	1,100.04	1,100.04	1.82	-1.74	0.56
11,970.00	87.92	89.250	10,783.45	-252.64	1,191.97	1,191.97	0.56	0.22	-0.52
12,062.00	90.10	89.120	10,785.04	-251.33	1,283.95	1,283.95	2.37	2.37	-0.14
12,154.00	90.37	88.950	10,784.66	-249.78	1,375.93	1,375.93	0.35	0.29	-0.18
12,246.00	90.57	89.590	10,783.91	-248.61	1,467.92	1,467.92	0.73	0.22	0.70
12,339.00	90.27	89.560	10,783.23	-247.92	1,560.92	1,560.92	0.32	-0.32	-0.03
12,433.00	88.73	89.720	10,784.05	-247.33	1,654.91	1,654.91	1.65	-1.64	0.17
12,528.00	89.50	90.280	10,785.51	-247.33	1,749.89	1,749.89	1.00	0.81	0.59
12,623.00	88.76	90.690	10,786.96	-248.13	1,844.88	1,844.88	0.89	-0.78	0.43
12,718.00	89.53	90.180	10,788.37	-248.85	1,939.87	1,939.87	0.97	0.81	-0.54
12,814.00	89.30	91.140	10,789.35	-249.96	2,035.85	2,035.85	1.03	-0.24	1.00
12,909.00	89.33	89.970	10,790.49	-250.88	2,130.84	2,130.84	1.23	0.03	-1.23
13,003.00	89.93	90.040	10,791.10	-250.89	2,224.84	2,224.84	0.64	0.64	0.07
13,098.00	89.06	90.590	10,791.93	-251.41	2,319.83	2,319.83	1.08	-0.92	0.58
13,193.00	90.10	90.580	10,792.63	-252.38	2,414.82	2,414.82	1.09	1.09	-0.01
13,288.00	88.39	90.300	10,793.88	-253.11	2,509.81	2,509.81	1.82	-1.80	-0.29
13,383.00	90.07	90.030	10,795.16	-253.38	2,604.80	2,604.80	1.79	1.77	-0.28
13,478.00	87.79	89.860	10,796.93	-253.29	2,699.77	2,699.77	2.41	-2.40	-0.18

Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Lewis Federal 5300 11-31 3B
Project:	McKenzie County, ND	TVD Reference:	GL 2110' & KB 25' @ 2135.00ft (Nabors B21)
Site:	Lewis Federal	MD Reference:	GL 2110' & KB 25' @ 2135.00ft (Nabors B21)
Well:	Lewis Federal 5300 11-31 3B	North Reference:	True
Wellbore:	OH	Survey Calculation Method:	Minimum Curvature
Design:	OH	Database:	Casper District

Survey									
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
13,573.00	87.99	89.440	10,800.43	-252.71	2,794.71	2,794.71	0.49	0.21	-0.44
13,668.00	90.00	87.910	10,802.10	-250.51	2,889.66	2,889.66	2.66	2.12	-1.61
13,762.00	91.37	88.720	10,800.97	-247.75	2,983.61	2,983.61	1.69	1.46	0.86
13,857.00	89.33	89.900	10,800.39	-246.61	3,078.59	3,078.59	2.48	-2.15	1.24
13,952.00	87.82	89.430	10,802.75	-246.05	3,173.56	3,173.56	1.66	-1.59	-0.49
14,046.00	88.29	90.740	10,805.94	-246.19	3,267.50	3,267.50	1.48	0.50	1.39
14,141.00	90.10	88.730	10,807.28	-245.75	3,362.48	3,362.48	2.85	1.91	-2.12
14,235.00	91.31	90.000	10,806.12	-244.71	3,456.47	3,456.47	1.87	1.29	1.35
14,330.00	89.43	91.470	10,805.51	-245.93	3,551.45	3,551.45	2.51	-1.98	1.55
14,424.00	86.75	90.190	10,808.64	-247.29	3,645.38	3,645.38	3.16	-2.85	-1.36
14,519.00	87.25	90.720	10,813.61	-248.04	3,740.24	3,740.24	0.77	0.53	0.56
14,582.00	86.45	89.900	10,817.08	-248.38	3,803.15	3,803.15	1.82	-1.27	-1.30
14,614.00	86.61	90.390	10,819.01	-248.46	3,835.09	3,835.09	1.61	0.50	1.53
14,645.00	88.69	89.940	10,820.28	-248.55	3,866.06	3,866.06	6.86	6.71	-1.45
14,676.00	91.21	89.820	10,820.31	-248.49	3,897.06	3,897.06	8.14	8.13	-0.39
14,708.00	91.67	89.550	10,819.51	-248.31	3,929.05	3,929.05	1.67	1.44	-0.84
14,803.00	91.78	89.680	10,816.65	-247.67	4,024.00	4,024.00	0.18	0.12	0.14
14,897.00	92.41	90.420	10,813.21	-247.76	4,117.94	4,117.94	1.03	0.67	0.79
14,992.00	91.37	89.760	10,810.08	-247.91	4,212.88	4,212.88	1.30	-1.09	-0.69
15,087.00	92.65	90.120	10,806.75	-247.81	4,307.82	4,307.82	1.40	1.35	0.38
15,181.00	87.02	91.740	10,807.02	-249.33	4,401.77	4,401.77	6.23	-5.99	1.72
15,276.00	87.32	90.770	10,811.71	-251.41	4,496.63	4,496.63	1.07	0.32	-1.02
15,370.00	90.67	91.480	10,813.36	-253.25	4,590.58	4,590.58	3.64	3.56	0.76
15,464.00	89.93	90.760	10,812.86	-255.09	4,684.56	4,684.56	1.10	-0.79	-0.77
15,558.00	89.16	91.320	10,813.61	-256.80	4,778.54	4,778.54	1.01	-0.82	0.60
15,653.00	87.96	89.350	10,816.00	-257.35	4,873.50	4,873.50	2.43	-1.26	-2.07
15,747.00	86.65	88.690	10,820.42	-255.75	4,967.38	4,967.38	1.56	-1.39	-0.70
15,842.00	88.96	87.930	10,824.06	-252.95	5,062.27	5,062.27	2.56	2.43	-0.80
15,937.00	90.20	89.500	10,824.75	-250.82	5,157.23	5,157.23	2.11	1.31	1.65
16,032.00	89.36	89.890	10,825.12	-250.31	5,252.23	5,252.23	0.97	-0.88	0.41
16,127.00	89.40	90.230	10,826.14	-250.41	5,347.23	5,347.23	0.36	0.04	0.36
16,222.00	88.86	89.420	10,827.59	-250.12	5,442.21	5,442.21	1.02	-0.57	-0.85
16,316.00	88.46	88.910	10,829.79	-248.75	5,536.18	5,536.18	0.69	-0.43	-0.54
16,411.00	89.63	87.540	10,831.37	-245.81	5,631.11	5,631.11	1.90	1.23	-1.44
16,505.00	91.00	89.290	10,830.85	-243.21	5,725.07	5,725.07	2.36	1.46	1.86
16,600.00	90.44	90.380	10,829.66	-242.94	5,820.06	5,820.06	1.29	-0.59	1.15
16,663.00	90.60	90.360	10,829.09	-243.34	5,883.06	5,883.06	0.26	0.25	-0.03
16,695.00	88.86	90.400	10,829.24	-243.55	5,915.05	5,915.05	5.44	-5.44	0.13
16,789.00	89.43	90.080	10,830.64	-243.95	6,009.04	6,009.04	0.70	0.61	-0.34
16,883.00	90.70	91.920	10,830.53	-245.59	6,103.02	6,103.02	2.38	1.35	1.96
16,914.00	91.11	92.100	10,830.04	-246.68	6,134.00	6,134.00	1.44	1.32	0.58
16,946.00	92.35	92.310	10,829.08	-247.91	6,165.96	6,165.96	3.93	3.88	0.66
16,978.00	91.41	93.400	10,828.03	-249.50	6,197.90	6,197.90	4.50	-2.94	3.41
17,009.00	88.16	93.240	10,828.14	-251.29	6,228.85	6,228.85	10.50	-10.48	-0.52

Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Lewis Federal 5300 11-31 3B
Project:	McKenzie County, ND	TVD Reference:	GL 2110' & KB 25' @ 2135.00ft (Nabors B21)
Site:	Lewis Federal	MD Reference:	GL 2110' & KB 25' @ 2135.00ft (Nabors B21)
Well:	Lewis Federal 5300 11-31 3B	North Reference:	True
Wellbore:	OH	Survey Calculation Method:	Minimum Curvature
Design:	OH	Database:	Casper District

Survey									
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
17,041.00	87.92	92.760	10,829.24	-252.97	6,260.78	6,260.78	1.68	-0.75	-1.50
17,073.00	88.63	93.090	10,830.20	-254.60	6,292.73	6,292.73	2.45	2.22	1.03
17,104.00	87.39	93.630	10,831.28	-256.42	6,323.65	6,323.65	4.36	-4.00	1.74
17,136.00	86.48	92.680	10,832.99	-258.18	6,355.56	6,355.56	4.11	-2.84	-2.97
17,167.00	85.34	93.250	10,835.20	-259.77	6,386.44	6,386.44	4.11	-3.68	1.84
17,199.00	84.02	92.710	10,838.17	-261.43	6,418.26	6,418.26	4.45	-4.13	-1.69
17,262.00	84.60	89.160	10,844.42	-262.45	6,480.93	6,480.93	5.68	0.92	-5.63
17,293.00	86.21	88.420	10,846.90	-261.80	6,511.82	6,511.82	5.71	5.19	-2.39
17,325.00	88.79	88.170	10,848.29	-260.85	6,543.77	6,543.77	8.10	8.06	-0.78
17,356.00	89.20	88.240	10,848.84	-259.88	6,574.75	6,574.75	1.34	1.32	0.23
17,451.00	91.34	88.190	10,848.39	-256.92	6,669.70	6,669.70	2.25	2.25	-0.05
17,545.00	91.14	89.420	10,846.36	-254.96	6,763.66	6,763.66	1.33	-0.21	1.31
17,640.00	90.27	89.590	10,845.19	-254.14	6,858.64	6,858.64	0.93	-0.92	0.18
17,735.00	89.60	88.890	10,845.29	-252.88	6,953.63	6,953.63	1.02	-0.71	-0.74
17,829.00	90.13	88.630	10,845.52	-250.84	7,047.61	7,047.61	0.63	0.56	-0.28
17,924.00	91.10	88.380	10,844.50	-248.37	7,142.57	7,142.57	1.05	1.02	-0.26
17,975.00	89.16	88.760	10,844.38	-247.09	7,193.55	7,193.55	3.88	-3.80	0.75
18,018.00	89.43	88.110	10,844.91	-245.92	7,236.53	7,236.53	1.64	0.63	-1.51
18,112.00	89.90	89.220	10,845.46	-243.73	7,330.51	7,330.51	1.28	0.50	1.18
18,208.00	88.66	90.300	10,846.67	-243.33	7,426.49	7,426.49	1.71	-1.29	1.13
18,302.00	88.42	89.220	10,849.06	-242.93	7,520.46	7,520.46	1.18	-0.26	-1.15
18,397.00	89.06	88.870	10,851.15	-241.35	7,615.42	7,615.42	0.77	0.67	-0.37
18,491.00	89.77	88.380	10,852.11	-239.09	7,709.39	7,709.39	0.92	0.76	-0.52
18,585.00	90.50	88.020	10,851.89	-236.14	7,803.34	7,803.34	0.87	0.78	-0.38
18,680.00	89.43	88.550	10,851.95	-233.30	7,898.30	7,898.30	1.26	-1.13	0.56
18,774.00	89.16	88.890	10,853.10	-231.20	7,992.27	7,992.27	0.46	-0.29	0.36
18,870.00	89.43	89.010	10,854.28	-229.44	8,088.25	8,088.25	0.31	0.28	0.13
18,964.00	88.69	90.130	10,855.83	-228.74	8,182.23	8,182.23	1.43	-0.79	1.19
19,058.00	89.80	94.140	10,857.07	-232.24	8,276.13	8,276.13	4.43	1.18	4.27
19,152.00	90.44	93.700	10,856.87	-238.66	8,369.91	8,369.91	0.83	0.68	-0.47
19,247.00	91.31	93.380	10,855.42	-244.53	8,464.72	8,464.72	0.98	0.92	-0.34
19,341.00	89.56	94.470	10,854.70	-250.96	8,558.49	8,558.49	2.19	-1.86	1.16
19,436.00	89.56	93.580	10,855.43	-257.63	8,653.25	8,653.25	0.94	0.00	-0.94
19,531.00	89.33	92.960	10,856.35	-263.05	8,748.09	8,748.09	0.70	-0.24	-0.65
19,626.00	88.69	91.720	10,858.00	-266.93	8,843.00	8,843.00	1.47	-0.67	-1.31
19,721.00	89.36	91.110	10,859.61	-269.27	8,937.95	8,937.95	0.95	0.71	-0.64
19,816.00	90.23	90.890	10,859.95	-270.93	9,032.94	9,032.94	0.94	0.92	-0.23
19,911.00	90.67	90.060	10,859.21	-271.72	9,127.93	9,127.93	0.99	0.46	-0.87
20,006.00	90.07	90.170	10,858.59	-271.91	9,222.93	9,222.93	0.64	-0.63	0.12
20,101.00	88.93	91.080	10,859.42	-272.94	9,317.92	9,317.92	1.54	-1.20	0.96
20,195.00	89.26	90.710	10,860.91	-274.41	9,411.89	9,411.89	0.53	0.35	-0.39
20,290.00	89.23	89.670	10,862.16	-274.73	9,506.88	9,506.88	1.10	-0.03	-1.09
20,385.00	90.00	88.750	10,862.80	-273.42	9,601.87	9,601.87	1.26	0.81	-0.97

Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Lewis Federal 5300 11-31 3B
Project:	McKenzie County, ND	TVD Reference:	GL 2110' & KB 25' @ 2135.00ft (Nabors B21)
Site:	Lewis Federal	MD Reference:	GL 2110' & KB 25' @ 2135.00ft (Nabors B21)
Well:	Lewis Federal 5300 11-31 3B	North Reference:	True
Wellbore:	OH	Survey Calculation Method:	Minimum Curvature
Design:	OH	Database:	Casper District

Survey										
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)	
20,479.00	91.11	88.620	10,861.89	-271.26	9,695.84	9,695.84	1.19	1.18	-0.14	
20,574.00	91.74	87.720	10,859.52	-268.23	9,790.76	9,790.76	1.16	0.66	-0.95	
20,669.00	90.47	87.800	10,857.69	-264.52	9,885.67	9,885.67	1.34	-1.34	0.08	
20,763.00	89.67	89.900	10,857.58	-262.63	9,979.64	9,979.64	2.39	-0.85	2.23	
20,785.00	87.15	90.310	10,858.19	-262.67	10,001.63	10,001.63	11.61	-11.45	1.86	
Last SDI MWD Survey										
20,860.00	87.15	90.310	10,861.92	-263.07	10,076.54	10,076.54	0.00	0.00	0.00	
Projection to TD										

Design Annotations					
Measured Depth (ft)	Vertical Depth (ft)	Local Coordinates			
		+N/-S (ft)	+E/-W (ft)	Comment	
168.00	168.00	0.44	-0.66	First SDI MWD Survey	
2,743.00	2,742.84	0.56	0.06	Last SDI MWD Surface Survey	
2,800.00	2,799.84	0.56	-0.21	First Gyro Survey	
3,545.00	3,544.82	-0.87	-5.71	Last Gyro Survey	
3,545.00	3,544.82	-0.87	-5.71	TIP to Gyro	
3,641.00	3,640.81	-0.97	-6.41	First SDI MWD Survey (Vertical Section)	
11,053.00	10,789.00	-241.45	278.04	Last SDI MWD Curve Survey	
11,140.00	10,791.98	-254.60	363.90	First SDI MWD Lateral Survey	
20,785.00	10,858.19	-262.67	10,001.63	Last SDI MWD Survey	
20,860.00	10,861.92	-263.07	10,076.54	Projection to TD	

Checked By: _____ Approved By: _____ Date: _____



AUTHORIZATION TO PURCHASE AND TRANSPORT OIL FROM LEASE – FORM 8
INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5698 (03-2000)

Received

NOV 12 2019

ND Oil & Gas Division

Well File No.

30188

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.

PLEASE SUBMIT THE ORIGINAL.

Well Name and Number LEWIS FEDERAL 5300 11-31 3B	Qtr-Qtr LOT1	Section 153	Township 100	Range 0	County McKenzie
Operator Oasis Petroleum North America LLC	Telephone Number (281) 404-9573		Field BAKER		
Address 1001 Fannin, Suite 1500	City Houston			State TX	Zip Code 77002

Name of First Purchaser Oasis Petroleum Marketing LLC	Telephone Number (281) 404-9627	% Purchased 100%	Date Effective August 4, 2019	
Principal Place of Business 1001 Fannin, Suite 1500	City Houston	State TX	Zip Code 77002	
Field Address	City	State	Zip Code	
Names of Transporter Hiland Crude, LLC	Telephone Number (918) 588-5000	% Transported 95%	Date Effective August 4, 2019	
Address 8811 South Yale Avenue, Suite 200	City Tulsa	State OK	Zip Code 74137	
The above named producer authorizes the above named purchaser to purchase the percentage of oil stated above which is produced from the lease designated above until further notice. The oil will be transported by the above named transporter.				

Other First Purchasers Purchasing From This Lease	% Purchased	Date Effective August 4, 2019
Other First Purchasers Purchasing From This Lease	% Purchased	Date Effective August 4, 2019
Other Transporters Transporting From This Lease Power Energy Logistics, LLC	% Transported 5%	Date Effective August 4, 2019
Other Transporters Transporting From This Lease	% Transported	Date Effective August 4, 2019
Comments		

I hereby swear or affirm that all transporters of Bakken Petroleum System oil, listed above implement or adhere to a tariff specification as stringent as the Commission's VPCR ₄ requirement. ☒ <u>13.7</u> VPCR ₄ Tariff Specification <u>DAPL</u> Tariff Authority

I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records.		Date November 5, 2019
Signature 	Printed Name Claudia Arguelles	Title Contracts Administrator
Above Signature Witnessed By:		
Signature 	Printed Name Kenzie Buchanan	Witness Title Scheduler

FOR STATE USE ONLY

Date Approved NOV 15 2019	NDIC CTB NO. 228190
By 	
Title Oil & Gas Production Analyst	



WELL COMPLETION OR RECOMPLETION REPORT - FORM 6

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 2468 (04-2010)



Well File No.
30188

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

Designate Type of Completion			
☒ Oil Well	☐ EOR Well	☐ Recompletion	☐ Deepened Well
☐ Gas Well	☐ SWD Well	☐ Water Supply Well	☐ Other:
Well Name and Number Lewis Federal 5300 11-31 3B		Spacing Unit Description Sec. 31/32 T153N R100W	
Operator Oasis Petroleum North America		Telephone Number (281) 404-9591	Field Baker
Address 1001 Fannin, Suite 1500		Pool Bakken	
City Houston	State TX	Zip Code 77002	Permit Type ☐ Wildcat ☒ Development ☐ Extension

LOCATION OF WELL

At Surface 1083 F N L	262 F WL	Qtr-Qtr Lot 1	Section 31	Township 153 N	Range 100 W	County McKenzie
Spud Date January 3, 2019	Date TD Reached January 22, 2019	Drilling Contractor and Rig Number Nabors B21		KB Elevation (Ft) 2135	Graded Elevation (Ft) 2110	
Type of Electric and Other Logs Run (See Instructions) MWD/GR from KOP to TD; CBL from int. TD to surface						

CASING & TUBULARS RECORD (Report all strings set in well)

Well Bore	String Type	Size (Inch)	Top Set (MD Ft)	Depth Set (MD Ft)	Hole Size (Inch)	Weight (Lbs/Ft)	Anchor Set (MD Ft)	Packer Set (MD Ft)	Sacks Cement	Top of Cement
Surface Hole	Surface	13 3/8	0	3605	17 1/2	54.5			2070	0
Vertical Hole	Intermediate	9 5/8	0	6076	13 1/2	36			1540	1999
Vertical Hole	Intermediate	7	0	11116	8 3/4	32			743	
Lateral1	Liner	4 1/2	10112	20855	6	13.5			490	

PERFORATION & OPEN HOLE INTERVALS

Well Bore	Well Bore TD Drillers Depth (MD Ft)	Completion Type	Open Hole/Perforated Interval (MD Ft)		Kick-off Point (MD Ft)	Top of Casing Window (MD Ft)	Date Per'd or Drilled	Date Isolated	Isolation Method	Sacks Cement
Lateral1	20860	Perforations	11184	20797	10190		06/20/2019			

PRODUCTION

Current Producing Open Hole or Perforated Interval(s), This Completion, Top and Bottom, (MD Ft) Lateral 1- 11184' to 20797'							Name of Zone (If Different from Pool Name)		
Date Well Completed (SEE INSTRUCTIONS) August 4, 2019			Producing Method flowing		Pumping-Size & Type of Pump		Well Status (Producing or Shut-In) producing		
Date of Test 08/09/2019	Hours Tested 24	Choke Size 18 /64	Production for Test		Oil (Bbls) 1569	Gas (MCF) 1979	Water (Bbls) 1141	Oil Gravity-API (Corr.) °	Disposition of Gas Sold
Flowing Tubing Pressure (PSI)		Flowing Casing Pressure (PSI) 1600		Calculated 24-Hour Rate	Oil (Bbls) 1569	Gas (MCF) 1979	Water (Bbls) 1141	Gas-Oil Ratio 1261	

GEOLOGICAL MARKERS

[illegible]

PLUG BACK INFORMATION

[illegible]

CORES CUT

Top (Ft)	Bottom (Ft)	Formation	Top (Ft)	Bottom (Ft)	Formation

Drill Stem Test

Drill Stem Test								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								

Well Specific Stimulations

Date Stimulated 06/20/2019	Stimulated Formation Bakken	Top (Ft) 11184	Bottom (Ft) 20797	Stimulation Stages 40	Volume 329963	Volume Units Barrels
Type Treatment Sand Frac	Acid %	Lbs Proppant 6006140	Maximum Treatment Pressure (PSI) 9015		Maximum Treatment Rate (BBLS/Min) 73.0	
Details 100 Mesh: 3588450 40/70 White: 1511380 40/70 CRC: 906310						
Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						
Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						
Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						
Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

ADDITIONAL INFORMATION AND/OR LIST OF ATTACHMENTS

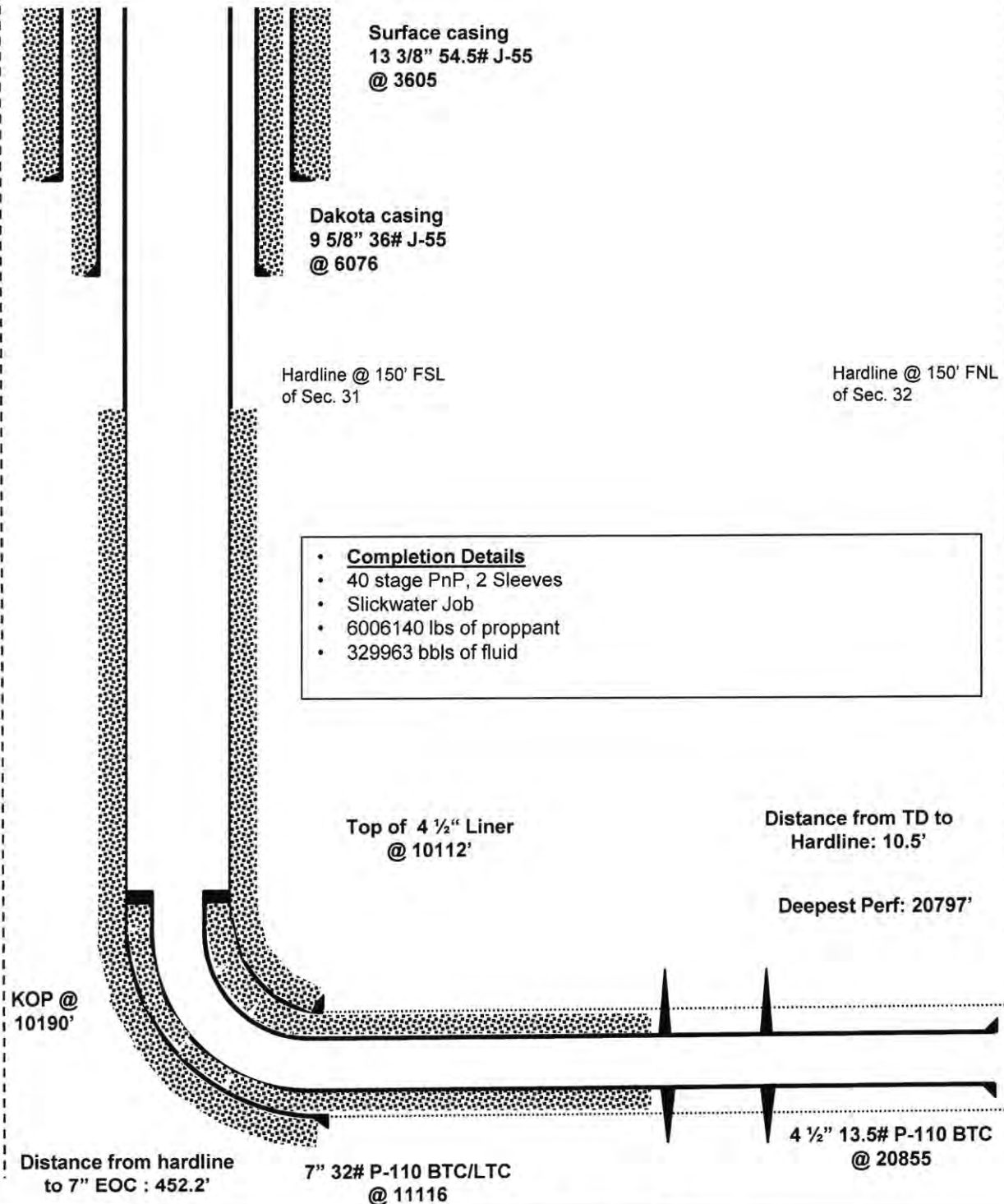
This report provides frac data.

I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records.	Email Address jswenson@oasispetroleum.com	Date 09/13/2019
	Signature 	Printed Name Jennifer Swenson

ELEVATION: 2110' GL

**Lewis Federal 5300 11-31 3B
Wellbore Schematic**

FORMATION: Bakken



OASIS PETROLEUM NA LLC

Lewis Federal 5300 11-31 3B

Wellbore: T153N-R100W Sec. 31 & 32

SHL: 1083' FNL & 262' FEL T153N-R100W Sec. 31

McKenzie County, North Dakota

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

Received

SEP 16 2019

Well File No.
30188

**ND Oil & Gas
Division**

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☐ Notice of Intent	Approximate Start Date	☐ Drilling Prognosis	☐ Spill Report
☒ Report of Work Done	Date Work Completed August 4, 2019	☐ Redrilling or Repair	☐ Shooting
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date	☐ Casing or Liner	☐ Acidizing
		☐ Plug Well	☐ Fracture Treatment
		☒ Supplemental History	☐ Change Production Method
		☐ Temporarily Abandon	☐ Reclamation
		☒ Other	Change well status to CONFIDENTIAL

Well Name and Number Lewis Federal 5300 11-31 3B							
Footages		Qtr-Qtr	Section	Township	Range		
1083 F N L 262 F W L		LOT1	31	153 N	100 W		
Field Baker		Pool Bakken		County McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Effective immediately, we request **CONFIDENTIAL STATUS** for the above referenced well.

Date of First Production August 4, 2019.

off confidential 2/4/20

Company Oasis Petroleum North America LLC		Telephone Number 713-770- 6570	
Address 1001 Fannin, Suite 1500			
City Houston		State TX	Zip Code 77002
Signature <i>Jasmine Crawford</i>		Printed Name Jasmine Crawford	
Title Regulatory Specialist		Date September 13, 2019	
Email Address jcrawford@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date 9/17/19	
By <i>Arthur Orselle</i>	
Title Petroleum Resource Specialist	

Industrial Commission of North Dakota
Oil and Gas Division
Verbal Approval To Purchase and Transport Oil

Well or Facility No
30188

Tight Hole **No**

OPERATOR

Operator	Representative	Rep Phone
OASIS PETROLEUM NORTH AMERICA LL	Mike Haase	(701) 570-6752

WELL INFORMATION

Well Name					Inspector
LEWIS FEDERAL 5300 11-31 3B					Richard Dunn
Well Location	QQ	Sec	Twp	Rng	County
	LOT1	31	153 N	100 W	MCKENZIE
Footages	1083	Feet From the N Line			Field
	262	Feet From the W Line			BAKER
					Pool
					BAKKEN
Date of First Production Through Permanent Wellhead				8/4/2019	This Is Not The First Sales

PURCHASER / TRANSPORTER

Purchaser	Transporter
OASIS PETROLEUM MARKETING LLC	HILAND CRUDE, LLC

TANK BATTERY

Single Well Tank Battery Number : 130188-01

SALES INFORMATION This Is Not The First Sales

ESTIMATED BARRELS TO BE SOLD		ACTUAL BARRELS SOLD		DATE
15000	BBLS	540	BBLS	8/7/2019
	BBLS		BBLS	
	BBLS		BBLS	
	BBLS		BBLS	
	BBLS		BBLS	
	BBLS		BBLS	
	BBLS		BBLS	
	BBLS		BBLS	
	BBLS		BBLS	

DETAILS

Must also forward Forms 6 & 8 to State prior to reaching 15000 Bbl estimate or no later than required time frame for submitting those forms.
--

Start Date	8/4/2019
Date Approved	8/22/2019
Approved By	Richard Dunn

**WELL COMPLETION OR RECOMPLETION REPORT - FORM 6**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 2468 (04-2010)



Well File No.
30188

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

Designate Type of Completion					
☒ Oil Well	☐ EOR Well	☐ Recompletion	☐ Deepened Well	☐ Added Horizontal Leg	☐ Extended Horizontal Leg
☐ Gas Well	☐ SWD Well	☐ Water Supply Well	☐ Other:		
Well Name and Number Lewis Federal 5300 11-31 3B				Spacing Unit Description Sec. 31/32 T153N R100W	
Operator Oasis Petroleum North America		Telephone Number (281) 404-9591		Field Baker	
Address 1001 Fannin, Suite 1500				Pool Bakken	
City Houston	State TX	Zip Code 77002		Permit Type ☐ Wildcat ☒ Development ☐ Extension	

LOCATION OF WELL

At Surface 1083 F N L	262 F W L	Qtr-Qtr Lot 1	Section 31	Township 153 N	Range 100 W	County McKenzie
Spud Date January 3, 2019	Date TD Reached January 22, 2019	Drilling Contractor and Rig Number Nabors B21		KB Elevation (Ft) 2135	Graded Elevation (Ft) 2110	
Type of Electric and Other Logs Run (See Instructions) MWD/GR from KOP to TD; CBL from int. TD to surface						

CASING & TUBULARS RECORD (Report all strings set in well)

Well Bore	String Type	String Size (Inch)	Top Set (MD Ft)	Depth Set (MD Ft)	Hole Size (Inch)	Weight (Lbs/Ft)	Anchor Set (MD Ft)	Packer Set (MD Ft)	Sacks Cement	Top of Cement
Surface Hole	Surface	13 3/8	0	3609	17 1/2	54.5			2070	
Vertical Hole	Intermediate	9 5/8	0	6075	13 1/2	36			1540	
Vertical Hole	Intermediate	7	0	11114	8 3/4	32			743	
Lateral1	Liner	4 1/2	10112	20855	6	13.5			490	

PERFORATION & OPEN HOLE INTERVALS

Well Bore	Well Bore TD Drillers Depth (MD Ft)	Completion Type	Open Hole/Perforated Interval (MD,Ft) Top Bottom	Kick-off Point (MD Ft)	Top of Casing Window (MD Ft)	Date Perf'd or Drilled	Date Isolated	Isolation Method	Sacks Cement
Lateral1	20860	Perforations		10190					

PRODUCTION

Current Producing Open Hole or Perforated Interval(s), This Completion, Top and Bottom, (MD Ft) Lateral 1-						Name of Zone (If Different from Pool Name)			
Date Well Completed (SEE INSTRUCTIONS)			Producing Method	Pumping-Size & Type of Pump			Well Status (Producing or Shut-In)		
Date of Test	Hours Tested	Choke Size /64	Production for Test	Oil (Bbls)	Gas (MCF)	Water (Bbls)	Oil Gravity-API (Corr.)	Disposition of Gas	
Flowing Tubing Pressure (PSI)		Flowing Casing Pressure (PSI)		Calculated 24-Hour Rate	Oil (Bbls)	Gas (MCF)	Water (Bbls)	Gas-Oil Ratio	

GEOLOGICAL MARKERS

[illegible]

PLUG BACK INFORMATION

[illegible]

CORES CUT

Top (Ft)	Bottom (Ft)	Formation	Top (Ft)	Bottom (Ft)	Formation

Drill Stem Test

Drill Stem Test								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								

Well Specific Stimulations

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units Barrels
Type Treatment Sand Frac	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

ADDITIONAL INFORMATION AND/OR LIST OF ATTACHMENTS

This is a preliminary completion report. A supplemental report will be filed upon first production of the well.

I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records.	Email Address jswenson@oasispetroleum.com		Date 02/20/2019
	Signature 	Printed Name Jennifer Swenson	Title Regulatory Specialist



Oasis Petroleum North America, LLC

Lewis Federal 5300 11-31 3B

1,083' FNL & 262' FWL

Lot 1 Section 31, T153N, R100W

Baker Field / Middle Bakken

McKenzie County, North Dakota

BOTTOM HOLE LOCATION:

263.07' south & 10,076.54' east of surface or approx.

1,346.07' FNL & 184.20' FEL; Lot 4 Sec. 32, T153N, R100W

Prepared for:

Shea Cook
Oasis Petroleum North America, LLC
1001 Fannin Suite 1500
Houston, TX 77002

Prepared by:

Dillon Johnson, Michelle Baker
PO Box 80507; Billings, MT 59108
(406) 259-4124
geology@sunburstconsulting.com
www.sunburstconsulting.com

Oasis Petroleum North America, LLC

Lewis Federal 5300 11-31 3B

Well Evaluation

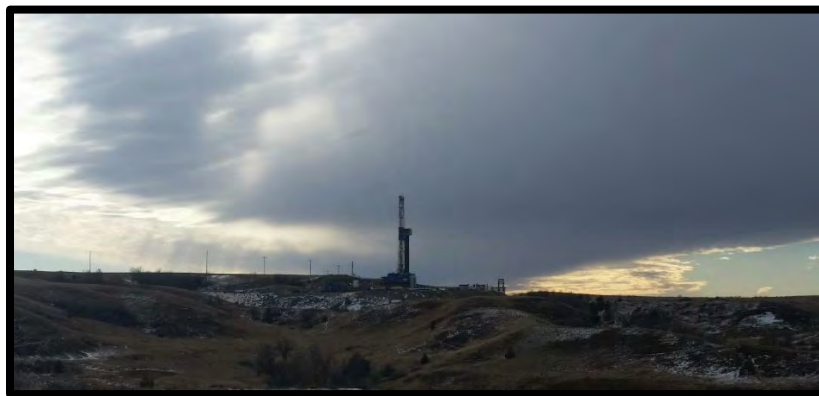


Figure 1. Nabors drilling rig #B21 at the Oasis Petroleum NA, LLC Lewis Federal 5300 11-31 3B; January 2019, McKenzie County, North Dakota (All photos by Dillon Johnson unless noted otherwise).

Introduction

The Oasis Petroleum NA, LLC Lewis Federal 5300 11-31 3B is located in Baker Field of the Williston Basin [Lot 1 Section 31, T153N, R100W]. The subject well lies approximately 8 miles south of the town of Williston, in the Indian Hills prospect of McKenzie County, North Dakota (**Figure 1**). The Lewis Federal 5300 11-31 3B is the second of three wells to be drilled on the Lewis Federal 5300 11-31 pad. Due to tight hole conditions the Lewis Federal 5300 11-31 4B was plugged and abandoned. This pad is set up as a 1,280-acre laydown spacing unit, with 500' N/S and 100' E/W drilling setbacks. The subject well is permitted to drill east from the surface location in Section 31 into Section 32. The well consists of a single Middle Bakken Member lateral, targeting a silty sandstone facies, with intent to intersect porosity and fracture trends enhancing reservoir quality.

Engineering Operations Overview

The Lewis Federal 5300 11-31 3B was spud on January 3, 2019. The surface hole was drilled with one 17.5" assembly to a depth of 3,605'. The complete BHA accomplishments can be found in an appendix to this report. The 13 3/8" surface casing was set to a depth of 3,600'. Due to the presence of several salt water disposal wells in the immediate area the decision was made to set a 9 5/8" isolation casing string through the Inyan Kara and into the Swift. The isolation portion was drilled without issue to a depth of 6,090'. The 9 5/8" casing was then set to a depth of 6,075'. The remainder of the vertical hole was completed with two 8.75" assemblies. The first vertical assembly drilled to a depth of 8,427' before being replaced due to low ROP, the second vertical assembly drilled to a depth of 10,190' (KOP). The curve assembly consisted of two 8.75" assemblies. The first assembly, a Reed TKC56 PDC bit (#5), attached to a 2.38° NOV mud motor and Scientific Drilling MWD tools drilled to a depth of 11,039' before a trip was required due to low build rates. The second curve BHA consisted of a Security F30T tri-cone bit, a 2.50° NOV mud motor and Scientific drilling MWD tools. The curve was successfully landed at 11,132' MD and 10,790' TVD, approximately 32' below the Upper Bakken Shale on 15 January 2019. Seven inch diameter 32# P-110 intermediate casing was set to 11,116' MD at landing. The lateral was completed using two 6" assemblies. The first lateral assembly drilled to a depth of 16,675' before a trip was required due to an MWD failure. The second lateral assembly drilled to a depth of 17,009' before another MWD failure occurred. Due to the short footage between the first and second lateral trips the bit and motor were not replaced after the second trip. The lateral reached a total depth of 20,860' on January 22, 2019.

Offset Control

Offset well data can be found in the 'Control Data' section appended to this report. Offset wells were essential in providing control, making it possible to develop a prognosis of formation tops and curve landing target depth. The three primary offsets were, the *Oasis Wade Federal 5300 41-30 8T3*, the *Oasis Lewis Federal 5300 31-31H*, and the *Oasis Gramma Federal 5300 41-31T*. By referencing the gamma signature of these offsets and using formation thicknesses, a model was formed for the target interval pinpointing a strategic landing. Formation thicknesses expressed by gamma ray signatures in these offset wells were compared to gamma data collected during drilling operations in to successfully land the curve.

Geology

Sample evaluation began in the Otter Formation at 8,200' measured depth (MD). Lagged samples were caught by Sunburst personnel in 30' intervals through the vertical and curve, and 50' intervals in the lateral. Rock samples were evaluated under wet and dry conditions using a stereo zoom binocular microscope for the identification of lithology including the presence of porosity and oil. Only observed prospective intervals are described here, but detailed lithological descriptions for all formations are provided in the 'Lithology' appendix.

The **Mission Canyon Formation** [Mississippian, Madison Group] was logged at 9,455' MD, 9,446' TVD (-7,311' MSL). The Mission Canyon Formation is described as cream, tan, light gray, and light brown gray in color. Samples are predominately microcrystalline and are a firm lime mudstone. The limestone has an earthy, rarely crystalline texture. Also noted in several samples are rare fossil fragments. The limestone is argillaceous in part throughout this interval. In certain areas possible intercrystalline porosity was noted but there was no significant oil staining observed in samples. Throughout the Mission Canyon gas shows are promising, with an average background gasses ~120u with gas shows peaking at 316u.



Figure 2. Wet sample cuttings of limestone from the Mission Canyon.

The Bakken Formation

The Upper Bakken Shale Member [Mississippian] was recorded at 10,897' MD, 10,745' TVD (-8,610' MSL). Entry into this member is characterized by high gamma counts (>300 API), elevated background gas and increased rates of penetration. While drilling through the Upper Bakken Shale, a background gas of 880u was observed, as well as a survey gas of 1917u. The distinct black shale is carbonaceous and *petroliferous*, as well as hard and platy. Minerals including disseminated/nodular pyrite and trace calcite fracture fill was observed.

The Middle Bakken Member [Mississippian-Devonian] was entered at 10,931' MD, 10,758' TVD (-8,623' MSL). Samples in the Middle Bakken are predominantly a light to medium gray, light brown, and occasionally cream silty sandstone. The silty sandstone is fine to very fine grained, friable to occasionally firm. The Middle Bakken typically contained sub-round to sub-angular grains. Samples are smooth, moderately sorted and poorly cemented by calcite. Rare to trace quantities of disseminated and nodular pyrite is present as was *trace to fair intergranular porosity*. Trace to rare *light-medium brown, spotty oil stain* was visible in many of these samples. While drilling the Middle Bakken background gasses ranged from ~300 to 4500 units while several shows exceeded 5000u. After the trip the trip at 16,675' gas was vented through the gas buster. This venting resulted in a constant 1'-8' flare until reaching TD.



Figure 3. Wet sample cuttings of silty sandstone from the Middle Bakken.

Geosteering

Structure maps provided by Oasis projected that the structure would be a consistent down dip averaging 0.55° down over the course of the lateral. The steering team anticipated the structure to be slightly steeper (-0.8°) in the first 3,000' and soften to a -0.4° for the remainder of the lateral. Due to the low build rates through the curve, the wellbore landed very low in the Middle Bakken and as a result the gamma signatures throughout the Middle Member were cataloged. Upon reaching casing point it became apparent that the gamma signature was very similar to that of the Lewis Federal 5300 31-31H. Since this offset well had the thinnest Middle Bakken (~36') of the three offsets, it was determined to steer ahead with an anticipated 36' Middle Bakken, and not a 40' interval that was previously anticipated. The target interval was defined by proximity to the shales and not by an ideal porosity interval. The 16' target began 10' below the Upper Bakken Shale and extended to 10' above the Lower Bakken Shale. Before drilling out of the 7" casing, it was determined that the cool C marker in the lower half of the target interval was going to be the most consistent steering guide (**Figure 4**). Due to the wellbore remaining in the upper portion of the target interval for a majority of the lateral, the cool C marker was rarely contacted. Over the course of the lateral the warm B marker and a slightly cool marker near the top of the target interval proved to be the most beneficial. After the trip at 16,675' the wellbore began to build inclination on rotation faster than anticipated and drilled within ~3' of the Upper Bakken Shale. To avoid contact with the Upper Member a stop card was pulled, and



Figure 4. Target definition for the Oasis Petroleum, Lewis Federal 5300 11-31 3B.

time drilling was utilized. Regardless of position in zone the assembly consistently built inclination on rotation. There did not appear to be any noticeable hard streaks or intervals that were more or less favorable in relation to ROP.

The Lewis Federal 5300 11-31 3B had an estimated overall formation dip of approximately -0.53° . Penetration rates, gas shows, gamma ray data, and sample observations were utilized to keep the wellbore in the preferred stratigraphic position within the target zone. Using offset well data provided by Oasis representatives, projected porosity zones were identified and used as preferred drilling areas.

The lateral was drilled in less than 5 days from casing exit to total depth, with two lateral assembly. A total depth of 20,860' MD was achieved at 05:30 hours CST on January 22, 2019. The wellbore was completed 84% within target, opening 9,744' (measurement taken from uncased lateral portion) of potentially productive reservoir rock.

Hydrocarbon Shows

Gas was continuously recorded from 103' to the completion of the lateral, along with the monitoring of free oil at the possum belly and shakers. In the closed mud system, hydrostatic conditions were maintained near balance, this allowed gas and fluid gains from the well to be evaluated. During the vertical, gas shows ranging from 10 to 316 units were noted, against a 10.25 to 12.85 pound per gallon (ppg) diesel-invert, mud weight. Background concentrations in the lateral ranged from 300 to 4500 units, against a 9.6-9.8 ppg saltwater gel drilling fluid (**Figure 6**). Due to the use of the gas buster from the first trip at 16,675' to 20,860' (TD) gas shows were significantly higher prior to 16,675'. There were two trips during the lateral. These trips resulted in trip gasses of 7438u and 2728u. Upon circulating a bottoms up after the trip, 20'-30' flares were observed. Chromatography of gas revealed typical concentrations of methane, ethane and propane characteristic of the Middle Bakken (**Figure 5**). Sample cuttings were examined for hydrocarbon "cut" by immersion of trichloroethylene and inspection under a UV fluoroscope. *Fluorescent cuts were generally pale yellow in color and had a diffuse habit at a slow to moderate speed.*

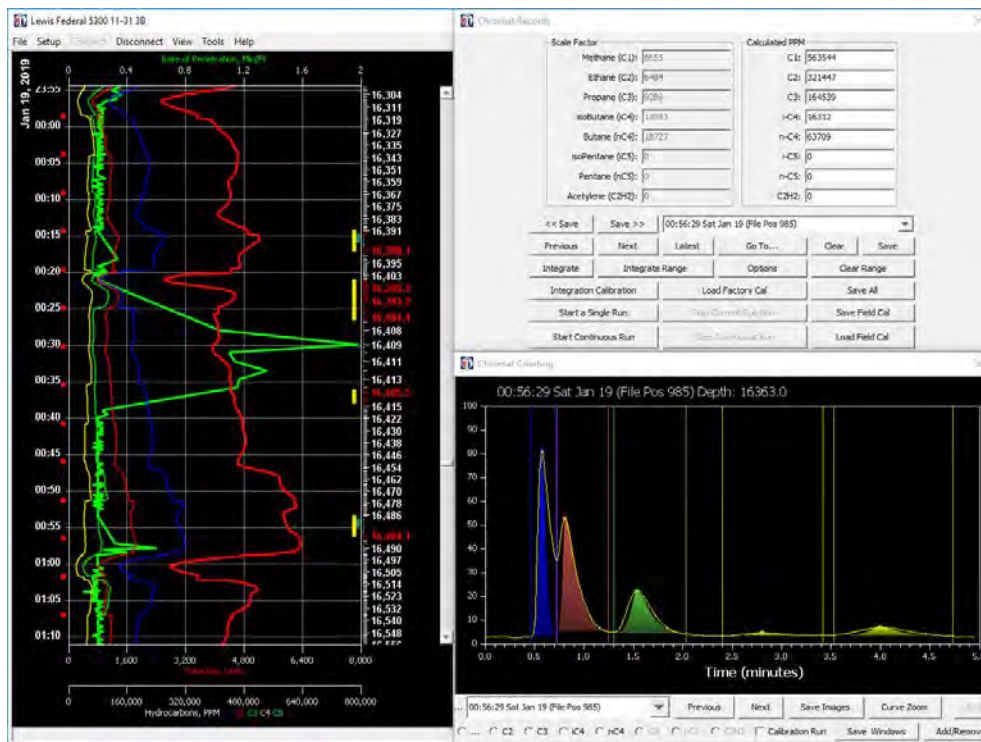


Figure 5. Screen shot of gas chromatography portraying total gas values and C₁-C₄ values, shown in parts per million.

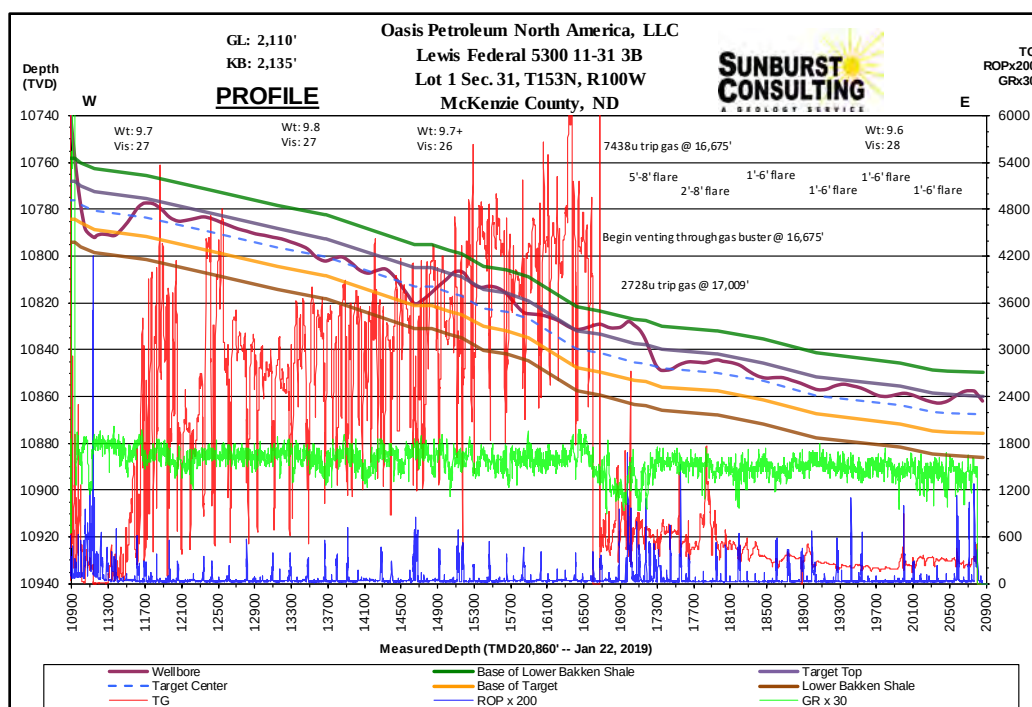


Figure 6. Cross-sectional profile of the Lewis Federal 5300 11-31 3B displaying stratigraphic position, total gas, rate of penetration and gamma values.

Summary

The *Lewis Federal 5300 11-31 3B* is a well in Oasis Petroleum's horizontal Middle Bakken Member development program, in the Indian Hill prospect of McKenzie County, North Dakota. The project was drilled from surface casing to total depth in 19 days. A total depth of 20,860' MD was achieved at 05:30 hours CST on 22 January 2019. The well-site team worked together to maintain the wellbore in the desired target interval for 84% within target, opening 9,744' of potentially productive reservoir rock.

Samples in the Middle Bakken Member are predominantly light to medium gray, light brown and occasionally cream silty sandstone. The silty sandstone is composed of fine to very fine grained, sub-rounded and occasionally sub-angular, moderately sorted quartz grains. Primary cement is calcite and rare quantities of disseminated and nodular pyrite are present. Poor to fair intergranular porosity contains trace to rare light-medium brown, spotty oil stain. The overall hydrocarbon "cuts", gas and hydrocarbon shows were encouraging and indicate an oil and gas rich system in the Middle Bakken Member.

The well should be regarded as an engineering and geological success based on the combination of:

- Maximum exposure to the target
- Minimal days from re-entry to total depth
- No side-tracks, shale strikes, or collision with existing wellbores

The *Oasis Petroleum North America, LLC, Lewis Federal 5300 11-31 3B* awaits completion operations to determine its ultimate production potential and commercial value.

Respectfully,
Dillon Johnson
 Lead Well Site Geologist & Geosteering Consultant
 Sunburst Consulting, Inc.
 22 January 2019

WELL DATA SUMMARY

<u>OPERATOR:</u>	Oasis Petroleum North America, LLC
<u>ADDRESS:</u>	1001 Fannin Suite 1500 Houston, TX 77002
<u>WELL NAME:</u>	Lewis Federal 5300 11-31 3B
<u>API #:</u>	33-053-06548
<u>WELL FILE #:</u>	30188
<u>SURFACE LOCATION:</u>	1,083' FNL & 262' FWL Lot 1 Section 31, T153N, R100W
<u>FIELD/ OBJECTIVE:</u>	Baker Field / Middle Bakken
<u>COUNTY, STATE:</u>	McKenzie County, North Dakota
<u>RESERVATION:</u>	N/A
<u>BASIN:</u>	Williston Basin
<u>WELL TYPE:</u>	Horizontal Development
<u>ELEVATION:</u>	GL: 2,110' KB: 2,135'
<u>SPUD DATE:</u>	January 3, 2019
<u>BOTTOM HOLE LOCATION:</u>	263.07' south & 10,076.54' east of surface or approx. 1,346.07' FNL & 184.20' FEL; Lot 4 Sec. 32, T153N, R100W
<u>CLOSURE COORDINATES:</u>	Closure Azimuth: 91.50° Closure Distance: 10,079.97'
<u>TOTAL DEPTH / DATE:</u>	20,860' on January, 2019 84% within target interval
<u>TOTAL DRILLING DAYS:</u>	19 days
<u>PUMP INFO:</u>	Stroke length - 12" Liner I.D. - 6" for surface; 5.0" for vertical, curve and lateral

COMPANY MEN: Doug Rakstad, Rudy Salivar, Ian Anderson, B. Houge, M. Ziegler

COMPANY GEOLOGIST: Shea Cook

WELLSITE GEOLOGISTS: Dillon Johnson, Michelle Baker

ROCK SAMPLING: 30' from 8,200' - 11,110'
50' from 11,150' - 20,800'

SAMPLE CUTS: Trichloroethylene

GAS DETECTION: Terra SLS, Inc. TGC - total gas w/ chromatograph
Serial Number(s): ML-466

DIRECTIONAL DRILLERS: RPM Consulting, Pat's Consulting
Christopher Bohn, Patrick Bidegaray, Mike Crow, Willem Zylstra

MWD: Scientific Drilling
James Swartz, Steve Gray

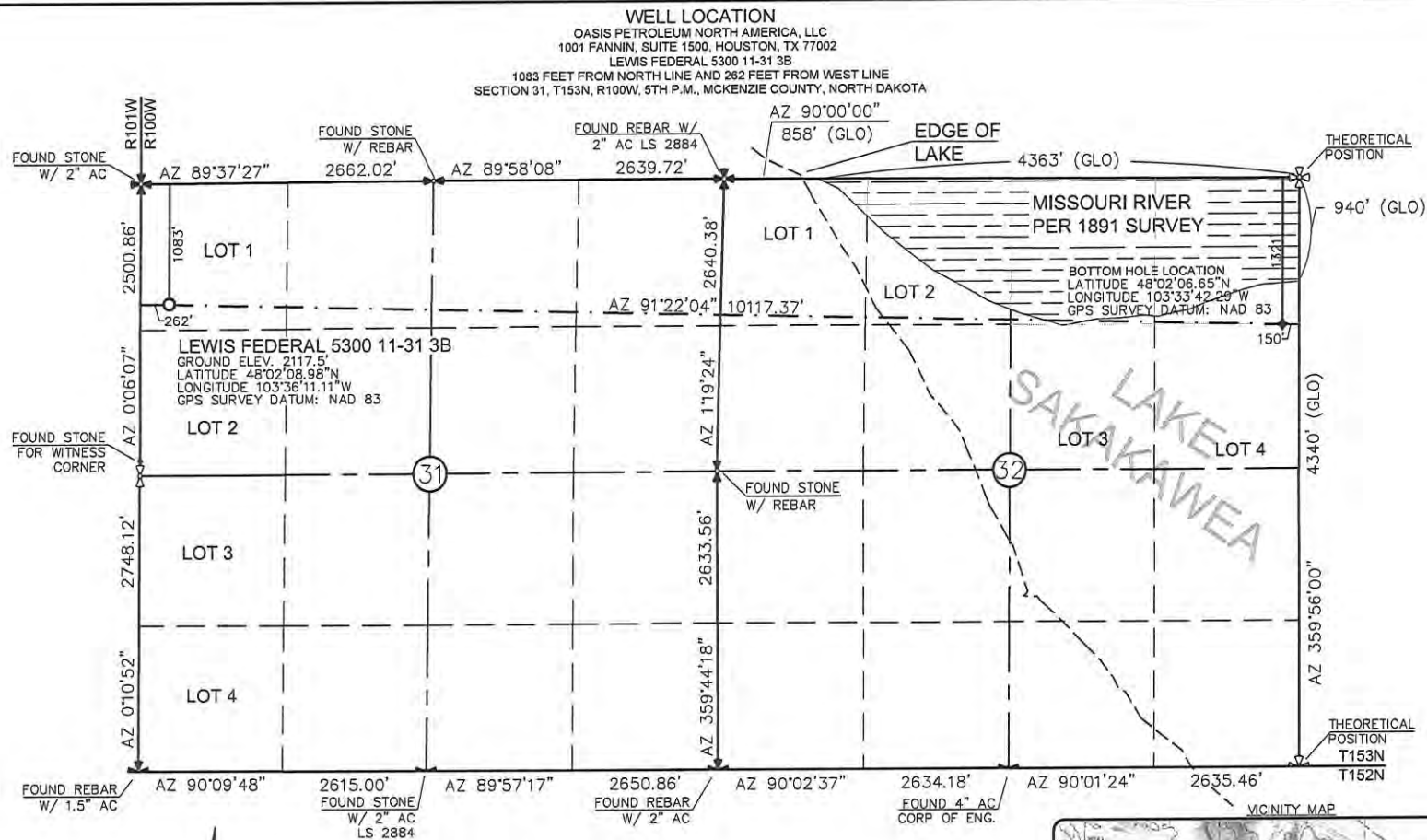
CASING: Surface: 13 3/8" 54.5# J-55 set to 3,600'
Isolation: 9 5/8" 36# J-55 set to 6,075'
Intermediate: 7" 32# set to 11,116'

KEY OFFSET WELLS:

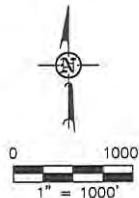
Oasis Petroleum North America, LLC
Wade Federal 5300 41-30 8T2
Lot 4 Sec. 30, T153N, R100W
McKenzie County, ND
NDIC: 28558
KB: 2,077'

Oasis Petroleum North America, LLC
Lewis Federal 5300 31-31H
Lot 3 Sec. 31, T153N, R100W
McKenzie County, ND
NDIC: 20314
KB: 2,185'

Oasis Petroleum North America, LLC
Gamma Federal 5300 41-31T
SW SW Sec. 31, T153N, R100W
McKenzie County, ND
NDIC: 23350
KB: 2,182'



THIS DOCUMENT WAS ORIGINALLY
 ISSUED AND SEALED BY DARYL D.
 KASEMAN, PLS, REGISTRATION
 NUMBER 3880 ON 3-27-18
 AND THE ORIGINAL DOCUMENTS
 ARE STORED AT THE OFFICES OF
 INTERSTATE ENGINEERING, INC.

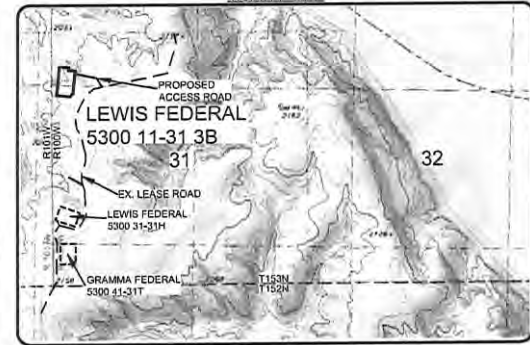
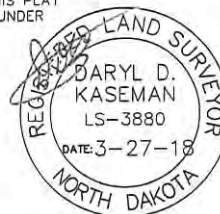


- MONUMENT - RECOVERED
 - MONUMENT - NOT RECOVERED

STAKED ON 3/26/18
 VERTICAL CONTROL DATUM WAS BASED UPON
 CONTROL POINT 705 WITH AN ELEVATION OF 2158.3'

THIS SURVEY AND PLAT IS BEING PROVIDED AT THE REQUEST
 OF JOHN LEE OF OASIS PETROLEUM. I CERTIFY THAT THIS PLAT
 CORRECTLY REPRESENTS WORK PERFORMED BY ME OR UNDER
 MY SUPERVISION AND IS TRUE AND CORRECT TO
 THE BEST OF MY KNOWLEDGE AND BELIEF.

[Signature]
 DARYL D. KASEMAN LS-3880



© 2018, INTERSTATE ENGINEERING, INC.

Revision	No.	Date	By	Description

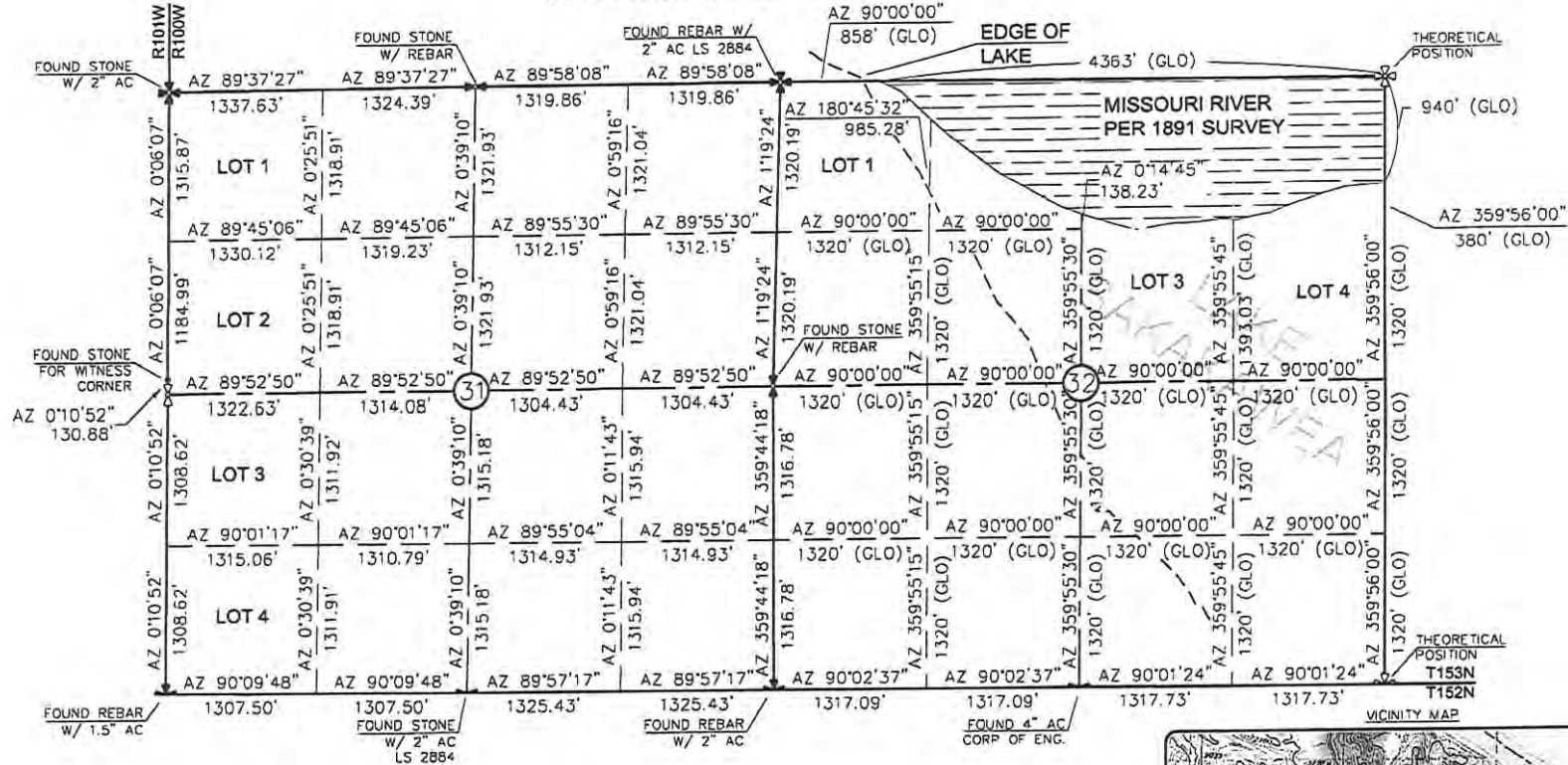
OASIS PETROLEUM NORTH AMERICA, LLC WELL LOCATION SECTION 31, T153N, R100W, 5TH P.M., MCKENZIE COUNTY, NORTH DAKOTA	Project No.: 181528183 Date: 03/28/2018
Drawn By: JDM Checked By: JDM	

Interstate Engineering, Inc.
 P.O. Box 648
 425 East Main Street
 Sully, ND 58570
 Phone: (701) 433-5876
 Fax: (701) 433-5818
 www.interstateeng.com
 Other offices in Minnesota, North Dakota and South Dakota

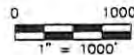


1/9
 SHEET NO.

SECTION BREAKDOWN
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "LEWIS FEDERAL 5300 11-31 3B"
 1005 FEET FROM NORTH LINE AND 233 FEET FROM WEST LINE
 SECTION 31, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

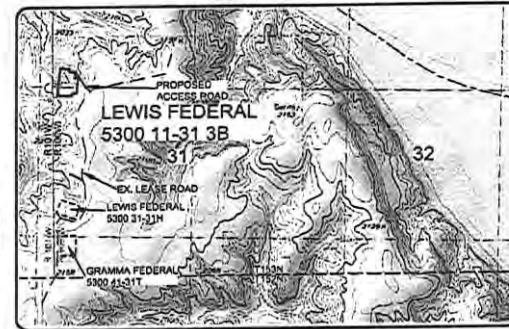


THIS DOCUMENT WAS ORIGINALLY
 ISSUED AND SEALED BY DARYL D.
 KASEMAN, PLS. REGISTRATION NUMBER
 3880 ON 1/21/14. AND THE
 ORIGINAL DOCUMENTS ARE STORED AT
 THE OFFICES OF INTERSTATE
 ENGINEERING, INC.



- ✕ - MONUMENT - RECOVERED
- ✕ - MONUMENT - NOT RECOVERED

ALL AZIMUTHS ARE BASED ON C.P.S.
 OBSERVATIONS. THE ORIGINAL SURVEY OF THIS
 AREA FOR THE GENERAL LAND OFFICE (G.L.O.)
 WAS 1897. THE CORNERS FOUND ARE AS
 INDICATED AND ALL OTHERS ARE COMPUTED FROM
 THOSE CORNERS FOUND AND BASED ON G.L.O.
 DATA. THE MAPPING ANGLE FOR THIS AREA IS
 APPROXIMATELY 0°03'.



© 2014, INTERSTATE ENGINEERING, INC.

Author	By	Date	Description

OASIS PETROLEUM NORTH AMERICA, LLC SECTION BREAKDOWN SECTIONS 31 & 32, T153N, R100W MCKENZIE COUNTY, NORTH DAKOTA	Project No.: 3152-2143 Drawn By: B.S.C. Checked By: J.D.K.
--	--

Interstate Engineering, Inc. P.O. Box 648 4325 Highway 10 Sifters, Montana 59270 Ph: (406) 432-5617 Fax: (406) 432-5618 www.interstate-engineering.com	2014-01-21 2014-01-21
--	--------------------------



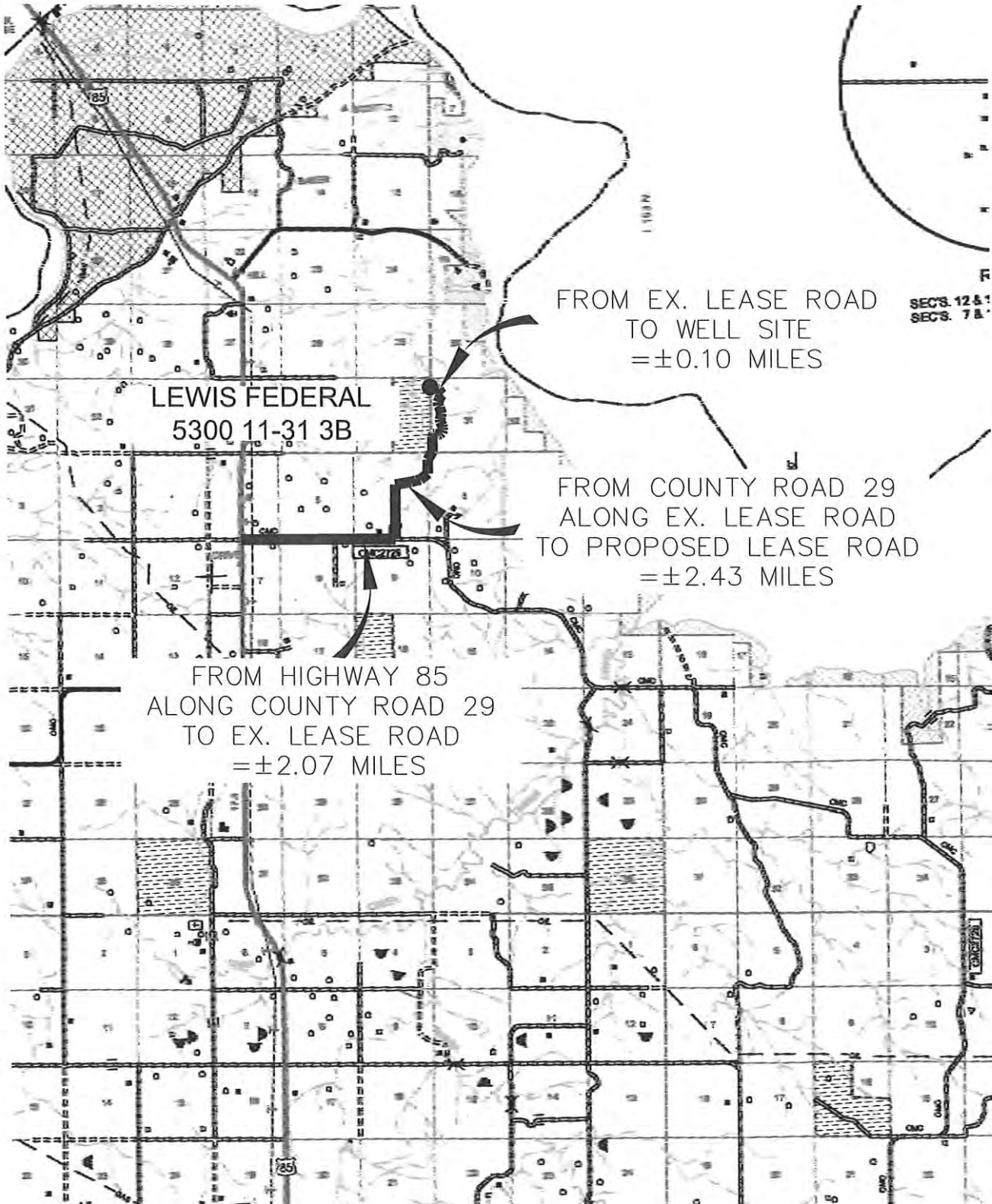
2/8
 SHEET NO.

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
LEWIS FEDERAL 5300 11-31 3B
1083 FEET FROM NORTH LINE AND 262 FEET FROM WEST LINE
SECTION 31, T153N, R100W, 5TH P.M., MCKENZIE COUNTY, NORTH DAKOTA

COUNTY ROAD MAP

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"LEWIS FEDERAL 5300 11-31 3B"

1005 FEET FROM NORTH LINE AND 233 FEET FROM WEST LINE
SECTION 31, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



© 2014, INTERSTATE ENGINEERING, INC.

SCALE: 1" = 2 MILE

7/9
SHEET NO.



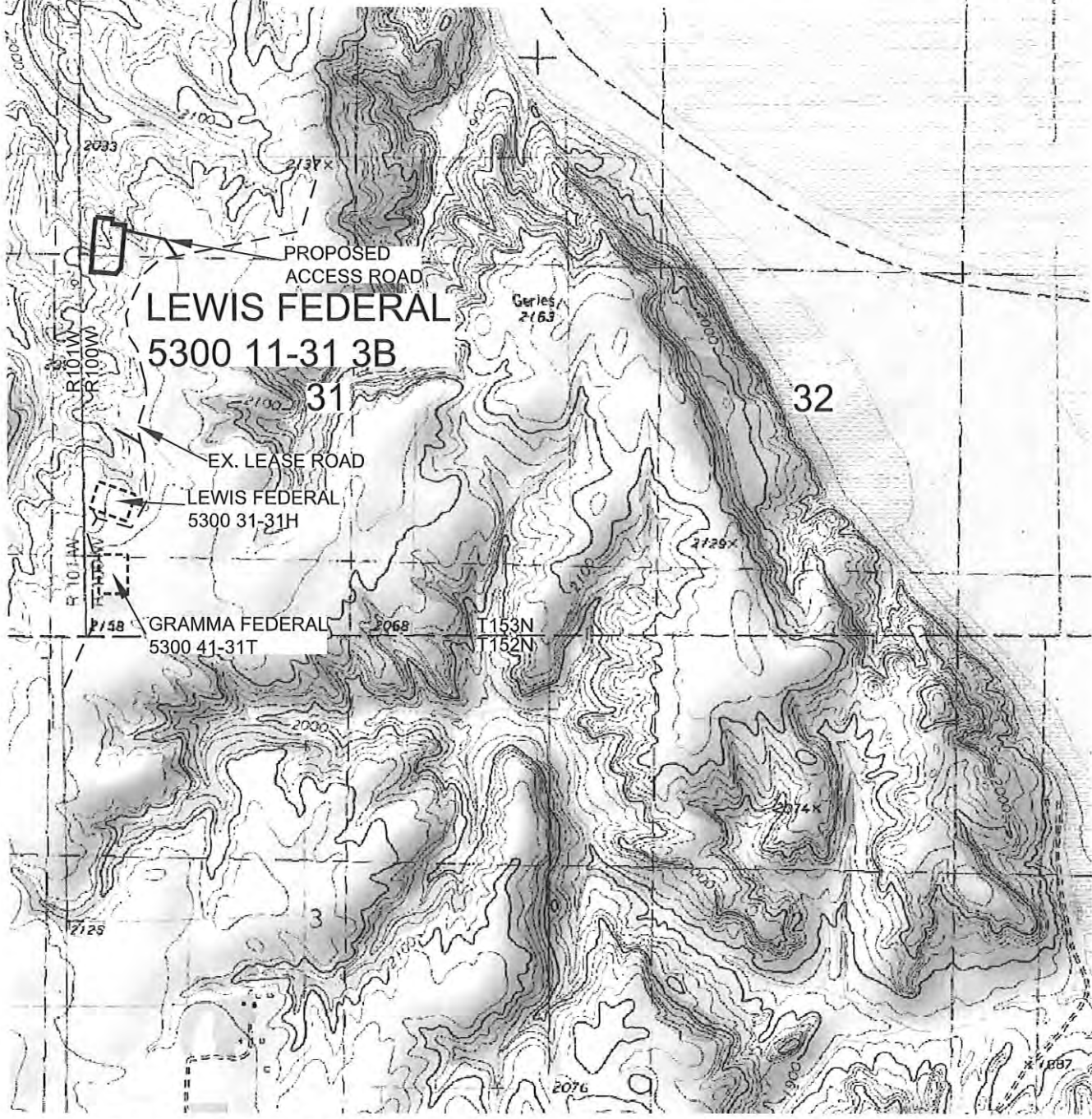
Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5817
Fax (406) 433-5618
www.interstateeng.com
Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
COUNTY ROAD MAP
SECTION 31, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S13-09-378.02
Checked By: D.D.K. Date: JAN. 2014

Revision No.	Date	By	Description
REV 1	12/3/14	BHH	REMOVED PIT FROM PAD

OASIS PETROLEUM NORTH AMERICA, LLC
 LEWIS FEDERAL 5300 11-31 3B
 1005' FNL/233' FWL
 QUAD LOCATION MAP
 SECTION 31, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA



© 2014, INTERSTATE ENGINEERING, INC.

6/9

SHEET NO.



Professionals you need, people you trust

Interstate Engineering, Inc.
 P.O. Box 648
 425 East Main Street
 Sidney, Montana 59270
 Ph (406) 433-5617
 Fax (406) 433-5618
 www.interstateeng.com

Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
 QUAD LOCATION MAP
 SECTION 31, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA

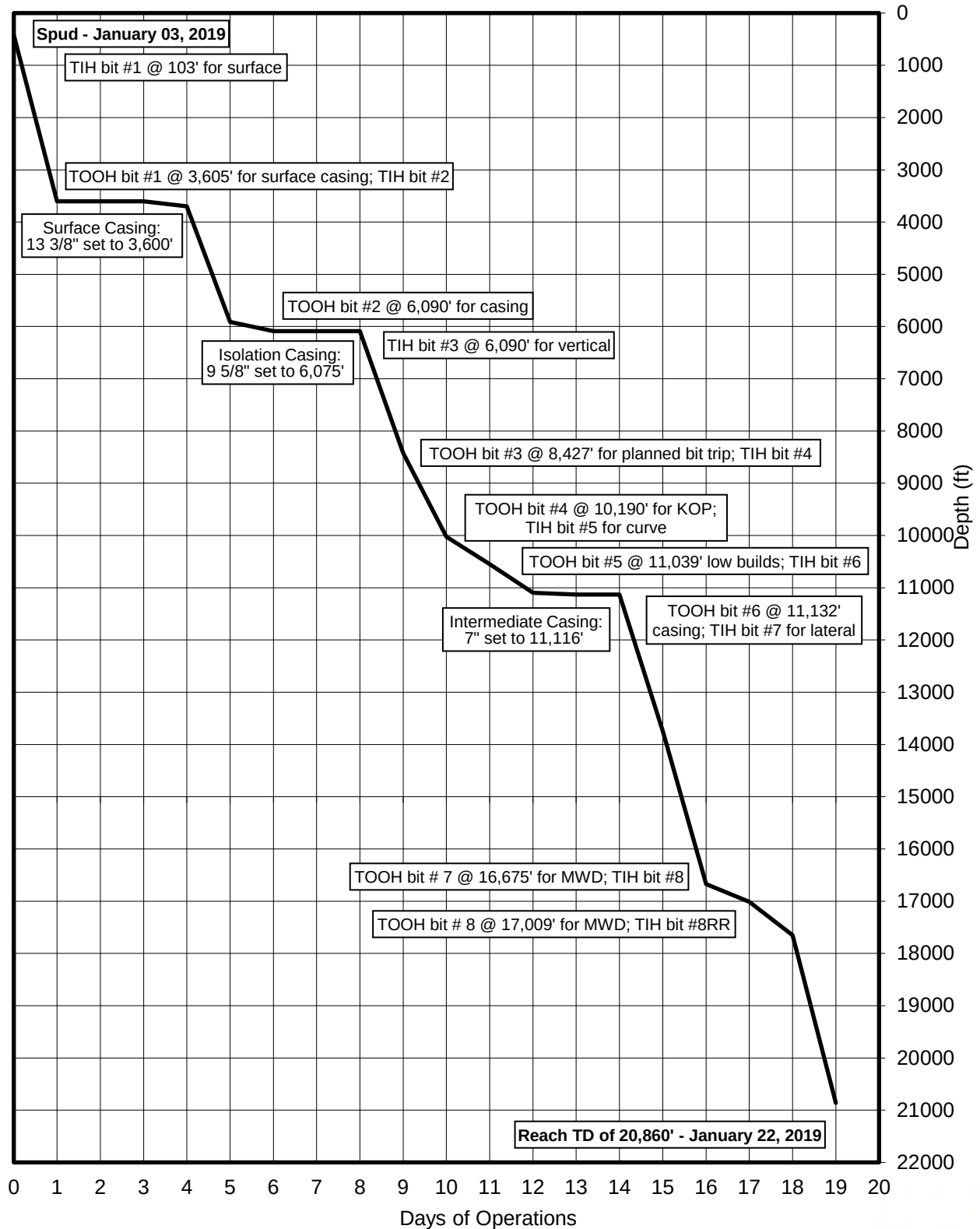
Drawn By: B.H.H. Project No.: S13-09-378.02
 Checked By: D.D.K. Date: JAN, 2014

Revision No.	Date	By	Description
REV 1	12/3/14	SHH	REMOVED PIT FROM PAD

TIME VS. DEPTH

Oasis Petroleum North America, LLC

Lewis Federal 5300 11-31 3B



MORNING REPORT SUMMARY

Rig Contractor: Nabors B21									Tool Pushers: Todd Miller, Matt Piehl					24 Hr Activity Summary	Formation
Day	Date 2019	Depth (0600 Hrs)	24 Hr Footage	Bit #	WOB (Klbs) RT	RPM (RT)	WOB (Klbs) MM	RPM (MM)	PP	SPM 1	SPM 2	SPM 3	GPM		
0	1/3	421'	318'	1	3	55	-	109	750	54	54	54	679	Skid rig from Lewis Federal 11-31 4B to Lewis Federal 11-31 3B; prepare rig for drilling; rig up catwalk, mud line, floor equipment, and cellar equipment; rig accepted at 13:00 hours on 01/02/2019; clean mud tanks, bear traps, dress shakers; rig up cellar; strap and gather BHA; pick up BHA; spud well on 01/03/2019; rotary drill from 103'-421'.	-
1	1/4	3,605'	3,184'	1	15	75	-	147	3850	73	73	73	917	Rotary drill from 421'-1,780'; circulate and condition mud ring; rig service; rotary drill from 1,780'-2,341'; circulate and condition mud ring; rig service; rotary drill from 2,341'-3,605' (casing depth); circulate and condition; TOOH.	Pierre
2	1/5	3,605'	0'	1	-	-	-	-	-	-	-	-	-	TOOH; rack back collars; inspect BHA; ream BHA down; TIH reaming; ream and wash tight spot; circulate and condition bottoms up 2x; displace 800 bbls of fresh water until hole and shakers come clean; TOOH; PJSA, lay down BHA; lay down drill collars; lay down BHA; PJSA to run casing; rig up to run casing; run casing.	Pierre
3	1/6	3,605'	0'	-	-	-	-	-	-	-	-	-	-	Run casing; rig down casing crew; circulate and condition bottoms until shakers are clean; prejob safety meeting with cementers; rig up Halliburton; primary cementing; PJSA; working as directed by operator, rig down cementers, clean cellar out, cut off casing and conductor, install well head.	Pierre
4	1/7	3,700'	95'	2	15	45	10	150	3600	75	75	75	654	Install well head; nipple up BOP; test BOP; pressure test casing shoe; rig down testers; install wear busing; safety meeting with wireline crew; rig up wireline gyro tool; shoot surveys; rig down wireline; pick up BHA; TIH; drill cement out of surface casing, tag float at 3,555' and shoe at 3,600'; rotate ahead to 3,620'; FIT using 14 EMW held 470 psi; rotary drilling from 3,620'-3,700'.	Pierre
5	1/8	5,910'	2,210'	2	20	45	10	158	4800	79	79	79	689	Rotary drilling from 3,700'-4,395'; service rig; rotary drilling from 4,395'-5,910'.	Swift
6	1/9	6,090'	180'	2	20	45	-	158	4800	79	79	79	689	Rotary drilling from 5,910'-6,090'; PJSA on tripping with high winds, flow check (no flow); short trip and back ream F/ 6,090'-5,400' and TIH f/ 5,400'-6,090'; PJSA on tripping with high winds, flow check (no flow); short trip and back ream F/ 6,090'-5,400'; spot pill, spot LCM sweep across Dakota; PJSA on TOOH in high winds; flow check, no flow, floor hands change clothes; TOOH F/ 4,200'-5,700'; service rig; flow check, no flow; safety stand down on service loop, looking up, proper hand signals; TOOH; lay down BHA; install/remove wear bushing, wear was broken; PJSA to rig up casers; rig up to run casing; install remove wear bushing; service top drive; run casing.	Swift
7	1/10	6,090'	0'	-	-	-	-	-	-	-	-	-	-	Run casing; PJSA with Noble for rig down; lay down 3rd party tools; circulate and condition bottoms up; wait on Haliburton; service rig; wait on Haliburton; PJSA with Haliburton cement; rig up cementers; primary cementing; safety inspection to rig down cementer; rig down cementers; pull landing joint, install packoff; install/remove wear bushing; trip in hole.	Swift
8	1/11	6,090'	0'	-	-	-	-	-	-	-	-	-	-	Trip in hole; cut drill line; service rig; wait on cement, circ and condition mud; Circulate and condition, get chemicals in and cut mud weight to 10 ppg; TOOH; pick up BHA; service top drive; PJSA to rig up wireline; rig up loggers; cased hole logs; PJSA rig down loggers; rig down loggers.	Swift

MORNING REPORT SUMMARY

Rig Contractor: Nabors B21									Tool Pushers: Todd Miller, Matt Piehl					24 Hr Activity Summary	Formation
Day	Date 2019	Depth (0600 Hrs)	24 Hr Footage	Bit #	WOB (Klbs) RT	RPM (RT)	WOB (Klbs) MM	RPM (MM)	PP	SPM 1	SPM 2	SPM 3	GPM		
9	1/12	8,427'	2,337'	3	50	45	25	153	4850	79	79	79	637	Prejob safety meeting; pressure test casing shoe to 1500 psi; pick up BHA; TIH; remove trip nipple, install rotating head rubber; test MWD tool; drill cement out of casing, tag float at 6,033' and shoe at 6,075'; rotary drill ahead to 6,090'; FIT; rotary drill from 6,090'-6,932'; rig service; rotary drill from 6,932'-8,427'; service rig; pump dry job.	Kibbey
10	1/13	10,022'	1,595'	4	45	45	25	176	4050	66	66	66	567	TOOH; lay down BHA; pick up BHA; TIH; rotary drill from 8,892'-10,022'; BOP drill; rig service.	Lodgepole
11	1/14	10,550'	528'	5	55	20	20	185	3950	73	73	73	637	Rotary drill from 10,022'-10,190'; TD vertical; TOOH, strap out; lay down BHA; pick up BHA; TIH; ream salts; rig service; orientate curve slide drilling, rotating as needed, from 10,190'-10,513'; Kelly hose blew; down time waiting on Nabors hotshot with replacement Kelly hose; TOOH to shoe; replace hose; test hose; TIH; slide drilling, rotating as needed, from 10,513'-10,550'.	Lodgepole
12	1/15	11,098'	548'	6	55	20	75	135	3950	80	80/-	465	-	Slide drilling, rotating as needed, from 10,550'-11,039'; rig service; TOOH due to insufficient build rates; laydown BHA; pick up new bit and 2.5 degree motor; TIH; slide drilling, rotating as needed, from 11,039'-11,098'.	Middle Bakken
13	1/16	11,132'	34'	6	43	15	-	160	2900	92	92/-	551	595	Drill F/11,098'-11,132'; short trip; orient the tool high side, 40'/min as per Oasis; circulate and condition 2 bottoms up; TOOH, orient tool high side through curve @ 40'/min; remove rotating head, install trip nipple; PJSA for BHA; lay down BHA; install/remove wear bushing; service rig; working as directed by operator, clean rig floor; PJSA with casers; rig up to run casing; run casing; service rig.	Middle Bakken
14	1/17	11,132'	0'	7	-	-	-	-	-	-	-	-	-	Pre job safety w/ Noble; lay down 2rd party tools R/D Noble; C&C; pre job safety with Halliburton; pic up 2rd party tools R/U Halliburton; primary cementing; pre job safety with Halliburton; lay down 3rd party tools R/D Halliburton; lay down 2rd party tools L/D landing JT/casing elevators/slips; working as directed by operator install Cactus packoff; install/remove wear bushing and change quill to 4"; working as directed by operator clean rig floor and get ready to P/U BHA; service rig obm yes; pre job safety to pick up BHA; pick up BHA; pick up drill pipe; pressure test CSG/Shoe/2500 PSI for 30 min; cut drilling live/15 wraps; working as directed by operator put 145 joints on catwalk/and strapped; pick up drill pipe	Middle Bakken
15	1/18	13,743'	2,611'	7	28	45	40	276	4500	55	-	55	325	Pick up drill pipe remove TN install RHR; drilling cement & plug @10,951' float @11,115'; C&C B/U for FIT; FIT @ 11,147' MW 9.7 PPG EMW 13ppg 1850 psi; rotary drilling F/11,132'-11,739'; service rig; rotary drilling F/11,739'-13,755'; service top drive.	Middle Bakken
16	1/19	16,675'	2,932'	7	28	50	40	276	4700	55	55	-	325	Rotary drilling F/13,743'-14,200'; safety stand down on oasis incidents; rotary drilling F/14,200'-15,256'; service rig; rotary drilling F/15,256'-16,675'; wait on 3rd party tools, trouble shoot MWD; TOOH, pull rotating head; TOOH; service rig.	Middle Bakken

MORNING REPORT SUMMARY

Rig Contractor: Nabors B21									Tool Pushers: Todd Miller, Matt Piehl					24 Hr Activity Summary	Formation
Day	Date 2019	Depth (0600 Hrs)	24 Hr Footage	Bit #	WOB (Klbs) RT	RPM (RT)	WOB (Klbs) MM	RPM (MM)	PP	SPM 1	SPM 2	SPM 3	GPM		
17	1/20	17,009'	334'	8	28	45	60	272	3900	55	-	55	325	TOOH; PJSA for BHA; lay down BHA; pick up BHA; TIH, remove trip nipple, install rotating head, 235'/min as per Oasis; Working as directed by operator, relog gamma; circulate gas through choke; relog gamma; service rig; Drill F/16,675'-16,974'; slide drilling/troughing F/16,964'-16,979'; slide drilling F/16979'-17009'; wait on 3rd party; troubleshoot MWD; circulate and condition, pump dry job; PJSA to TOOH; TOOH; PJSA, lay down BHA; lay down BHA; wait on 3rd party, wait on MWD tools.	Middle Bakken
18	1/21	17,650'	641'	8RR	22	45	32	275	4700	55	-	55	325	Waiting on 3rd party tools MWD; pick up BHA; TIH remove TN, Install RHR, test MWD tool; cut drilling line 11 wraps; TIH; service rig; working as directed by operator clean out mud leg and gas buster; C&C B/U for gas;C&C; working as directed by operator/relog gamma/shoot surveys, slide drilling F/17,009-17,025'; rotary drilling F/17,025'-17,100' slide drilling F/17,100'-17,115' drill F/17,115'-17,650'; service top drive.	Middle Bakken
19	1/22	20,860'	3,210'	8RR	28	50	45	276	4500	55	-	55	325	Rotary drilling F/17,620'-19,416'; service rig; rotary drilling F/19,416'-21,860'; service rig; circulate and condition.	Middle Bakken

DAILY MUD SUMMARY

Chemical Company: Reliable Drilling Fluids LLC							Mud Engineer: S. Zimmermann					Fresh water in surface; Diesel invert in vertical/curve; Salt water in lateral									
Date 2019	Mud Depth	Mud WT (ppg)	VIS (sec/qt)	PV (cP)	YP (lbs/100 ft ²)	Gels (lbs/100 ft ²)	600/300	Oil/ H ₂ O (ratio)	Oil/ H ₂ O (%)	Cake (API/HTHP)	Solids (%)	Cor. Solids (%)	Alk	pH	Excess Lime (lb/bbl)	Cl ⁻ (mg/L)	LGS/ HGS (%)	Salinity (ppm)	Electrical Stability	Mud loss (bbls)	Mud Gain (bbls)
01/03	103'	8.3	25	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
01/04	3,605'	9.85	47	6	26	16/22/29	38/32	-	0/86	-	14	13.95	7.5	7.5	0	1.1k	13.95/0	-	-	-	-
01/05	3,605'	9.9	47	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
01/06	3,605'	9.85	47	6	26	16/22/29	38/32	-	0/86	-	14	13.95	7.5	7.5	0	1.1k	13.95/0	-	-	-	-
01/07	5,340'	12.1	53	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
01/08	6,090'	12.85	54	30	13	11/17/21	73/43	71/29	55/22	3	23	20.27	1.9	-	2.46	45k	3.89/16.38	252108	359	-	-
01/09	6,090'	12.85	54	30	13	11/17/21	73/43	71/29	55/22	3	23	20.27	1.9	-	2.46	45k	3.89/16.38	252108	359	-	-
01/10	6,090'	11.15	41	12	7	7/10/14	31/19	78/22	66/19	2	15	13.01	1.7	-	2.2	33k	2.22/10.79	222535	388	-	-
01/11	6,090'	10.3	42	10	7	7/10/13	27/17	79/21	68/18	2	14	11.94	1.9	-	2.46	34k	2.95/8.99	237392	559	-	-
01/12	8,427'	10.4	45	14	9	9/13/16	37/23	80/20	68/17	3	15	12.94	2.4	-	3.11	34k	4.21/8.73	247894	924	-	-
01/13	10,190'	10.25	44	11	8	8/11/15	30/19	85/15	72/13	3	15	13.3	2.1	-	2.72	28k	4.7/8.6	261967	580	-	-
01/14	10,840'	10.45	46	9	8	9/13/16	26/17	82/18	69/15	3	16	14.12	1.9	-	2.46	31k	5.43/8.69	254058	498	-	-
01/15	11,132'	10.25	40	11	8	8/12/16	30/19	82/18	69/15	3	16	14.12	1.8	-	2.33	31k	5.64/8.48	302663	529	-	-
01/16	Change from OBM to salt water																				
01/17	11,185'	9.8	28	2	1	1/1/1	5/3	-	1/90	-	9	0.64	0.1	8.5	0	171k	0.64/0	-	-	-	-
01/18	13,743'	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
01/19	16,675'	9.6	28	2	1	1/1/1	5/3	-	0/91	-	9	0.19	0.1	8.5	0	178k	1.73/-	-	-	-	-
01/20	17,009'	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
01/21	19,000'	9.6	28	2	1	1/1/1	5/3	-	0/91	-	9	0.19	0.1	8.5	0	178k	0.19/0	-	-	-	-

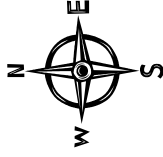
BOTTOM HOLE ASSEMBLY RECORD

Bit Data											Motor Data						Reason For Removal
Bit #	Size (in.)	Type	Make	Model	Depth In	Depth Out	Footage	Hours	Σ hrs	Vert. Dev.	Make	Model	Lobe	Stage	Bend	Rev/Gal	
1	17 1/2	PDC	Ultrerra	36992	103'	3,605'	3,502'	24.0	24	Surface	Stickman	Predator	5/6	6.0	2.0°	0.16	TD surface
2	12 1/4	PDC	Ultrerra	41164	3,605'	6,090'	2,485'	30.0	54	Vertical	Stickman	Predator	5/6	6.0	2.0°	0.16	Run Isolation casing
3	8 3/4	PDC	Smith	XS616	6,090'	8,427'	2,337'	15.0	69	Vertical	NOV	-	7/8	5.7	1.50°	0.24	Planned bit trip
4	8 3/4	PDC	Varel	FY616P-F	8,427'	10,190'	1,763'	16.0	85	Vertical	Predator	-	7/8	5.7	1.50°	0.31	TD vertical
5	8 3/4	PDC	Reed	TK 56	10,190'	11,039'	849'	14.5	99.5	Curve	NOV	-	7/8	5.0	2.12°	0.29	Low build rates
6	8 3/4	Tri-cone	Smith	F30T	11,039'	11,132'	93'	5.5	105	Curve	NOV	-	7/8	5.0	2.50°	0.28	TD curve
7	6	PDC	Security	GTD54HE	11,132'	16,675'	5,543'	38.0	143	Lateral	Discovery	-	7/8	10.6	1.50°	0.85	MWD failure
8	6	PDC	Reed	TKC 53	16,675'	17,009'	334'	4.0	147	Lateral	Discovery	-	7/8	10.6	1.50°	0.85	MWD failure
8RR	6	PDC	Reed	TKC 53	17,009'	20,860'	3,851'	26.0	173	Lateral	Discovery	-	7/8	10.6	1.50°	0.85	TD lateral

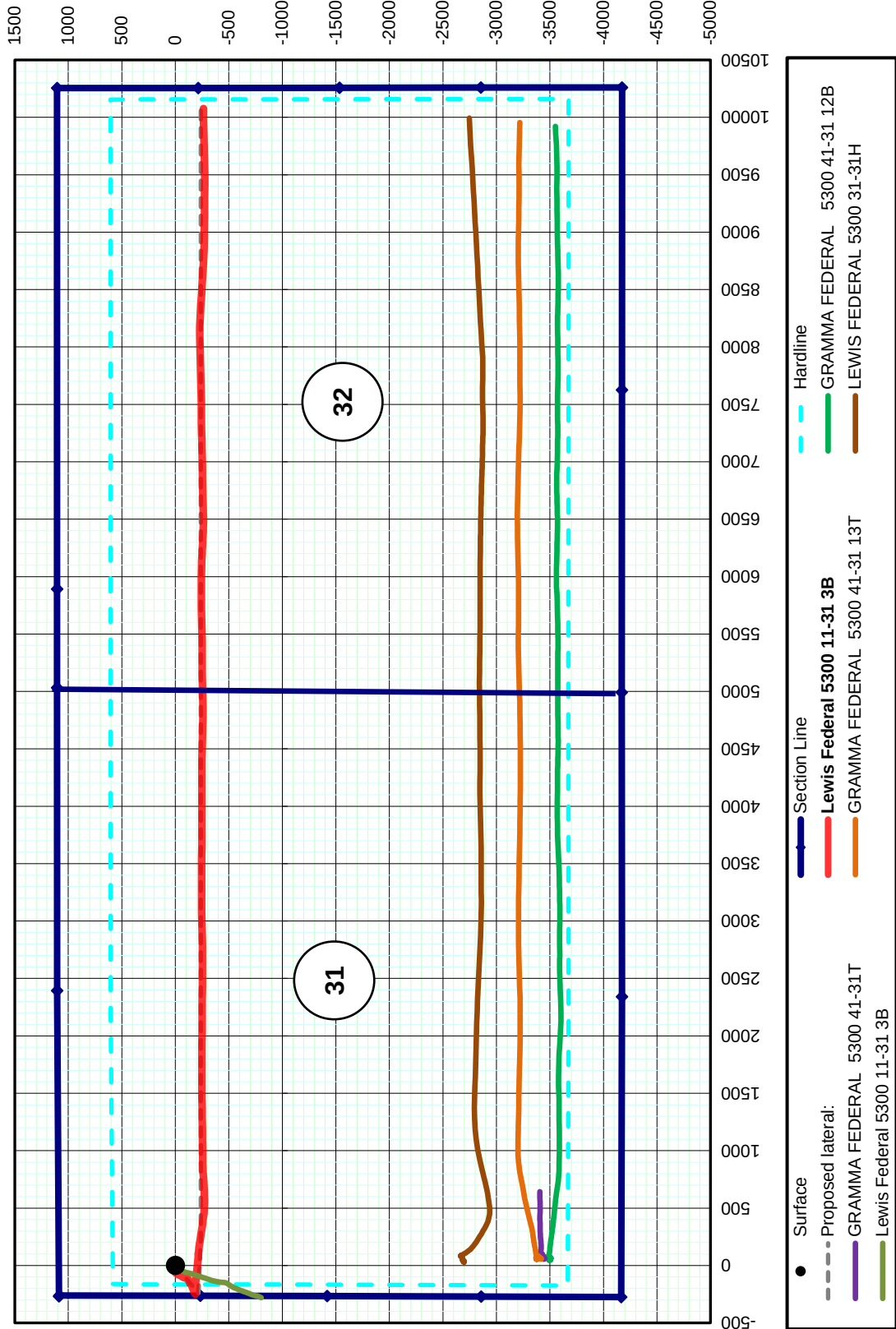
Note: 1,280 acre laydown spacing unit with 500' N/S & 100' EW setbacks

PLAN VIEW

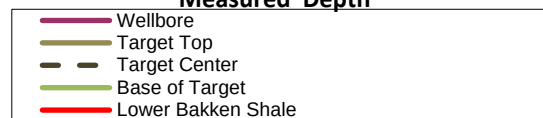
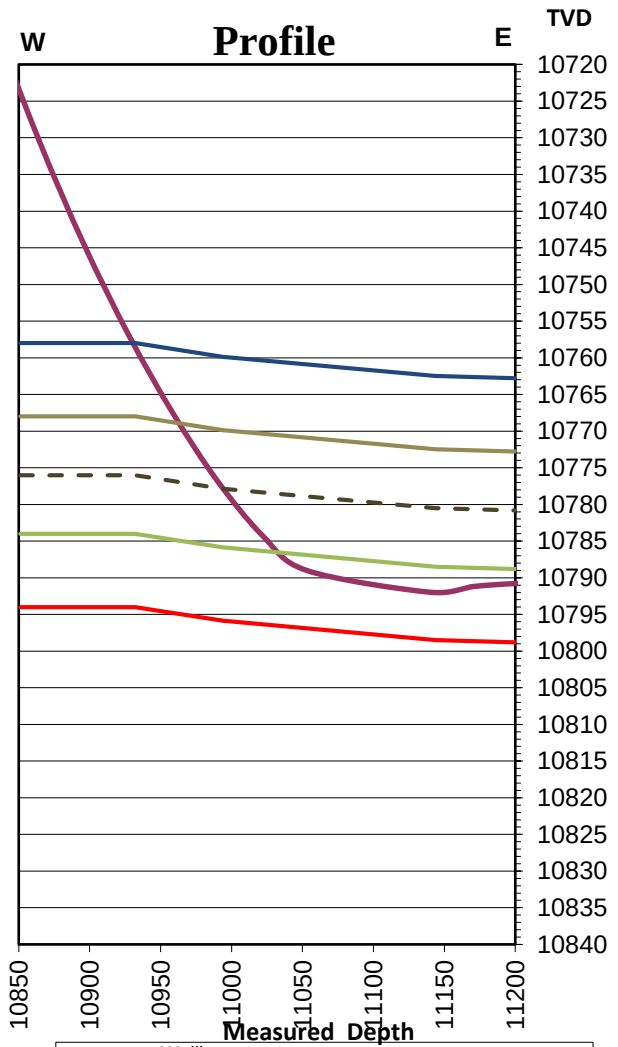
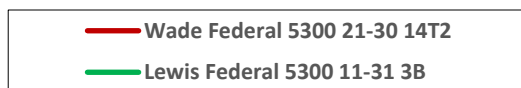
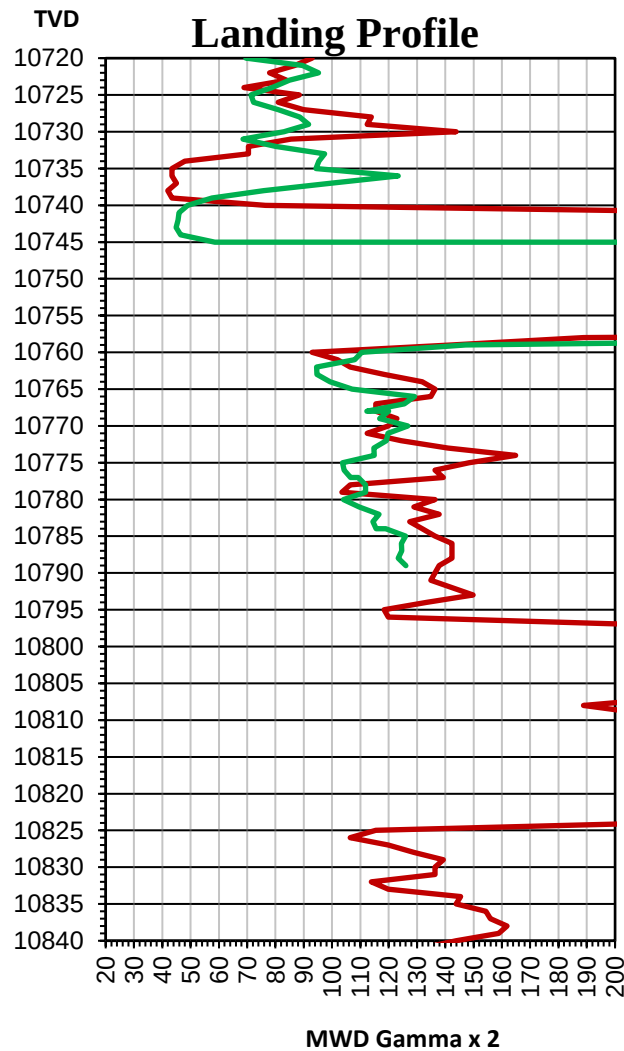
Oasis Petroleum North America, LLC
Lewis Federal 5300 11-31 3B
1,083' FNL & 262' FWL
Lot 1 Sec. 31, T153N, R100W



Bottom Hole Location
236.07' S & 10,076.54' E
of surface location or
1,346.07' FNL & 184.20' FEL
Lot 4 Sec. 32, T153N, R100W



Lewis Federal 5300 11-31 3B
257' FNL & 1,817' FEL
Lot 1 Sec. 31, T153N, R100W
McKenzie County, ND



LANDING PROJECTION

Formation/ Zone:	Proposed Landing Target From:			
	Wade Federal 5300 41-30 8T2	Lewis Federal 5300 31-31H	Gramma Federal 5300 41-31T	Average of Offset Wells
Pierre	-	10,744'	-	10,744'
Greenhorn	10,767'	10,801'	-	10,784'
Mowry (Dakota Group)	10,764'	10,787'	-	10,776'
Inyan Kara (Dakota Group)	10,733'	10,747'	-	10,740'
Swift (Base Dakota Group)	10,754'	10,802'	-	10,778'
Rierdon	10,740'	10,859'	-	10,799'
Dunham Salt	10,741'	10,857'	-	10,799'
Dunham Salt Base	10,734'	10,793'	-	10,763'
Pine Salt	10,765'	10,768'	-	10,767'
Pine Salt Base	10,783'	10,749'	-	10,766'
Opeche Salt	10,786'	10,764'	-	10,775'
Opeche Salt Base	10,773'	10,749'	-	10,761'
Amsden	10,788'	10,758'	-	10,773'
Tyler	10,774'	10,748'	-	10,761'
Otter/Base Minnelusa	10,777'	10,751'	-	10,764'
Kibbey "Lime"	10,774'	10,753'	10,765'	10,764'
Charles Salt	10,781'	10,753'	10,767'	10,767'
Base Last Salt	10,778'	10,764'	10,781'	10,774'
Mission Canyon	10,769'	10,761'	10,773'	10,768'
Lodgepole	10,770'	10,768'	10,781'	10,773'
Lodgepole A	10,772'	10,774'	10,788'	10,778'
Lodgepole B	10,764'	10,756'	10,765'	10,762'
Lodgepole C	10,774'	10,776'	10,795'	10,782'
Lodgepole D	10,773'	10,783'	10,780'	10,779'
Lodgepole E	10,778'	10,782'	10,784'	10,781'
Lodgepole F	10,778'	10,783'	10,783'	10,782'
False Bakken	10,781'	10,784'	10,782'	10,782'
Upper Bakken	10,782'	10,781'	10,783'	10,782'
Middle Bakken	10,779'	10,779'	10,779'	10,779'

Current Landing Target (21' below the base of the UBS):

10,779'

GL: 2,110'

KB: 2,135'

Oasis Petroleum North America, LLC

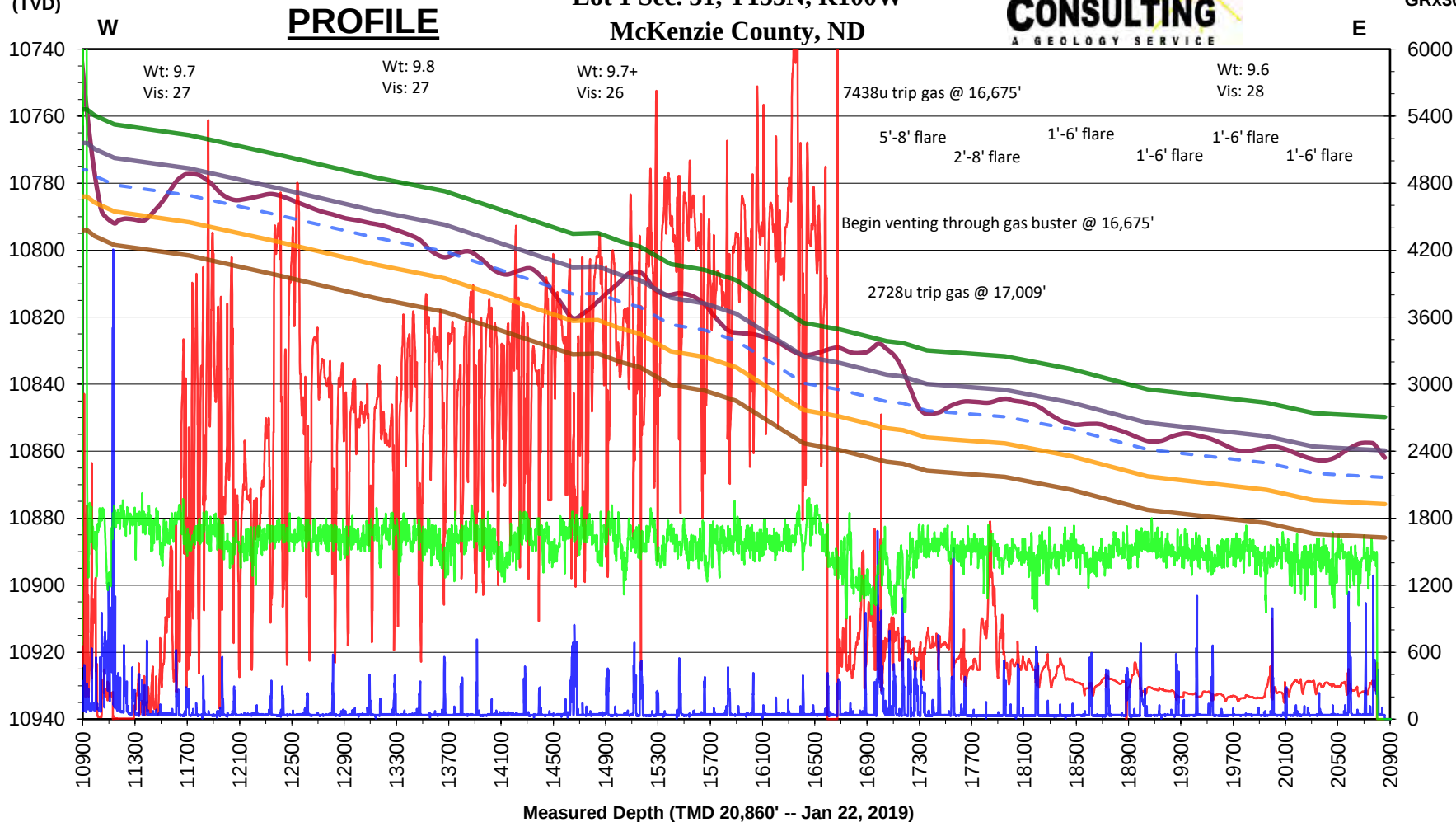
Lewis Federal 5300 11-31 3B

Lot 1 Sec. 31, T153N, R100W

McKenzie County, ND



TG
ROPx200
GRx30



- | | | |
|---------------|----------------------------|--------------------|
| Wellbore | Base of Upper Bakken Shale | Target Top |
| Target Center | Base of Target | Lower Bakken Shale |
| TG | ROP x 200 | GR x 30 |

WELL

Lewis Federal 5300 11-31 3B



API
33-053-06548

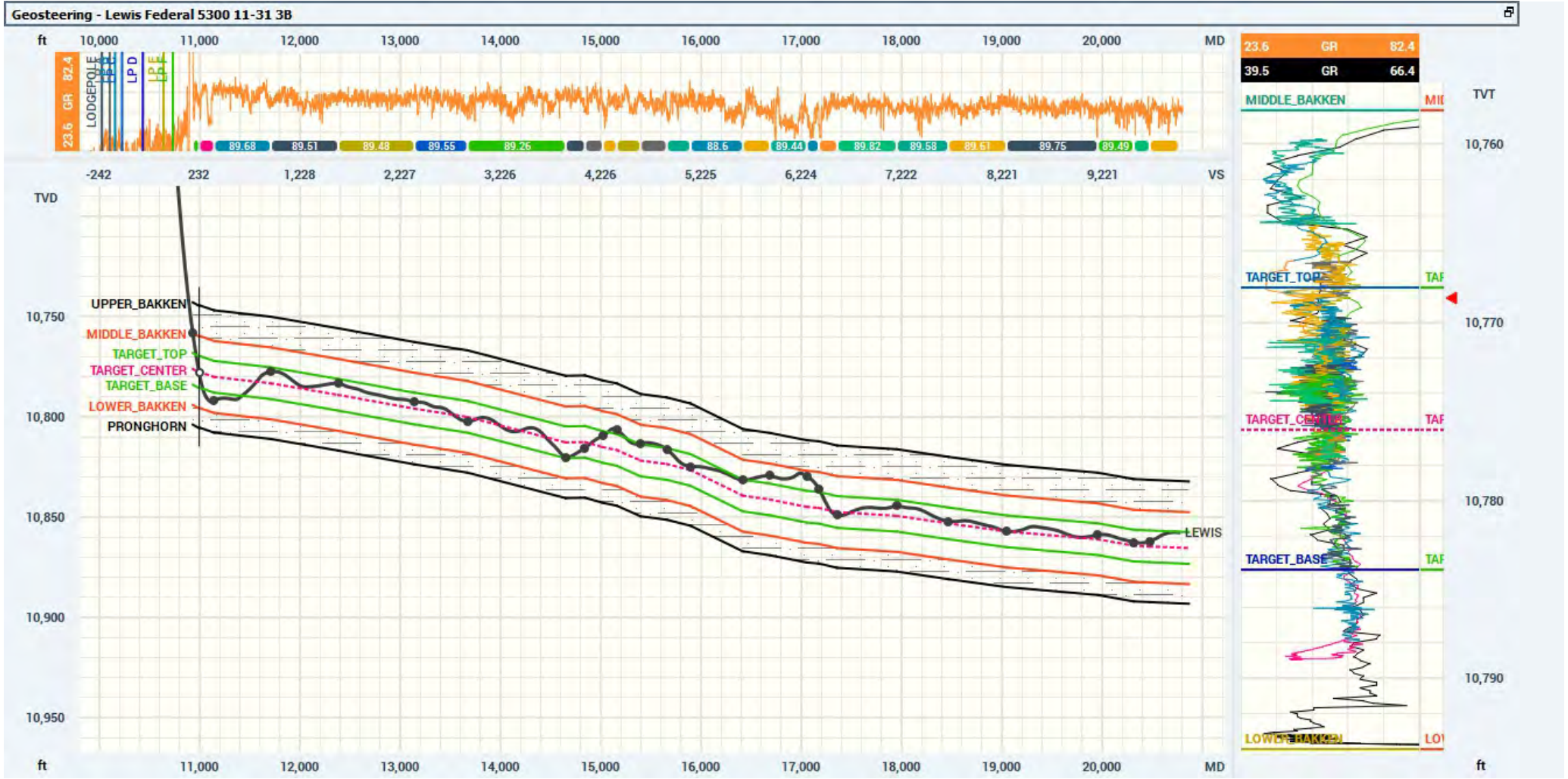
TYPEWELL
Lewis Federal 5300 31-31H

FIELD
Baker

INTERPRETER
D. Johnson, C. Kyler

DATE
1/22/2019 6:44 AM

VS AZIMUTH
91.35°



Last Surveys					
MD	INCL	AZIM	TVD	VS	DLS
20,669.0	90.47	87.8	10,857.7	9,889.2	1.3
20,763.0	89.67	89.9	10,857.6	9,983.1	2.4
Last Segment Dip					
89.88°					

COMMENTS

Subsurface Structure Points - Lewis Federal 5300 11-31 3B

GL: 2,110'

KB: 2,135'

Measured Depth	Base of Upper Bakken Shale (TVD)	Target Top (TVD)	Target Center (TVD)	Base of Target (TVD)	Lower Bakken Shale (TVD)	Dip	Up/Down
10,931.00'	10758.00	10768.00	10776.00	10784.00	10784.00		
10,995.00'	10759.90	10769.90	10777.90	10785.90	10785.90	-1.70°	Down
11,144.00'	10762.50	10772.50	10780.50	10788.50	10788.50	-1.00°	Down
11,709.00'	10765.60	10775.60	10783.60	10791.60	10791.60	-0.31°	Down
12,382.00'	10771.40	10781.40	10789.40	10797.40	10797.40	-0.49°	Down
13,142.00'	10778.30	10788.30	10796.30	10804.30	10804.30	-0.52°	Down
13,670.00'	10782.40	10792.40	10800.40	10808.40	10808.40	-0.44°	Down
14,649.00'	10795.10	10805.10	10813.10	10821.10	10821.10	-0.74°	Down
14,842.00'	10794.90	10804.90	10812.90	10820.90	10820.90	0.06°	Up
15,020.00'	10797.50	10807.50	10815.50	10823.50	10823.50	-0.84°	Down
15,158.00'	10798.90	10808.90	10816.90	10824.90	10824.90	-0.58°	Down
15,397.00'	10804.20	10814.20	10822.20	10830.20	10830.20	-1.27°	Down
15,657.00'	10805.90	10815.90	10823.90	10831.90	10831.90	-0.37°	Down
15,894.00'	10808.90	10818.90	10826.90	10834.90	10834.90	-0.73°	Down
16,416.00'	10821.70	10831.70	10839.70	10847.70	10847.70	-1.40°	Down
16,688.00'	10823.60	10833.60	10841.60	10849.60	10849.60	-0.40°	Down
17,053.00'	10827.20	10837.20	10845.20	10853.20	10853.20	-0.57°	Down
17,174.00'	10827.70	10837.70	10845.70	10853.70	10853.70	-0.24°	Down
17,358.00'	10829.90	10839.90	10847.90	10855.90	10855.90	-0.69°	Down
17,951.00'	10831.70	10841.70	10849.70	10857.70	10857.70	-0.17°	Down
18,465.00'	10835.50	10845.50	10853.50	10861.50	10861.50	-0.42°	Down
19,045.00'	10841.50	10851.50	10859.50	10867.50	10867.50	-0.59°	Down
19,955.00'	10845.50	10855.50	10863.50	10871.50	10871.50	-0.25°	Down
20,313.00'	10848.60	10858.60	10866.60	10874.60	10874.60	-0.50°	Down
20,474.00'	10849.00	10859.00	10867.00	10875.00	10875.00	-0.14°	Down
20,860.00'	10849.80	10859.80	10867.80	10875.80	10875.80	-0.12°	Down

GL: 2,110'

KB: 2,135'

Oasis Petroleum North America, LLC

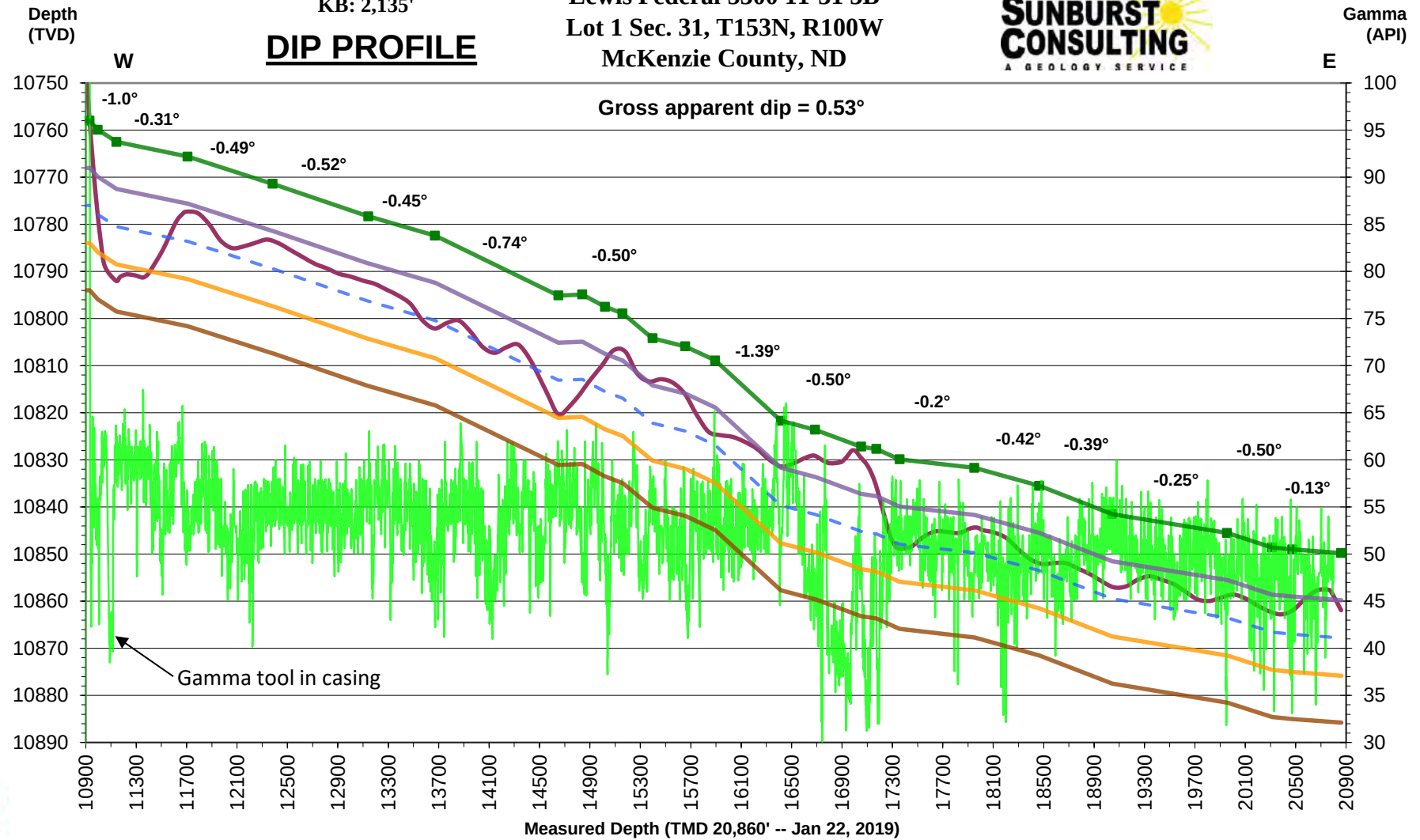
Lewis Federal 5300 11-31 3B

Lot 1 Sec. 31, T153N, R100W

McKenzie County, ND



DIP PROFILE



wellbore Base of Upper Bakken Shale Target Top Target Center Base of Target Lower Bakken Shale Gamma

Operator: Oasis Petroleum North America, LLC
 Well: Lewis Federal 5300 11-31 3B
 Surface Coordinates: 1,083' FNL & 262' FWL
 Surface Location: Lot 1 Sec. 31, T153N, R100W
 County, State: McKenzie County, ND

Bottom Hole Location: 236.07' S & 10,076.54' E
 of surface location or
 1,346.07' FNL & 184.20' FEL
 Lot 4 Sec. 32, T153N, R100W



Kick-off: 1/13/2019
 Finish: 1/22/2019

Directional Supervision:
 Scientific Drilling
 RPM Consulting

GL: 2,110'
 KB: 2,135'

Minimum Curvature Method (SPE-3362)

[North and East are positive and South and West are negative]

Vertical Section Azimuth: 91.35

No.	MD	INC	TRUE AZM	TVD	N-S	E-W	SECT	DLS/ 100
Tie	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1	168.00	0.54	304.14	168.00	0.44	-0.66	-0.67	0.32
2	261.00	0.29	4.55	261.00	0.92	-1.00	-1.02	0.51
3	355.00	0.61	316.07	354.99	1.52	-1.33	-1.36	0.50
4	447.00	0.80	325.27	446.99	2.40	-2.03	-2.09	0.24
5	537.00	0.85	323.54	536.98	3.46	-2.79	-2.87	0.06
6	626.00	0.24	338.37	625.97	4.16	-3.25	-3.35	0.70
7	725.00	0.62	348.62	724.97	4.88	-3.43	-3.55	0.39
8	815.00	0.79	346.47	814.96	5.96	-3.67	-3.81	0.19
9	905.00	0.81	323.76	904.95	7.08	-4.19	-4.36	0.35
10	996.00	0.96	344.77	995.94	8.33	-4.77	-4.97	0.39
11	1086.00	0.84	322.19	1085.93	9.58	-5.38	-5.60	0.41
12	1176.00	0.07	75.77	1175.93	10.11	-5.73	-5.96	0.97
13	1266.00	0.67	154.08	1265.93	9.65	-5.44	-5.67	0.73
14	1357.00	0.70	123.65	1356.92	8.87	-4.75	-4.96	0.40
15	1447.00	0.98	127.92	1446.91	8.09	-3.68	-3.87	0.32
16	1539.00	0.99	134.82	1538.90	7.05	-2.50	-2.67	0.13
17	1629.00	0.87	147.21	1628.89	5.92	-1.58	-1.72	0.26
18	1719.00	1.33	137.89	1718.87	4.57	-0.51	-0.62	0.55
19	1810.00	0.75	144.37	1809.85	3.31	0.55	0.47	0.65
20	1903.00	0.24	97.42	1902.85	2.79	1.09	1.03	0.66
21	1996.00	0.20	174.02	1995.85	2.60	1.31	1.24	0.30
22	2089.00	0.26	120.44	2088.85	2.33	1.50	1.45	0.23
23	2183.00	0.34	205.73	2182.85	1.97	1.57	1.52	0.44
24	2276.00	0.30	175.01	2275.85	1.48	1.47	1.43	0.19
25	2370.00	0.39	219.14	2369.84	0.99	1.29	1.26	0.29
26	2463.00	0.14	209.48	2462.84	0.64	1.03	1.02	0.27
27	2557.00	0.03	250.32	2556.84	0.53	0.95	0.94	0.13
28	2650.00	0.38	267.07	2649.84	0.51	0.62	0.61	0.38
29	2743.00	0.32	283.78	2742.84	0.56	0.06	0.05	0.13
30	2800.00	0.25	252.81	2799.84	0.56	-0.21	-0.23	0.29
31	2900.00	0.48	279.36	2899.84	0.56	-0.83	-0.85	0.28
32	3000.00	0.67	258.42	2999.83	0.51	-1.82	-1.83	0.28
33	3100.00	0.44	255.96	3099.83	0.30	-2.77	-2.77	0.23
34	3200.00	0.42	237.60	3199.83	0.01	-3.45	-3.45	0.14
35	3300.00	0.42	253.99	3299.82	-0.29	-4.11	-4.10	0.12



Operator: Oasis Petroleum North America, LLC
 Well: Lewis Federal 5300 11-31 3B
 Surface Coordinates: 1,083' FNL & 262' FWL
 Surface Location: Lot 1 Sec. 31, T153N, R100W
 County, State: McKenzie County, ND

Bottom Hole Location: 236.07' S & 10,076.54' E
 of surface location or
 1,346.07' FNL & 184.20' FEL
 Lot 4 Sec. 32, T153N, R100W



Kick-off: 1/13/2019
 Finish: 1/22/2019

Directional Supervision:
 Scientific Drilling
 RPM Consulting

GL: 2,110'
 KB: 2,135'

Minimum Curvature Method (SPE-3362)

[North and East are positive and South and West are negative]

Vertical Section Azimuth: 91.35

No.	MD	INC	TRUE AZM	TVD	N-S	E-W	SECT	DLS/ 100
36	3400.00	0.33	248.64	3399.82	-0.49	-4.73	-4.72	0.10
37	3500.00	0.47	245.50	3499.82	-0.77	-5.37	-5.35	0.14
38	3545.00	0.44	259.59	3544.82	-0.87	-5.71	-5.69	0.26
39	3641.00	0.41	265.66	3640.81	-0.97	-6.41	-6.39	0.06
40	3672.00	0.35	227.31	3671.81	-1.04	-6.59	-6.57	0.83
41	3766.00	1.41	265.05	3765.80	-1.33	-7.96	-7.92	1.23
42	3859.00	2.07	263.16	3858.76	-1.63	-10.77	-10.72	0.71
43	3953.00	2.65	271.61	3952.68	-1.77	-14.62	-14.58	0.72
44	4046.00	2.50	287.08	4045.59	-1.12	-18.71	-18.68	0.76
45	4140.00	2.33	287.92	4139.50	0.07	-22.49	-22.48	0.18
46	4233.00	2.79	274.45	4232.41	0.83	-26.54	-26.56	0.81
47	4326.00	3.79	266.05	4325.26	0.79	-31.87	-31.88	1.19
48	4420.00	3.52	280.15	4419.07	1.09	-37.81	-37.82	1.00
49	4513.00	3.32	296.15	4511.90	2.78	-43.03	-43.09	1.04
50	4608.00	3.47	289.23	4606.74	4.94	-48.22	-48.32	0.46
51	4703.00	3.99	255.25	4701.55	5.04	-54.13	-54.23	2.35
52	4798.00	4.11	230.69	4796.32	2.04	-59.96	-59.99	1.82
53	4893.00	3.76	212.64	4891.10	-2.74	-64.28	-64.19	1.35
54	4988.00	4.17	206.21	4985.87	-8.46	-67.48	-67.26	0.64
55	5083.00	3.49	210.52	5080.66	-14.05	-70.47	-70.12	0.78
56	5178.00	3.11	213.80	5175.50	-18.68	-73.38	-72.92	0.45
57	5272.00	3.68	210.86	5269.33	-23.39	-76.34	-75.77	0.63
58	5367.00	4.27	209.40	5364.10	-29.09	-79.64	-78.94	0.63
59	5462.00	3.68	203.48	5458.88	-34.96	-82.59	-81.75	0.76
60	5557.00	3.35	202.51	5553.70	-40.32	-84.87	-83.90	0.35
61	5652.00	3.92	199.74	5648.51	-45.95	-87.03	-85.92	0.63
62	5746.00	3.49	206.21	5742.31	-51.54	-89.38	-88.14	0.64
63	5841.00	3.24	212.42	5837.15	-56.40	-92.10	-90.74	0.46
64	5935.00	3.53	209.59	5930.98	-61.16	-94.95	-93.48	0.36
65	6021.00	4.32	212.13	6016.78	-66.20	-97.98	-96.39	0.94
66	6116.00	4.88	210.58	6111.47	-72.71	-101.94	-100.20	0.60
67	6210.00	3.85	199.95	6205.20	-79.12	-105.05	-103.16	1.39
68	6303.00	3.13	185.94	6298.03	-84.58	-106.38	-104.35	1.19
69	6396.00	3.29	184.17	6390.88	-89.77	-106.83	-104.69	0.20
70	6490.00	3.02	191.33	6484.74	-94.88	-107.52	-105.25	0.51
71	6583.00	2.96	193.64	6577.62	-99.62	-108.56	-106.19	0.14



Operator: Oasis Petroleum North America, LLC
 Well: Lewis Federal 5300 11-31 3B
 Surface Coordinates: 1,083' FNL & 262' FWL
 Surface Location: Lot 1 Sec. 31, T153N, R100W
 County, State: McKenzie County, ND

Bottom Hole Location: 236.07' S & 10,076.54' E
 of surface location or
 1,346.07' FNL & 184.20' FEL
 Lot 4 Sec. 32, T153N, R100W



Kick-off: 1/13/2019
 Finish: 1/22/2019

Directional Supervision:
 Scientific Drilling
 RPM Consulting

GL: 2,110'
 KB: 2,135'

Minimum Curvature Method (SPE-3362)

[North and East are positive and South and West are negative]

Vertical Section Azimuth: 91.35

No.	MD	INC	TRUE AZM	TVD	N-S	E-W	SECT	DLS/ 100
72	6677.00	2.41	211.57	6671.51	-103.66	-110.17	-107.70	1.06
73	6770.00	2.82	216.37	6764.42	-107.17	-112.55	-109.99	0.50
74	6863.00	2.73	212.46	6857.31	-110.88	-115.10	-112.45	0.23
75	6957.00	3.13	218.71	6951.18	-114.77	-117.90	-115.17	0.54
76	7050.00	3.18	213.88	7044.04	-118.90	-120.93	-118.09	0.29
77	7143.00	3.19	223.59	7136.90	-122.91	-124.15	-121.22	0.58
78	7237.00	2.70	248.74	7230.78	-125.61	-128.02	-125.02	1.46
79	7330.00	2.71	249.53	7323.68	-127.17	-132.12	-129.09	0.04
80	7424.00	2.57	246.15	7417.58	-128.80	-136.13	-133.06	0.22
81	7517.00	1.97	255.09	7510.50	-130.05	-139.58	-136.48	0.75
82	7611.00	2.10	249.72	7604.44	-131.07	-142.76	-139.63	0.25
83	7704.00	3.18	242.57	7697.34	-132.85	-146.64	-143.47	1.21
84	7798.00	2.91	244.25	7791.21	-135.08	-151.11	-147.88	0.30
85	7891.00	2.74	239.41	7884.10	-137.24	-155.15	-151.87	0.31
86	7985.00	1.96	235.94	7978.02	-139.28	-158.41	-155.09	0.84
87	8078.00	1.52	230.92	8070.98	-140.95	-160.69	-157.32	0.50
88	8171.00	1.44	235.24	8163.95	-142.40	-162.61	-159.21	0.15
89	8265.00	2.04	229.53	8257.90	-144.16	-164.85	-161.41	0.66
90	8358.00	2.01	225.21	8350.84	-146.38	-167.27	-163.77	0.17
91	8451.00	2.57	234.41	8443.77	-148.74	-170.12	-166.57	0.72
92	8544.00	3.68	242.80	8536.63	-151.32	-174.47	-170.86	1.29
93	8638.00	4.30	246.64	8630.40	-154.10	-180.39	-176.71	0.72
94	8731.00	4.39	247.40	8723.13	-156.85	-186.87	-183.13	0.11
95	8824.00	4.07	244.47	8815.88	-159.64	-193.14	-189.32	0.42
96	8918.00	3.95	241.70	8909.65	-162.61	-199.00	-195.11	0.24
97	9011.00	3.82	243.10	9002.44	-165.53	-204.58	-200.63	0.17
98	9105.00	3.49	246.34	9096.25	-168.09	-210.00	-205.98	0.41
99	9198.00	3.01	248.91	9189.10	-170.11	-214.87	-210.80	0.54
100	9292.00	3.59	247.44	9282.94	-172.13	-219.89	-215.77	0.62
101	9386.00	3.30	248.32	9376.77	-174.25	-225.12	-220.95	0.31
102	9481.00	2.90	252.10	9471.63	-176.00	-229.95	-225.74	0.47
103	9575.00	2.31	249.48	9565.53	-177.40	-233.98	-229.74	0.64
104	9670.00	2.13	251.57	9660.46	-178.63	-237.45	-233.18	0.21
105	9765.00	2.30	232.60	9755.39	-180.34	-240.64	-236.32	0.79
106	9860.00	2.46	219.19	9850.31	-183.08	-243.44	-239.06	0.61
107	9954.00	1.65	225.41	9944.25	-185.60	-245.68	-241.24	0.89



Operator: Oasis Petroleum North America, LLC
 Well: Lewis Federal 5300 11-31 3B
 Surface Coordinates: 1,083' FNL & 262' FWL
 Surface Location: Lot 1 Sec. 31, T153N, R100W
 County, State: McKenzie County, ND

Bottom Hole Location: 236.07' S & 10,076.54' E
 of surface location or
 1,346.07' FNL & 184.20' FEL
 Lot 4 Sec. 32, T153N, R100W



Kick-off: 1/13/2019
 Finish: 1/22/2019

Directional Supervision:
 Scientific Drilling
 RPM Consulting

GL: 2,110'
 KB: 2,135'

Minimum Curvature Method (SPE-3362)

[North and East are positive and South and West are negative]

Vertical Section Azimuth: 91.35

No.	MD	INC	TRUE AZM	TVD	N-S	E-W	SECT	DLS/ 100
108	10049.00	0.90	230.90	10039.23	-187.03	-247.23	-242.76	0.80
109	10143.00	0.45	331.38	10133.22	-187.17	-247.98	-243.51	1.15
110	10206.00	2.74	83.57	10196.20	-186.78	-246.61	-242.14	4.67
111	10237.00	6.77	90.40	10227.09	-186.71	-244.04	-239.58	13.10
112	10268.00	11.63	90.87	10257.68	-186.77	-239.09	-234.62	15.68
113	10299.00	15.88	90.19	10287.78	-186.83	-231.72	-227.25	13.72
114	10331.00	19.88	89.68	10318.23	-186.82	-221.90	-217.43	12.51
115	10362.00	24.28	89.89	10346.95	-186.78	-210.25	-205.79	14.20
116	10394.00	27.62	91.43	10375.72	-186.95	-196.25	-191.79	10.65
117	10425.00	27.59	91.40	10403.19	-187.30	-181.89	-177.42	0.11
118	10457.00	30.13	93.17	10431.21	-187.93	-166.46	-161.98	8.37
119	10488.00	32.93	96.17	10457.64	-189.26	-150.31	-145.81	10.35
120	10520.00	32.84	96.43	10484.51	-191.17	-133.04	-128.50	0.52
121	10552.00	35.04	95.64	10511.06	-193.05	-115.27	-110.69	7.01
122	10583.00	37.03	97.85	10536.12	-195.20	-97.17	-92.54	7.67
123	10615.00	36.54	97.65	10561.75	-197.78	-78.18	-73.50	1.58
124	10646.00	38.79	96.62	10586.29	-200.13	-59.38	-54.65	7.54
125	10678.00	41.36	97.13	10610.77	-202.60	-38.93	-34.15	8.10
126	10709.00	41.92	96.37	10633.94	-205.02	-18.48	-13.64	2.43
127	10741.00	45.61	95.14	10657.05	-207.23	3.54	8.42	11.83
128	10772.00	49.29	94.94	10678.01	-209.23	26.29	31.21	11.88
129	10803.00	53.11	94.53	10697.43	-211.22	50.36	55.32	12.37
130	10834.00	57.69	93.17	10715.03	-212.93	75.81	80.81	15.21
131	10865.00	61.64	94.97	10730.68	-214.84	102.49	107.52	13.69
132	10896.00	65.17	95.68	10744.56	-217.41	130.09	135.17	11.57
133	10928.00	68.76	98.02	10757.08	-220.93	159.32	164.48	13.08
134	10959.00	71.32	99.85	10767.66	-225.46	188.10	193.36	9.95
135	10990.00	74.45	100.00	10776.78	-230.56	217.28	222.65	10.11
136	11022.00	78.44	100.05	10784.28	-235.98	247.90	253.40	12.47
137	11053.00	84.04	100.52	10789.00	-241.45	278.04	283.65	18.13
138	11140.00	92.04	96.91	10791.98	-254.60	363.90	369.80	10.09
139	11171.00	90.90	98.88	10791.18	-258.86	394.59	400.58	7.34
140	11202.00	90.64	97.42	10790.76	-263.25	425.28	431.36	4.78
141	11233.00	90.00	96.17	10790.59	-266.92	456.06	462.22	4.53
142	11294.00	89.53	91.98	10790.84	-271.25	516.89	523.14	6.91
143	11325.00	89.46	91.59	10791.11	-272.22	547.87	554.14	1.28



Operator: Oasis Petroleum North America, LLC
 Well: Lewis Federal 5300 11-31 3B
 Surface Coordinates: 1,083' FNL & 262' FWL
 Surface Location: Lot 1 Sec. 31, T153N, R100W
 County, State: McKenzie County, ND

Bottom Hole Location: 236.07' S & 10,076.54' E
 of surface location or
 1,346.07' FNL & 184.20' FEL
 Lot 4 Sec. 32, T153N, R100W



Kick-off: 1/13/2019
 Finish: 1/22/2019

Directional Supervision:
 Scientific Drilling
 RPM Consulting

GL: 2,110'
 KB: 2,135'

Minimum Curvature Method (SPE-3362)

[North and East are positive and South and West are negative]

Vertical Section Azimuth: 91.35

No.	MD	INC	TRUE AZM	TVD	N-S	E-W	SECT	DLS/ 100
144	11356.00	89.93	87.91	10791.28	-272.09	578.87	585.12	11.97
145	11387.00	92.04	87.64	10790.75	-270.88	609.84	616.05	6.86
146	11418.00	92.28	87.55	10789.58	-269.58	640.79	646.96	0.83
147	11510.00	92.84	86.34	10785.47	-264.68	732.56	738.60	1.45
148	11602.00	93.72	86.03	10780.20	-258.57	824.21	830.07	1.01
149	11633.00	91.61	88.55	10778.76	-257.11	855.14	860.96	10.60
150	11664.00	91.68	87.64	10777.87	-256.08	886.11	891.89	2.94
151	11695.00	90.37	89.51	10777.32	-255.31	917.09	922.85	7.36
152	11787.00	89.30	89.22	10777.58	-254.29	1009.08	1014.79	1.21
153	11878.00	87.72	89.73	10779.95	-253.46	1100.04	1105.71	1.82
154	11970.00	87.92	89.25	10783.45	-252.64	1191.97	1197.60	0.56
155	12062.00	90.10	89.12	10785.04	-251.33	1283.95	1289.51	2.37
156	12154.00	90.37	88.95	10784.66	-249.78	1375.93	1381.43	0.35
157	12246.00	90.57	89.59	10783.91	-248.61	1467.92	1473.37	0.73
158	12339.00	90.27	89.56	10783.23	-247.92	1560.92	1566.32	0.32
159	12433.00	88.73	89.72	10784.05	-247.33	1654.91	1660.27	1.65
160	12528.00	89.50	90.28	10785.51	-247.33	1749.89	1755.24	1.00
161	12623.00	88.76	90.69	10786.96	-248.13	1844.88	1850.21	0.89
162	12718.00	89.53	90.18	10788.37	-248.85	1939.87	1945.19	0.97
163	12814.00	89.30	91.14	10789.35	-249.96	2035.85	2041.18	1.03
164	12909.00	89.33	89.97	10790.49	-250.88	2130.84	2136.16	1.23
165	13003.00	89.93	90.04	10791.10	-250.89	2224.84	2230.13	0.64
166	13098.00	89.06	90.59	10791.93	-251.41	2319.83	2325.11	1.08
167	13193.00	90.10	90.58	10792.63	-252.38	2414.82	2420.10	1.09
168	13288.00	88.39	90.30	10793.88	-253.11	2509.81	2515.07	1.82
169	13383.00	90.07	90.03	10795.16	-253.38	2604.80	2610.04	1.79
170	13478.00	87.79	89.86	10796.93	-253.29	2699.77	2704.99	2.41
171	13573.00	87.99	89.44	10800.43	-252.71	2794.71	2799.88	0.49
172	13668.00	90.00	87.91	10802.10	-250.51	2889.66	2894.76	2.66
173	13762.00	91.37	88.72	10800.97	-247.75	2983.61	2988.62	1.69
174	13857.00	89.33	89.90	10800.39	-246.61	3078.59	3083.55	2.48
175	13952.00	87.82	89.43	10802.75	-246.05	3173.56	3178.47	1.66
176	14046.00	88.29	90.74	10805.94	-246.19	3267.50	3272.39	1.48
177	14141.00	90.10	88.73	10807.28	-245.75	3362.48	3367.34	2.85
178	14235.00	91.31	90.00	10806.12	-244.71	3456.47	3461.27	1.87
179	14330.00	89.43	91.47	10805.51	-245.93	3551.45	3556.26	2.51



Operator: Oasis Petroleum North America, LLC

Well: Lewis Federal 5300 11-31 3B

Surface Coordinates: 1,083' FNL & 262' FWL

Surface Location: Lot 1 Sec. 31, T153N, R100W

County, State: McKenzie County, ND

Bottom Hole Location: 236.07' S & 10,076.54' E
of surface location or1,346.07' FNL & 184.20' FEL
Lot 4 Sec. 32, T153N, R100W

Kick-off: 1/13/2019

Finish: 1/22/2019

Directional Supervision:

Scientific Drilling

RPM Consulting

GL: 2,110'

KB: 2,135'

Minimum Curvature Method (SPE-3362)

[North and East are positive and South and West are negative]

Vertical Section Azimuth: 91.35

No.	MD	INC	TRUE AZM	TVD	N-S	E-W	SECT	DLS/ 100
180	14424.00	86.75	90.19	10808.64	-247.29	3645.38	3650.19	3.16
181	14519.00	87.25	90.72	10813.61	-248.04	3740.24	3745.05	0.77
182	14582.00	86.45	89.90	10817.08	-248.38	3803.15	3807.94	1.82
183	14614.00	86.61	90.39	10819.01	-248.46	3835.09	3839.88	1.61
184	14645.00	88.69	89.94	10820.28	-248.55	3866.06	3870.84	6.86
185	14676.00	91.21	89.82	10820.31	-248.49	3897.06	3901.83	8.14
186	14708.00	91.67	89.55	10819.51	-248.31	3929.05	3933.81	1.67
187	14803.00	91.78	89.68	10816.65	-247.67	4024.00	4028.72	0.18
188	14897.00	92.41	90.42	10813.21	-247.76	4117.94	4122.63	1.03
189	14992.00	91.37	89.76	10810.08	-247.91	4212.88	4217.55	1.30
190	15087.00	92.65	90.12	10806.75	-247.81	4307.82	4312.46	1.40
191	15181.00	87.02	91.74	10807.02	-249.33	4401.77	4406.42	6.23
192	15276.00	87.32	90.77	10811.71	-251.41	4496.63	4501.30	1.07
193	15370.00	90.67	91.48	10813.36	-253.25	4590.58	4595.27	3.64
194	15464.00	89.93	90.76	10812.86	-255.09	4684.56	4689.27	1.10
195	15558.00	89.16	91.32	10813.61	-256.80	4778.54	4783.27	1.01
196	15653.00	87.96	89.35	10816.00	-257.35	4873.50	4878.21	2.43
197	15747.00	86.65	88.69	10820.42	-255.75	4967.38	4972.03	1.56
198	15842.00	88.96	87.93	10824.06	-252.95	5062.27	5066.82	2.56
199	15937.00	90.20	89.50	10824.75	-250.82	5157.23	5161.71	2.11
200	16032.00	89.36	89.89	10825.12	-250.31	5252.23	5256.67	0.97
201	16127.00	89.40	90.23	10826.14	-250.41	5347.23	5351.64	0.36
202	16222.00	88.86	89.42	10827.59	-250.12	5442.21	5446.59	1.02
203	16316.00	88.46	88.91	10829.78	-248.75	5536.18	5540.50	0.69
204	16411.00	89.63	87.54	10831.37	-245.81	5631.11	5635.34	1.90
205	16505.00	91.00	89.29	10830.85	-243.21	5725.07	5729.21	2.36
206	16600.00	90.44	90.38	10829.66	-242.94	5820.06	5824.17	1.29
207	16663.00	90.60	90.36	10829.09	-243.34	5883.06	5887.16	0.26
208	16695.00	88.86	90.40	10829.24	-243.55	5915.05	5919.15	5.44
209	16789.00	89.43	90.08	10830.64	-243.95	6009.04	6013.12	0.70
210	16883.00	90.70	91.92	10830.53	-245.59	6103.02	6107.11	2.38
211	16914.00	91.11	92.10	10830.04	-246.68	6134.00	6138.11	1.44
212	16946.00	92.35	92.31	10829.08	-247.91	6165.96	6170.09	3.93
213	16978.00	91.41	93.40	10828.03	-249.50	6197.90	6202.06	4.50
214	17009.00	88.16	93.24	10828.14	-251.29	6228.85	6233.04	10.50
215	17041.00	87.92	92.76	10829.24	-252.97	6260.78	6265.01	1.68



Operator: Oasis Petroleum North America, LLC
 Well: Lewis Federal 5300 11-31 3B
 Surface Coordinates: 1,083' FNL & 262' FWL
 Surface Location: Lot 1 Sec. 31, T153N, R100W
 County, State: McKenzie County, ND

Bottom Hole Location: 236.07' S & 10,076.54' E
 of surface location or
 1,346.07' FNL & 184.20' FEL
 Lot 4 Sec. 32, T153N, R100W



Kick-off: 1/13/2019
 Finish: 1/22/2019

Directional Supervision:
 Scientific Drilling
 RPM Consulting

GL: 2,110'
 KB: 2,135'

Minimum Curvature Method (SPE-3362)

[North and East are positive and South and West are negative]

Vertical Section Azimuth: 91.35

No.	MD	INC	TRUE AZM	TVD	N-S	E-W	SECT	DLS/ 100
216	17073.00	88.63	93.09	10830.20	-254.60	6292.73	6296.98	2.45
217	17104.00	87.39	93.63	10831.28	-256.42	6323.65	6327.94	4.36
218	17136.00	86.48	92.68	10832.99	-258.18	6355.56	6359.88	4.11
219	17167.00	85.34	93.25	10835.20	-259.77	6386.44	6390.79	4.11
220	17199.00	84.02	92.71	10838.17	-261.43	6418.26	6422.63	4.45
221	17262.00	84.60	89.16	10844.41	-262.45	6480.93	6485.31	5.68
222	17293.00	86.21	88.42	10846.90	-261.80	6511.82	6516.18	5.71
223	17325.00	88.79	88.17	10848.29	-260.85	6543.77	6548.10	8.10
224	17356.00	89.20	88.24	10848.84	-259.88	6574.75	6579.05	1.34
225	17451.00	91.34	88.19	10848.39	-256.92	6669.70	6673.90	2.25
226	17545.00	91.14	89.42	10846.36	-254.96	6763.66	6767.79	1.33
227	17640.00	90.27	89.59	10845.19	-254.14	6858.64	6862.73	0.93
228	17735.00	89.60	88.89	10845.29	-252.88	6953.63	6957.66	1.02
229	17829.00	90.13	88.63	10845.52	-250.84	7047.61	7051.57	0.63
230	17924.00	91.10	88.38	10844.50	-248.37	7142.57	7146.44	1.05
231	17975.00	89.16	88.76	10844.38	-247.09	7193.55	7197.38	3.88
232	18018.00	89.43	88.11	10844.91	-245.92	7236.53	7240.32	1.64
233	18112.00	89.90	89.22	10845.46	-243.73	7330.51	7334.21	1.28
234	18208.00	88.66	90.30	10846.67	-243.33	7426.49	7430.17	1.71
235	18302.00	88.42	89.22	10849.06	-242.93	7520.46	7524.10	1.18
236	18397.00	89.06	88.87	10851.15	-241.35	7615.42	7619.00	0.77
237	18491.00	89.77	88.38	10852.11	-239.09	7709.39	7712.88	0.92
238	18585.00	90.50	88.02	10851.89	-236.14	7803.34	7806.74	0.87
239	18680.00	89.43	88.55	10851.94	-233.30	7898.30	7901.60	1.26
240	18774.00	89.16	88.89	10853.10	-231.20	7992.27	7995.50	0.46
241	18870.00	89.43	89.01	10854.28	-229.44	8088.25	8091.41	0.31
242	18964.00	88.69	90.13	10855.82	-228.74	8182.23	8185.35	1.43
243	19058.00	89.80	94.14	10857.06	-232.24	8276.13	8279.31	4.43
244	19152.00	90.44	93.70	10856.87	-238.66	8369.91	8373.21	0.83
245	19247.00	91.31	93.38	10855.42	-244.53	8464.72	8468.13	0.98
246	19341.00	89.56	94.47	10854.70	-250.96	8558.49	8562.03	2.19
247	19436.00	89.56	93.58	10855.43	-257.63	8653.25	8656.92	0.94
248	19531.00	89.33	92.96	10856.35	-263.05	8748.09	8751.86	0.70
249	19626.00	88.69	91.72	10857.99	-266.93	8843.00	8846.83	1.47
250	19721.00	89.36	91.11	10859.61	-269.27	8937.95	8941.82	0.95
251	19816.00	90.23	90.89	10859.95	-270.93	9032.94	9036.81	0.94



Operator: Oasis Petroleum North America, LLC
 Well: Lewis Federal 5300 11-31 3B
 Surface Coordinates: 1,083' FNL & 262' FWL
 Surface Location: Lot 1 Sec. 31, T153N, R100W
 County, State: McKenzie County, ND

Bottom Hole Location: 236.07' S & 10,076.54' E
 of surface location or
 1,346.07' FNL & 184.20' FEL
 Lot 4 Sec. 32, T153N, R100W



Kick-off: 1/13/2019
 Finish: 1/22/2019

Directional Supervision:
 Scientific Drilling
 RPM Consulting

GL: 2,110'
 KB: 2,135'

Minimum Curvature Method (SPE-3362)

[North and East are positive and South and West are negative]

Vertical Section Azimuth: 91.35

No.	MD	INC	TRUE AZM	TVD	N-S	E-W	SECT	DLS/ 100
252	19911.00	90.67	90.06	10859.20	-271.72	9127.93	9131.80	0.99
253	20006.00	90.07	90.17	10858.59	-271.91	9222.93	9226.77	0.64
254	20101.00	88.93	91.08	10859.42	-272.94	9317.92	9321.76	1.54
255	20195.00	89.26	90.71	10860.90	-274.41	9411.89	9415.75	0.53
256	20290.00	89.23	89.67	10862.16	-274.73	9506.88	9510.72	1.10
257	20385.00	90.00	88.75	10862.79	-273.42	9601.87	9605.65	1.26
258	20479.00	91.11	88.62	10861.88	-271.26	9695.84	9699.54	1.19
259	20574.00	91.74	87.72	10859.52	-268.23	9790.76	9794.36	1.16
260	20669.00	90.47	87.80	10857.69	-264.52	9885.67	9889.16	1.34
261	20763.00	89.67	89.90	10857.57	-262.63	9979.64	9983.06	2.39
262	20785.00	87.15	90.31	10858.18	-262.67	10001.63	10005.05	11.61
PTB	20860.00	87.15	90.31	10861.91	-263.07	10076.54	10079.94	0.00



FORMATION TOPS & STRUCTURAL RELATIONSHIPS

Operator: Well Name: Location: Elevation:	Subject Well:								Offset Wells:		
	Oasis Petroleum North America, LLC Lewis Federal 5300 11-31 3B 1,083' FNL & 262' FWL Lot 1 Section 31, T153N, R100W GL: 2,110' Sub: 25' KB: 2,135'								Dip To Wade Federal 5300 41-30 8T2	Dip To Lewis Federal 5300 31-31H	Dip To Gramma Federal 5300 41-31T
Formation/ Marker	Prog. Top	Prog. Datum (MSL)	Driller's Depth Top (MD)	Driller's Depth Top (TVD)	Datum (MSL)	Interval Thickness	Thickness to Target	Dip To Prog.			
Pierre	1,985'	150'	1,929'	1,929'	206'	2,697'	8,847'	56'	-	24'	-
Greenhorn	4,617'	-2,482'	4,628'	4,626'	-2,491'	401'	6,150'	-9'	4'	-33'	-
Mowry (Dakota Group)	5,025'	-2,890'	5,029'	5,027'	-2,892'	399'	5,749'	-2'	7'	-19'	-
Inyan Kara (Dakota Group)	5,449'	-3,314'	5,429'	5,426'	-3,291'	482'	5,350'	23'	38'	21'	-
Swift (Base Dakota Group)	5,895'	-3,760'	5,912'	5,908'	-3,773'	535'	4,868'	-13'	17'	-34'	-
Rierdon	6,414'	-4,279'	6,448'	6,443'	-4,308'	486'	4,333'	-29'	31'	-91'	-
Dunham Salt	6,927'	-4,792'	6,935'	6,929'	-4,794'	49'	3,847'	-2'	30'	-89'	-
Dunham Salt Base	6,963'	-4,828'	6,984'	6,978'	-4,843'	284'	3,798'	-15'	37'	-25'	-
Pine Salt	7,259'	-5,124'	7,268'	7,262'	-5,127'	51'	3,514'	-3'	6'	0'	-
Pine Salt Base	7,318'	-5,183'	7,319'	7,313'	-5,178'	123'	3,463'	5'	-12'	19'	-
Opeche Salt	7,437'	-5,302'	7,442'	7,436'	-5,301'	15'	3,340'	1'	-15'	4'	-
Opeche Salt Base	7,462'	-5,327'	7,457'	7,451'	-5,316'	242'	3,325'	11'	-2'	19'	-
Amsden	7,658'	-5,523'	7,700'	7,693'	-5,558'	143'	3,083'	-35'	-17'	10'	-
Tyler	7,843'	-5,708'	7,843'	7,836'	-5,701'	230'	2,940'	7'	-3'	20'	-
Otter/Base Minnelusa	8,060'	-5,925'	8,073'	8,066'	-5,931'	344'	2,710'	-6'	-6'	17'	-
Kibbey "Lime"	8,412'	-6,277'	8,417'	8,410'	-6,275'	150'	2,366'	2'	-3'	15'	5'
Charles Salt	8,558'	-6,423'	8,567'	8,560'	-6,425'	681'	2,216'	-2'	-10'	15'	3'
Base Last Salt	9,240'	-7,105'	9,250'	9,241'	-7,106'	205'	1,535'	-1'	-7'	4'	-11'
Mission Canyon	9,446'	-7,311'	9,455'	9,446'	-7,311'	569'	1,330'	0'	2'	7'	-3'
Lodgepole	10,012'	-7,877'	10,025'	10,015'	-7,880'	78'	761'	-3'	1'	0'	-11'
Lodgepole A	10,085'	-7,950'	10,103'	10,093'	-7,958'	53'	683'	-8'	-1'	-6'	-18'
Lodgepole B	10,159'	-8,024'	10,156'	10,146'	-8,011'	70'	630'	13'	7'	12'	5'
Lodgepole C	10,209'	-8,074'	10,226'	10,216'	-8,081'	194'	560'	-7'	-3'	-8'	-25'
Lodgepole D	10,403'	-8,268'	10,433'	10,410'	-8,275'	171'	366'	-7'	-2'	-15'	-10'
Lodgepole E	10,539'	-8,404'	10,640'	10,581'	-8,446'	71'	195'	-42'	-7'	-14'	-14'
Lodgepole F	10,641'	-8,506'	10,733'	10,652'	-8,517'	83'	124'	-11'	-7'	-15'	-13'
False Bakken	10,719'	-8,584'	10,874'	10,735'	-8,600'	10'	41'	-16'	-10'	-16'	-12'
Upper Bakken	10,727'	-8,592'	10,897'	10,745'	-8,610'	13'	31'	-18'	-11'	-13'	-13'
Middle Bakken	10,743'	-8,608'	10,931'	10,758'	-8,623'	10'	18'	-15'	-8'	-11'	-9'
Target Top	10,755'	-8,620'	10,961'	10,768'	-8,633'	8'	8'	-13'	-6'	-9'	-7'
Target Landing	10,764'	-8,629'	10,988'	10,776'	-8,641'	8'	0'	-12'	-5'	-8'	-6'
Target Base	10,773'	-8,638'	11,021'	10,784'	-8,649'	-	-8'	-11'	-4'	-7'	-5'
Lower Bakken	10,783'	-8,648'	-	-	-	-	-	-	-	-	-

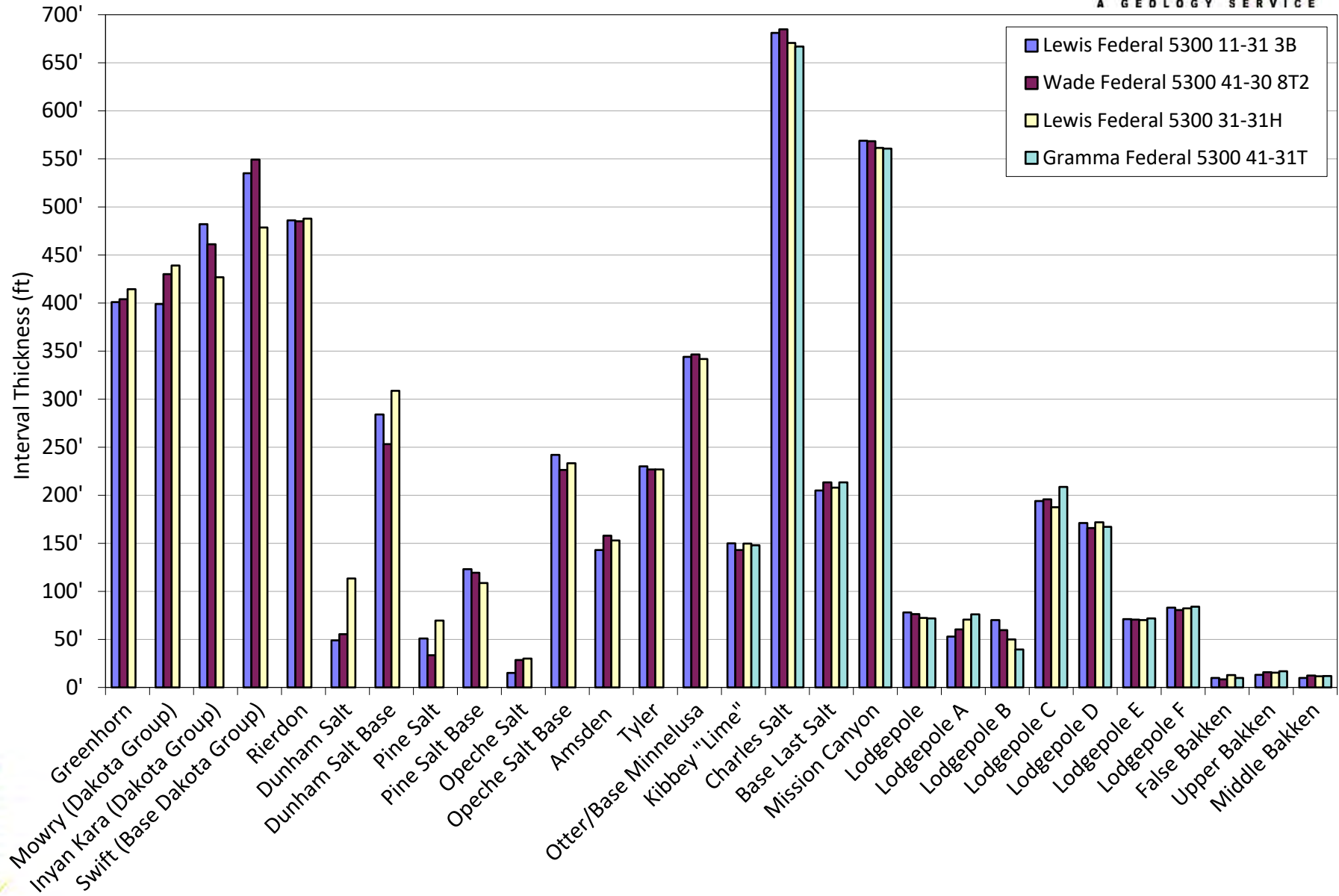
CONTROL DATA

Operator:	Oasis Petroleum North America, LLC				Oasis Petroleum North America, LLC				Oasis Petroleum North America, LLC			
Well Name:	Wade Federal 5300 41-30 8T2				Lewis Federal 5300 31-31H				Gramma Federal 5300 41-31T			
Location:	Lot 4 Sec. 30, T153N, R100W McKenzie County, ND 0.4 miles N of subject well				Lot 3 Sec. 31, T153N, R100W McKenzie County, ND 0.5 miles S of subject well				SW SW Sec. 31, T153N, R100W McKenzie County, ND 0.7 miles S of subject well			
Elevation:	KB: 2,077' NDIC: 28558				KB: 2,185' NDIC: 20314				KB: 2,182' NDIC: 23350			
Formation/ Zone	Driller's (TVD)	Datum (MSL)	Interval Thickness	Thickness to Target	Driller's (TVD)	Datum (MSL)	Interval Thickness	Thickness to Target	E-Log (TVD)	Datum (MSL)	Interval Thickness	Thickness to Target
Pierre	-	-	-	-	2,003'	182'	2,640'	8,815'	-	-	-	-
Greenhorn	4,572'	-2,495'	404'	6,141'	4,643'	-2,458'	414'	6,175'	-	-	-	-
Mowry (Dakota Group)	4,976'	-2,899'	430'	5,737'	5,058'	-2,873'	439'	5,760'	-	-	-	-
Inyan Kara (Dakota Group)	5,406'	-3,329'	461'	5,307'	5,497'	-3,312'	427'	5,321'	-	-	-	-
Swift (Base Dakota Group)	5,867'	-3,790'	549'	4,846'	5,924'	-3,739'	478'	4,894'	-	-	-	-
Rierdon	6,416'	-4,339'	485'	4,297'	6,402'	-4,217'	488'	4,416'	-	-	-	-
Dunham Salt	6,901'	-4,824'	55'	3,812'	6,890'	-4,705'	113'	3,928'	-	-	-	-
Dunham Salt Base	6,957'	-4,880'	253'	3,756'	7,003'	-4,818'	309'	3,815'	-	-	-	-
Pine Salt	7,210'	-5,133'	33'	3,503'	7,312'	-5,127'	70'	3,506'	-	-	-	-
Pine Salt Base	7,243'	-5,166'	119'	3,470'	7,382'	-5,197'	109'	3,436'	-	-	-	-
Opeche Salt	7,363'	-5,286'	29'	3,350'	7,490'	-5,305'	30'	3,328'	-	-	-	-
Opeche Salt Base	7,391'	-5,314'	226'	3,322'	7,520'	-5,335'	233'	3,298'	-	-	-	-
Amsden	7,618'	-5,541'	158'	3,095'	7,753'	-5,568'	153'	3,065'	-	-	-	-
Tyler	7,775'	-5,698'	227'	2,938'	7,906'	-5,721'	227'	2,912'	-	-	-	-
Otter/Base Minnelusa	8,002'	-5,925'	346'	2,711'	8,133'	-5,948'	342'	2,685'	-	-	-	-
Kibbey "Lime"	8,349'	-6,272'	143'	2,364'	8,475'	-6,290'	150'	2,343'	8,462'	-6,280'	148'	2,355'
Charles Salt	8,492'	-6,415'	685'	2,221'	8,625'	-6,440'	671'	2,193'	8,610'	-6,428'	667'	2,207'
Base Last Salt	9,176'	-7,099'	213'	1,537'	9,295'	-7,110'	208'	1,523'	9,277'	-7,095'	214'	1,540'
Mission Canyon	9,390'	-7,313'	568'	1,323'	9,503'	-7,318'	561'	1,315'	9,490'	-7,308'	561'	1,327'
Lodgepole	9,958'	-7,881'	76'	755'	10,065'	-7,880'	72'	753'	10,051'	-7,869'	72'	766'
Lodgepole A	10,034'	-7,957'	60'	679'	10,137'	-7,952'	71'	681'	10,122'	-7,940'	76'	695'
Lodgepole B	10,095'	-8,018'	60'	618'	10,208'	-8,023'	50'	610'	10,198'	-8,016'	39'	619'
Lodgepole C	10,155'	-8,078'	196'	558'	10,258'	-8,073'	187'	560'	10,238'	-8,056'	209'	579'
Lodgepole D	10,350'	-8,273'	166'	363'	10,445'	-8,260'	172'	373'	10,447'	-8,265'	167'	370'
Lodgepole E	10,516'	-8,439'	71'	197'	10,617'	-8,432'	70'	201'	10,614'	-8,432'	72'	203'
Lodgepole F	10,587'	-8,510'	80'	126'	10,687'	-8,502'	82'	131'	10,686'	-8,504'	84'	131'
False Bakken	10,667'	-8,590'	8'	46'	10,769'	-8,584'	13'	49'	10,770'	-8,588'	10'	47'
Upper Bakken	10,676'	-8,599'	16'	37'	10,782'	-8,597'	15'	36'	10,779'	-8,597'	17'	38'
Middle Bakken	10,692'	-8,615'	12'	21'	10,797'	-8,612'	12'	21'	10,796'	-8,614'	12'	21'
Target Top	10,704'	-8,627'	9'	9'	10,809'	-8,624'	9'	9'	10,808'	-8,626'	9'	9'
Target Landing	10,713'	-8,636'	9'	0'	10,818'	-8,633'	9'	0'	10,817'	-8,635'	9'	0'
Target Base	10,722'	-8,645'	14'	-9'	10,827'	-8,642'	11'	-9'	10,826'	-8,644'	7'	-9'
Lower Bakken	10,736'	-8,659'	-	-23'	10,838'	-8,653'	-	-20'	10,833'	-8,651'	-	-16'

Projected Depth

INTERVAL THICKNESS

Oasis Petroleum North America, LLC - Lewis Federal 5300 11-31 3B



LITHOLOGY

Oasis Petroleum North America, LLC

Lewis Federal 5300 11-31 3B

Sunburst geologists caught 20' sample intervals from 3,470'-4,780'; 30' sample intervals from 8,200' to 11,110'; and 50' sample intervals to the TD of the lateral at 20,860'. 10' spot samples were caught through the vertical, curve, and lateral as needed. Formation tops and lithologic markers have been inserted into the sample descriptions below for reference. Samples were examined wet and dry under a binocular microscope. Sample fluorescent cuts are masked by invert mud through intermediate casing. Quantifiers in order of increasing abundance are trace, rare, occasional, common and abundant.

Vertical Log Descriptions:

MD / TVD (MSL Datum)

Drilling in the Pierre Formation [Upper Montana Group]

3,470-3,480 SHALE: gray, firm, blocky, earthy, no visible porosity, no visible oil stain

4,580-4,600 SHALE: light-medium gray, firm, occasional soft, sub platy, rare sub blocky, earthy texture, trace disseminated pyrite, no visible porosity, no visible oil stain

4,600-4,620 SHALE: light-medium gray, firm, occasional soft, sub platy, rare sub blocky, earthy texture, trace disseminated pyrite, no visible porosity, no visible oil stain

Greenhorn [Upper Colorado Group]

4,628' MD / 4,626' TVD (-2,491')

4,620-4,640 SHALE: medium gray, firm, rare soft, sub platy, rare sub blocky, earthy texture, trace disseminated pyrite, no visible porosity, no visible oil stain; trace LIMESTONE: mudstone, light gray, cream, rare medium gray, trace dark gray, microcrystalline, earthy texture, firm, occasional friable, earthy texture, calcareous, trace intercrystalline porosity, no visible oil stain

4,640-4,660 SHALE: medium-dark gray, friable, occasional firm, sub platy, common sub blocky, earthy texture, trace disseminated pyrite, no visible porosity, no visible oil stain; trace LIMESTONE: mudstone, off white, cream, light gray, microcrystalline, earthy texture, firm, occasional friable, earthy texture, calcareous, trace intercrystalline porosity, no visible oil stain

4,660-4,680 SHALE: medium gray, friable, sub platy, common sub blocky, earthy texture, no visible porosity, no visible oil stain; very trace LIMESTONE: mudstone, cream, light-medium gray, microcrystalline, earthy texture, firm, trace friable, earthy texture, calcareous, trace intercrystalline porosity, no visible oil stain

4,680-4,700 SHALE: medium gray, firm, sub platy, rare sub blocky, earthy texture, no visible porosity, no visible oil stain; very trace LIMESTONE: mudstone, light gray, microcrystalline, earthy texture, firm, trace friable, earthy texture, calcareous, occasional disseminated pyrite, possible intercrystalline porosity, no visible oil stain

4,700-4,720 SHALE: light-medium gray, firm, occasional friable, trace hard, sub platy, rare sub blocky, earthy texture, trace disseminated pyrite, no visible porosity, no visible oil stain

4,720-4,740 SHALE: medium gray, firm, sub platy, rare sub blocky, earthy texture, no visible porosity, no visible oil stain; trace LIMESTONE: mudstone, off white, cream, microcrystalline, earthy texture, firm, earthy texture, calcareous, rare disseminated pyrite, possible intercrystalline porosity, no visible oil stain

4,740-4,760 SHALE: light-medium gray, firm, sub platy, rare sub blocky, earthy texture, no visible porosity, no visible oil stain; trace LIMESTONE: mudstone, cream, occasional light gray, rare light-medium brown gray, microcrystalline, earthy texture, firm, rare friable, earthy texture, calcareous, rare disseminated pyrite, possible intercrystalline porosity, no visible oil stain

4,760-4,780 SHALE: medium gray, firm, sub platy, common sub blocky, earthy texture, trace disseminated pyrite, no visible porosity, no visible oil stain

Drilling in the Otter Formation [Mississippian Big Snowy Group]

8,200-8,230 SILTSTONE: light red orange, gray, firm, sub blocky, calcite cement, well cemented, no visible porosity, no visible oil stain; common SILTY SANDSTONE: gray, trace off white, very fine-fine grained, sub rounded, moderately sorted, calcite cement, poorly cemented, no visible porosity, no visible oil stain

8,230-8,260 SILTSTONE: light red orange, gray, firm, sub blocky, calcite cement, well cemented, no visible porosity, no visible oil stain; common SILTY SANDSTONE: gray-light gray, trace off white, very fine-fine grained, sub rounded, moderately sorted, calcite cement, poorly cemented, no visible porosity, no visible oil stain

8,260-8,290 SILTSTONE: light red orange, gray, firm, sub blocky, calcite cement, well cemented, no visible porosity, no visible oil stain; common SILTY SANDSTONE: light gray-gray, trace off white, very fine-grained, sub rounded, moderately sorted, calcite cement, poorly cemented, no visible porosity, no visible oil stain

8,290-8,320 SILTSTONE: light red orange, gray, firm, sub blocky, calcite cement, well cemented, no visible porosity, no visible oil stain; common SILTY SANDSTONE: gray, trace off white, very fine-grained, sub rounded, moderately sorted, calcite cement, poorly cemented, no visible porosity, no visible oil stain

8,320-8,350 SILTSTONE: light red orange, gray, firm, sub blocky, calcite cement, well cemented, no visible porosity, no visible oil stain; common SILTY SANDSTONE: gray, trace off white, very fine-grained, sub rounded, moderately sorted, calcite cement, poorly cemented, no visible porosity, no visible oil stain

8,350-8,380 SILTSTONE: light red orange, gray, firm, sub blocky, calcite cement, well cemented, no visible porosity, no visible oil stain; common SILTY SANDSTONE: gray, trace off white, very fine-grained, sub rounded, moderately sorted, calcite cement, poorly cemented, no visible porosity, no visible oil stain

8,380-8,410 No sample

Kibbey "Lime" [Mississippian Big Snowy Group]

8,417' MD / 8,410' TVD (-6,275')

8,410-8,440 LIMESTONE: mudstone, light-medium gray, common cream, trace dark gray, firm, rare friable, microcrystalline, dense, earthy texture, no visible porosity

8,440-8,470 SILTSTONE: dark orange, firm, sub blocky, calcite cement, poorly-moderately cemented, no visible porosity; rare SILTY SANDSTONE: light gray, cream, trace medium gray, very fine grained, firm, calcareous, poorly-moderately cemented, no visible porosity

8,470-8,500 SILTSTONE: dark orange, firm, trace friable, sub blocky, calcite cement, poorly-moderately cemented, no visible porosity; rare SILTY SANDSTONE: light-medium gray, very fine grained, friable, calcareous, poorly-moderately cemented, no visible porosity, no visible oil stain; trace ANHYDRITE off white, microcrystalline, soft, chalky, massive, anhedral, amorphous; no visible porosity, no visible oil stain

8,500-8,530 SILTSTONE: red brown, dark rounded, firm, sub blocky, calcite cement, poorly-moderately cemented, no visible porosity; trace SILTY SANDSTONE: light-medium gray, very fine grained, friable, calcareous, poorly-moderately cemented, no visible porosity, no visible oil stain

8,530-8,560 SILTSTONE: orange brown, soft, blocky-sub platy, calcite cement, well sorted, no visible porosity

Charles Formation [Mississippian Madison Group]

8,567' MD / 8,560' TVD (-6,425')

8,560-8,590 LIMESTONE: mudstone, cream, light gray, microcrystalline, firm, earthy texture, no visible porosity, no visible oil stain; rare SALT: translucent, common milky white, firm, hard, crystalline, massive, anhedral, no visible porosity

8,590-8,620 SALT: translucent, common milky white, firm, hard, crystalline, massive, anhedral, no visible porosity; rare LIMESTONE: mudstone, light-medium gray, common tan, microcrystalline, firm, trace friable, earthy texture, no visible porosity, no visible oil stain

8,620-8,650 SALT: translucent, common milky white, firm, hard, crystalline, massive, anhedral, no visible porosity

8,650-8,680 SALT: translucent, common milky white, firm, hard, crystalline, massive, anhedral, no visible porosity; rare LIMESTONE: mudstone, cream, rare light-medium gray, trace light brown gray, microcrystalline, firm, earthy texture, no visible porosity, no visible oil stain

8,680-8,710 SALT: milky white, rare translucent firm, hard, crystalline, massive, anhedral, no visible porosity

8,710-8,740 SALT: translucent, common milky white, firm, hard, crystalline, massive, anhedral, no visible porosity

8,740-8,770 SALT: translucent, common milky white, firm, hard, crystalline, massive, anhedral, no visible porosity; rare LIMESTONE: mudstone, light-medium gray, common tan, microcrystalline, firm, trace friable, earthy texture, no visible porosity, no visible oil stain

8,770-8,800 ARGILLACEOUS LIMESTONE: mudstone, medium-dark gray, light gray, rare cream, microcrystalline, firm, earthy texture, occasional crystalline texture, no visible porosity; occasional SALT: translucent-transparent, common milky white, firm, hard, crystalline, massive, anhedral, no visible porosity

8,800-8,830 SALT: translucent-transparent, common milky white, firm, hard, crystalline, massive, anhedral, no visible porosity; ARGILLACEOUS LIMESTONE: mudstone, medium-dark gray, light gray, rare cream, microcrystalline, firm, earthy texture, occasional crystalline texture, no visible porosity

8,830-8,860 SALT: translucent-transparent, common milky white, firm, hard, crystalline, massive, anhedral, no visible porosity; ARGILLACEOUS LIMESTONE: mudstone, medium-dark gray, light gray, rare cream, microcrystalline, firm, earthy texture, occasional crystalline texture, no visible porosity

8,860-8,890 LIMESTONE: mudstone, medium-light gray, light brown gray, tan, earthy texture, firm, dolomitic in part, argillaceous in part, no visible porosity; rare ANHYDRITE: milky white, cream, rare light gray, microcrystalline, soft, amorphous, no visible porosity; rare SALT: as above

8,890-8,920 SALT: translucent-transparent, common milky white, firm, hard, crystalline, massive, anhedral, no visible porosity; LIMESTONE: mudstone, medium gray, light gray, earthy texture, firm, argillaceous in part, no visible porosity; rare ANHYDRITE: milky white, cream, rare light gray, microcrystalline, soft, amorphous, no visible porosity

8,920-8,950 SALT: translucent-transparent, rare milky white, firm, hard, crystalline, massive, anhedral, no visible porosity; very trace ANHYDRITE: as above

8,950-8,980 SALT: translucent-transparent, common milky white, firm, hard, crystalline, massive, anhedral, no visible porosity; trace LIMESTONE: mudstone, medium gray, light gray, earthy texture, firm, argillaceous in part, no visible porosity; very trace ANHYDRITE: milky white, cream, rare light gray, microcrystalline, soft, amorphous, no visible porosity

8,980-9,010 ARGILLACEOUS LIMESTONE: mudstone, light brown gray-light gray, medium gray, cream, microcrystalline, firm, dense, earthy texture, rare crystalline texture, no visible porosity; DOLOMITE: mudstone, tan, light brown gray rare light-medium gray, friable, common firm, crystalline texture, no visible porosity, no visible oil stain; SALT: translucent-transparent, white-clear, firm, hard, crystalline, massive, anhedral, no visible porosity

9,010-9,040 DOLOMITE: mudstone, tan, light brown gray rare light-medium gray, friable, common firm, crystalline texture, no visible porosity, no visible oil stain; ANHYDRITE: milky white, microcrystalline, soft, amorphous, no visible porosity, no visible oil stain; rare SALT: as above; rare ARGILLACEOUS LIMESTONE: as above

9,040-9,070 SALT: translucent-transparent, white-clear, firm, hard, crystalline, massive, anhedral, no visible porosity; rare ARGILLACEOUS LIMESTONE: mudstone, light brown gray-light gray, cream, microcrystalline, firm, dense, earthy texture, rare crystalline texture, no visible porosity; trace ANHYDRITE: milky white, cream, rare light gray, microcrystalline, soft, amorphous, no visible porosity

9,070-9,100 LIMESTONE: mudstone, medium-light gray, light brown gray, tan, earthy texture, friable, common firm, no visible porosity; occasional ANHYDRITE: milky white, cream, rare light gray, microcrystalline, soft, amorphous, no visible porosity; rare SALT: translucent-transparent, white-clear, firm, hard, crystalline, massive, anhedral, no visible porosity

9,100-9,130 LIMESTONE: mudstone, light brown gray, gray, tan, earthy-crystalline texture, friable-firm, no visible porosity; occasional ANHYDRITE: milky white, cream, rare light gray, microcrystalline, soft, amorphous, no visible porosity

9,130-9,160 LIMESTONE: mudstone, tan, cream, light-medium gray, microcrystalline, firm, dense-banded, crystalline texture, argillaceous in part, no visible porosity; occasional ANHYDRITE: off white, tan, microcrystalline, firm-soft, massive, amorphous, no visible porosity; rare DOLOMITE: mudstone, tan, light brown gray, microcrystalline, firm-hard, dense-banded, crystalline-sucrosic texture, argillaceous in part, no visible porosity

9,160-9,190 LIMESTONE: mudstone, tan, cream, light-medium gray, microcrystalline, firm, dense-banded, crystalline texture, argillaceous in part, no visible porosity; occasional ANHYDRITE: off white, tan, microcrystalline, firm-soft, massive, amorphous, no visible porosity

9,190-9,220 SALT: translucent-transparent, white-clear, firm, hard, crystalline, massive, anhedral, no visible porosity; rare LIMESTONE: mudstone, light brown gray-light gray, cream, microcrystalline, firm, dense, earthy texture, argillaceous in part, crystalline texture, no visible porosity

9,220-9,250 SALT: translucent-transparent, white-clear, firm, hard, crystalline, massive, anhedral, no visible porosity

Base of Last Salt [Charles Formation]

9,250' MD / 9,241' TVD (-7,106')

9,250-9,280 LIMESTONE-ARGILLACEOUS LIMESTONE: mudstone, tan, light-medium gray, rare dark gray, microcrystalline, firm, trace friable, dense, earthy texture, rare crystalline texture, no visible porosity, no visible oil stain

9,280-9,310 LIMESTONE: mudstone, tan, cream, light-medium gray, microcrystalline, firm, trace friable, dense, earthy texture, common crystalline texture, argillaceous in part, no visible porosity, no visible oil stain

9,310-9,340 LIMESTONE: mudstone, light-medium gray, occasional cream, rare tan, microcrystalline, firm, trace friable, dense, earthy-crystalline texture, argillaceous in part, no visible porosity, no visible oil stain

9,340-9,370 LIMESTONE: mudstone, light-medium gray, occasional cream, rare tan, microcrystalline, firm, trace friable, dense, earthy-crystalline texture, argillaceous in part, no visible porosity, no visible oil stain

9,370-9,400 LIMESTONE: mudstone, light-medium gray, occasional cream, rare tan, microcrystalline, firm, trace friable, dense, earthy-crystalline texture, argillaceous in part, no visible porosity, no visible oil stain

9,400-9,430 LIMESTONE: mudstone, tan, cream, microcrystalline, firm, trace friable, dense, earthy texture, common crystalline texture, argillaceous in part, no visible porosity, no visible oil stain

9,430-9,460 LIMESTONE: mudstone, off white-light gray, microcrystalline, firm-friable, dense, earthy-crystalline texture, argillaceous in part, no visible porosity, no visible oil stain

Mission Canyon [Mississippian Madison Group]

9,455' MD / 9,446' TVD (-7,311')

9,460-9,490 LIMESTONE: mudstone, dark cream-off white, common tan-light brown, microcrystalline, firm, dense, earthy-crystalline texture, argillaceous in part, trace free clear calcite, possible intercrystalline porosity, no visible oil stain

9,490-9,520 LIMESTONE: mudstone, dark cream-off white, common tan-light brown, microcrystalline, firm, dense, earthy-crystalline texture, argillaceous in part, trace free clear calcite, possible intercrystalline porosity, no visible oil stain

9,520-9,550 LIMESTONE: mudstone, tan, cream, light-medium gray, microcrystalline, firm, trace friable, dense, earthy texture, common crystalline texture, argillaceous in part, no visible porosity, no visible oil stain

9,550-9,580 LIMESTONE: mudstone, tan, cream, light-medium gray, microcrystalline, firm, trace friable, dense, earthy texture, common crystalline texture, argillaceous in part, no visible porosity, no visible oil stain

9,580-9,610 LIMESTONE: mudstone, cream-gray, gray-brown, microcrystalline, firm-hard, dense-banded, crystalline texture, argillaceous in part, no visible porosity, no visible oil stain

9,610-9,640 LIMESTONE: mudstone, dark cream-off white, common tan-light brown, microcrystalline, firm, dense, earthy-crystalline texture, argillaceous in part, trace free clear calcite, possible intercrystalline porosity, no visible oil stain

9,640-9,670 LIMESTONE: mudstone, cream, tan-light brown, microcrystalline, firm, dense, earthy-crystalline texture, argillaceous in part, trace free clear calcite, possible intercrystalline porosity, no visible oil stain

9,670-9,700 LIMESTONE: mudstone, cream-gray, light brown, microcrystalline, firm, trace friable, dense-banded, earthy-crystalline texture, argillaceous in part, no visible porosity, no visible oil stain

9,700-9,730 LIMESTONE: mudstone, cream-gray, light brown, microcrystalline, firm, trace friable, dense-banded, earthy-crystalline texture, argillaceous in part, no visible porosity, no visible oil stain

9,730-9,760 LIMESTONE: mudstone, brown-gray brown, rare gray, microcrystalline, firm, dense, earthy-crystalline texture, no visible porosity, no visible oil stain

9,760-9,790 LIMESTONE: mudstone, brown-gray brown, rare gray, microcrystalline, firm, dense, earthy-crystalline texture, no visible porosity, no visible oil stain

9,790-9,820 LIMESTONE: mudstone, cream-gray, light brown, microcrystalline, firm, trace friable, dense-banded, earthy-crystalline texture, argillaceous in part, no visible porosity, no visible oil stain

9,820-9,850 LIMESTONE: mudstone, cream-gray, light brown, microcrystalline, firm, trace friable, dense-banded, earthy-crystalline texture, argillaceous in part, no visible porosity, no visible oil stain

9,850-9,880 LIMESTONE: mudstone, cream-tan, rare light brown, trace light gray, microcrystalline, firm, trace friable, dense-banded, crystalline texture, no visible porosity, no visible oil stain

9,880-9,910 LIMESTONE: mudstone, cream-tan, rare light brown, trace light gray, microcrystalline, firm, trace friable, dense-banded, crystalline texture, no visible porosity, no visible oil stain

9,910-9,940 LIMESTONE: mudstone, cream, light brown, microcrystalline, firm, trace friable, dense, crystalline texture, argillaceous in part, no visible porosity, no visible oil stain

9,940-9,970 LIMESTONE: mudstone, light-medium, cream, tan, microcrystalline, firm, trace friable, dense, crystalline texture, occasional earthy texture, no visible porosity, no visible oil stain

9,970-10,000 Limestone: mudstone, light-medium, cream, tan, microcrystalline, firm, trace friable, dense, crystalline texture, occasional earthy texture, no visible porosity, no visible oil stain

10,000-10,030 Limestone-Argillaceous Limestone: mudstone, medium-light gray, light brown, rare tan, dark gray, microcrystalline, firm, occasional friable dense, earthy texture, trace disseminated pyrite, no visible porosity, no visible oil stain

Lodgepole Formation [Mississippian Madison Group]

10.025' MD / 10.015' TVD (-7.880')

10,030-10,060 ARGILLACEOUS Limestone: mudstone, trace wackestone, medium-dark gray, common light gray, rare cream, microcrystalline, firm, trace friable, dense, earthy texture, rare disseminated pyrite, no visible porosity, no visible oil stain

10,060-10,090 ARGILLACEOUS Limestone: mudstone, occasional wackestone, medium-dark gray, common light brown gray, rare cream, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, no visible porosity, no visible oil stain

10,090-10,190 No sample

Horizontal Log Descriptions:

MD / TVD (MSL Datum)

10,190-10,210 ARGILLACEOUS Limestone: mudstone, occasional wackestone, medium-light gray, rare light brown gray, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, no visible porosity, no visible oil stain

10,210-10,240 ARGILLACEOUS Limestone: mudstone, occasional wackestone, light-medium gray, rare light brown gray, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, no visible porosity, no visible oil stain

10,240-10,270 ARGILLACEOUS Limestone: mudstone, occasional wackestone, medium-light gray, rare light brown gray, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, no visible porosity, no visible oil stain

10,270-10,300 ARGILLACEOUS Limestone: mudstone, rare wackestone, light gray, rare light brown gray, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, no visible porosity, no visible oil stain

10,300-10,330 ARGILLACEOUS Limestone: mudstone, occasional wackestone, medium-light gray, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, no visible porosity, no visible oil stain

10,330-10,360 ARGILLACEOUS Limestone: mudstone, occasional wackestone, medium-light gray, rare light brown gray, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, no visible porosity, no visible oil stain

10,360-10,390 ARGILLACEOUS Limestone: mudstone, rare wackestone, light-medium gray, trace light brown gray, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, no visible porosity, no visible oil stain

10,390-10,420 ARGILLACEOUS Limestone: mudstone, occasional wackestone, medium-light gray, rare light brown gray, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, no visible porosity, no visible oil stain

10,420-10,450 ARGILLACEOUS Limestone: mudstone, rare wackestone, light-medium gray, trace light brown gray, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, no visible porosity, no visible oil stain

10,450-10,480 ARGILLACEOUS Limestone: mudstone, rare wackestone, light-medium gray, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, no visible porosity, no visible oil stain

10,480-10,510 No sample

10,510-10,540 ARGILLACEOUS Limestone: mudstone, occasional wackestone, dark gray, occasional light brown gray, rare light-medium gray, trace cream, microcrystalline, firm, trace hard, dense, earthy texture, trace disseminated pyrite, no visible porosity, no visible oil stain

10,540-10,570 ARGILLACEOUS Limestone: mudstone, occasional wackestone, dark gray, occasional light brown gray, rare light-medium gray, trace cream, microcrystalline, firm, trace hard, dense, earthy texture, trace disseminated pyrite, no visible porosity, no visible oil stain

10,570-10,600 ARGILLACEOUS LIMESTONE: mudstone, occasional wackestone, light brown gray, medium-dark gray, rare cream, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, no visible porosity, no visible oil stain

10,600-10,630 ARGILLACEOUS LIMESTONE: mudstone, occasional wackestone, light brown gray, medium-dark gray, rare cream, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, no visible porosity, no visible oil stain

10,630-10,660 ARGILLACEOUS LIMESTONE: mudstone, rare wackestone, cream, tan, light brown gray, rare, medium-light gray, microcrystalline, firm, rare friable, dense, earthy texture, trace disseminated pyrite, no visible porosity, no visible oil stain

10,660-10,690 ARGILLACEOUS LIMESTONE: mudstone, rare wackestone, cream, tan, light brown gray, rare, medium-light gray, microcrystalline, firm, rare friable, dense, earthy texture, trace disseminated pyrite, no visible porosity, no visible oil stain

10,690-10,720 ARGILLACEOUS LIMESTONE: mudstone, rare wackestone, light gray, medium-dark gray, rare light brown gray, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, no visible porosity, no visible oil stain

10,720-10,750 ARGILLACEOUS LIMESTONE: mudstone, rare wackestone, light gray, medium-dark gray, rare light brown gray, microcrystalline, firm, dense, earthy texture, trace disseminated pyrite, no visible porosity, no visible oil stain

10,750-10,780 ARGILLACEOUS LIMESTONE: mudstone, rare wackestone, medium-dark gray, light brown gray, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, no visible porosity, no visible oil stain

10,780-10,810 ARGILLACEOUS LIMESTONE: mudstone, rare wackestone, medium-dark gray, light brown gray, microcrystalline, firm, dense, earthy texture, rare disseminated pyrite, no visible porosity, no visible oil stain

10,810-10,840 ARGILLACEOUS LIMESTONE: mudstone, occasional wackestone, dark gray, medium gray brown, medium-light gray, trace cream, microcrystalline, firm, rare hard, dense, earthy texture, rare disseminated pyrite, no visible porosity, no visible oil stain

10,840-10,870 ARGILLACEOUS LIMESTONE: mudstone, trace wackestone, cream, tan, light brown gray, microcrystalline, firm, occasional friable, dense, earthy texture, no visible porosity, no visible oil stain

False Bakken [Lodgepole Formation] 10.874' MD / 10.735' TVD (-8.600')

10,870-10,900 SHALE: black, hard, brittle, blocky, carbonaceous, petroliferous, pyritic, no visible porosity, possible fracture porosity, dark brown even oil stain

Upper Bakken Shale Member [Mississippian-Bakken Formation] 10.897' MD / 10.745' TVD (-8.610')

10,900-10,930 SHALE: black, hard, brittle, blocky, carbonaceous, petroliferous, pyritic, no visible porosity, possible fracture porosity, dark brown even oil stain

Middle Bakken Member [Mississippian-Devonian Bakken Formation] 10.931' MD / 10.758' TVD (-8.623')

10,930-10,960 SILTY SANDSTONE: light-medium gray, light brown gray, occasional cream, rare dark gray, firm, friable, fine grained, sub rounded, occasional sub angular, moderately poorly sorted, calcite cement, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain

10,960-10,990 SILTY SANDSTONE: light gray, light brown gray, light brown, firm, friable, fine grained, sub rounded, moderately poorly sorted, calcite cement, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain

10,990-11,020 No sample

11,039-11,050 SILTY SANDSTONE: light-medium brown, gray, firm, friable, fine grained, sub rounded, moderately poorly sorted, calcite cement, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain

11,050-11,080 SILTY SANDSTONE: light-medium brown, gray, firm-friable, fine grained, sub rounded, moderately poorly sorted, calcite cement, moderately cemented, common disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain

111,080-11,110 SILTY SANDSTONE: light gray, light gray brown, friable, fine grained, sub rounded, moderately-poorly sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain

11,132-11,150 SILTY SANDSTONE: light gray, medium-dark gray, rare light brown gray, firm, trace hard, fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

11,150-11,200 SILTY SANDSTONE: light gray, medium-dark gray, rare light brown gray, firm, trace hard, fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

11,200-11,250 SILTY SANDSTONE: light-medium gray, occasional light brown gray, rare dark gray, firm, trace hard, fine grained, sub rounded, trace sub angular, moderately-poorly sorted, calcite cement, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

11,250-11,300 SILTY SANDSTONE: light-medium gray, occasional dark gray, firm, fine grained, sub rounded, trace sub angular, moderately-poorly sorted, calcite cement, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

11,300-11,350 SILTY SANDSTONE: light-medium gray, occasional dark gray, firm, fine grained, sub rounded, trace sub angular, moderately-poorly sorted, calcite cement, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

11,350-11,400 SILTY SANDSTONE: light-medium gray, rare off white, trace dark gray, firm, fine grained, sub rounded, occasional sub angular, moderately-poorly sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

11,400-11,450 SILTY SANDSTONE: light-medium gray, rare off white, trace dark gray, firm, fine grained, sub rounded, occasional sub angular, moderately-poorly sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

11,450-11,500 SILTY SANDSTONE: off white, light-medium gray, light brown gray, firm, fine grained, sub rounded, rare sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

11,500-11,550 SILTY SANDSTONE: off white, light-medium gray, light brown gray, firm, fine grained, sub rounded, rare sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

11,550-11,600 SILTY SANDSTONE: light gray, medium-dark gray, rare light brown gray, firm, trace hard, fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

11,600-11,650 SILTY SANDSTONE: light brown gray, light brown, medium-light gray, rare cream, firm, trace hard, fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

11,650-11,700 SILTY SANDSTONE: off white, light-medium gray, light brown gray, firm, fine grained, sub rounded, rare sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

11,700-11,750 SILTY SANDSTONE: light gray, medium-dark gray, rare light brown gray, firm, trace hard, fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

11,750-11,800 SILTY SANDSTONE: light brown gray, light brown, medium-light gray, rare cream, firm, trace hard, fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

11,800-11,850 SILTY SANDSTONE: light brown gray, light brown, medium-light gray, rare cream, firm, trace hard, fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

11,850-11,900 SILTY SANDSTONE: light gray, medium-dark gray, rare light brown gray, firm, trace hard, fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

11,900-11,950 SILTY SANDSTONE: light gray, medium-dark gray, rare light brown gray, firm, trace hard, fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

11,950-12,000 SILTY SANDSTONE: light gray, medium-dark gray, rare light brown gray, firm, trace hard, fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

12,000-12,050 SILTY SANDSTONE: light gray, medium-dark gray, rare light brown gray, firm, trace hard, fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

12,050-12,100 SILTY SANDSTONE: light gray, medium-dark gray, rare light brown gray, firm, trace hard, fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

12,100-12,150 SILTY SANDSTONE: light gray, medium-dark gray, rare light brown gray, firm, trace hard, fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

12,150-12,200 SILTY SANDSTONE: light gray, medium-dark gray, rare light brown gray, firm, trace hard, fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

12,200-12,250 SILTY SANDSTONE: light brown gray, light brown, medium-light gray, rare cream, firm, trace hard, fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

12,250-12,300 SILTY SANDSTONE: light brown gray, light brown, medium-light gray, rare cream, firm, trace hard, fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

12,300-12,350 SILTY SANDSTONE: light brown gray, light brown, medium-light gray, cream, firm, hard, fine grained, sub rounded, rare sub angular, moderately sorted, calcite cement, moderately-poorly cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace brown spotty oil stain; slow pale green-yellow diffuse cut fluorescence

12,350-12,400 SILTY SANDSTONE: light brown gray, light brown, medium-light gray, cream, firm, hard, fine grained, sub rounded, rare sub angular, moderately sorted, calcite cement, moderately-poorly cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace brown spotty oil stain; slow pale green-yellow diffuse cut fluorescence

12,400-12,450 SILTY SANDSTONE: light gray, medium-dark gray, rare light brown gray, firm, trace hard, fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

12,450-12,500 SILTY SANDSTONE: light gray, medium-dark gray, rare light brown gray, firm, trace hard, fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

12,500-12,550 SILTY SANDSTONE: light gray, medium-dark gray, rare light brown gray, firm, trace hard, fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

12,550-12,600 SILTY SANDSTONE: light gray, medium-dark gray, rare light brown gray, firm, trace hard, fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

12,600-12,650 SILTY SANDSTONE: light brown gray, light brown, medium-light gray, cream, firm, hard, fine grained, sub rounded, rare sub angular, moderately sorted, calcite cement, moderately-poorly cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace brown spotty oil stain; slow pale green-yellow diffuse cut fluorescence

12,650-12,700 SILTY SANDSTONE: light brown gray, light brown, medium-light gray, cream, firm, hard, fine grained, sub rounded, rare sub angular, moderately sorted, calcite cement, moderately-poorly cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace brown spotty oil stain; slow pale green-yellow diffuse cut fluorescence

12,700-12,750 SILTY SANDSTONE: light brown gray, light brown, medium-light gray, cream, firm, hard, fine grained, sub rounded, rare sub angular, moderately sorted, calcite cement, moderately-poorly cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace brown spotty oil stain; slow pale green-yellow diffuse cut fluorescence

12,750-12,800 SILTY SANDSTONE: light brown gray, light brown, medium-light gray, cream, firm, hard, fine grained, sub rounded, rare sub angular, moderately sorted, calcite cement, moderately-poorly cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace brown spotty oil stain; slow pale green-yellow diffuse cut fluorescence

12,800-12,850 SILTY SANDSTONE: light brown gray, light brown, medium-light gray, rare cream, firm, trace hard, fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately cemented, trace

disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

12,850-12,900 SILTY SANDSTONE: light brown gray, light brown, medium-light gray, rare cream, firm, trace hard, fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

12,900-12,950 SILTY SANDSTONE: light brown gray, light brown, medium-light gray, cream, firm, hard, fine grained, sub rounded, rare sub angular, moderately sorted, calcite cement, moderately-poorly cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace brown spotty oil stain; slow pale green-yellow diffuse cut fluorescence

12,950-13,000 SILTY SANDSTONE: light brown gray, light brown, medium-light gray, rare cream, firm, trace hard, fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

13,000-13,050 SILTY SANDSTONE: light brown gray, light brown, medium-light gray, rare cream, firm, trace hard, fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

13,050-13,100 SILTY SANDSTONE: light brown gray, light brown, medium-light gray, rare cream, firm, trace hard, fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

13,100-13,150 SILTY SANDSTONE: light gray, medium-dark gray, rare light brown gray, firm, trace hard, fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

13,150-13,200 SILTY SANDSTONE: light gray, medium-dark gray, rare light brown gray, firm, trace hard, fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

13,200-13,250 SILTY SANDSTONE: light brown-gray, medium-dark gray, rare light brown gray, firm, trace hard, fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

13,250-13,300 SILTY SANDSTONE: light brown-gray, medium-dark gray, rare light brown gray, firm, trace hard, fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

13,300-13,350 SILTY SANDSTONE: light brown-gray, medium-dark gray, rare light brown gray, firm, trace hard, fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

13,350-13,400 SILTY SANDSTONE: light brown-gray, medium-dark gray, rare light brown gray, firm, trace hard, fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

13,400-13,450 SILTY SANDSTONE: medium-dark gray, rare light brown gray, firm, trace hard, fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

13,450-13,500 SILTY SANDSTONE: medium-dark gray, rare light brown gray, firm, trace hard, fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

13,500-13,550 SILTY SANDSTONE: light brown-gray, medium-dark gray, rare light brown gray, firm, trace hard, fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

13,550-13,600 SILTY SANDSTONE: light brown-gray, medium-dark gray, rare light brown gray, firm, trace hard, fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

13,600-13,650 SILTY SANDSTONE: light brown, light brown gray, light-medium gray, off white, cream, firm, trace hard, fine grained, sub rounded, occasional sub angular, moderately-poorly sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

13,650-13,700 SILTY SANDSTONE: light brown, light brown gray, light-medium gray, off white, cream, firm, trace hard, fine grained, sub rounded, occasional sub angular, moderately-poorly sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

13,700-13,750 SILTY SANDSTONE: light brown gray, occasional medium-light gray, rare light brown gray, trace dark gray, firm, fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

13,750-13,800 SILTY SANDSTONE: light brown gray, occasional light brown gray, rare medium-light gray, trace dark gray, firm, fine grained, sub rounded, moderately-poorly sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

13,800-13,850 SILTY SANDSTONE: light brown gray, occasional light brown gray, rare medium-light gray, trace dark gray, firm, fine grained, sub rounded, moderately-poorly sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

13,850-13,900 SILTY SANDSTONE: light brown gray, occasional light-medium gray, firm, fine grained, sub rounded, moderately-poorly sorted, calcite cement, moderately-poorly cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; slow-moderate pale yellow diffuse cut fluorescence

13,900-13,950 SILTY SANDSTONE: light brown gray, occasional light-medium gray, firm, fine grained, sub rounded, moderately-poorly sorted, calcite cement, moderately-poorly cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; slow-moderate pale yellow diffuse cut fluorescence

13,950-14,000 SILTY SANDSTONE: medium-light gray, common light brown, rare cream, firm, occasional hard, fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately-poorly cemented,

trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; slow-moderate pale yellow diffuse cut fluorescence

14,000-14,050 SILTY SANDSTONE: medium-light gray, common light brown, rare cream, firm, occasional hard, fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately-poorly cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; slow-moderate pale yellow diffuse cut fluorescence

14,050-14,100 SILTY SANDSTONE: light gray, off white, rare light brown gray, firm, rare hard, fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately-poorly cemented, trace disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow-light green diffuse cut fluorescence

14,100-14,150 SILTY SANDSTONE: light gray, off white, rare light brown gray, firm, rare hard, fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately-poorly cemented, trace disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow-light green diffuse cut fluorescence

14,150-14,200 SILTY SANDSTONE: light gray, off white, rare light brown gray, firm, rare hard, fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately-poorly cemented, trace disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow-light green diffuse cut fluorescence

14,200-14,250 SILTY SANDSTONE: light brown gray, light brown, rare medium-light gray, trace off white, firm, fine-very fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately-poorly cemented, trace disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow-light green diffuse cut fluorescence

14,250-14,300 SILTY SANDSTONE: light brown gray, light brown, rare medium-light gray, trace off white, firm, fine-very fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately-poorly cemented, trace disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow-light green diffuse cut fluorescence

14,300-14,350 SILTY SANDSTONE: light brown gray, light brown, rare medium-light gray, trace off white, firm, fine-very fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately-poorly cemented, trace disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow-light green diffuse cut fluorescence

14,350-14,400 SILTY SANDSTONE: light-medium gray, occasional light brown gray, rare dark gray, firm, trace hard, fine grained, sub rounded, trace sub angular, moderately-poorly sorted, calcite cement, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

14,400-14,450 SILTY SANDSTONE: light-medium gray, occasional light brown gray, rare dark gray, firm, trace hard, fine grained, sub rounded, trace sub angular, moderately-poorly sorted, calcite cement, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

14,450-14,500 SILTY SANDSTONE: off white, light brown, rare light-medium gray, firm, fine grained, sub rounded, trace sub angular, moderately-poorly sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; slow-moderate pale yellow diffuse cut fluorescence

14,500-14,550 SILTY SANDSTONE: off white, light brown, rare light-medium gray, firm, fine grained, sub rounded, trace sub angular, moderately-poorly sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; slow-moderate pale yellow diffuse cut fluorescence

14,550-14,600 SILTY SANDSTONE: light brown, occasional light brown gray, rare medium-light gray, trace cream, firm, fine grained, sub rounded, occasional sub angular, moderately-poorly sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; slow-moderate pale yellow diffuse cut fluorescence

14,600-14,650 SILTY SANDSTONE: light brown, occasional light brown gray, rare medium-light gray, trace cream, firm, fine grained, sub rounded, occasional sub angular, moderately-poorly sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; slow-moderate pale yellow diffuse cut fluorescence

14,650-14,700 SILTY SANDSTONE: light gray, light brown gray, firm, fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

14,700-14,750 SILTY SANDSTONE: light gray, light brown gray, firm, fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

14,750-14,800 SILTY SANDSTONE: light gray, light brown gray, firm, fine grained, sub rounded, rare sub angular, moderately-poorly sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

14,800-14,850 SILTY SANDSTONE: off white, light gray, occasional light brown, trace medium-dark gray, firm, fine grained, sub rounded, occasional sub angular, moderately-poorly sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

14,850-14,900 SILTY SANDSTONE: off white, light gray, occasional light brown, trace medium-dark gray, firm, fine grained, sub rounded, occasional sub angular, moderately-poorly sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

14,900-14,950 SILTY SANDSTONE: off white, light gray, occasional light brown, trace medium-dark gray, firm, fine grained, sub rounded, occasional sub angular, moderately-poorly sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; slow pale yellow diffuse cut fluorescence

14,950-15,000 SILTY SANDSTONE: light-medium gray, light brown gray, firm, fine grained, sub rounded, occasional sub angular, moderately-poorly sorted, calcite cement, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; slow-moderate pale yellow diffuse cut fluorescence

15,000-15,050 SILTY SANDSTONE: light-medium gray, light brown gray, firm, fine grained, sub rounded, occasional sub angular, moderately-poorly sorted, calcite cement, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; slow-moderate pale yellow diffuse cut fluorescence

15,050-15,100 SILTY SANDSTONE: light brown, tan, light-medium gray, rare off white, firm, fine grained, sub rounded, trace sub angular, moderately-poorly sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; slow-moderate pale yellow diffuse cut fluorescence

15,100-15,150 SILTY SANDSTONE: light brown, tan, light-medium gray, rare off white, firm, fine grained, sub rounded, trace sub angular, moderately-poorly sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; slow-moderate pale yellow diffuse cut fluorescence

15,150-15,200 SILTY SANDSTONE: light brown, tan, light-medium gray, rare off white, firm, fine grained, sub rounded, trace sub angular, moderately-poorly sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; slow-moderate pale yellow diffuse cut fluorescence

15,200-15,250 SILTY SANDSTONE: light brown, tan, light-medium gray, rare off white, firm, fine grained, sub rounded, trace sub angular, moderately-poorly sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; slow-moderate pale yellow diffuse cut fluorescence

15,250-15,300 SILTY SANDSTONE: light-medium gray, light brown gray, firm, fine grained, sub rounded, occasional sub angular, moderately-poorly sorted, calcite cement, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; slow-moderate pale yellow diffuse cut fluorescence

15,300-15,350 SILTY SANDSTONE: light-medium gray, light brown gray, firm, fine grained, sub rounded, occasional sub angular, moderately-poorly sorted, calcite cement, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; slow-moderate pale yellow diffuse cut fluorescence

15,350-15,400 SILTY SANDSTONE: light-medium gray, light brown gray, firm, fine grained, sub rounded, occasional sub angular, moderately-poorly sorted, calcite cement, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; slow-moderate pale yellow diffuse cut fluorescence

15,400-15,450 SILTY SANDSTONE: light-dark gray, gray, firm, fine grained, sub rounded, occasional sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; slow-moderate pale yellow diffuse cut fluorescence

15,450-15,500 SILTY SANDSTONE: light-dark gray, gray, firm, fine grained, sub rounded, occasional sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; slow-moderate pale yellow diffuse cut fluorescence

15,500-15,550 SILTY SANDSTONE: light-dark gray, gray, firm, fine grained, sub rounded, occasional sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; slow-moderate pale yellow diffuse cut fluorescence

15,550-15,600 SILTY SANDSTONE: light-dark gray, gray, firm, fine grained, sub rounded, occasional sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; slow-moderate pale yellow diffuse cut fluorescence

15,600-15,650 SILTY SANDSTONE: light-medium brown-gray, gray, firm, fine grained, sub rounded, occasional sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; moderate pale yellow diffuse cut fluorescence

15,650-15,700 SILTY SANDSTONE: light-medium brown-gray, gray, firm, fine grained, sub rounded, occasional sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; moderate pale yellow diffuse cut fluorescence

15,700-15,750 SILTY SANDSTONE: light-medium brown-gray, gray, firm, fine grained, sub rounded, occasional sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; moderate pale yellow diffuse cut fluorescence

15,750-15,800 SILTY SANDSTONE: light-medium brown-gray, gray, firm, fine grained, sub rounded, occasional sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; moderate pale yellow diffuse cut fluorescence

trace intergranular porosity, trace light-medium brown spotty oil stain; moderate pale yellow-grain diffuse cut fluorescence

16,500-16,550 SILTY SANDSTONE: light-medium gray, gray, firm, fine grained, sub rounded, occasional sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; moderate pale yellow-grain diffuse cut fluorescence

16,550-16,600 SILTY SANDSTONE: light-medium gray, gray, firm, fine grained, sub rounded, occasional sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; moderate pale yellow-grain diffuse cut fluorescence

16,600-16,650 No sample

16,650-16,700 No sample

16,700-16,750 SILTY SANDSTONE: medium gray, brown-gray, firm, fine grained, sub rounded, occasional sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; moderate pale yellow-grain diffuse cut fluorescence

16,750-16,800 SILTY SANDSTONE: medium gray, brown-gray, firm, fine grained, sub rounded, occasional sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; moderate pale yellow-grain diffuse cut fluorescence

16,800-16,850 SILTY SANDSTONE: medium gray, brown-gray, firm, fine grained, sub rounded, occasional sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; moderate pale yellow-grain diffuse cut fluorescence

16,800-16,850 SILTY SANDSTONE: medium gray, brown-gray, firm, fine grained, sub rounded, occasional sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; moderate pale yellow-grain diffuse cut fluorescence

16,900-16,950 SILTY SANDSTONE: dark gray, brown-gray, firm, fine grained, sub rounded, occasional sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; moderate pale yellow-grain diffuse cut fluorescence

16,950-17,000 SILTY SANDSTONE: dark gray, brown-gray, firm, fine grained, sub rounded, occasional sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; moderate pale yellow-grain diffuse cut fluorescence

17,020-17,040 SILTY SANDSTONE: dark gray, brown-gray, tan, firm, fine grained, sub rounded, occasional sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; moderate pale yellow-grain diffuse cut fluorescence

17,040-17,060 SILTY SANDSTONE: dark gray, brown-gray, firm, fine grained, sub rounded, occasional sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; moderate pale yellow-grain diffuse cut fluorescence

17,000-17,050 SILTY SANDSTONE: dark gray, brown-gray, firm, fine grained, sub rounded, occasional sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; moderate pale yellow-grain diffuse cut fluorescence

17,060-17,080 SILTY SANDSTONE: dark gray, dark brown-gray, firm, fine grained, sub rounded, occasional sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; moderate pale yellow-grain diffuse cut fluorescence

17,080-17,100 SILTY SANDSTONE: dark gray, dark brown-gray, firm, fine grained, sub rounded, occasional sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; moderate pale yellow-grain diffuse cut fluorescence

17,050-17,100 SILTY SANDSTONE: dark gray, brown-gray, firm, fine grained, sub rounded, occasional sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; moderate pale yellow-grain diffuse cut fluorescence

17,100-17,120 SILTY SANDSTONE: dark gray, brown-gray, firm, fine grained, sub rounded, occasional sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; moderate pale yellow-grain diffuse cut fluorescence

17,120-17,150 SILTY SANDSTONE: dark gray, brown-gray, firm, fine grained, sub rounded, occasional sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; moderate pale yellow-grain diffuse cut fluorescence

17,150-17,200 SILTY SANDSTONE: dark brown-gray, tan-gray, firm, fine grained, sub rounded, occasional sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; slow-moderate pale yellow-grain diffuse cut fluorescence

17,200-17,250 SILTY SANDSTONE: dark brown-gray, tan-gray, firm, fine grained, sub rounded, occasional sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; slow-moderate pale yellow-grain diffuse cut fluorescence

17,250-17,300 SILTY SANDSTONE: light-dark gray, gray, firm, fine grained, sub rounded, occasional sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; slow-moderate pale yellow diffuse cut fluorescence

17,300-17,350 SILTY SANDSTONE: light-dark gray, gray, firm, fine grained, sub rounded, occasional sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; slow-moderate pale yellow diffuse cut fluorescence

17,350-17,400 SILTY SANDSTONE: light-dark gray, gray, firm, fine grained, sub rounded, occasional sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; slow-moderate pale yellow diffuse cut fluorescence

17,400-17,450 SILTY SANDSTONE: light-dark gray, gray, firm, fine grained, sub rounded, occasional sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; slow-moderate pale yellow diffuse cut fluorescence

17,450-17,500 SILTY SANDSTONE: light-dark gray, gray, firm, fine grained, sub rounded, occasional sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; slow-moderate pale yellow diffuse cut fluorescence

17,500-17,550 SILTY SANDSTONE: light-medium gray, light brown gray, occasional off white, rare dark gray, firm, fine grained, sub rounded, trace sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; slow-moderate pale yellow diffuse cut fluorescence

17,550-17,600 SILTY SANDSTONE: light-medium gray, light brown gray, occasional off white, rare dark gray, firm, fine grained, sub rounded, trace sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; slow-moderate pale yellow diffuse cut fluorescence

17,600-17,650 Sample moderately contaminated with lube; SILTY SANDSTONE: light brown gray, light-medium gray, firm, fine-very fine grained, sub rounded, occasional sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

17,650-17,700 Sample moderately contaminated with lube; SILTY SANDSTONE: light brown gray, light-medium gray, firm, fine-very fine grained, sub rounded, occasional sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

17,700-17,750 Sample moderately contaminated with lube; SILTY SANDSTONE: light brown gray, light-medium gray, firm, fine-very fine grained, sub rounded, occasional sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

17,750-17,800 Sample moderately contaminated with lube; SILTY SANDSTONE: off white, light gray, rare light brown gray, trace dark gray, firm, fine-very fine grained, sub rounded, occasional sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

17,800-17,850 Sample moderately contaminated with lube; SILTY SANDSTONE: off white, light gray, rare light brown gray, trace dark gray, firm, fine-very fine grained, sub rounded, occasional sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

17,850-17,900 Sample moderately contaminated with lube; SILTY SANDSTONE: off white, light gray, rare light brown gray, trace dark gray, firm, fine-very fine grained, sub rounded, occasional sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

17,900-17,950 Sample moderately contaminated with lube; SILTY SANDSTONE: light brown, light brown gray, occasional light-medium gray, rare off white, trace dark gray, firm, fine-very fine grained, sub rounded, rare sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

17,950-18,000 Sample moderately contaminated with lube; SILTY SANDSTONE: light brown, light brown gray, occasional light-medium gray, rare off white, trace dark gray, firm, fine-very fine grained, sub rounded, rare sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

18,000-18,050 Sample moderately contaminated with lube; SILTY SANDSTONE: light brown, light brown gray, occasional light-medium gray, rare off white, trace dark gray, firm, fine-very fine grained, sub rounded, rare sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

18,050-18,100 Sample moderately contaminated with lube; SILTY SANDSTONE: light brown gray, light brown, firm, fine-very fine grained, sub rounded, rare sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

18,100-18,150 Sample moderately contaminated with lube; SILTY SANDSTONE: light brown gray, light brown, firm, fine-very fine grained, sub rounded, rare sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

18,150-18,200 Sample moderately contaminated with lube; SILTY SANDSTONE: light brown gray, light brown, firm, fine-very fine grained, sub rounded, rare sub angular, moderately sorted, calcite cement, moderately cemented,

rare disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

18,200-18,250 Sample moderately contaminated with lube; SILTY SANDSTONE: light brown gray, occasional light brown, rare medium-light gray, trace off white, firm, fine-very fine grained, sub rounded, rare sub angular, moderately sorted, calcite cement, moderately-poorly cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

18,250-18,300 Sample moderately contaminated with lube; SILTY SANDSTONE: light brown gray, occasional light brown, rare medium-light gray, trace off white, firm, fine-very fine grained, sub rounded, rare sub angular, moderately sorted, calcite cement, moderately-poorly cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

18,300-18,350 Sample moderately contaminated with lube; SILTY SANDSTONE: light brown gray, occasional light brown, rare medium-light gray, trace off white, firm, fine-very fine grained, sub rounded, rare sub angular, moderately sorted, calcite cement, moderately-poorly cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

18,350-18,400 Sample moderately contaminated with lube; SILTY SANDSTONE: medium-light gray, occasional light brown gray, trace dark gray, firm, fine-very fine grained, sub rounded, rare sub angular, moderately sorted, calcite cement, moderately-poorly cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

18,400-18,450 Sample moderately contaminated with lube; SILTY SANDSTONE: medium-light gray, occasional light brown gray, trace dark gray, firm, fine-very fine grained, sub rounded, rare sub angular, moderately sorted, calcite cement, moderately-poorly cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

18,450-18,500 Sample moderately contaminated with lube; SILTY SANDSTONE: medium-light gray, occasional light brown gray, trace dark gray, firm, fine-very fine grained, sub rounded, rare sub angular, moderately sorted, calcite cement, moderately-poorly cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

18,500-18,550 Sample moderately contaminated with lube; SILTY SANDSTONE: off white, tan, light gray, rare light brown gray, trace dark gray, firm, fine-very fine grained, sub rounded, occasional sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

18,550-18,600 Sample moderately contaminated with lube; SILTY SANDSTONE: off white, tan, light gray, rare light brown gray, trace dark gray, firm, fine-very fine grained, sub rounded, occasional sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

18,600-18,650 Sample moderately contaminated with lube; SILTY SANDSTONE: off white, tan, light gray, rare light brown gray, trace dark gray, firm, fine-very fine grained, sub rounded, occasional sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

18,650-18,700 Sample moderately contaminated with lube; SILTY SANDSTONE: light brown, medium-light gray, rare off white, firm, fine grained, sub rounded, occasional sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

18,700-18,750 Sample moderately contaminated with lube; SILTY SANDSTONE: light brown, medium-light gray, rare off white, firm, fine grained, sub rounded, occasional sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

18,750-18,800 Sample moderately contaminated with lube; SILTY SANDSTONE: light brown, medium-light gray, rare off white, firm, fine grained, sub rounded, occasional sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

18,800-18,850 Sample moderately contaminated with lube; SILTY SANDSTONE: light brown gray, occasional light brown, rare medium-dark gray, firm, fine grained, sub rounded, rare sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

18,850-18,900 Sample moderately contaminated with lube; SILTY SANDSTONE: medium-light gray, light brown, rare cream, firm, fine grained, sub rounded, rare sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; contaminated cut fluorescence

18,900-18,950 Sample moderately contaminated with lube; SILTY SANDSTONE: medium-light gray, light brown, rare cream, firm, fine grained, sub rounded, rare sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; contaminated cut fluorescence

18,950-19,000 Sample moderately contaminated with lube; SILTY SANDSTONE: medium-light gray, light brown, rare cream, firm, fine grained, sub rounded, rare sub angular, moderately sorted, calcite cement, moderately cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; contaminated cut fluorescence

19,000-19,050 Sample moderately contaminated with lube; SILTY SANDSTONE: light brown gray, common light brown, rare medium gray, firm, fine grained, sub rounded, rare sub angular, moderately sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; contaminated cut fluorescence

19,050-19,100 Sample moderately contaminated with lube; SILTY SANDSTONE: light brown gray, common light brown, rare medium gray, firm, fine grained, sub rounded, rare sub angular, moderately sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; contaminated cut fluorescence

19,100-19,150 Sample moderately contaminated with lube; SILTY SANDSTONE: light brown gray, occasional light brown, rare medium-light gray, trace off white, firm, fine-very fine grained, sub rounded, rare sub angular, moderately sorted, calcite cement, moderately-poorly cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

19,150-19,200 Sample moderately contaminated with lube; SILTY SANDSTONE: light brown gray, common light brown, rare medium gray, firm, fine grained, sub rounded, rare sub angular, moderately sorted, calcite cement, moderately cemented, trace disseminated pyrite, trace nodular pyrite, trace intergranular porosity, trace light-medium brown spotty oil stain; contaminated cut fluorescence

19,200-19,250 Sample moderately contaminated with lube; SILTY SANDSTONE: light-medium gray, off white, firm, fine-very fine grained, sub rounded, moderately sorted, calcite cement, moderately-poorly cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

19,250-19,300 Sample moderately contaminated with lube; SILTY SANDSTONE: light-medium gray, off white, firm, fine-very fine grained, sub rounded, moderately sorted, calcite cement, moderately-poorly cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

19,300-19,350 Sample moderately contaminated with lube; SILTY SANDSTONE: light-medium gray, off white, firm, fine-very fine grained, sub rounded, moderately sorted, calcite cement, moderately-poorly cemented, rare

disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

19,350-19,400 Sample moderately contaminated with lube; SILTY SANDSTONE: light-medium gray, off white, firm, fine-very fine grained, sub rounded, moderately sorted, calcite cement, moderately-poorly cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

19,400-19,450 Sample moderately contaminated with lube; SILTY SANDSTONE: light-medium gray, off white, firm, fine-very fine grained, sub rounded, moderately sorted, calcite cement, moderately-poorly cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

19,450-19,500 Sample moderately contaminated with lube; SILTY SANDSTONE: light-medium gray, off white, firm, fine-very fine grained, sub rounded, moderately sorted, calcite cement, moderately-poorly cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

19,500-19,550 Sample moderately contaminated with lube; SILTY SANDSTONE: light-medium gray, off white, firm, fine-very fine grained, sub rounded, moderately sorted, calcite cement, moderately-poorly cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

19,550-19,600 Sample moderately contaminated with lube; SILTY SANDSTONE: light-medium gray, off white, firm, fine-very fine grained, sub rounded, moderately sorted, calcite cement, moderately-poorly cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

19,600-19,650 Sample moderately contaminated with lube; SILTY SANDSTONE: light-medium gray, off white, firm, fine-very fine grained, sub rounded, moderately sorted, calcite cement, moderately-poorly cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

19,650-19,700 Sample moderately contaminated with lube; SILTY SANDSTONE: light-medium gray, off white, firm, fine-very fine grained, sub rounded, moderately sorted, calcite cement, moderately-poorly cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

19,700-19,750 Sample moderately contaminated with lube; SILTY SANDSTONE: light-medium gray, off white, firm, fine-very fine grained, sub rounded, moderately sorted, calcite cement, moderately-poorly cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

19,750-19,800 Sample moderately contaminated with lube; SILTY SANDSTONE: light-medium gray, off white, firm, fine-very fine grained, sub rounded, moderately sorted, calcite cement, moderately-poorly cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

19,800-19,850 Sample moderately contaminated with lube; SILTY SANDSTONE: light-medium gray, off white, firm, fine-very fine grained, sub rounded, moderately sorted, calcite cement, moderately-poorly cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

19,850-19,900 Sample moderately contaminated with lube; SILTY SANDSTONE: light-medium gray, off white, firm, fine-very fine grained, sub rounded, moderately sorted, calcite cement, moderately-poorly cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

19,900-19,950 Sample moderately contaminated with lube; SILTY SANDSTONE: light-medium gray, off white, firm, fine-very fine grained, sub rounded, moderately sorted, calcite cement, moderately-poorly cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

19,950-20,000 Sample moderately contaminated with lube; SILTY SANDSTONE: light-medium gray, off white, firm, fine-very fine grained, sub rounded, moderately sorted, calcite cement, moderately-poorly cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

20,000-20,050 Sample moderately contaminated with lube; SILTY SANDSTONE: light-medium gray, off white, firm, fine-very fine grained, sub rounded, moderately sorted, calcite cement, moderately-poorly cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

20,050-20,100 Sample moderately contaminated with lube; SILTY SANDSTONE: light-medium gray, off white, firm, fine-very fine grained, sub rounded, moderately sorted, calcite cement, moderately-poorly cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

20,100-20,150 Sample moderately contaminated with lube; SILTY SANDSTONE: light-medium gray, off white, firm, fine-very fine grained, sub rounded, moderately sorted, calcite cement, moderately-poorly cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

20,150-20,200 Sample moderately contaminated with lube; SILTY SANDSTONE: light-medium gray, off white, firm, fine-very fine grained, sub rounded, moderately sorted, calcite cement, moderately-poorly cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

20,200-20,250 Sample moderately contaminated with lube; SILTY SANDSTONE: light-medium gray, off white, firm, fine-very fine grained, sub rounded, moderately sorted, calcite cement, moderately-poorly cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

20,250-20,300 Sample moderately contaminated with lube; SILTY SANDSTONE: light-medium gray, off white, firm, fine-very fine grained, sub rounded, moderately sorted, calcite cement, moderately-poorly cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

20,300-20,350 Sample moderately contaminated with lube; SILTY SANDSTONE: light-medium gray, off white, firm, fine-very fine grained, sub rounded, moderately sorted, calcite cement, moderately-poorly cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

20,350-20,400 Sample moderately contaminated with lube; SILTY SANDSTONE: light-medium gray, off white, firm, fine-very fine grained, sub rounded, moderately sorted, calcite cement, moderately-poorly cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

20,400-20,450 Sample moderately contaminated with lube; SILTY SANDSTONE: light-medium gray, off white, firm, fine-very fine grained, sub rounded, moderately sorted, calcite cement, moderately-poorly cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

20,450-20,500 Sample moderately contaminated with lube; SILTY SANDSTONE: light-medium gray, off white, firm, fine-very fine grained, sub rounded, moderately sorted, calcite cement, moderately-poorly cemented, rare

disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

20,500-20,550 Sample moderately contaminated with lube; SILTY SANDSTONE: light-medium gray, off white, firm, fine-very fine grained, sub rounded, moderately sorted, calcite cement, moderately-poorly cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

20,550-20,600 Sample moderately contaminated with lube; SILTY SANDSTONE: light-medium gray, off white, firm, fine-very fine grained, sub rounded, moderately sorted, calcite cement, moderately-poorly cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

20,600-20,650 Sample moderately contaminated with lube; SILTY SANDSTONE: light-medium gray, off white, firm, fine-very fine grained, sub rounded, moderately sorted, calcite cement, moderately-poorly cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

20,650-20,700 Sample moderately contaminated with lube; SILTY SANDSTONE: light-medium gray, off white, firm, fine-very fine grained, sub rounded, moderately sorted, calcite cement, moderately-poorly cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

20,700-20,750 Sample moderately contaminated with lube; SILTY SANDSTONE: light-medium gray, off white, firm, fine-very fine grained, sub rounded, moderately sorted, calcite cement, moderately-poorly cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

20,750-20,800 Sample moderately contaminated with lube; SILTY SANDSTONE: light-medium gray, off white, firm, fine-very fine grained, sub rounded, moderately sorted, calcite cement, moderately-poorly cemented, rare disseminated pyrite, rare nodular pyrite, trace intergranular porosity, trace light brown spotty oil stain; contaminated cut fluorescence

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

RECEIVED

JAN - 4 2019

Well File No.

30188

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date January 3, 2019
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other	Waiver from tubing/packer requirement

Well Name and Number Lewis Federal 5300 11-31 3B					
Footages 1003	262	Qtr-Qtr	Section	Township	Range
1005 F N L	233 F W L	LOT 1	31	153 N	100 W
Field Baker	Pool Bakken	County McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum North America LLC requests a variance to NDAC 43-02-03-21 for the tubing/packer requirement: Casing, tubing, and cementing requirements during the completion period immediately following the upcoming fracture stimulation.

The following assurances apply:

1. the well is equipped with new 29# and 32# casing at surface with an API burst rating of 11,220 psi;
2. The Frac design will use a safety factor of 0.85 API burst rating to determine the maximum pressure;
3. Damage to the casing during the frac would be detected immediately by monitoring equipment;
4. The casing is exposed to significantly lower rates and pressures during flowback than during the frac job;
5. The frac fluid and formation fluids have very low corrosion and erosion rates;
6. Production equipment will be installed as soon as possible after the well ceases flowing;
7. A 300# gauge will be installed on the surface casing during the flowback period

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9436	
Address 1001 Fannin, Suite 1500			
City Houston	State TX	Zip Code 77002	
Signature 	Printed Name Jennifer Swenson		
Title Regulatory Specialist	Date January 3, 2019		
Email Address jswenson@oasispetroleum.com			

FOR STATE USE ONLY

☐ Received	☒ Approved
Date January 10, 2019	
By 	
Title PETROLEUM ENGINEER	

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

Received

Well File No.
30188

JAN - 3 2019

**ND Oil & Gas
Division**

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date December 15, 2018
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☒ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☐ Other	

Well Name and Number

Lewis Federal 5300 11-31 3B

Footages	Qtr-Qtr	Section	Township	Range
1005 F N L 233 F W L	Lot 1	31	153 N	100 W
Field	Pool	County		
Baker	Bakken	McKenzie		

24-HOUR PRODUCTION RATE

Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)

Address	City	State	Zip Code
---------	------	-------	----------

DETAILS OF WORK

Oasis Petroleum respectfully requests to amend the surface casing depth to 3600' MD (previously 4900') due to potential flowback in the Dakota formation. In order to counteract projected flowback at surface, Oasis needs to increase mud weight, and lowering the surface casing point will allow this to be performed safely.

Attached are revised drill plans reflecting the amended casing points, cement volumes, and other related volumes.

No DU tool will be used per Jennifer.

Company	Telephone Number	
Oasis Petroleum North America LLC	281-404-9436	
Address		
1001 Fannin St, Suite 1500		
City	State	Zip Code
Houston	TX	77002
Signature	Printed Name	
	Jennifer Swenson	
Title	Date	
Regulatory Specialist	January 3, 2019	
Email Address		
jswenson@oasispetroleum.com		

FOR STATE USE ONLY

☐ Received	☒ Approved
Date	1-3-2019
By	
Title	DAVID BURNS Engineering Technician

Oasis Petroleum
Well Summary
Lewis Federal 5300 11-31 3B
Section 31 T153N R100W
McKenzie County, ND

SURFACE CASING AND CEMENT DESIGN

Size	Interval	Weight	Grade	Coupling	I.D.	Drift
13-3/8"	0' - 3600'	54.5	J-55	STC	12.615"	12.459"

Interval	Description	Collapse	Burst	Tension
		(psi) / a	(psi) / b	(1000 lbs) / c
0' - 3600'	13-3/8", 54.5#, J-55, LTC, 8rd	1130 / 0.67	2730 / 1.12	514 / 1.90

API Rating & Safety Factor

- a) Collapse based on full casing evacuation with 9 ppg fluid on backside (3600' setting depth).
- b) Burst pressure based on 13 ppg fluid with no fluid on backside (3600' setting depth).
- c) Based on string weight in 9 ppg fluid at 3600' TVD plus 100k# overpull. (Buoyed weight equals 169k lbs.)

Cement volumes are based on 13-3/8" casing set in 17-1/2 " hole with 60% excess to circulate cement back to surface.
Mix and pump the following slurry.

Pre-flush (Spacer): **20 bbls** fresh water

Lead Slurry: **1281 sks** (662 bbls), 11.5 lb/gal, 2.97 cu. Ft./sk Varicem Cement with 0.125 il/sk Lost Circulation Additive

Tail Slurry: **300 sks** (62 bbls), 13.0 lb/gal, 2.01 cu.ft./sk Varicem with .125 lb/sk Lost Circulation Agent

Oasis Petroleum
Well Summary
Lewis Federal 5300 11-31 3B
Section 31 T153N R100W
McKenzie County, ND

Contingency INTERMEDIATE CASING AND CEMENT DESIGN

Size	Interval	Weight	Grade	Coupling	I.D.	Drift
9-5/8"	0' - 6016'	36	J-55	LTC	8.921"	8.765"

Interval	Description	Collapse	Burst	Tension
		(psi) / a	(psi) / b	(1000 lbs) / c
0' - 6016'	9-5/8", 36#, J-55, LTC, 8rd	2020 / 2.25	3520 / 1.36	453 / 1.60

API Rating & Safety Factor

- a) Collapse based on full casing evacuation with 10.4 ppg fluid on backside (6016' setting depth).
- b) Burst pressure calculated from a gas kick coming from the production zone (Bakken Pool) at 9,000psi and a subsequent breakdown at the 9-5/8" shoe, based on a 15.2#/ft fracture gradient. Backup of 9 ppg fluid..
- c) Tension based on string weight in 10.4 ppg fluid at 6016' TVD plus 100k# overpull. (Buoyed weight equals 182k lbs.)

Cement volumes are based on 9-5/8" casing set in 12-1/4 " hole with 10% excess to circulate cement back to surface.

Pre-flush (Spacer): **20 bbls** Chem wash

Lead Slurry: **545 sks** (281 bbls), 2.90 ft³/sk, 11.5 lb/gal Conventional system with 94 lb/sk cement, 4% D079 extender, 2% D053 expanding agent, 2% CaCl₂ and 0.250 lb/sk D130 lost circulation control agent.

Tail Slurry: **612 sks** (126 bbls), 1.16 ft³/sk 15.8 lb/gal Conventional system with 94 lb/sk cement, 0.25% CaCl₂, and 0.250 lb/sk lost circulation control agent

INTERMEDIATE CASING AND CEMENT DESIGN

****Special Drift 7"32# to 6.0"**

API Rating & Safety Factor

- Tail Slurry:** **568 sks** (166 bbls), 14.0 ppg, 1.55 cu. ft./sk Extendcem System with .2% HR-5 Retarder and .25 lb/sk Lost Circulation Additive

Oasis Petroleum
Well Summary
Lewis Federal 5300 11-31 3B
Section 31 T153N R100W
McKenzie County, ND

PRODUCTION LINER

Size	Interval	Weight	Grade	Coupling	I.D.	Drift
4-1/2"	10226' - 20895'	13.5	P-110	GB CD BTC	3.920"	3.795"

Interval	Length	Description	Collapse	Burst	Tension
		4-1/2", 13.5 lb, P-110, GB CD BTC	(psi) a	(psi) b	(1000 lbs) c
10226' - 20895'	10669		10670 / 1.99	12410 / 1.28	443 / 1.98

API Rating & Safety Factor

- a) Based on full casing evacuation with 9.5 ppg fluid on backside @ 10841' TVD.
- b) Burst pressure based on 9000 psi treating pressure with 10.2 ppg internal fluid gradient and 9 ppg external fluid gradient @ 10841' TVD.
- c) Based on string weight in 9.5 ppg fluid (Buoyed weight: 123k lbs.) plus 100k lbs overpull.

Cement volumes are estimates based on 4-1/2" casing hung from 7" casing, and into 6" OH. 20% excess.
Mix and pump the cement slurry. Follow cement with liner dart and then saltwater displacement

Pre-flush (Spacer): **20 bbls** Viscous spacer

Cement Slurry: **726 sks** (197 bbls), 14.3ppg, 1.52 cu/ft/sk conventional system with 20% silica flour

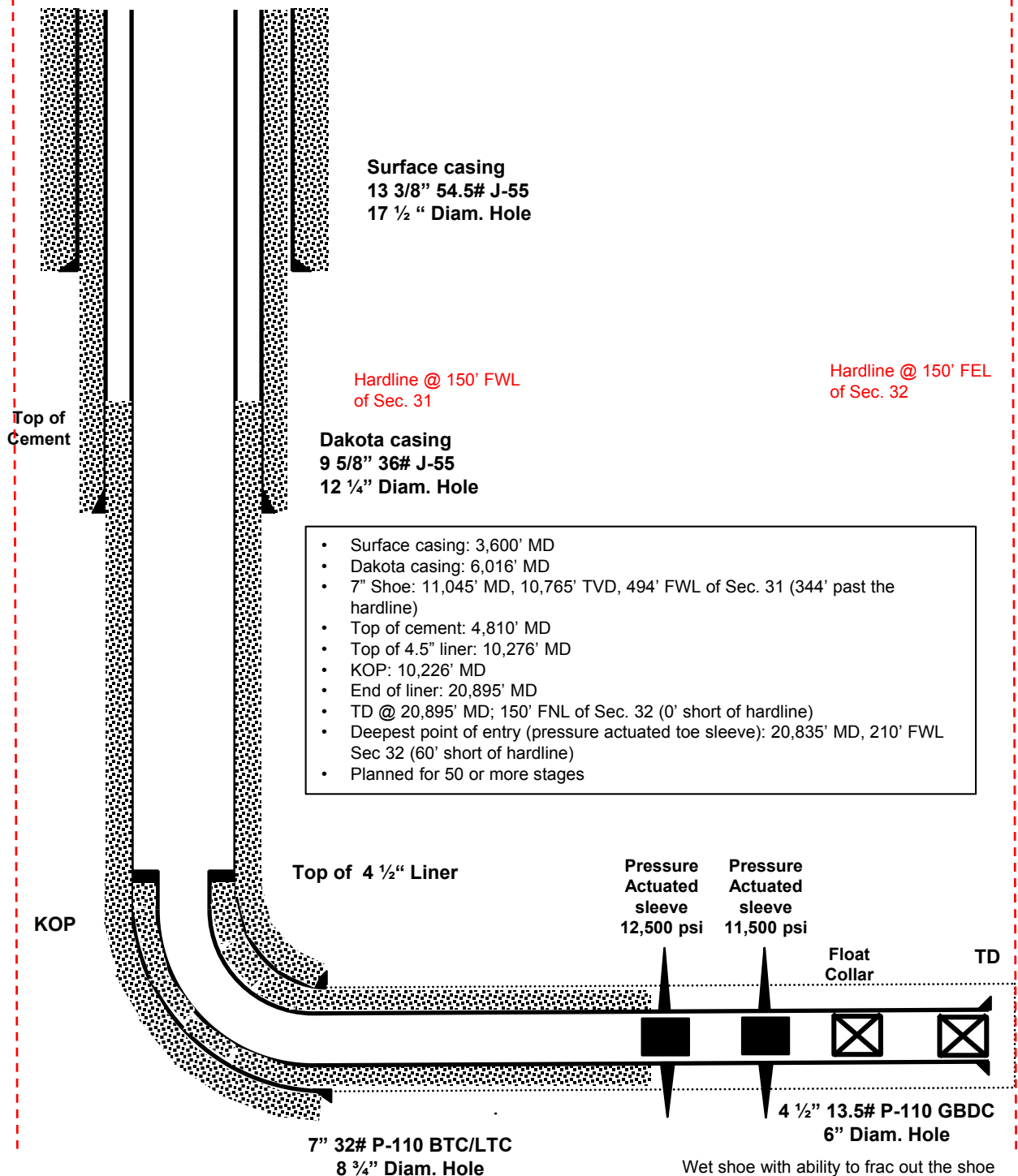
Displacement **272 bbls** Based on 53 ft shoe track and 4" drill pipe from surface to top of liner
4" DP: 0ft to 10226ft @ 0.011bbl/ft
4.5" casing: 10226ft to 20842ft; 0.0149bbl/ft

DRILLING PLAN									
OPERATOR		Oasis Petroleum			COUNTY/STATE		McKenzie Co., ND		
WELL NAME		Lewis Federal 5300 11-31 3B			RIG		-		
WELL TYPE		Middle Bakken							
LOCATION		T153N R100W S31 NWNW			Surface Location (survey plat):			1083' FNL 262' FWL	
EST. T.D.		20,895'			FINISH PAD ELEV:		2,110'		Sub Height: 25'
TOTAL LATERAL:		9,850'			KB ELEV:		2,135'		
MARKER		TVD		Subsea TVD		LOGS:		Type Interval	
Pierre NDIC MAP		1,985		150'		OH Logs: Triple Combo		KOP to Kibbey (or min run of 1800' whichever is greater)	
Greenhorn		4,596		-2,461'		GR/Resistivity		Bottom of surface casing	
Mowry (Dakota Group)		5,010		-2,875'		GR		To surface	
Inyan Kara (Dakota Group)		5,432		-3,297'		CND		Through Dakota Group (Inyan Kara Sands)	
Swift (Base Dakota Group)		5,855		-3,720'		CBL/GR:		Above top of cement/GR to base of casing	
Rierdon		6,371		-4,236'		MWD GR:		KOP to lateral TD	
Dunham Salt		6,898		-4,763'		DEVIATION:			
Dunham Salt Base		6,955		-4,820'					
Pine Salt		7,262		-5,127'					
Pine Salt Base		7,323		-5,188'		Surf:		3 deg. max., 1 deg / 100'; srvy every 500'	
Opeche Salt		7,441		-5,306'		Prod:		5 deg. max., 1 deg / 100'; srvy every 100'	
Opeche Salt Base		7,467		-5,332'		DST'S:			
Amsden		7,663		-5,528'					
Tyler		7,851		-5,716'					
Otter/Base Minnelusa		8,069		-5,934'		CORES:			
Kibbey Lime		8,416		-6,281'					
Charles Salt		8,562		-6,427'					
Base Last Salt		9,233		-7,098'		Core Planned?		NO	
Mission Canyon		9,447		-7,312'		Core Type:		-	
Lodgepole		10,006		-7,871'		Formations/Depths:			
False Bakken		10,720		-8,585'					
Upper Bakken Shale		10,730		-8,595'					
Middle Bakken		10,746		-8,611'		MUDLOGGING:			
Target Top		10,756		-8,621'					
Target Landing		10,765		-8,630'					
Target Base		10,774		-8,639'		Company:		TBD	
Lower Bakken		10,784		-8,649'		Starting Depth:		Begin 200' above Kibbey	
-		-		-		Sample Protocol:		30' samples in curve, 50' samples in lateral	
-		-		-		BOP:			
-		-		-					
-		-		-					
-		-		-		11" 5000 psi blind, pipe & annular			
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					

ELEVATION: 2,110' SL

Lewis Federal 5300 11-31 3B Proposed Wellbore Schematic

FORMATION: Bakken



OASIS PETROLEUM NA LLC

Lewis Federal 5300 11-31 3B

Wellbore: T153N-R100W Sec. 31 & 32

SHL: 1083' FNL & 262' FWL T153N-R100W Sec. 31

Williams County, North Dakota

Updated: 4-12-2018 TR



SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4.
INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

Received

Well File No.
30188

OCT 18 2018

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

ND Oil & Gas
Division

☒ Notice of Intent	Approximate Start Date December 15, 2018
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☒ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☐ Other	

Well Name and Number Lewis Federal 5300 11-31 3B					
Footages	Qtr-Qtr	Section	Township	Range	
1005 F N L 233 F W L Lot 1		31	153 N	100 W	
Field Baker	Pool Bakken	County McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum respectfully requests to amend the surface casing depth to 4900' MD (previously 2135') due to potential flowback in the Dakota formation. In order to counteract projected flowback at surface, Oasis needs to increase mud weight, and lowering the surface casing point will allow this to be performed safely. Oasis will also perform a two-stage cementing job using a DV tool set at approximately 2200'. A CBL will be run on 13-3/8" casing.

Attached are revised drill plans reflecting the amended casing points, cement volumes, and other related volumes.

** * OASIS May not drill into and/or set 13 3/8" casing into the Dakota Group.*

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9436	
Address 1001 Fannin St, Suite 1500			
City Houston		State TX	Zip Code 77002
Signature 	Printed Name Jennifer Swenson		
Title Regulatory Specialist	Date December 17, 2018		
Email Address jswenson@oasispetroleum.com			

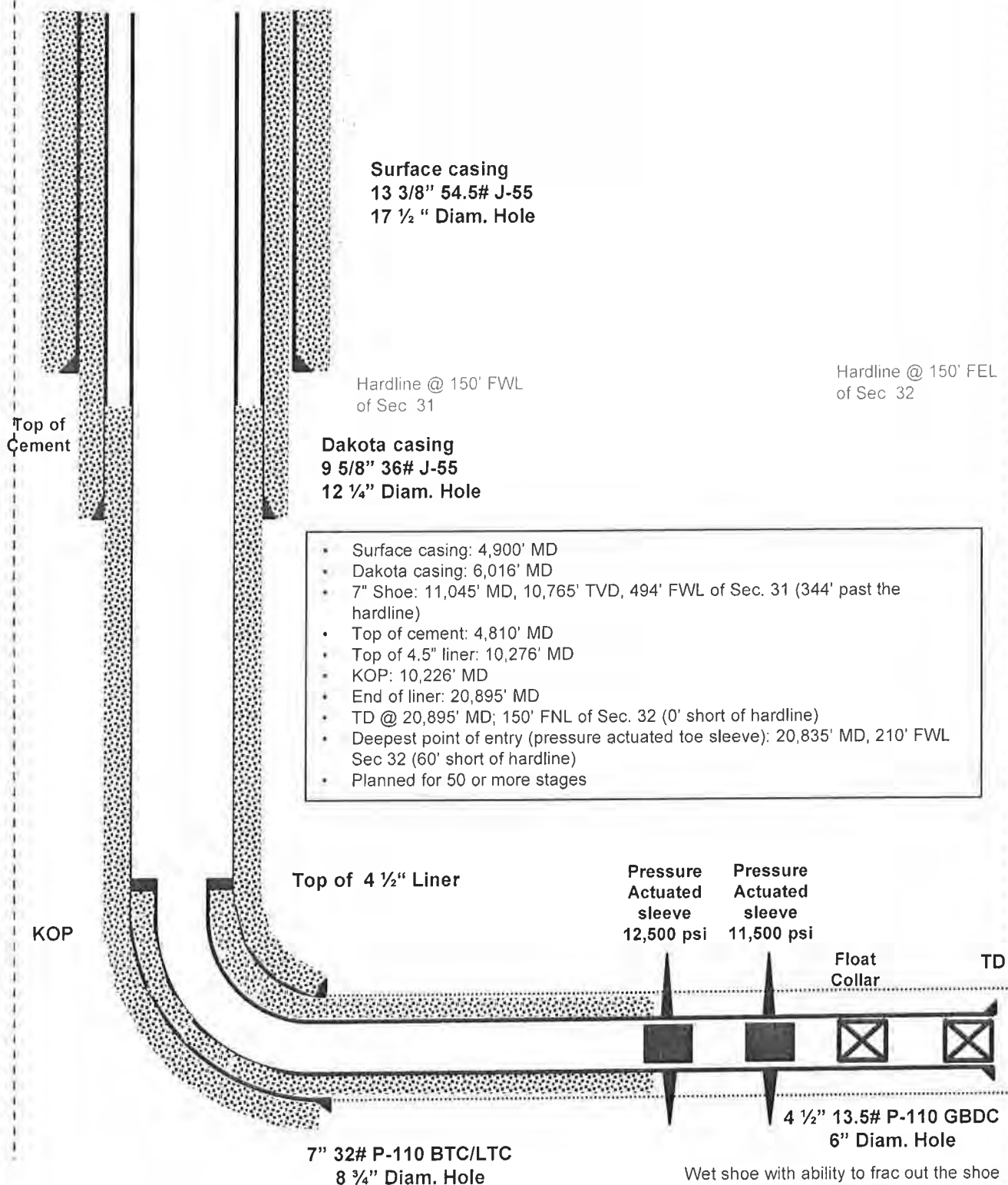
FOR STATE USE ONLY	
☐ Received	☒ Approved <i>***</i>
Date 12/18/18	
By 	
Title Mineral Resources Permit Manager	

DRILLING PLAN									
OPERATOR		Oasis Petroleum		COUNTY/STATE		McKenzie Co ND			
WELL NAME		Lewis Federal 5300 31-31H 2B		RIG					
WELL TYPE		Middle Bakken							
LOCATION		T153N R100W S31 N44W		Surface Location (survey plat): 1053' FNL		100' FWL			
EST. T.D.		20,895'		FINISH PAD ELEV:		2,110'		Sub Height: 25'	
TOTAL LATERAL:		9,850'		KB ELEV:		2,135'			
MARKER		TVD		Subsea TVD		LOGS:		Type Interval	
Pierre NDIC MAP		1,985		150'		OH Logs: Triple Combo		KOP to Kibbey (or min run of 1800' whichever is greater)	
Greenhorn		4,596		-2,481'		GR/Resistivity		Bottom of surface casing	
Mowry (Dakota Group)		5,010		-2,875'		GR		To surface	
Inyan Kara (Dakota Group)		5,432		-3,297'		CND		Through Dakota Group (Inyan Kara Sands)	
Swift (Base Dakota Group)		5,855		-3,720'		CBL/GR:		Above top of cement/GR to base of casing	
Rierdon		6,371		-4,230'		MWD GR:		KOP to lateral TD	
Dunham Salt		6,898		-4,763'		DEVIATION:			
Dunham Salt Base		6,955		-4,820'		Surf		3 deg max, 1 deg / 100'; srvy every 500'	
Pine Salt		7,262		-5,127'		Prod:		5 deg max, 1 deg / 100'; srvy every 100'	
Pine Salt Base		7,323		-5,188'		DST'S:			
Opeche Salt		7,441		-5,306'		Cores:			
Opeche Salt Base		7,467		-5,332'		Core Planned?		NO	
Amsden		7,663		-5,528'		Core Type:			
Tyler		7,851		-5,716'		Formations/Depths:			
Otter Base Minnelusa		8,069		-5,934'		MUDLOGGING:			
Kibbey Lime		8,416		-6,281'		Company:		TBD	
Charles Salt		8,562		-6,427'		Starting Depth:		Begin 200' above Kibbey	
Base Last Salt		9,233		-7,098'		Sample Protocol		30' samples in curve, 50' samples in lateral	
Mission Canyon		9,447		-7,312'		BOP:			
Lodgepole		10,006		-7,871'		11" 5000 psi blind, pipe & annular			
False Bakken		10,720		-8,585'		Est. Average Dip Rate:		0.0°/100'	
Upper Bakken Shale		10,730		-8,595'		Max. Anticipated BHP:		4,638'	
Middle Bakken		10,746		-8,611'		Surface Formation:		Glacial fill	
Target Top		10,756		-8,621'		MUD:			
Target Landing		10,765		-8,630'		Interval			
Target Base		10,774		-8,639'		Surface:		4,900' 4,900' FW/Gel Lime Sweeps	
Lower Bakken		10,784		-8,649'		Intermediate:		11,045' 11,045' Invert	
						Lateral:		20,895' 20,895' Salt Water	
						Casing:			
						Size			
						WT ppg			
						Hole			
						Depth			
						Cement			
						WOC			
						Remarks			
						150' into Pierre			
						Set Casing across Dakota			
						200' above Mowry			
						50' above KOP			
PROBABLE PLUGS, IF REQ'D:									
OTHER:									
		MD		TVD		FNL/FSL		FEL/FWL	
Surface:		4,900'		4,900'		1083 FNL		262 FWL	
KOP:		10,276'		10,266'		1300 FNL		21 FWL	
EOC:		11,045'		10,765'		1321 FNL		494 FWL	
Casing Point:		11,045'		10,765'		1321 FNL		494 FWL	
TD:		20,895'		10,641'		1321 FNL		150 FEL	
								S-T-R	
								AZI	
								Survey Company:	
								Build Rate: 12 deg / 100'	
Comments:									
Request waiver of open hole logs. Justification well: Lewis Federal 5300 31-31H (33053034330000) ~0.54 miles S of SHL									
The above open hole logs will be run if Oasis does not submit and receive an approved logging waiver from the NDIC									
Currently planned for 50 stages. No frac string planned. 4-1/2" cemented liner completed using plug & perf method									
Oasis Petroleum does not use Diesel Fuel, as defined by the US EPA in the list below, in our hydraulic fracture operations.									
68334-30-5 (Primary Name: Fuels, diesel) 68476-34-6 (Primary Name: Fuels, diesel, No. 2) 68476-30-2 (Primary Name: Fuel oil No. 2)									
68476-31-3 (Primary Name: Fuel oil, No. 4) 8008-20-6 (Primary Name: Kerosene)									
Geology:		LRH		4/3/2016		Engineering:		TR 4/12/16	
Revision:		Revision 2				Revision:		Revision 2	

ELEVATION: 2,110' SL

Lewis Federal 5300 11-31 3B Proposed Wellbore Schematic

FORMATION: Bakken



OASIS PETROLEUM NA LLC

Lewis Federal 5300 11-31 3B
Wellbore: T153N-R100W Sec. 31 & 32
SHL: 1083' FNL & 262' FWL T153N-R100W Sec. 31

Williams County, North Dakota
Updated: 4-12-2018 TR

**Oasis Petroleum
Well Summary
Lewis Federal 5300 11-31 3B
Section 31 T153N R100W
McKenzie County, ND**

SURFACE CASING AND CEMENT DESIGN

Size	Interval	Weight	Grade	Coupling	I.D.	Drift
13-3/8"	0' - 4900'	54.5	J-55	STC	12.615"	12.459"

Interval	Description	Collapse	Burst	Tension
		(psi) / a	(psi) / b	(1000 lbs) / c
0' - 4900'	13-3/8", 54.5#, J-55, LTC, 8rd	1130 / 0.49	2730 / 0.82	514 / 1.55

API Rating & Safety Factor

- a) Collapse based on full casing evacuation with 9 ppg fluid on backside (4900' setting depth)
- b) Burst pressure based on 13 ppg fluid with no fluid on backside (4900' setting depth)
- c) Based on string weight in 9 ppg fluid at 4900' TVD plus 100k# overpull (Buoyed weight equals 230k lbs)

Cement volumes are based on 13-3/8" casing set in 17-1/2" hole with 60% excess to circulate cement back to surface.
Mix and pump the following slurry

Pre-flush (Spacer): 20 bbls fresh water

Lead Slurry: 1779 sks (919 bbls), 11.5 lb/gal, 2.97 cu ft/sk Varicem Cement with 0.125 il/sk Lost Circulation Additive

Tail Slurry: 300 sks (62 bbls), 13.0 lb/gal, 2.01 cu ft/sk Varicem with .125 lb/sk Lost Circulation Agent

**Oasis Petroleum
Well Summary
Lewis Federal 5300 11-31 3B
Section 31 T153N R100W
McKenzie County, ND**

Contingency INTERMEDIATE CASING AND CEMENT DESIGN

Size	Interval	Weight	Grade	Coupling	I.D.	Drift
9-5/8"	0' - 6016'	36	J-55	LTC	8 921"	8.765"

Interval	Description	Collapse	Burst	Tension
		(psi) / a	(psi) / b	(1000 lbs) / c
0' - 6016'	9-5/8", 36#, J-55, LTC, 8rd	2020 / 2 25	3520 / 1 36	453 / 1 60

API Rating & Safety Factor

- a) Collapse based on full casing evacuation with 10.4 ppg fluid on backside (6016' setting depth)
- b) Burst pressure calculated from a gas kick coming from the production zone (Bakken Pool) at 9,000psi and a subsequent breakdown at the 9-5/8" shoe, based on a 15 2#/ft fracture gradient Backup of 9 ppg fluid
- c) Tension based on string weight in 10.4 ppg fluid at 6016' TVD plus 100k# overpull. (Buoyed weight equals 182k lbs)

Cement volumes are based on 9-5/8" casing set in 12-1/4 " hole with 10% excess to circulate cement back to surface

Pre-flush (Spacer): 20 bbls Chem wash

Lead Slurry: 569 sks (294 bbls), 2.90 ft³/sk, 11 5 lb/gal Conventional system with 94 lb/sk cement, 4% D079 extender, 2% D053 expanding agent, 2% CaCl₂ and 0 250 lb/sk D130 lost circulation control agent

Tail Slurry: 612 sks (126 bbls), 1.16 ft³/sk 15 8 lb/gal Conventional system with 94 lb/sk cement, 0.25% CaCl₂, and 0.250 lb/sk lost circulation control agent

**Oasis Petroleum
Well Summary
Lewis Federal 5300 11-31 3B
Section 31 T153N R100W
McKenzie County, ND**

PRODUCTION LINER

Size	Interval	Weight	Grade	Coupling	I.D.	Drift
4-1/2"	10226' - 20895'	13.5	P-110	GB CD BTC	3.920"	3.795"

Interval	Length	Description	Collapse	Burst	Tension
			(psi) a	(psi) b	(1000 lbs) c
10226' - 20895'	10669	4-1/2", 13.5 lb, P-110, GB CD BTC	10670 / 1.99	12410 / 1.28	443 / 1.98

API Rating & Safety Factor

- a) Based on full casing evacuation with 9.5 ppg fluid on backside @ 10841' TVD
- b) Burst pressure based on 9000 psi treating pressure with 10.2 ppg internal fluid gradient and 9 ppg external fluid gradient @ 10841' TVD.
- c) Based on string weight in 9.5 ppg fluid (Buoyed weight: 123k lbs) plus 100k lbs overpull.

Cement volumes are estimates based on 4-1/2" casing hung from 7" casing, and into 6" OH 20% excess
Mix and pump the cement slurry Follow cement with liner dart and then saltwater displacement

Pre-flush (Spacer): 20 bbls Viscous spacer

Cement Slurry: 726 sks (197 bbls), 14.3ppg, 1.52 cu/ft/sk conventional system with 20% silica flour

Displacement 272 bbls Based on 53 ft shoe track and 4" drill pipe from surface to top of liner
4" DP: 0ft to 10226ft @ 0.011 bbl/ft
4.5" casing: 10226ft to 20842ft; 0.0149bbl/ft

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

Received

Well File No.
30188

DEC 12 2018

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date December 11, 2018
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

ND Oil & Gas Division	
☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other APD Renewal	

Well Name and Number Lewis Federal 5300 11-31 3B							
Footages 1083		262		Qtr-Qtr	Section	Township	Range
1005 F N L		233 F W L		Lot 1	31	153 N	100 W
Field Baker		Pool Bakken		County McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum requests the above referenced well be renewed. There are no planned changes to the current drill plan.

Please use the credit card on file for the \$100.00 application processing fee.

Permit Expires 12/11/19.

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9436	
Address 1001 Fannin St, Suite 1500			
City Houston		State TX	Zip Code 77002
Signature 	Printed Name Jennifer Swenson		
Title Regulatory Specialist	Date December 11, 2018		
Email Address jswenson@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date 12/12/18	
By 	
Title Engineering Technician	



Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.dmr.nd.gov/oilgas/

30188

November 13, 2018

OASIS PETRO NO AMER
1001 FANNIN STE 1500
HOUSTON, TX 77002

RE: LEWIS FEDERAL 5300 11-31 3B
LOT1 Sec. 31-153N-100W
MCKENZIE COUNTY
WELL FILE NO. 30188

Gentlemen:

The records and files of the Industrial Commission indicate that the above referenced permit will expire December 11, 2018.

Permits to drill are only valid for one year in the State of North Dakota. If you would like to renew for another year, please submit a Form 4 along with the \$100.00 filing fee. Alternatively, you may elect to send in a Form 4 cancelling the permit. If you have any questions, please contact Todd Holweger.

Sincerely,


Rachel Morris
Administrative Assistant

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

Received

APR 18 2018

Well File No.

30188**ND Oil & Gas
Division**

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date April 30, 2018
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☒ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☐ Other	SHL and BHL Changes

Well Name and Number Lewis Federal 5300 11-31 3B					
Footages		Qtr-Qtr	Section	Township	Range
1005 F N L 233 F W L		LOT1	31	153 N	100 W
Field Baker	Pool Bakken	County McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum respectfully requests the following changes to the above referenced permitted well:

SHL change: 1083' FNL & 262' FWL Lot 1 Sec. 31 T153N R100W (Previously 1005' FNL & 233' FWL Lot 1 Sec. 31 T153N R100W)

BHL change: 1321' FNL & 150' FEL LOT4 Sec. 32 T153N R100W (Previously 998' FNL & 210' FEL NENE Sec. 32 T153N R100W)

Oasis must submit plat of production CTB within 30 days of 4-24-2018.

TD Change: 20895' MD / 10841' TVD (Previously 20610' MD / 10844' TVD)

Please see attached supporting documents.

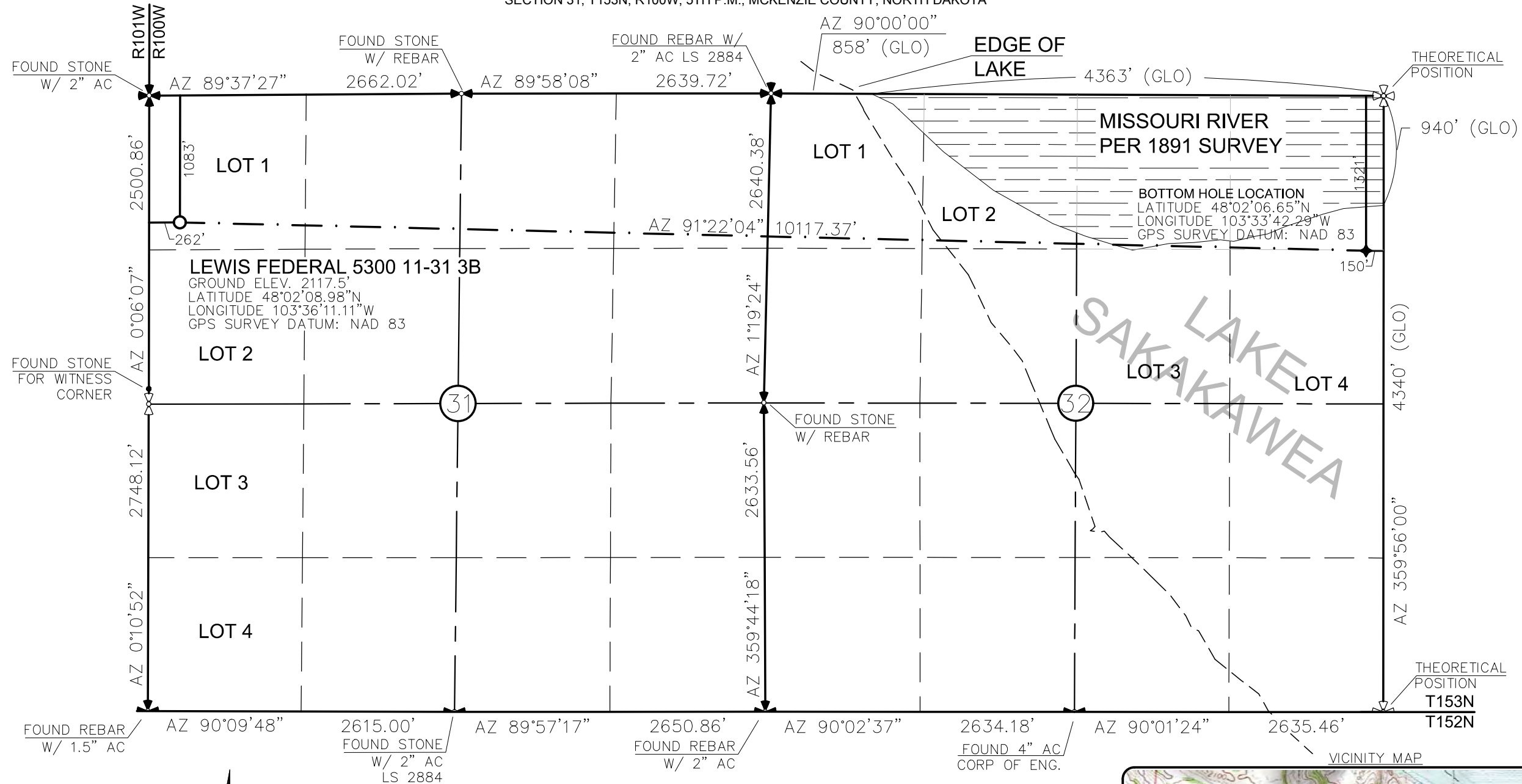
Must run a CBL on the 9-5/8" intermediate string which is proposed to isolate the Dakota Group prior to running 7" casing.

The East 150' setback is based on a production liner cemented in the lateral with a wet shoe and the ability to frac out the shoe.

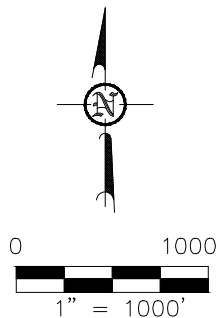
Company Oasis Petroleum North America LLC		Telephone Number 281-404-9494	
Address 1001 Fannin, Suite 1500			
City Houston	State TX	Zip Code 77002	
Signature <i>Sadie Goodrum</i>	Printed Name Sadie Goodrum		
Title Regulatory Specialist II	Date April 18, 2018		
Email Address sgoodrum@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date 4/24/2018	
By <i>[Signature]</i>	
Title Engineering Tech.	

WELL LOCATION
OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
LEWIS FEDERAL 5300 11-31 3B
1083 FEET FROM NORTH LINE AND 262 FEET FROM WEST LINE
SECTION 31, T153N, R100W, 5TH P.M., MCKENZIE COUNTY, NORTH DAKOTA



THIS DOCUMENT WAS ORIGINALLY
ISSUED AND SEALED BY DARYL D.
KASEMAN, PLS, REGISTRATION
NUMBER 3880 ON 3-27-18
AND THE ORIGINAL DOCUMENTS
ARE STORED AT THE OFFICES OF
INTERSTATE ENGINEERING, INC.

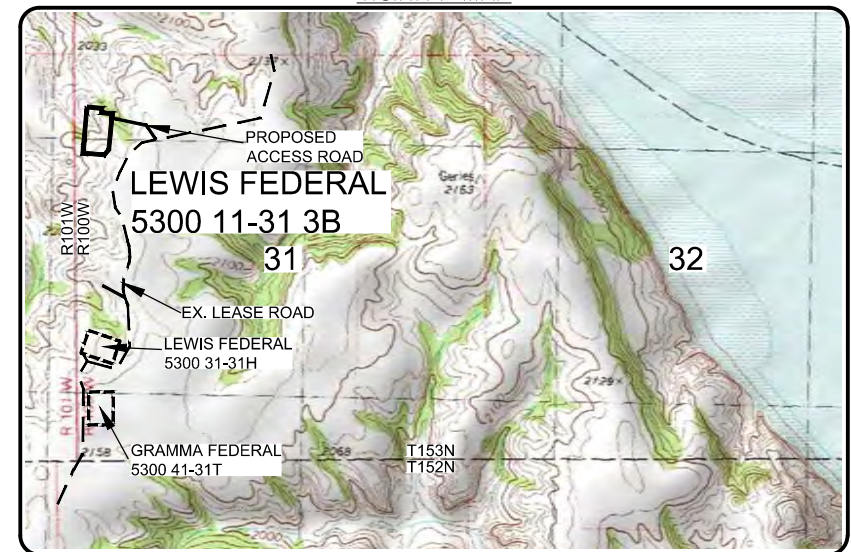
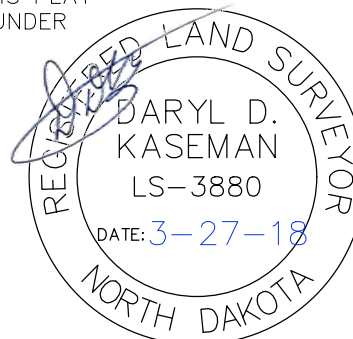


- ⊕ — MONUMENT — RECOVERED
⊗ — MONUMENT — NOT RECOVERED

STAKED ON 3/26/18
VERTICAL CONTROL DATUM WAS BASED UPON
CONTROL POINT 705 WITH AN ELEVATION OF 2158.3'

THIS SURVEY AND PLAT IS BEING PROVIDED AT THE REQUEST
OF JOHN LEE OF OASIS PETROLEUM. I CERTIFY THAT THIS PLAT
CORRECTLY REPRESENTS WORK PERFORMED BY ME OR UNDER
MY SUPERVISION AND IS TRUE AND CORRECT TO
THE BEST OF MY KNOWLEDGE AND BELIEF.

[Signature]
DARYL D. KASEMAN LS-3880

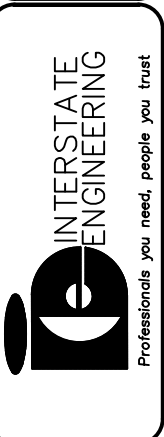


© 2018, INTERSTATE ENGINEERING, INC.

Revision No.	Date	By	Description

OASIS PETROLEUM NORTH AMERICA, LLC WELL LOCATION SECTION 31, T153N, R100W, 5TH P.M., MCKENZIE COUNTY, NORTH DAKOTA	Project No.: S17-09-183 Date: MARCH 2018
Drawn By: J.D.M. Checked By: D.D.K.	

Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617
Fax (406) 433-5618
www.interstateeng.com
Other offices in Minnesota, North Dakota and South Dakota



DRILLING PLAN									
OPERATOR		Oasis Petroleum			COUNTY/STATE		McKenzie Co., ND		
WELL NAME		Lewis Federal 5300 11-31 3B			RIG		-		
WELL TYPE		Middle Bakken							
LOCATION		T153N R100W S31 NWNW			Surface Location (survey plat):		1083' FNL 262' FWL		
EST. T.D.		20,895'			FINISH PAD ELEV:		2,110'		Sub Height: 25'
TOTAL LATERAL:		9,850'			KB ELEV:		2,135'		
MARKER		TVD		Subsea TVD	LOGS:		Type	Interval	
Pierre	NDIC MAP	1,985		150'	OH Logs:		Triple Combo	KOP to Kibbey (or min run of 1800' whichever is greater)	
Greenhorn		4,596		-2,461'			GR/Resistivity	Bottom of surface casing	
Mowry (Dakota Group)		5,010		-2,875'			GR	To surface	
Inyan Kara (Dakota Group)		5,432		-3,297'			CND	Through Dakota Group (Inyan Kara Sands)	
Swift (Base Dakota Group)		5,855		-3,720'	CBL/GR:			Above top of cement/GR to base of casing	
Rierdon		6,371		-4,236'	MWD GR:			KOP to lateral TD	
Dunham Salt		6,898		-4,763'	DEVIATION:				
Dunham Salt Base		6,955		-4,820'	Surf: 3 deg. max., 1 deg / 100'; srvy every 500'				
Pine Salt		7,262		-5,127'	Prod: 5 deg. max., 1 deg / 100'; srvy every 100'				
Pine Salt Base		7,323		-5,188'	DST'S:				
Opeche Salt		7,441		-5,306'					
Opeche Salt Base		7,467		-5,332'					
Amsden		7,663		-5,528'	CORES:				
Tyler		7,851		-5,716'	Core Planned? NO				
Otter/Base Minnelusa		8,069		-5,934'	Core Type: -				
Kibbey Lime		8,416		-6,281'	Formations/Depths:				
Charles Salt		8,562		-6,427'					
Base Last Salt		9,233		-7,098'					
Mission Canyon		9,447		-7,312'	MUDLOGGING:				
Lodgepole		10,006		-7,871'	Company: TBD				
False Bakken		10,720		-8,585'	Starting Depth: Begin 200' above Kibbey				
Upper Bakken Shale		10,730		-8,595'	Sample Protocol: 30' samples in curve, 50' samples in lateral				
Middle Bakken		10,746		-8,611'					
Target Top		10,756		-8,621'					
Target Landing		10,765		-8,630'					
Target Base		10,774		-8,639'	BOP:				
Lower Bakken		10,784		-8,649'	11" 5000 psi blind, pipe & annular				
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
-		-		-					
Est. Average Dip Rate:		89.58							
Max. Anticipated BHP:		4,698'			Surface Formation: Glacial till				
MUD:		Interval	Type	WT	Vis	WL	Remarks		
Surface:	0' -	2,135'	FW/Gel Lime Sweeps	8.4-9.0	28-32	NC	Circ Mud Tanks		
Intermediate:	2,135' -	11,045'	Invert	9.5-10.4	40-50	30+HtHp	Circ Mud Tanks		
Lateral:	11,045' -	20,895'	Salt Water	9.8-10.2	28-32	NC	Circ Mud Tanks		
CASING:		Size	Wt ppf	Hole	Depth	Cement	WOC	Remarks	
Surface:	13-3/8"	54.5#	17-1/2"	2,135'		To Surface	12 hours	150' into Pierre	
Intermediate: (Dakota)	9-5/8"	36#	12-1/4"	5,955'		To Surface	24 hours	Set Casing across Dakota	
Intermediate:	7"	32#	8-3/4"	11,045'		4810	24 hours	200' above Mowry	
Production Liner:	4.5"	13.5#	6"	20,895'		10226		50' above KOP	
PROBABLE PLUGS, IF REQ'D:									
OTHER:		MD	TVD	FNL/FSL	FEL/FWL	S-T-R	AZI		
Surface:	2,135'	2,135'	1083 FNL	262 FWL	Sec 31 T153N R100W	-	-	Survey Company:	
KOP:	10,276'	10,266'	1300 FNL	21 FWL	Sec 31 T153N R100W	-	-	Build Rate:	12 deg /100'
EOC:	11,045'	10,765'	1321 FNL	494 FWL	Sec 31 T153N R100W	90.0	90.0		
Casing Point:	11,045'	10,765'	1321 FNL	494 FWL	Sec 31 T153N R100W	90.0	90.0		
TD:	20,895'	10,841'	1321 FNL	150 FEL	Sec 32 T153N R100W	90.0	90.0		
Comments:									
Request waiver of open hole logs. Justification well: Lewis Federal 5300 31-31H (33053034330000) ~0.54 miles S of SHL									
The above open hole logs will be run if Oasis does not submit and receive an approved logging waiver from the NDIC.									
Currently planned for 50 stages. No frac string planned. 4-1/2" cemented liner completed using plug & perf method									
Oasis Petroleum does not use Diesel Fuel, as defined by the US EPA in the list below, in our hydraulic fracture operations.									
68334-30-5 (Primary Name: Fuels, diesel) 68476-34-6 (Primary Name: Fuels, diesel, No. 2) 68476-30-2 (Primary Name: Fuel oil No. 2)									
68476-31-3 (Primary Name: Fuel oil, No. 4) 8008-20-6 (Primary Name: Kerosene)									
Geology:		LRH		4/3/2018		Engineering:		TR 4/12/18	
Revision:						Revision:			
Revision 2:						Revision 2:			

Oasis Petroleum
Well Summary
Lewis Federal 5300 11-31 3B
Section 31 T153N R100W
McKenzie County, ND

SURFACE CASING AND CEMENT DESIGN

Size	Interval	Weight	Grade	Coupling	I.D.	Drift
13-3/8"	0' - 2135'	54.5	J-55	STC	12.615"	12.459"

Interval	Description	Collapse	Burst	Tension
		(psi) / a	(psi) / b	(1000 lbs) / c
0' - 2135'	13-3/8", 54.5#, J-55, LTC, 8rd	1130 / 1.13	2730 / 1.89	514 / 2.56

API Rating & Safety Factor

- a) Collapse based on full casing evacuation with 9 ppg fluid on backside (2135' setting depth).
- b) Burst pressure based on 13 ppg fluid with no fluid on backside (2135' setting depth).
- c) Based on string weight in 9 ppg fluid at 2135' TVD plus 100k# overpull. (Buoyed weight equals 100k lbs.)

Cement volumes are based on 13-3/8" casing set in 17-1/2 " hole with 60% excess to circulate cement back to surface.
Mix and pump the following slurry.

Pre-flush (Spacer): **20 bbls** fresh water

Lead Slurry: **720 sks** (372 bbls), 11.5 lb/gal, 2.97 cu. Ft./sk Varicem Cement with 0.125 il/sk Lost Circulation Additive

Tail Slurry: **300 sks** (62 bbls), 13.0 lb/gal, 2.01 cu.ft./sk Varicem with .125 lb/sk Lost Circulation Agent

Oasis Petroleum
Well Summary
Lewis Federal 5300 11-31 3B
Section 31 T153N R100W
McKenzie County, ND

Contingency INTERMEDIATE CASING AND CEMENT DESIGN

Size	Interval	Weight	Grade	Coupling	I.D.	Drift
9-5/8"	0' - 5955'	36	J-55	LTC	8.921"	8.765"

Interval	Description	Collapse	Burst	Tension
		(psi) / a	(psi) / b	(1000 lbs) / c
0' - 5955'	9-5/8", 36#, J-55, LTC, 8rd	2020 / 2.29	3520 / 1.38	453 / 1.61

API Rating & Safety Factor

- a) Collapse based on full casing evacuation with 10.4 ppg fluid on backside (5955' setting depth).
- b) Burst pressure calculated from a gas kick coming from the production zone (Bakken Pool) at 9,000psi and a subsequent breakdown at the 9-5/8" shoe, based on a 15.2#/ft fracture gradient. Backup of 9 ppg fluid..
- c) Tension based on string weight in 10.4 ppg fluid at 5955' TVD plus 100k# overpull. (Buoyed weight equals 180k lbs.)

Cement volumes are based on 9-5/8" casing set in 12-1/4 " hole with 10% excess to circulate cement back to surface.

Pre-flush (Spacer): **20 bbls** Chem wash

Lead Slurry: **517 sks** (267 bbls), 2.90 ft³/sk, 11.5 lb/gal Conventional system with 94 lb/sk cement, 4% D079 extender, 2% D053 expanding agent, 2% CaCl₂ and 0.250 lb/sk D130 lost circulation control agent.

Tail Slurry: **594 sks** (123 bbls), 1.16 ft³/sk 15.8 lb/gal Conventional system with 94 lb/sk cement, 0.25% CaCl₂, and 0.250 lb/sk lost circulation control agent

INTERMEDIATE CASING AND CEMENT DESIGN

****Special Drift 7"32# to 6.0"**

API Rating & Safety Factor

- Tail Slurry:** **568 sks** (166 bbls), 14.0 ppg, 1.55 cu. ft./sk Extendercem System with .2% HR-5 Retarder and .25 lb/sk Lost Circulation Additive

Oasis Petroleum
Well Summary
Lewis Federal 5300 11-31 3B
Section 31 T153N R100W
McKenzie County, ND

PRODUCTION LINER

Size	Interval	Weight	Grade	Coupling	I.D.	Drift
4-1/2"	10226' - 20895'	13.5	P-110	GB CD BTC	3.920"	3.795"

Interval	Length	Description	Collapse	Burst	Tension
		4-1/2", 13.5 lb, P-110, GB CD BTC	(psi) a	(psi) b	(1000 lbs) c
10226' - 20895'	10669		10670 / 1.99	12410 / 1.28	443 / 1.98

API Rating & Safety Factor

- a) Based on full casing evacuation with 9.5 ppg fluid on backside @ 10841' TVD.
- b) Burst pressure based on 9000 psi treating pressure with 10.2 ppg internal fluid gradient and 9 ppg external fluid gradient @ 10841' TVD.
- c) Based on string weight in 9.5 ppg fluid (Buoyed weight: 123k lbs.) plus 100k lbs overpull.

Cement volumes are estimates based on 4-1/2" casing hung from 7" casing, and into 6" OH. 20% excess.
Mix and pump the cement slurry. Follow cement with liner dart and then saltwater displacement

Pre-flush (Spacer): **20 bbls** Viscous spacer

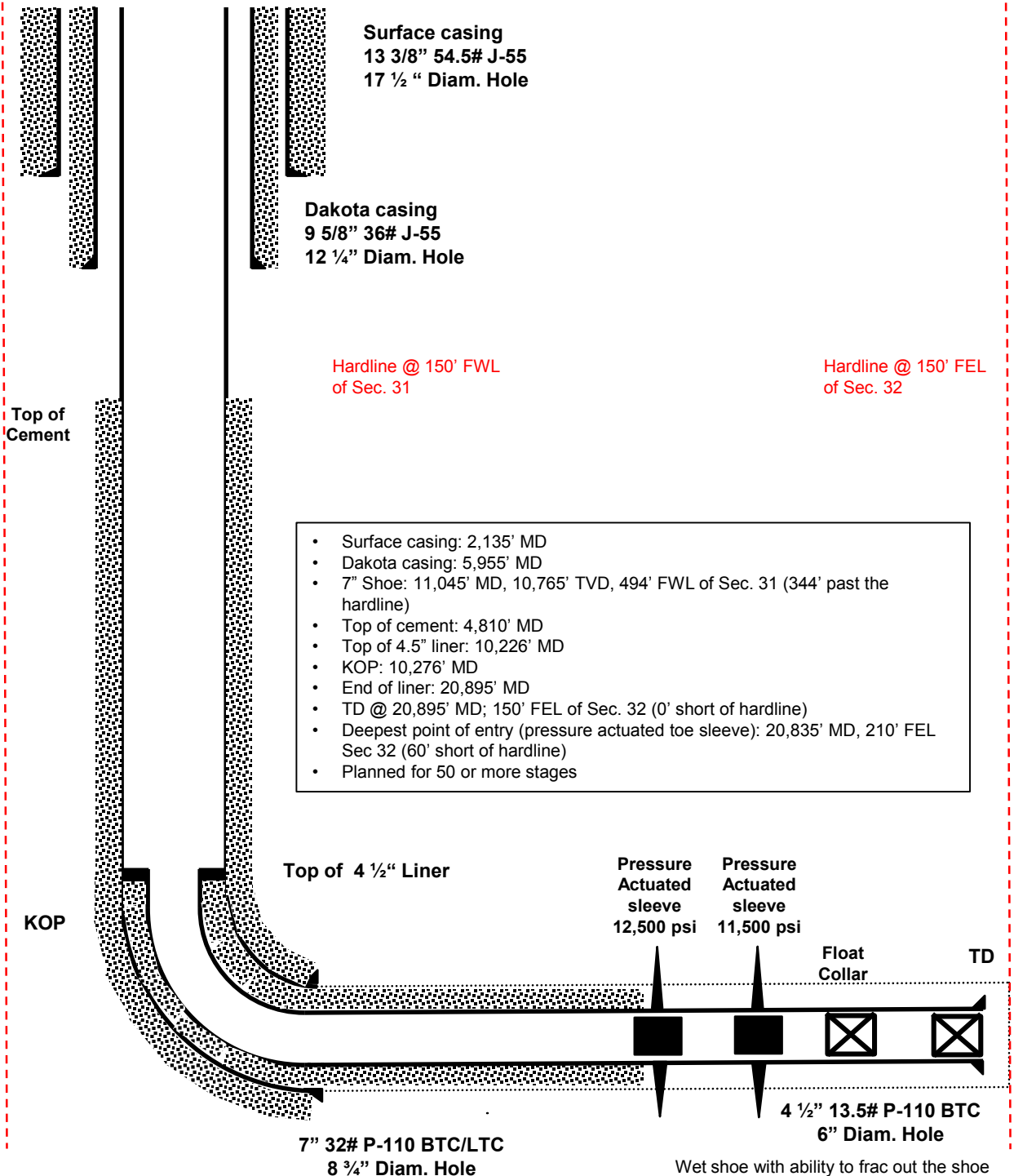
Cement Slurry: **726 sks** (197 bbls), 14.3ppg, 1.52 cu/ft/sk conventional system with
20% silica flour

Displacement **272 bbls** Based on 53 ft shoe track and 4" drill pipe from surface to top of liner
4" DP: 0ft to 10226ft @ 0.011bbl/ft
4.5" casing: 10226ft to 20842ft; 0.0149bbl/ft

ELEVATION: 2,110' SL

Lewis Federal 5300 11-31 3B Proposed Wellbore Schematic

FORMATION: Bakken



OASIS PETROLEUM NA LLC

Lewis Federal 5300 11-31 3B

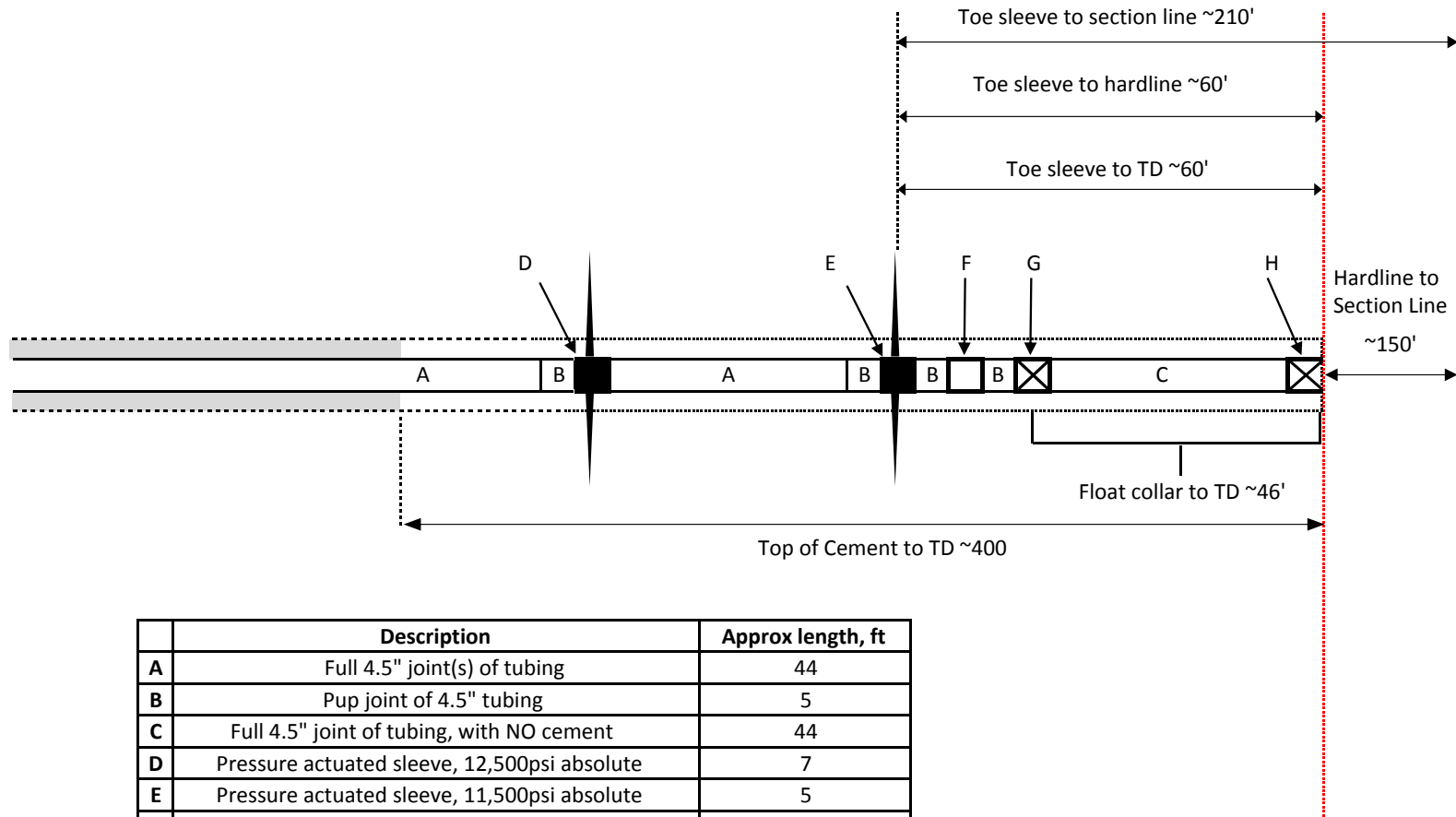
Wellbore: T153N-R100W Sec. 31 & 32

SHL: 1083' FNL & 262' FWL T153N-R100W Sec. 31

Williams County, North Dakota

Updated: 4-12-2018 TR

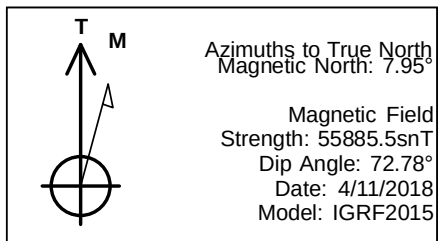
Lewis Federal 5300 11-31 3B planned toe completion



*First stage to be pumped out of sleeves labeled as "D" and "E." Acid used as necessary to break down formation

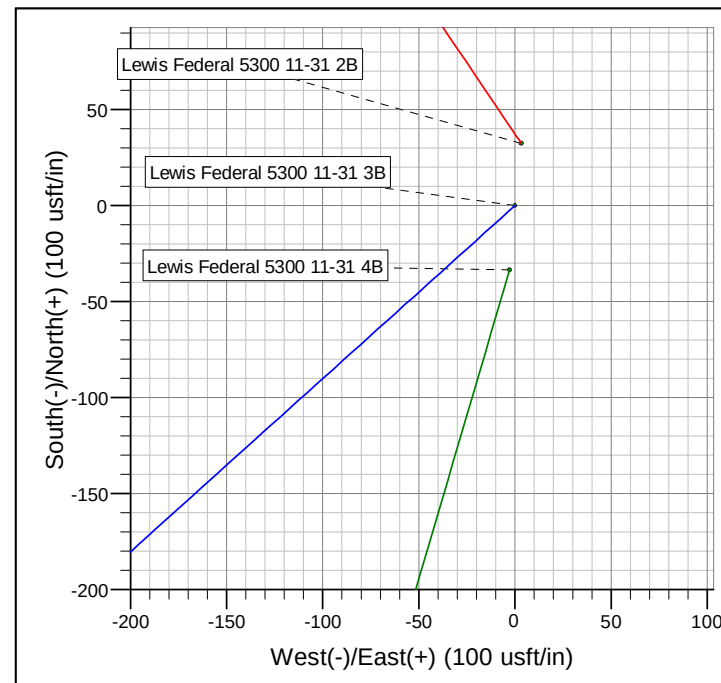
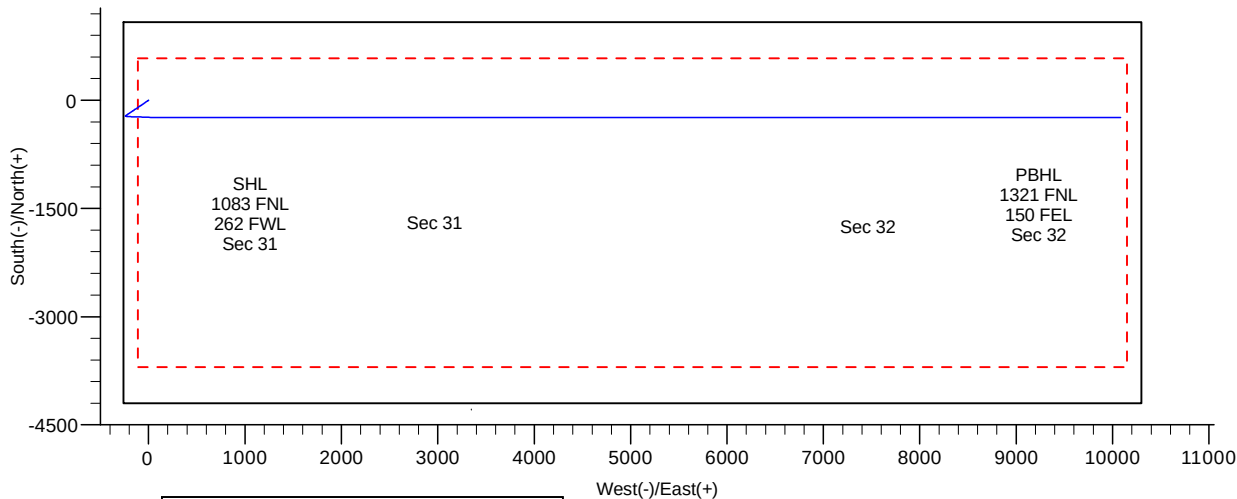
*Diagram not to scale

Project: Indian Hills
 Site: 153N-100W-31/32
 Well: Lewis Federal 5300 11-31 3B
 Wellbore: Lewis Federal 5300 11-31 3B
 Design: Design #1



WELL DETAILS: Lewis Federal 5300 11-31 3B

Ground Level: 2110.0
 Northing 393129.76 Easting 1209541.15 Latitude 48° 2' 8.980 N Longitude 103° 36' 11.110 W

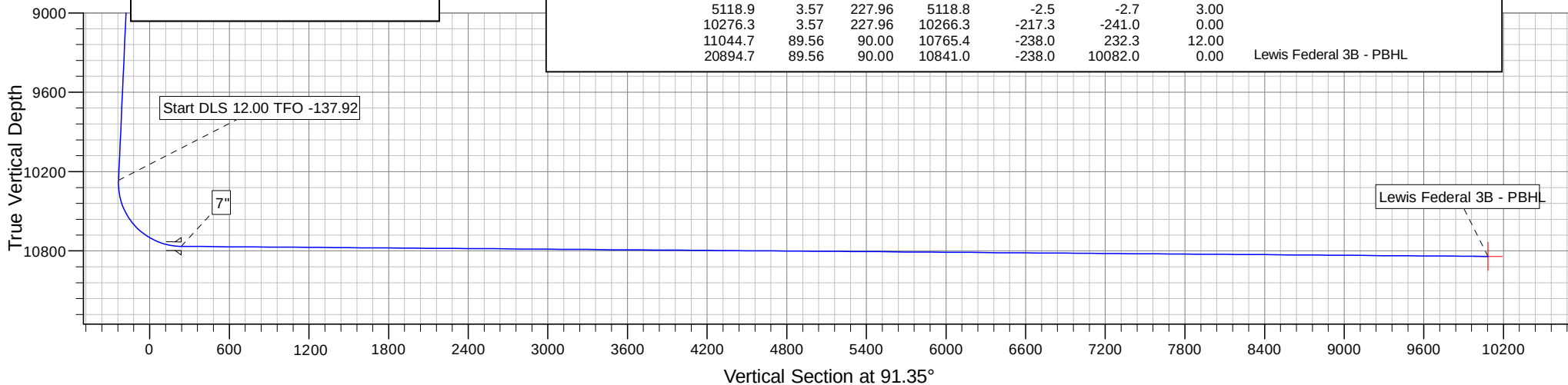


CASING DETAILS

TVD	MD	Name	Size
2135.0	2135.0	13 3/8"	13.375
5953.3	5955.0	9 5/8"	9.625
10765.4	11044.7	7"	7.000

SECTION DETAILS

MD	Inc	Azi	TVD	+N/-S	+E/-W	Dleg	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00	
5000.0	0.00	0.00	5000.0	0.0	0.0	0.00	
5118.9	3.57	227.96	5118.8	-2.5	-2.7	3.00	
10276.3	3.57	227.96	10266.3	-217.3	-241.0	0.00	
11044.7	89.56	90.00	10765.4	-238.0	232.3	12.00	
20894.7	89.56	90.00	10841.0	-238.0	10082.0	0.00	Lewis Federal 3B - PBHL



Vertical Section at 91.35°

Oasis

Indian Hills

153N-100W-31/32

Lewis Federal 5300 11-31 3B

Lewis Federal 5300 11-31 3B

Lewis Federal 5300 11-31 3B

Plan: Design #1

Standard Planning Report

12 April, 2018

Oasis Petroleum

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well Lewis Federal 5300 11-31 3B
Company:	Oasis	TVD Reference:	WELL @ 2135.0usft
Project:	Indian Hills	MD Reference:	WELL @ 2135.0usft
Site:	153N-100W-31/32	North Reference:	True
Well:	Lewis Federal 5300 11-31 3B	Survey Calculation Method:	Minimum Curvature
Wellbore:	Lewis Federal 5300 11-31 3B		
Design:	Design #1		

Project	Indian Hills		
Map System:	US State Plane 1983	System Datum:	Mean Sea Level
Geo Datum:	North American Datum 1983		
Map Zone:	North Dakota Northern Zone		

Site		153N-100W-31/32				
Site Position:		Northing:	390,397.86 usft	Latitude:	48° 1' 42.010 N	
From:	Lat/Long	Easting:	1,209,464.32 usft	Longitude:	103° 36' 10.620 W	
Position Uncertainty:		0.0 usft	Slot Radius:	13.200 in	Grid Convergence:	-2.31 °

Well	Lewis Federal 5300 11-31 3B, DEV					
Well Position	+N/-S	2,732.8 usft	Northing:	393,129.76 usft	Latitude:	48° 2' 8.980 N
	+E/-W	-33.3 usft	Easting:	1,209,541.15 usft	Longitude:	103° 36' 11.110 W
Position Uncertainty	2.0 usft		Wellhead Elevation:		Ground Level:	2,110.0 usft

Wellbore	Lewis Federal 5300 11-31 3B				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	IGRF2015	4/11/2018	7.95	72.78	55,886

Design	Design #1			
Audit Notes:				
Version:	Phase:	PROTOTYPE	Tie On Depth:	0.0
Vertical Section:	Depth From (TVD) (usft)	+N/-S (usft)	+E/-W (usft)	Direction (°)
	0.0	0.0	0.0	91.35

Plan Sections										
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)	TFO (°)	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00	0.00	0.00	0.00	
5,000.0	0.00	0.00	5,000.0	0.0	0.0	0.00	0.00	0.00	0.00	
5,118.9	3.57	227.96	5,118.8	-2.5	-2.7	3.00	3.00	0.00	227.96	
10,276.3	3.57	227.96	10,266.3	-217.3	-241.0	0.00	0.00	0.00	0.00	
11,044.7	89.56	90.00	10,765.4	-238.0	232.3	12.00	11.19	-17.95	-137.92	
20,894.7	89.56	90.00	10,841.0	-238.0	10,082.0	0.00	0.00	0.00	0.00	Lewis Federal 3B - PE

Oasis Petroleum

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well Lewis Federal 5300 11-31 3B
Company:	Oasis	TVD Reference:	WELL @ 2135.0usft
Project:	Indian Hills	MD Reference:	WELL @ 2135.0usft
Site:	153N-100W-31/32	North Reference:	True
Well:	Lewis Federal 5300 11-31 3B	Survey Calculation Method:	Minimum Curvature
Wellbore:	Lewis Federal 5300 11-31 3B		
Design:	Design #1		

Planned Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
5,000.0	0.00	0.00	5,000.0	0.0	0.0	0.0	0.00	0.00	0.00
Start Build 3.00									
5,100.0	3.00	227.96	5,100.0	-1.8	-1.9	-1.9	3.00	3.00	0.00
5,118.9	3.57	227.96	5,118.8	-2.5	-2.7	-2.7	3.00	3.00	0.00
Start 5157.5 hold at 5118.9 MD									
5,200.0	3.57	227.96	5,199.8	-5.9	-6.5	-6.4	0.00	0.00	0.00
5,300.0	3.57	227.96	5,299.6	-10.0	-11.1	-10.9	0.00	0.00	0.00
5,400.0	3.57	227.96	5,399.4	-14.2	-15.7	-15.4	0.00	0.00	0.00
5,500.0	3.57	227.96	5,499.2	-18.4	-20.4	-19.9	0.00	0.00	0.00
5,600.0	3.57	227.96	5,599.0	-22.5	-25.0	-24.4	0.00	0.00	0.00
5,700.0	3.57	227.96	5,698.8	-26.7	-29.6	-29.0	0.00	0.00	0.00
5,800.0	3.57	227.96	5,798.6	-30.9	-34.2	-33.5	0.00	0.00	0.00
5,900.0	3.57	227.96	5,898.4	-35.0	-38.8	-38.0	0.00	0.00	0.00
5,955.0	3.57	227.96	5,953.3	-37.3	-41.4	-40.5	0.00	0.00	0.00
9 5/8"									
6,000.0	3.57	227.96	5,998.2	-39.2	-43.5	-42.5	0.00	0.00	0.00
6,100.0	3.57	227.96	6,098.0	-43.3	-48.1	-47.0	0.00	0.00	0.00
6,200.0	3.57	227.96	6,197.8	-47.5	-52.7	-51.6	0.00	0.00	0.00
6,300.0	3.57	227.96	6,297.6	-51.7	-57.3	-56.1	0.00	0.00	0.00
6,400.0	3.57	227.96	6,397.4	-55.8	-61.9	-60.6	0.00	0.00	0.00
6,500.0	3.57	227.96	6,497.2	-60.0	-66.5	-65.1	0.00	0.00	0.00
6,600.0	3.57	227.96	6,597.1	-64.2	-71.2	-69.6	0.00	0.00	0.00
6,700.0	3.57	227.96	6,696.9	-68.3	-75.8	-74.2	0.00	0.00	0.00
6,800.0	3.57	227.96	6,796.7	-72.5	-80.4	-78.7	0.00	0.00	0.00
6,900.0	3.57	227.96	6,896.5	-76.7	-85.0	-83.2	0.00	0.00	0.00
7,000.0	3.57	227.96	6,996.3	-80.8	-89.6	-87.7	0.00	0.00	0.00
7,100.0	3.57	227.96	7,096.1	-85.0	-94.3	-92.2	0.00	0.00	0.00
7,200.0	3.57	227.96	7,195.9	-89.2	-98.9	-96.8	0.00	0.00	0.00
7,300.0	3.57	227.96	7,295.7	-93.3	-103.5	-101.3	0.00	0.00	0.00
7,400.0	3.57	227.96	7,395.5	-97.5	-108.1	-105.8	0.00	0.00	0.00
7,500.0	3.57	227.96	7,495.3	-101.7	-112.7	-110.3	0.00	0.00	0.00
7,600.0	3.57	227.96	7,595.1	-105.8	-117.4	-114.8	0.00	0.00	0.00
7,700.0	3.57	227.96	7,694.9	-110.0	-122.0	-119.4	0.00	0.00	0.00
7,800.0	3.57	227.96	7,794.7	-114.2	-126.6	-123.9	0.00	0.00	0.00
7,900.0	3.57	227.96	7,894.5	-118.3	-131.2	-128.4	0.00	0.00	0.00
8,000.0	3.57	227.96	7,994.3	-122.5	-135.8	-132.9	0.00	0.00	0.00
8,100.0	3.57	227.96	8,094.1	-126.7	-140.5	-137.4	0.00	0.00	0.00
8,200.0	3.57	227.96	8,194.0	-130.8	-145.1	-142.0	0.00	0.00	0.00
8,300.0	3.57	227.96	8,293.8	-135.0	-149.7	-146.5	0.00	0.00	0.00
8,400.0	3.57	227.96	8,393.6	-139.2	-154.3	-151.0	0.00	0.00	0.00
8,500.0	3.57	227.96	8,493.4	-143.3	-158.9	-155.5	0.00	0.00	0.00
8,600.0	3.57	227.96	8,593.2	-147.5	-163.6	-160.0	0.00	0.00	0.00
8,700.0	3.57	227.96	8,693.0	-151.7	-168.2	-164.6	0.00	0.00	0.00
8,800.0	3.57	227.96	8,792.8	-155.8	-172.8	-169.1	0.00	0.00	0.00
8,900.0	3.57	227.96	8,892.6	-160.0	-177.4	-173.6	0.00	0.00	0.00
9,000.0	3.57	227.96	8,992.4	-164.2	-182.0	-178.1	0.00	0.00	0.00
9,100.0	3.57	227.96	9,092.2	-168.3	-186.7	-182.6	0.00	0.00	0.00
9,200.0	3.57	227.96	9,192.0	-172.5	-191.3	-187.2	0.00	0.00	0.00
9,300.0	3.57	227.96	9,291.8	-176.7	-195.9	-191.7	0.00	0.00	0.00
9,400.0	3.57	227.96	9,391.6	-180.8	-200.5	-196.2	0.00	0.00	0.00
9,500.0	3.57	227.96	9,491.4	-185.0	-205.1	-200.7	0.00	0.00	0.00
9,600.0	3.57	227.96	9,591.2	-189.2	-209.8	-205.2	0.00	0.00	0.00
9,700.0	3.57	227.96	9,691.1	-193.3	-214.4	-209.8	0.00	0.00	0.00

Oasis Petroleum

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well Lewis Federal 5300 11-31 3B
Company:	Oasis	TVD Reference:	WELL @ 2135.0usft
Project:	Indian Hills	MD Reference:	WELL @ 2135.0usft
Site:	153N-100W-31/32	North Reference:	True
Well:	Lewis Federal 5300 11-31 3B	Survey Calculation Method:	Minimum Curvature
Wellbore:	Lewis Federal 5300 11-31 3B		
Design:	Design #1		

Planned Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
9,800.0	3.57	227.96	9,790.9	-197.5	-219.0	-214.3	0.00	0.00	0.00
9,900.0	3.57	227.96	9,890.7	-201.7	-223.6	-218.8	0.00	0.00	0.00
10,000.0	3.57	227.96	9,990.5	-205.8	-228.2	-223.3	0.00	0.00	0.00
10,100.0	3.57	227.96	10,090.3	-210.0	-232.9	-227.8	0.00	0.00	0.00
10,200.0	3.57	227.96	10,190.1	-214.1	-237.5	-232.4	0.00	0.00	0.00
10,276.3	3.57	227.96	10,266.3	-217.3	-241.0	-235.8	0.00	0.00	0.00
Start DLS 12.00 TFO -137.92									
10,300.0	2.40	175.43	10,289.9	-218.3	-241.5	-236.3	12.00	-4.94	-222.05
10,400.0	12.42	100.90	10,389.1	-222.4	-230.7	-225.4	12.00	10.02	-74.53
10,500.0	24.30	95.28	10,483.8	-226.4	-199.6	-194.2	12.00	11.88	-5.62
10,600.0	36.26	93.24	10,570.0	-230.0	-149.4	-143.9	12.00	11.96	-2.04
10,700.0	48.23	92.12	10,643.9	-233.0	-82.3	-76.8	12.00	11.98	-1.13
10,800.0	60.22	91.35	10,702.2	-235.4	-1.4	4.2	12.00	11.98	-0.77
10,900.0	72.21	90.75	10,742.5	-237.1	89.9	95.5	12.00	11.99	-0.60
11,000.0	84.20	90.22	10,762.9	-237.9	187.6	193.2	12.00	11.99	-0.52
11,044.7	89.55	90.00	10,765.4	-238.0	232.2	237.8	12.00	11.99	-0.50
Start 9850.0 hold at 11044.7 MD - 7"									
11,100.0	89.56	90.00	10,765.8	-238.0	287.5	293.1	0.01	0.01	0.00
11,200.0	89.56	90.00	10,766.6	-238.0	387.5	393.1	0.00	0.00	0.00
11,300.0	89.56	90.00	10,767.3	-238.0	487.5	493.0	0.00	0.00	0.00
11,400.0	89.56	90.00	10,768.1	-238.0	587.5	593.0	0.00	0.00	0.00
11,500.0	89.56	90.00	10,768.9	-238.0	687.5	693.0	0.00	0.00	0.00
11,600.0	89.56	90.00	10,769.6	-238.0	787.5	792.9	0.00	0.00	0.00
11,700.0	89.56	90.00	10,770.4	-238.0	887.5	892.9	0.00	0.00	0.00
11,800.0	89.56	90.00	10,771.2	-238.0	987.5	992.9	0.00	0.00	0.00
11,900.0	89.56	90.00	10,771.9	-238.0	1,087.5	1,092.8	0.00	0.00	0.00
12,000.0	89.56	90.00	10,772.7	-238.0	1,187.5	1,192.8	0.00	0.00	0.00
12,100.0	89.56	90.00	10,773.5	-238.0	1,287.5	1,292.8	0.00	0.00	0.00
12,200.0	89.56	90.00	10,774.2	-238.0	1,387.5	1,392.7	0.00	0.00	0.00
12,300.0	89.56	90.00	10,775.0	-238.0	1,487.5	1,492.7	0.00	0.00	0.00
12,400.0	89.56	90.00	10,775.8	-238.0	1,587.5	1,592.7	0.00	0.00	0.00
12,500.0	89.56	90.00	10,776.5	-238.0	1,687.5	1,692.7	0.00	0.00	0.00
12,600.0	89.56	90.00	10,777.3	-238.0	1,787.5	1,792.6	0.00	0.00	0.00
12,700.0	89.56	90.00	10,778.1	-238.0	1,887.5	1,892.6	0.00	0.00	0.00
12,800.0	89.56	90.00	10,778.8	-238.0	1,987.5	1,992.6	0.00	0.00	0.00
12,900.0	89.56	90.00	10,779.6	-238.0	2,087.5	2,092.5	0.00	0.00	0.00
13,000.0	89.56	90.00	10,780.4	-238.0	2,187.5	2,192.5	0.00	0.00	0.00
13,100.0	89.56	90.00	10,781.1	-238.0	2,287.5	2,292.5	0.00	0.00	0.00
13,200.0	89.56	90.00	10,781.9	-238.0	2,387.5	2,392.4	0.00	0.00	0.00
13,300.0	89.56	90.00	10,782.7	-238.0	2,487.5	2,492.4	0.00	0.00	0.00
13,400.0	89.56	90.00	10,783.4	-238.0	2,587.5	2,592.4	0.00	0.00	0.00
13,500.0	89.56	90.00	10,784.2	-238.0	2,687.5	2,692.3	0.00	0.00	0.00
13,600.0	89.56	90.00	10,785.0	-238.0	2,787.5	2,792.3	0.00	0.00	0.00
13,700.0	89.56	90.00	10,785.7	-238.0	2,887.5	2,892.3	0.00	0.00	0.00
13,800.0	89.56	90.00	10,786.5	-238.0	2,987.5	2,992.2	0.00	0.00	0.00
13,900.0	89.56	90.00	10,787.3	-238.0	3,087.5	3,092.2	0.00	0.00	0.00
14,000.0	89.56	90.00	10,788.1	-238.0	3,187.5	3,192.2	0.00	0.00	0.00
14,100.0	89.56	90.00	10,788.8	-238.0	3,287.5	3,292.2	0.00	0.00	0.00
14,200.0	89.56	90.00	10,789.6	-238.0	3,387.5	3,392.1	0.00	0.00	0.00
14,300.0	89.56	90.00	10,790.4	-238.0	3,487.5	3,492.1	0.00	0.00	0.00
14,400.0	89.56	90.00	10,791.1	-238.0	3,587.4	3,592.1	0.00	0.00	0.00
14,500.0	89.56	90.00	10,791.9	-238.0	3,687.4	3,692.0	0.00	0.00	0.00
14,600.0	89.56	90.00	10,792.7	-238.0	3,787.4	3,792.0	0.00	0.00	0.00
14,700.0	89.56	90.00	10,793.4	-238.0	3,887.4	3,892.0	0.00	0.00	0.00

Oasis Petroleum

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well Lewis Federal 5300 11-31 3B
Company:	Oasis	TVD Reference:	WELL @ 2135.0usft
Project:	Indian Hills	MD Reference:	WELL @ 2135.0usft
Site:	153N-100W-31/32	North Reference:	True
Well:	Lewis Federal 5300 11-31 3B	Survey Calculation Method:	Minimum Curvature
Wellbore:	Lewis Federal 5300 11-31 3B		
Design:	Design #1		

Planned Survey										
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)	
14,800.0	89.56	90.00	10,794.2	-238.0	3,987.4	3,991.9	0.00	0.00	0.00	
14,900.0	89.56	90.00	10,795.0	-238.0	4,087.4	4,091.9	0.00	0.00	0.00	
15,000.0	89.56	90.00	10,795.7	-238.0	4,187.4	4,191.9	0.00	0.00	0.00	
15,100.0	89.56	90.00	10,796.5	-238.0	4,287.4	4,291.8	0.00	0.00	0.00	
15,200.0	89.56	90.00	10,797.3	-238.0	4,387.4	4,391.8	0.00	0.00	0.00	
15,300.0	89.56	90.00	10,798.0	-238.0	4,487.4	4,491.8	0.00	0.00	0.00	
15,400.0	89.56	90.00	10,798.8	-238.0	4,587.4	4,591.8	0.00	0.00	0.00	
15,500.0	89.56	90.00	10,799.6	-238.0	4,687.4	4,691.7	0.00	0.00	0.00	
15,600.0	89.56	90.00	10,800.3	-238.0	4,787.4	4,791.7	0.00	0.00	0.00	
15,700.0	89.56	90.00	10,801.1	-238.0	4,887.4	4,891.7	0.00	0.00	0.00	
15,800.0	89.56	90.00	10,801.9	-238.0	4,987.4	4,991.6	0.00	0.00	0.00	
15,900.0	89.56	90.00	10,802.6	-238.0	5,087.4	5,091.6	0.00	0.00	0.00	
16,000.0	89.56	90.00	10,803.4	-238.0	5,187.4	5,191.6	0.00	0.00	0.00	
16,100.0	89.56	90.00	10,804.2	-238.0	5,287.4	5,291.5	0.00	0.00	0.00	
16,200.0	89.56	90.00	10,804.9	-238.0	5,387.4	5,391.5	0.00	0.00	0.00	
16,300.0	89.56	90.00	10,805.7	-238.0	5,487.4	5,491.5	0.00	0.00	0.00	
16,400.0	89.56	90.00	10,806.5	-238.0	5,587.4	5,591.4	0.00	0.00	0.00	
16,500.0	89.56	90.00	10,807.3	-238.0	5,687.4	5,691.4	0.00	0.00	0.00	
16,600.0	89.56	90.00	10,808.0	-238.0	5,787.4	5,791.4	0.00	0.00	0.00	
16,700.0	89.56	90.00	10,808.8	-238.0	5,887.4	5,891.4	0.00	0.00	0.00	
16,800.0	89.56	90.00	10,809.6	-238.0	5,987.4	5,991.3	0.00	0.00	0.00	
16,900.0	89.56	90.00	10,810.3	-238.0	6,087.4	6,091.3	0.00	0.00	0.00	
17,000.0	89.56	90.00	10,811.1	-238.0	6,187.4	6,191.3	0.00	0.00	0.00	
17,100.0	89.56	90.00	10,811.9	-238.0	6,287.4	6,291.2	0.00	0.00	0.00	
17,200.0	89.56	90.00	10,812.6	-238.0	6,387.4	6,391.2	0.00	0.00	0.00	
17,300.0	89.56	90.00	10,813.4	-238.0	6,487.4	6,491.2	0.00	0.00	0.00	
17,400.0	89.56	90.00	10,814.2	-238.0	6,587.4	6,591.1	0.00	0.00	0.00	
17,500.0	89.56	90.00	10,814.9	-238.0	6,687.4	6,691.1	0.00	0.00	0.00	
17,600.0	89.56	90.00	10,815.7	-238.0	6,787.4	6,791.1	0.00	0.00	0.00	
17,700.0	89.56	90.00	10,816.5	-238.0	6,887.4	6,891.0	0.00	0.00	0.00	
17,800.0	89.56	90.00	10,817.2	-238.0	6,987.3	6,991.0	0.00	0.00	0.00	
17,900.0	89.56	90.00	10,818.0	-238.0	7,087.3	7,091.0	0.00	0.00	0.00	
18,000.0	89.56	90.00	10,818.8	-238.0	7,187.3	7,191.0	0.00	0.00	0.00	
18,100.0	89.56	90.00	10,819.5	-238.0	7,287.3	7,290.9	0.00	0.00	0.00	
18,200.0	89.56	90.00	10,820.3	-238.0	7,387.3	7,390.9	0.00	0.00	0.00	
18,300.0	89.56	90.00	10,821.1	-238.0	7,487.3	7,490.9	0.00	0.00	0.00	
18,400.0	89.56	90.00	10,821.8	-238.0	7,587.3	7,590.8	0.00	0.00	0.00	
18,500.0	89.56	90.00	10,822.6	-238.0	7,687.3	7,690.8	0.00	0.00	0.00	
18,600.0	89.56	90.00	10,823.4	-238.0	7,787.3	7,790.8	0.00	0.00	0.00	
18,700.0	89.56	90.00	10,824.1	-238.0	7,887.3	7,890.7	0.00	0.00	0.00	
18,800.0	89.56	90.00	10,824.9	-238.0	7,987.3	7,990.7	0.00	0.00	0.00	
18,900.0	89.56	90.00	10,825.7	-238.0	8,087.3	8,090.7	0.00	0.00	0.00	
19,000.0	89.56	90.00	10,826.4	-238.0	8,187.3	8,190.6	0.00	0.00	0.00	
19,100.0	89.56	90.00	10,827.2	-238.0	8,287.3	8,290.6	0.00	0.00	0.00	
19,200.0	89.56	90.00	10,828.0	-238.0	8,387.3	8,390.6	0.00	0.00	0.00	
19,300.0	89.56	90.00	10,828.8	-238.0	8,487.3	8,490.6	0.00	0.00	0.00	
19,400.0	89.56	90.00	10,829.5	-238.0	8,587.3	8,590.5	0.00	0.00	0.00	
19,500.0	89.56	90.00	10,830.3	-238.0	8,687.3	8,690.5	0.00	0.00	0.00	
19,600.0	89.56	90.00	10,831.1	-238.0	8,787.3	8,790.5	0.00	0.00	0.00	
19,700.0	89.56	90.00	10,831.8	-238.0	8,887.3	8,890.4	0.00	0.00	0.00	
19,800.0	89.56	90.00	10,832.6	-238.0	8,987.3	8,990.4	0.00	0.00	0.00	
19,900.0	89.56	90.00	10,833.4	-238.0	9,087.3	9,090.4	0.00	0.00	0.00	
20,000.0	89.56	90.00	10,834.1	-238.0	9,187.3	9,190.3	0.00	0.00	0.00	
20,100.0	89.56	90.00	10,834.9	-238.0	9,287.3	9,290.3	0.00	0.00	0.00	

Oasis Petroleum

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well Lewis Federal 5300 11-31 3B
Company:	Oasis	TVD Reference:	WELL @ 2135.0usft
Project:	Indian Hills	MD Reference:	WELL @ 2135.0usft
Site:	153N-100W-31/32	North Reference:	True
Well:	Lewis Federal 5300 11-31 3B	Survey Calculation Method:	Minimum Curvature
Wellbore:	Lewis Federal 5300 11-31 3B		
Design:	Design #1		

Planned Survey										
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)	
20,200.0	89.56	90.00	10,835.7	-238.0	9,387.3	9,390.3	0.00	0.00	0.00	
20,300.0	89.56	90.00	10,836.4	-238.0	9,487.3	9,490.2	0.00	0.00	0.00	
20,400.0	89.56	90.00	10,837.2	-238.0	9,587.3	9,590.2	0.00	0.00	0.00	
20,500.0	89.56	90.00	10,838.0	-238.0	9,687.3	9,690.2	0.00	0.00	0.00	
20,600.0	89.56	90.00	10,838.7	-238.0	9,787.3	9,790.2	0.00	0.00	0.00	
20,700.0	89.56	90.00	10,839.5	-238.0	9,887.3	9,890.1	0.00	0.00	0.00	
20,800.0	89.56	90.00	10,840.3	-238.0	9,987.3	9,990.1	0.00	0.00	0.00	
20,894.7	89.56	90.00	10,841.0	-238.0	10,082.0	10,084.8	0.00	0.00	0.00	
TD at 20894.7										

Design Targets										
Target Name	Dip Angle (°)	Dip Dir. (°)	TVD (usft)	+N/-S (usft)	+E/-W (usft)	Northing (usft)	Easting (usft)	Latitude	Longitude	
- hit/miss target										
- Shape										
Lewis Federal 3B - PBH	0.00	0.00	10,841.0	-238.0	10,082.0	392,485.74	1,219,605.37	48° 2' 6.604 N	103° 33' 42.755 W	
- plan hits target center										
- Point										

Casing Points					
Measured Depth (usft)	Vertical Depth (usft)	Name	Casing Diameter (in)	Hole Diameter (in)	
2,135.0	2,135.0	13 3/8"	13.375	17.500	
5,955.0	5,953.3	9 5/8"	9.625	12.250	
11,044.7	10,765.4	7"	7.000	8.750	

Oasis Petroleum

Planning Report

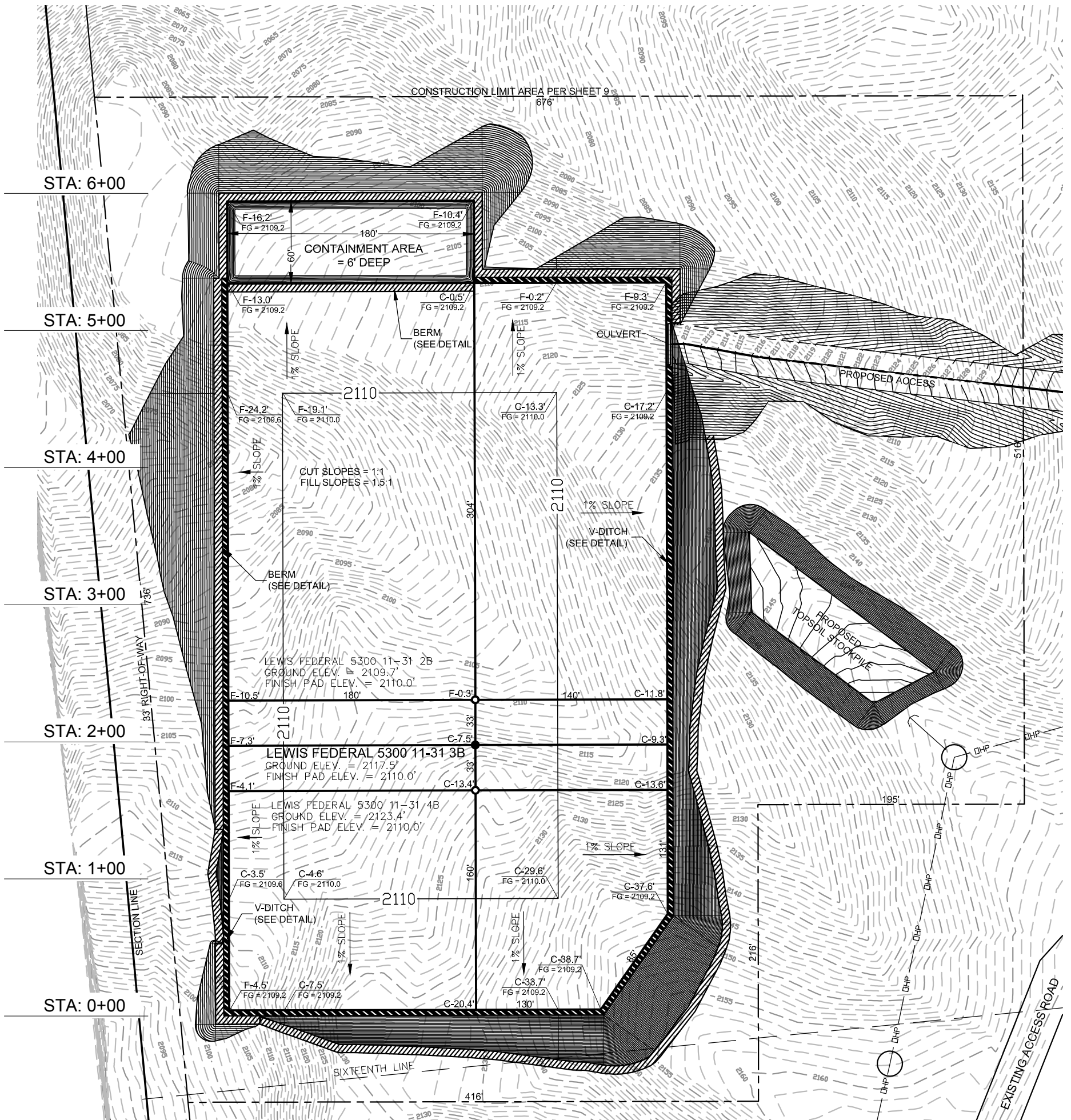
Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well Lewis Federal 5300 11-31 3B
Company:	Oasis	TVD Reference:	WELL @ 2135.0usft
Project:	Indian Hills	MD Reference:	WELL @ 2135.0usft
Site:	153N-100W-31/32	North Reference:	True
Well:	Lewis Federal 5300 11-31 3B	Survey Calculation Method:	Minimum Curvature
Wellbore:	Lewis Federal 5300 11-31 3B		
Design:	Design #1		

Formations						
Measured Depth (usft)	Vertical Depth (usft)	Name	Lithology	Dip (°)	Dip Direction (°)	
1,985.0	1,985.0	Pierre				
4,596.0	4,596.0	Greenhorn				
5,010.0	5,010.0	Mowry (Dakota Group)				
5,432.7	5,432.0	Inyan Kara (Dakota Group)				
5,856.5	5,855.0	Swift (Base Dakota Group)				
6,373.5	6,371.0	Rierdon				
6,901.5	6,898.0	Dunham Salt				
6,958.6	6,955.0	Dunham Salt Base				
7,266.2	7,262.0	Pine Salt				
7,327.4	7,323.0	Pine Salt Base				
7,445.6	7,441.0	Opeche Salt				
7,471.6	7,467.0	Opeche Salt Base				
7,668.0	7,663.0	Amsden				
7,856.4	7,851.0	Tyler				
8,074.8	8,069.0	Otter/Base Minnelusa				
8,422.5	8,416.0	Kibbey Lime				
8,568.8	8,562.0	Charles Salt				
9,241.1	9,233.0	Base Last Salt				
9,455.5	9,447.0	Mission Canyon				
10,015.6	10,006.0	Lodgepole				
10,838.5	10,720.0	False Bakken				
10,863.4	10,730.0	Upper Bakken Shale				
10,911.9	10,746.0	Middle Bakken				
10,953.6	10,756.0	Target Top				
11,029.5	10,765.0	Target Landing				
12,170.1	10,774.0	Target Base				
13,472.3	10,784.0	Lower Bakken				

Plan Annotations					
Measured Depth (usft)	Vertical Depth (usft)	Local Coordinates			
		+N/-S (usft)	+E/-W (usft)	Comment	
5,000.0	5,000.0	0.0	0.0	Start Build 3.00	
5,118.9	5,118.8	-2.5	-2.7	Start 5157.5 hold at 5118.9 MD	
10,276.3	10,266.3	-217.3	-241.0	Start DLS 12.00 TFO -137.92	
11,044.7	10,765.4	-238.0	232.3	Start 9850.0 hold at 11044.7 MD	
20,894.7	10,841.0	-238.0	10,082.0	TD at 20894.7	

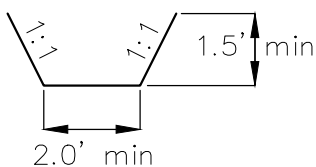
PAD LAYOUT
OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
LEWIS FEDERAL 5300 11-31 3B
1083 FEET FROM NORTH LINE AND 262 FEET FROM WEST LINE
SECTION 31, T153N, R100W, 5TH P.M., MCKENZIE COUNTY, NORTH DAKOTA

THIS DOCUMENT WAS ORIGINALLY
ISSUED AND SEALED BY DARYL D.
KASEMAN, PLS, REGISTRATION
NUMBER 3880 ON 3-27-18
AND THE ORIGINAL DOCUMENTS
ARE STORED AT THE OFFICES OF
INTERSTATE ENGINEERING, INC.



NOTE: Pad dimensions shown are to
usable area, the v-ditch and berm
areas shall be built to the outside of
the pad dimensions.

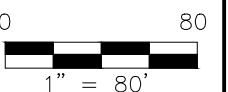
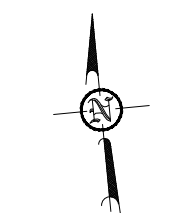
V-DITCH DETAIL



- BERM
- DITCH
- Proposed Contours
- Original Contours

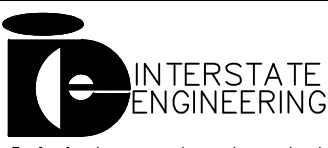
NOTE: All utilities shown are preliminary only, a complete
utilities location is recommended before construction.

© 2018, INTERSTATE ENGINEERING, INC.



3/9

SHEET NO.



Professionals you need, people you trust

Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617
Fax (406) 433-5618
www.interstateeng.com
Other offices in Minnesota, North Dakota and South Dakota

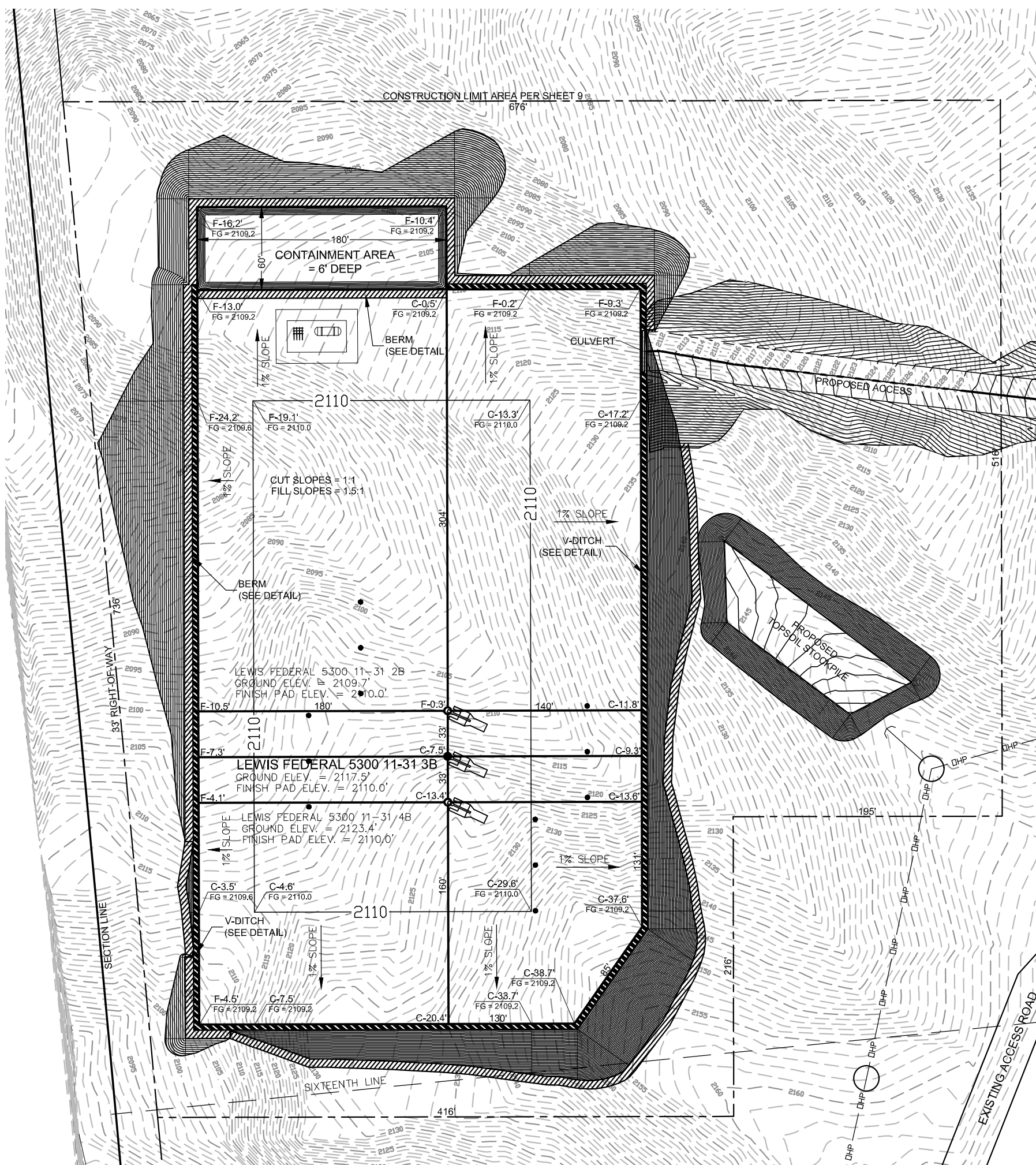
OASIS PETROLEUM NORTH AMERICA, LLC
PAD LAYOUT
SECTION 31, T153N, R100W, 5TH P.M.,
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: J.D.M. Project No.: S17-09-183
Checked By: D.D.K. Date: MARCH 2018

Revision No.	Date	By	Description

5300 11-31 Quad Pad to New Standards\CADD\LEWIS FEDERAL 5300 11-31 3B\LEWIS
FEDERAL 5300 11-31 3B.dwg - 3/27/2018 3:43 PM jocke miller

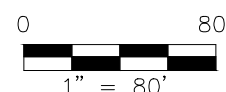
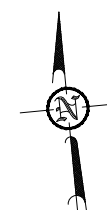
PRODUCTION LAYOUT
OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
LEWIS FEDERAL 5300 11-31 3B
1083 FEET FROM NORTH LINE AND 262 FEET FROM WEST LINE
SECTION 31, T153N, R100W, 5TH P.M., MCKENZIE COUNTY, NORTH DAKOTA



 — BERM
 — DITCH
 Proposed Contours
 Original Contours

NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.

© 2018, INTERSTATE ENGINEERING, INC.



**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

Received

Well File No.
30188

DEC 19 2017

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

ND Oil & Gas**Division**☒ Notice of Intent

Approximate Start Date

November 22, 2018☐ Report of Work Done

Date Work Completed

☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.

Approximate Start Date

☐ Drilling Prognosis☐ Spill Report☐ Redrilling or Repair☐ Shooting☐ Casing or Liner☐ Acidizing☐ Plug Well☐ Fracture Treatment☐ Supplemental History☐ Change Production Method☐ Temporarily Abandon☐ Reclamation☒ Other**APD Renewal**

Well Name and Number

Lewis Federal 5300 11-31 3B

Footages

1005 F N L**233 F W L**

Qtr-Qtr

Lot 1

Section

31

Township

153 N

Range

100 W

Field

Baker

Pool

Bakken

County

McKenzie**24-HOUR PRODUCTION RATE**

Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)

Address

City

State

Zip Code

DETAILS OF WORK

Oasis Petroleum requests the above referenced well be renewed. There are no planned changes to the current drill plan.

Please use the credit card on file for the \$100.00 application processing fee.

See attached gas capture plan.

Permit Expires 12/11/18

cc \$100.00 12/21/17 the
INV#58318

Company

Oasis Petroleum North America LLC

Telephone Number

281-404-9436

Address

1001 Fannin St, Suite 1500

City

Houston

State

TX

Zip Code

77002

Signature

Printed Name

Jennifer Swenson

Title

Regulatory Specialist

Date

December 19, 2017

Email Address

iswenson@oasisnetroleum.com**FOR STATE USE ONLY**☐ Received☒ Approved

Date

12/19/17

By

Alicia D. Wehler

Title

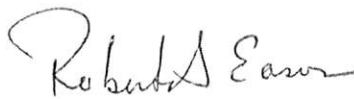
Engineering Technician

GAS CAPTURE PLAN AFFIDAVIT

STATE OF TEXAS §
 §
COUNTY OF HARRIS §

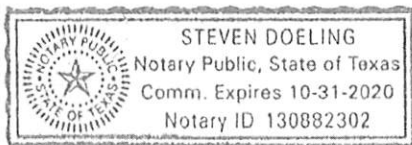
Robert Eason, being duly sworn, states as follows:

1. I am employed by Oasis Petroleum North America LLC ("Oasis") as Marketing Manager, I'm over the age of 21 and have personal knowledge of the matters set forth in this affidavit.
2. This affidavit is submitted in conjunction with the Application for Permit to Drill for the Lewis Federal 5300 11-31 3B well, with a surface location in Lot 1 of Section 31, Township 153 North, Range 100 West, McKenzie County, North Dakota (the "Well").
3. Oasis currently anticipates that gas to be produced from the Well will be gathered by Hiland Partners (the "Gathering Company"). Oasis has advised the Gathering Company of its intent to drill the Well and has advised the Gathering Company that it currently anticipates that the Well will be completed in ~ December 2018, with an initial gas production rate of approximately 1,200 mcf/day.



Robert H. Eason
Marketing Manager

Subscribed and sworn to before me this 7th day of December, 2017.




Notary Public in and for the State of Texas

GAS CAPTURE PLAN – OASIS PETROLEUM

Lewis Federal 5300 11-31 3B

Section 31-T153N-R100W

Baker Field

McKenzie County, North Dakota

Anticipated first flow date ~ December 2018
 Gas Gatherer: Kinder Morgan
 Gas to be processed at*: Hiland Operated Watford City Plant

Maximum Daily Capacity of Existing Gas Line*:	36,000 MCFD
Current Throughput of Existing Gas Line*:	26,600 MCFD
Anticipated Daily Capacity of Existing Gas Line at Date of First Gas Sales*:	36,000 MCFD
Anticipated Throughput of Existing Gas Line at Date of First Gas Sales*:	24,500 MCFD
Gas Gatherer's Issues or Expansion Plans for the Area*:	No expansion plans in this area.
Gas Plant Capacity:	95,000 MCFD
Map:	Attached
Affidavit:	Attached
*Provided by Gatherer	

Flowback Strategy

Total Number of Wells at Location: 10
 Multi-Well Start-up Plan: Initial production from the 1st new well at the CTB is anticipated in December 2017 with each following well making 1st production every 5th day thereafter.

Estimated Flow Rate:	Lewis Federal 5300 11-31 3B		Lewis 5300 11-31 CTB (10 wells)	
	MCFD	BOPD	MCFD	BOPD
30 Days:	743	743	4,703	4,851
60 Days:	564	564	6,422	6,609
180 Days:	324	324	3,175	3,277

Oasis Flaring Percentage

	Statewide	Baker Field
Oasis % of Gas Flared:	13%	1%

*Flared percentage reflects November 2017 with estimated 14 day flowback exemptions

Alternatives to Flaring

The installation of a temporary gas liquification unit recovering >50% of C3+ NGLs for an estimated reduction in flared volumes of ~30%.

Gas Capture Plan - Detail View
LEWIS FEDERAL 5300 11-31 3B
Section 31 T153N R100W
McKenzie County, North Dakota

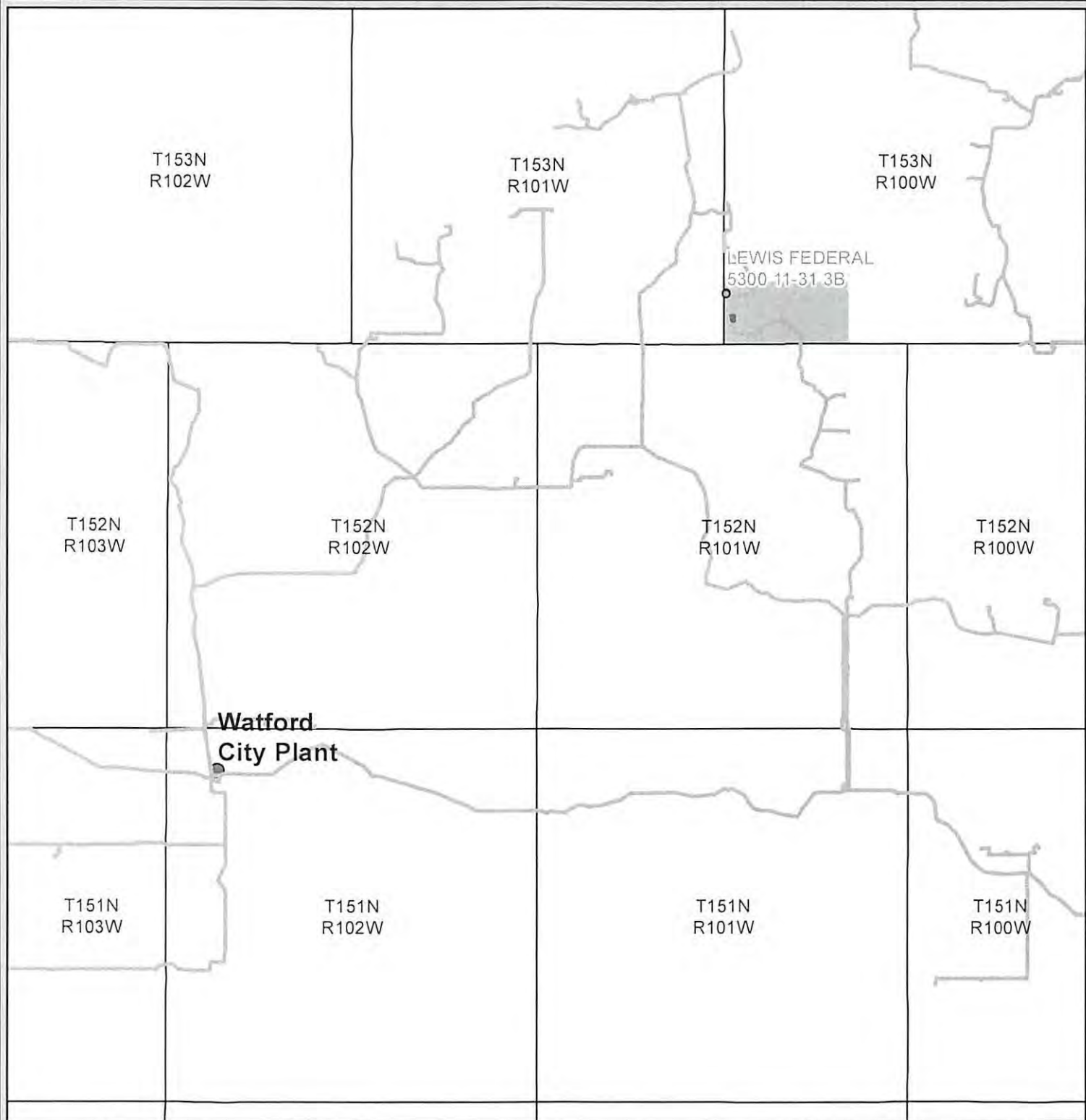


- CTB Outline
- Proposed Well
- Hiland Gas Line
- Oneok Gas Line

Gas Gatherer: Hiland Partners, LP
Gas to be processed at: Watford City Plant



Gas Capture Plan - Overview
LEWIS FEDERAL 5300 11-31 3B
Section 31 T153N R100W
McKenzie County, North Dakota



-  CTB Outline
-  Hiland Gas Line
-  Processing Plant

Gas Gatherer: Hiland Partners, LP
Gas to be processed at: Watford City Plant





Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.dmr.nd.gov/oilgas/

30188

November 13, 2017

OASIS PETRO NO AMER
1001 FANNIN STE 1500
HOUSTON, TX 77002

RE: LEWIS FEDERAL 5300 11-31 3B
LOT1 Sec. 31-153N-100W
MCKENZIE COUNTY
WELL FILE NO. 30188

Gentlemen:

The records and files of the Industrial Commission indicate that the above referenced permit will expire December 12, 2017.

Permits to drill are only valid for one year in the State of North Dakota. If you would like to renew for another year, please submit a Form 4 along with the \$100.00 filing fee. Alternatively, you may elect to send in a Form 4 cancelling the permit. If you have any questions, please contact Todd Holweger.

Sincerely,

Rachel Morris

Administrative Assistant

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

Received**NOV 23 2016**

Well File No.

30188

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

ND Oil & Gas Division☒ Notice of Intent

Approximate Start Date

December 11, 2016☐ Report of Work Done

Date Work Completed

☐ Notice of Intent to Begin a Workover Project that may Qualify
for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.

Approximate Start Date

☐ Drilling Prognosis☐ Spill Report☐ Redrilling or Repair☐ Shooting☐ Casing or Liner☐ Acidizing☐ Plug Well☐ Fracture Treatment☐ Supplemental History☐ Change Production Method☐ Temporarily Abandon☐ Reclamation☒ Other**APD Renewal**

Well Name and Number

Lewis Federal 5300 11-31 3B

Footages

1005 F N L**233 F W L**

Qtr-Qtr

Lot 1

Section

31

Township

153 N

Range

100 W

Field

Baker

Pool

Bakken

County

McKenzie**24-HOUR PRODUCTION RATE**

Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)

Address

City

State

Zip Code

DETAILS OF WORK

Oasis Petroleum requests the above referenced well be renewed. There are no planned changes to the current drill plan.

Please use the credit card on file for the \$100.00 application processing fee.

See attached gas capture plan.

DMV# 50042
PERMIT EXPIRES 12/11/17. *CC \$100.00 12/14/16 th*

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9436	
Address 1001 Fannin St, Suite 1500			
City Houston		State TX	Zip Code 77002
Signature <i>[Signature]</i>		Printed Name Jennifer Swenson	
Title Regulatory Specialist		Date November 22, 2016	
Email Address jswenson@oasispetroleum.com			

FOR STATE USE ONLY

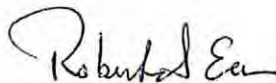
☐ Received	☒ Approved
Date 12/12/16	
By <i>[Signature]</i>	
Title Engineering Technician	

GAS CAPTURE PLAN AFFIDAVIT

STATE OF TEXAS §
 §
COUNTY OF HARRIS §

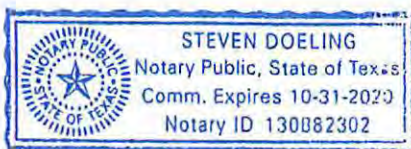
Robert Eason, being duly sworn, states as follows:

1. I am employed by Oasis Petroleum North America LLC ("Oasis") as Marketing Manager, I'm over the age of 21 and have personal knowledge of the matters set forth in this affidavit.
2. This affidavit is submitted in conjunction with the Application for Permit to Drill for the Lewis Federal 5300 11-31 3B well, with a surface location in Lot 1 of Section 31, Township 153 North, Range 100 West, McKenzie County, North Dakota (the "Well").
3. Oasis currently anticipates that gas to be produced from the Well will be gathered by Hiland Partners (the "Gathering Company"). Oasis has advised the Gathering Company of its intent to drill the Well and has advised the Gathering Company that it currently anticipates that the Well will be completed in ~ December 2017, with an initial gas production rate of approximately 1200 mcf/day.



Robert H. Eason
Marketing Manager

Subscribed and sworn to before me this 15th day of November, 2016.





Notary Public in and for the State of Texas

GAS CAPTURE PLAN – OASIS PETROLEUM

Lewis Federal 5300 11-31 3B

Section 31-T153N-R100W

Baker Field

McKenzie County, North Dakota

Anticipated first flow date ~ December 2017
 Gas Gatherer: Kinder Morgan
 Gas to be processed at*: Hiland Operated Watford City Plant

Maximum Daily Capacity of Existing Gas Line*: 36,000 MCFD

Current Throughput of Existing Gas Line*: 36,000 MCFD

Anticipated Daily Capacity of Existing Gas Line at Date of First Gas Sales*: 36,000 MCFD

Anticipated Throughput of Existing Gas Line at Date of First Gas Sales*: 31,275 MCFD

Gas Gatherer's Issues or Expansion Plans for the Area*: No expansion plans in this area.

Gas Plant Capacity: 95,000 MCFD

Map: Attached

Affidavit: Attached

*Provided by Gatherer

Flowback Strategy

Total Number of Wells at Location: 10

Multi-Well Start-up Plan: Initial production from the 1st new well at the CTB is anticipated in December 2017 with each following well making 1st production every 5th day thereafter.

Estimated Flow Rate:	Lewis Federal 5300 11-31 3B		Lewis 5300 11-31 CTB (10 wells)	
	MCFD	BOPD	MCFD	BOPD
30 Days:	743	743	4,703	4,851
60 Days:	564	564	6,422	6,609
180 Days:	324	324	3,175	3,277

Oasis Flaring Percentage

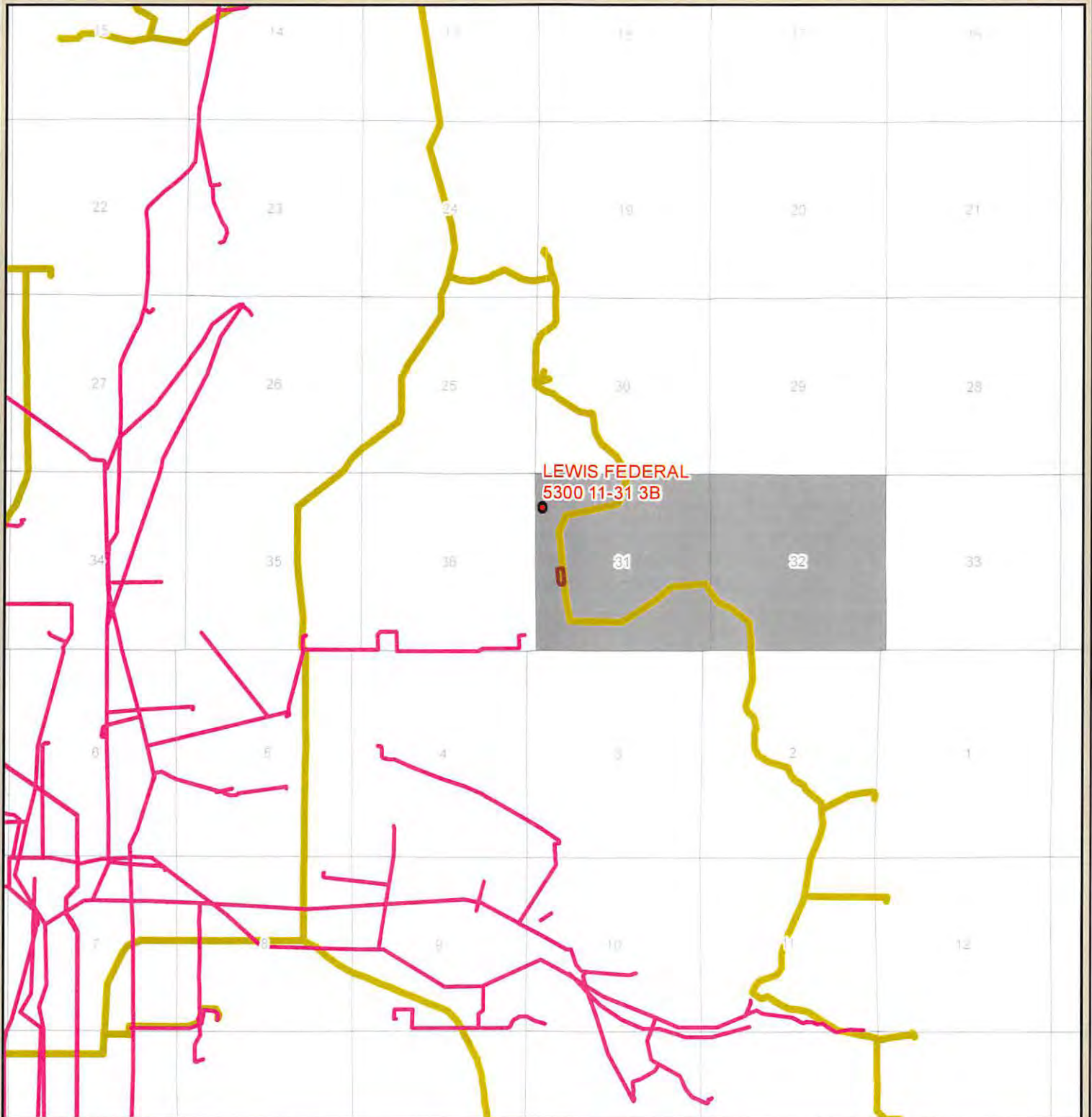
	Statewide	Baker Field
Oasis % of Gas Flared:	12%	10%

*Flared percentage reflects October 2016

Alternatives to Flaring

The installation of a temporary gas liquification unit recovering >50% of C3+ NGLs for an estimated reduction in flared volumes of ~30%.

Gas Capture Plan - Detail View
LEWIS FEDERAL 5300 11-31 3B
Section 31 T153N R100W
McKenzie County, North Dakota

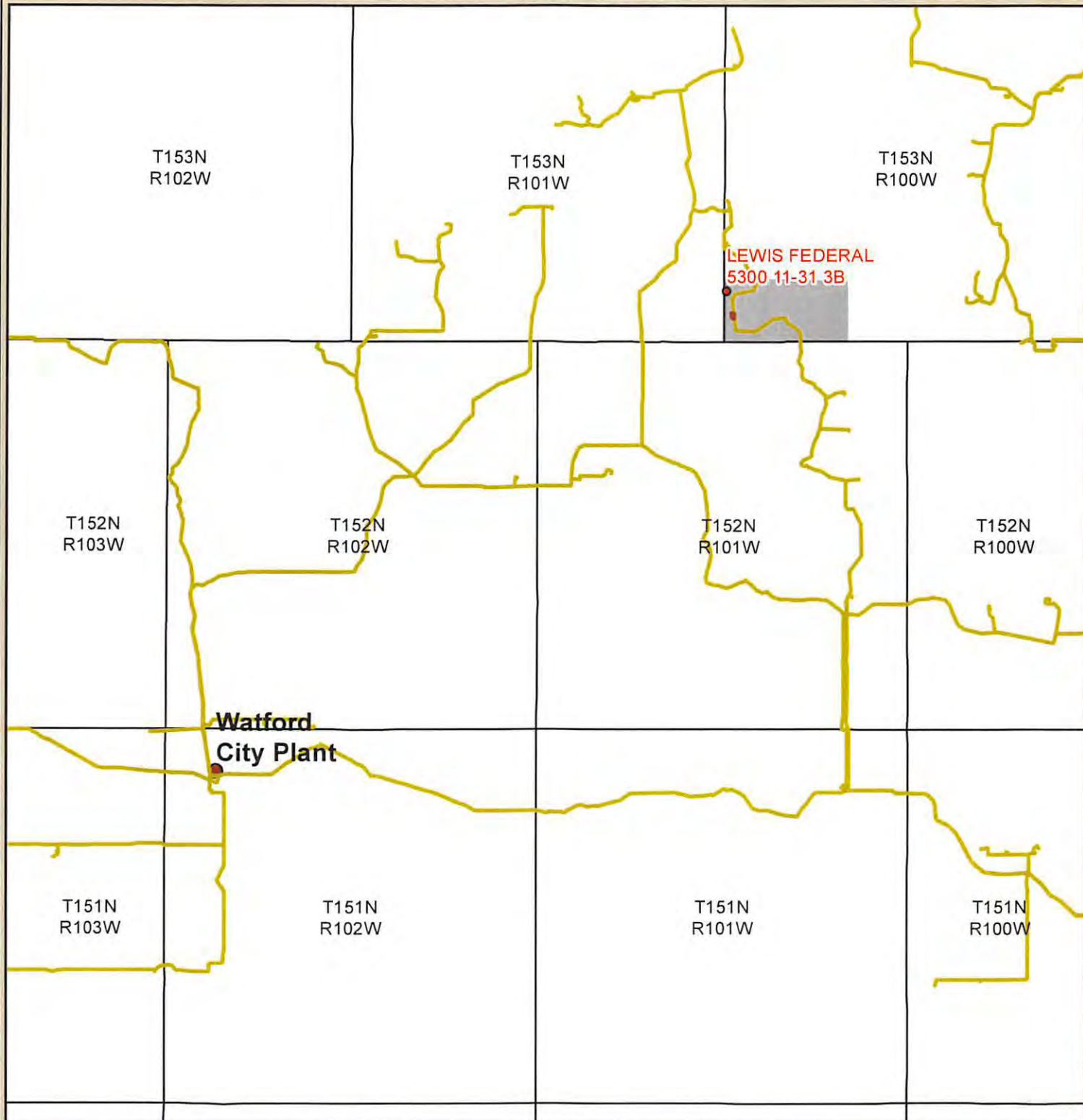


- CTB Outline
- Proposed Well
- Hiland Gas Line
- Oneok Gas Line

Gas Gatherer: Hiland Partners, LP
Gas to be processed at: Watford City Plant



Gas Capture Plan - Overview
LEWIS FEDERAL 5300 11-31 3B
Section 31 T153N R100W
McKenzie County, North Dakota



-  CTB Outline
-  Hiland Gas Line
-  Processing Plant

Gas Gatherer: Hiland Partners, LP
Gas to be processed at: Watford City Plant





Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.dmr.nd.gov/oilgas/

30188

November 7, 2016

OASIS PETRO NO AMER
1001 FANNIN STE 1500
HOUSTON, TX 77002

RE: LEWIS FEDERAL 5300 11-31 3B
LOT1 Sec. 31-153N-100W
MCKENZIE COUNTY
WELL FILE NO. 30188

Gentlemen:

The records and files of the Industrial Commission indicate that the above referenced permit will expire December 11, 2016.

Permits to drill are only valid for one year in the State of North Dakota. If you would like to renew for another year, please submit a Form 4 along with the \$100.00 filing fee. Alternatively, you may elect to send in a Form 4 cancelling the permit. If you have any questions, please contact Todd Holweger.

Sincerely,

Rachel Morris
Administrative Assistant



SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2008)



Well File No.
30188

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date December 11, 2015
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other APD Renewal	

Well Name and Number Lewis Federal 5300 11-31 3B					
Footages	Qtr-Qtr	Section	Township	Range	
1005 F N L	233 F W L	Lot 1	31	153 N	100 W
Field Baker	Pool Bakken	County McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum requests the above referenced well be renewed. There are no planned changes to the current drill plan.

Please use the credit card on file for the \$100.00 application processing fee.

See attached gas capture plan.

Permit Expires 12/11/16.

CC 100.00 12-14-15 KB

Company Oasis Petroleum North America LLC		Telephone Number (281) 404-9652	
Address 1001 Fannin St, Suite 1500			
City Houston	State TX	Zip Code 77002	
Signature <i>[Signature]</i>	Printed Name Victoria Siemieniowski		
Title Regulatory Specialist	Date December 1, 2015		
Email Address vsiemieniowski@oasispetroleum.com			

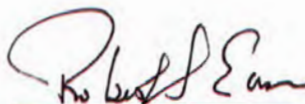
FOR STATE USE ONLY	
☐ Received	☒ Approved
Date 12/11/15	
By <i>[Signature]</i>	
Title Engineering Technician	

GAS CAPTURE PLAN AFFIDAVIT

STATE OF TEXAS §
 §
COUNTY OF HARRIS §

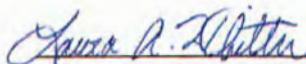
Robert Eason, being duly sworn, states as follows:

1. I am employed by Oasis Petroleum North America LLC ("Oasis") as Marketing Manager, I'm over the age of 21 and have personal knowledge of the matters set forth in this affidavit.
2. This affidavit is submitted in conjunction with the Application for Permit to Drill for the Lewis Federal 5300 11-31 3B well, with a surface location in the NW NW of Section 31, Township 153 North, Range 100 West, McKenzie County, North Dakota (the "Well").
3. Oasis currently anticipates that gas to be produced from the Well will be gathered by Hiland Partners (the "Gathering Company"). Oasis has advised the Gathering Company of its intent to drill the Well and has advised the Gathering Company that it currently anticipates that the Well will be completed in November 2016, with an initial gas production rate of approximately 983 mcf/day.



Robert H. Eason
Marketing Manager

Subscribed and sworn to before me this 1st day of December, 2015.



Notary Public in and for the State of Texas

GAS CAPTURE PLAN – OASIS PETROLEUM**Lewis Federal 5300 11-31 3B****Section 31-T153N-R100W****Baker Field****McKenzie County, North Dakota**

Anticipated first flow date Nov-16
Gas Gatherer: Hilland Partners
Gas to be processed at*: Watford City Plant

Maximum Daily Capacity of Existing Gas Line*:	91,500 MCFD
Current Throughput of Existing Gas Line*:	70,000 MCFD
Anticipated Daily Capacity of Existing Gas Line at Date of First Gas Sales*:	91,500 MCFD
Anticipated Throughput of Existing Gas Line at Date of First Gas Sales*:	82,000 MCFD
Gas Gatherer's Issues or Expansion Plans for the Area*:	There are no plans for expansion at this time.

Map: Attached

Affidavit: Attached

*Provided by Gatherer

Flowback Strategy

Total Number of Wells at Location: 7
Multi-Well Start-up Plan: Initial production from the 1st new well at the CTB is anticipated November 2016 with each following well making 1st production every 5th day thereafter

Estimated Flow Rate:	<u>Lewis Federal 5300 11-31 3B</u>		<u>Lewis Federal 5300 31-32 CTB (7 wells)</u>	
	<u>MCFD</u>	<u>BOPD</u>	<u>MCFD</u>	<u>BOPD</u>
30 Days:	705	783	3,695	4,106
60 Days:	565	627	4,026	4,473
180 Days:	345	384	2,265	2,517

Oasis Flaring Percentage

	Statewide	Baker Field
Oasis % of Gas Flared:	9%	25%

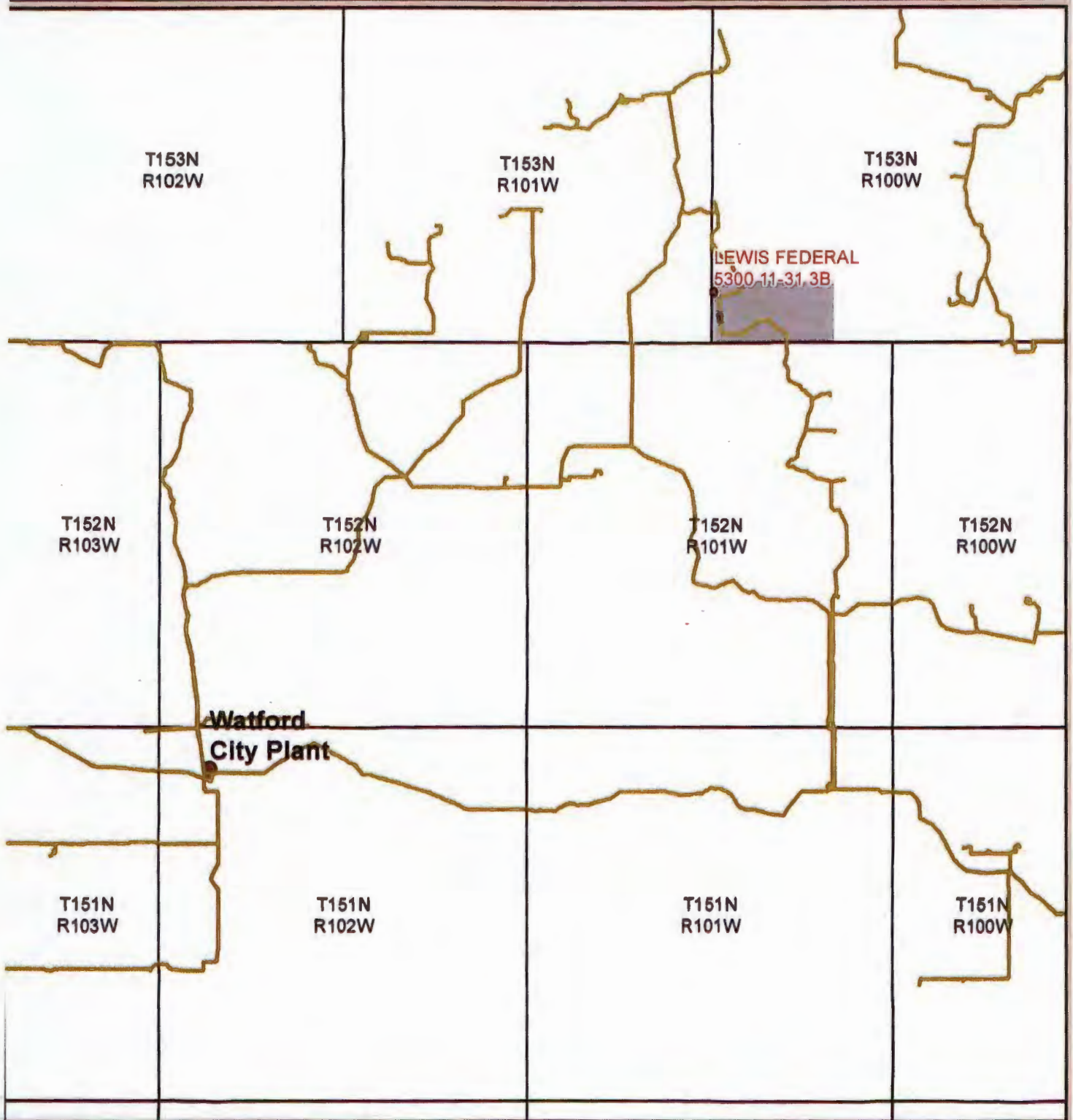
*Flared percentage reflects October 2015

Alternatives to Flaring

The installation of a temporary gas liquification unit recovering >50% of C3+ NGLs for an estimated reduction in flared volumes of ~30%

Source: Oasis Marketing (281) 404-9661

Gas Capture Plan - Overview
LEWIS FEDERAL 5300 11-31 3B
Section 31 T153N R100W
McKenzie County, North Dakota



- CTB Outline
- Hiland Gas Line
- Processing Plant

Gas Gatherer: Hiland Partners, LP
Gas to be processed at: Watford City Plant



Gas Capture Plan - Detail View
LEWIS FEDERAL 5300 11-31 3B
Section 31 T153N R100W
McKenzie County, North Dakota



- CTB Outline
- Proposed Well
- Hiland Gas Line
- Oneok Gas Line

Gas Gatherer: Hiland Partners, LP
 Gas to be processed at: Watford City Plant





Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.dmr.nd.gov/oilgas

30188

November 9, 2015

OASIS PETRO NO AMER
1001 FANNIN STE 1500
HOUSTON, TX 77002

RE: LEWIS FEDERAL 5300 11-31 3B
LOT1 Sec. 31-153N-100W
MCKENZIE COUNTY
WELL FILE NO. 30188

Gentlemen:

The records and files of the Industrial Commission indicate that the above referenced permit will expire December 11, 2015.

Permits to drill are only valid for one year in the State of North Dakota. If you would like to renew for another year, please submit a Form 4 along with the \$100.00 filing fee. Alternatively, you may elect to send in a Form 4 cancelling the permit. If you have any questions, please contact Todd Holweger.

Sincerely,

Rachel Morris

Administrative Assistant



Oil and Gas Division 30188

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.dmr.nd.gov/oilgas/

BRANDI TERRY
OASIS PETROLEUM NORTH AMERICA LLC
1001 FANNIN STE 1500
HOUSTON, TX 77002 USA

Date: 12/15/2014

RE: CORES AND SAMPLES

Well Name: **LEWIS FEDERAL 5300 11-31 3B** Well File No.: **30188**
Location: **LOT1 31-153-100** County: **MCKENZIE**
Permit Type: **Development - HORIZONTAL**
Field: **BAKER** Target Horizon: **BAKKEN**

Dear BRANDI TERRY:

North Dakota Century Code Section 38-08-04 provides for the preservation of cores and samples and their shipment to the State Geologist when requested. The following is required on the above referenced well:

- 1) All cores, core chips and samples must be submitted to the State Geologist as provided for under North Dakota Century Code: Section 38-08-04 and North Dakota Administrative Code: Section 43-02-03-38.1.
- 2) Samples: The Operator is to begin collecting sample drill cuttings no lower than the:
Base of the Last Charles Salt
 - Sample cuttings shall be collected at:
 - o 30' maximum intervals through all vertical and build sections.
 - o 100' maximum intervals through any horizontal sections.
 - Samples must be washed, dried, placed in standard sample envelopes (3" x 4.5"), packed in the correct order into standard sample boxes (3.5" x 5.25" x 15.25").
 - Samples boxes are to be carefully identified with a label that indicates the operator, well name, well file number, American Petroleum Institute (API) number, location and depth of samples; and forwarded in to the state core and sample library within 30 days of the completion of drilling operations.
- 3) Cores: Any cores cut shall be preserved in correct order, boxed in standard core boxes (4.5", 4.5", 35.75"), and the entire core forwarded to the state core and samples library within 180 days of completion of drilling operations. Any extension of time must have approval on a Form 4 Sundry Notice.

All cores, core chips, and samples must be shipped, prepaid, to the state core and samples library at the following address:

**ND Geological Survey Core Library
2835 Campus Road, Stop 8156
Grand Forks, ND 58202**

North Dakota Century Code Section 38-08-16 allows for a civil penalty for any violation of Chapter 38 08 not to exceed \$12,500 for each offense, and each day's violation is a separate offense.

Sincerely


Stephen Fried
Geologist

**SUNDRY NOTICES AND REPORTS ON WELLS**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.
30188

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date November 1, 2014
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other	Waiver to rule Rule 43-02-03-31

Well Name and Number Lewis Federal 5300 11-31 3B					
Footages 1005 F N L 233 F W L		Qtr-Qtr NW1/4	Section 31	Township 153 N	Range 100 W
Field Baker	Pool Bakken		County McKenzie		

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum respectfully requests a waiver to Rule 43-02-03-31 in regards to running open hole logs for the above referenced well. Justification for this request is as follows:

#20314
The **Lewis Federal 5300 31-31** 2,850' S of surface location located within a mile of the subject well.

If this exception is approved, Oasis Petroleum will run a CBL on the intermediate string, and we will also run GR to surface. Oasis Petroleum will also submit two digital copies of each cased hole log and a copy of the mud log containing MWD gamma ray.

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9563	
Address 1001 Fannin, Suite 1500			
City Houston	State TX	Zip Code 77002	
Signature <i>Heather McCowan</i>	Printed Name Heather McCowan		
Title Heather McCowan	Date March 25, 2014		
Email Address hmccowan@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date 12-11-2014	
By <i>Stephen Fried</i>	
Title Stephen Fried Geologist	



SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.
30188

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date December 16, 2014	☐ Drilling Log/Logbook	☐ Spill Report
☐ Report of Work Done	Date Work Completed	☐ Redrilling or Repair	☐ Shooting
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date	☐ Casing or Liner	☐ Acidizing
		☐ Plug Well	☐ Fracture Treatment
		☐ Supplemental History	☐ Change Production Method
		☐ Temporarily Abandon	☐ Reclamation
		☒ Other	Offsite Cutting Pit Request

Well Name and Number Lewis Federal 5300 11-31 3B							
Footages		Qtr-Qtr	Section	Township	Range		
1005 F N L 233 F W L		NWNW	31	153 N	100 W		
Field		Pool		County			
Baker		Bakken		McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

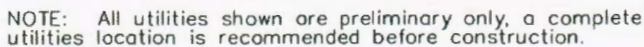
Oasis Petroleum respectfully requests approval to utilize an off-site drill cuttings pit for the above well. See attached off-site cutting pit plat and landowner agreement. Cuttings from the following wells are also planned to be disposed of in the same pit:

Lewis Federal 5300 11-31 2T (File No. 30189)
Lewis Federal 5300 11-31 4T2 (File No. 30187)
Lewis Federal 5300 11-31 5T (File No. 30186)

Company Oasis Petroleum North America LLC		Telephone Number (281) 404-9500	
Address 1001 Fannin Suite 1500			
City Houston		State TX	Zip Code 77002
Signature 		Printed Name Victoria Siemieniewski	
Title Regulatory Specialist		Date December 15, 2014	
Email Address vsiemieniewski@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date 12-18-14	
By 	
Title 	

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
AS-STAKED OFFSITE PIT FOR LEWIS FEDERAL 5300 11-31 2T, 3B, 4T2, 5T
SECTION 31, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



THIS DOCUMENT WAS ORIGINALLY ISSUED
AND SEALED BY DARYL D. KASEMAN, PLS,
REGISTRATION NUMBER 3880
ON 11-25-14 AND THE ORIGINAL
DOCUMENTS ARE STORED AT THE OFFICES
OF INTERSTATE ENGINEERING, INC.

[illegible]

OFF-SITE PIT AGREEMENT

In consideration of the sum of [REDACTED] paid by Oasis Petroleum North America LLC ("Oasis") the undersigned surface owners, Wesley and Barbara Lindvig, husband and wife, whose address is 14075 41st St. NW, Alexander, ND 5883 for themselves and their heirs, successors, administrators and assigns, hereby acknowledge the receipt and sufficiency of said payment in full and complete settlement for and as a release of all claim for loss, damage or injury to the hereafter described surface property arising out of the off-site cuttings pit, in which the cuttings from the Lewis Federal 5300 11-31 2T, Lewis Federal 5300 11-31 3B, Lewis Federal 5300 11-31 4T2, Lewis Federal 5300 11-31 5T wells will be buried, located on the approximately two and one half (2.5) acre tract of land identified on the plat attached hereto as Exhibit "A" and which is situated on the following described real property located in McKenzie County, State of North Dakota, to-wit:

Township 153 North, Range 100 West, 5th P.M.
Section 31: SE/4NW/4

The undersigned knows that Oasis Petroleum North America LLC is the operator and will be drilling the Lewis Federal 5300 11-31 2T, Lewis Federal 5300 11-31 3B, Lewis Federal 5300 11-31 4T2, Lewis Federal 5300 11-31 5T wells. The undersigned further states that they are fully aware that the cuttings generated from the drilling of the Lewis Federal 5300 11-31 2T, Lewis Federal 5300 11-31 3B, Lewis Federal 5300 11-31 4T2, Lewis Federal 5300 11-31 5T wells will be buried in the pit on the above described location.

Dated this 9th day of December, 2014.

SURFACE OWNER(S)

Wesley Lindvig
Wesley Lindvig

Barbara Lindvig
Barbara Lindvig

ACKNOWLEDGMENT INDIVIDUAL

State of North Dakota)

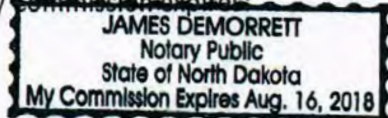
)

County of McKenzie)

BE IT REMEMBERED, That on this 9 day of ^{December}~~November~~, 2014 before me, a Notary Public, in and for said County and State, personally appeared Wesley and Barbara Lindvig, husband and wife, to me known to be the identical persons described in and who executed the within and foregoing instrument and acknowledged to me to that they executed the same as their free and voluntary act and deed for the uses and purposes therein set forth.

IN WITNESS WHEREOF, I have hereunto set my official signature and affixed my notarial seal, the day and year last above written.

My Commission expires:



A handwritten signature in blue ink, appearing to read "James Demorrett", written over a horizontal line.

Notary Public

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

Well File No. 30188

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date November 1, 2014
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other	Suspension of Drilling

Well Name and Number Lewis Federal 5300 11-31 3B							
Footages		Qtr-Qtr	Section	Township	Range		
1005 F N L 233 F WL		NWNW	31	153 N	100 W		
Field		Pool		County			
		Bakken		McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s) Advanced Energy Services			
Address	City	State	Zip Code

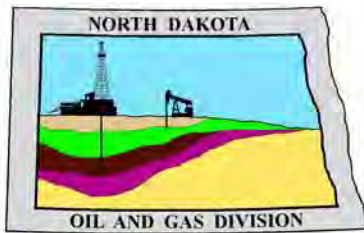
DETAILS OF WORK

Oasis Petroleum North America LLC requests permission for suspension of drilling for up to 90 days for the referenced well under NDAC 43-02-03-55. Oasis Petroleum North America LLC intends to drill the surface hole with freshwater based drilling mud and set surface casing with a small drilling rig and move off within 3 to 5 days. The casing will be set at a depth pre-approved by the NDIC per the Application for Permit to Drill NDAC 43-02-03-21. No saltwater will be used in the drilling and cementing operations of the surface casing. Once the surface casing is cemented, a plug or mechanical seal will be placed at the top of the casing to prevent any foreign matter from getting into the well. A rig capable of drilling to TD will move onto the location within the 90 days previously outlined to complete the drilling and casing plan as per the APD. The undersigned states that this request for suspension of drilling operations in accordance with the Subsection 4 of Section 43-02-03-55 of the NDAC, is being requested to take advantage of the cost savings and time savings of using an initial rig that is smaller than the rig necessary to drill a well to total depth but is not intended to alter or extend the terms and conditions of, or suspend any obligation under, any oil and gas lease with acreage in or under the spacing or drilling unit for the above-referenced well. Oasis Petroleum North America LLC understands NDAC 43-02-03-31 requirements regarding confidentiality pertaining to this permit. The drilling pit will be fenced immediately after construction if the well pad is located in a pasture (NDAC 43-02-03-19 & 19.1). Oasis Petroleum North America LL will plug and abandon the well and reclaim the well site if the well is not drilled by the larger rotary rig within 90 days after spudding the well with the smaller drilling rig.

Oasis must notify NDIC Field Inspector Richard Dunn @701-770-3554 with spud and TD.

Company Oasis Petroleum North America LLC		Telephone Number (281) 404-9563	
Address 1001 Fannin, Suite 1500			
City Houston		State TX	Zip Code 77002
Signature 		Printed Name HEATHER MCCOWAN	
Title REGULATORY ASSISTANT		Date March 25, 2014	
Email Address hmccowan@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date	12/11/2014
By	David Burns
Title	Engineering Tech.



Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.oilgas.nd.gov

December 11, 2014

Heather McCowan
Regulatory Assistant
OASIS PETROLEUM NORTH AMERICA LLC
1001 Fannin Suite 1500
Houston, TX 77002

**RE: HORIZONTAL WELL
LEWIS FEDERAL 5300 11-31 3B
LOT1 Section 31-153N-100W
McKenzie County
Well File # 30188**

Dear Heather:

Pursuant to Commission Order No. 23752, approval to drill the above captioned well is hereby given. The approval is granted on the condition that all portions of the well bore not isolated by cement, be no closer than the **500' setback** from the north & south boundaries and **200' setback** from the east & west boundaries within the 1280 acre spacing unit consisting of Sections 31 & 32 T153N R100W.

PERMIT STIPULATIONS: Effective June 1, 2014, a covered leak-proof container (with placard) for filter sock disposal must be maintained on the well site beginning when the well is spud, and must remain on-site during clean-out, completion, and flow-back whenever filtration operations are conducted. OASIS PETRO NO AMER must contact NDIC Field Inspector Richard Dunn at 701-770-3554 prior to location construction.

Drilling pit

NDAC 43-02-03-19.4 states that "a pit may be utilized to bury drill cuttings and solids generated during well drilling and completion operations, providing the pit can be constructed, used and reclaimed in a manner that will prevent pollution of the land surface and freshwaters. Reserve and circulation of mud system through earthen pits are prohibited. All pits shall be inspected by an authorized representative of the director prior to lining and use. Drill cuttings and solids must be stabilized in a manner approved by the director prior to placement in a cuttings pit."

Form 1 Changes & Hard Lines

Any changes, shortening of casing point or lengthening at Total Depth must have prior approval by the NDIC. The proposed directional plan is at a legal location. Based on the azimuth of the proposed lateral the maximum legal coordinate from the well head is: 10089 E.

Location Construction Commencement (Three Day Waiting Period)

Operators shall not commence operations on a drill site until the 3rd business day following publication of the approved drilling permit on the NDIC - OGD Daily Activity Report. If circumstances require operations to commence before the 3rd business day following publication on the Daily Activity Report, the waiting period may be waived by the Director. Application for a waiver must be by sworn affidavit providing the information necessary to evaluate the extenuating circumstances, the factors of NDAC 43-02-03-16.2 (1), (a)-(f), and any other information that would allow the Director to conclude that in the event another owner seeks revocation of the drilling permit, the applicant should retain the permit.

Permit Fee & Notification

Payment was received in the amount of \$100 via credit card .The permit fee has been received. It is requested that notification be given immediately upon the spudding of the well. This information should be relayed to the Oil & Gas Division, Bismarck, via telephone. The following information must be included: Well name, legal location, permit number, drilling contractor, company representative, date and time of spudding. Office hours are 8:00 a.m. to 12:00 p.m. and 1:00 p.m. to 5:00 p.m. Central Time. Our telephone number is (701) 328-8020, leave a message if after hours or on the weekend.

Survey Requirements for Horizontal, Horizontal Re-entry, and Directional Wells

NDAC Section 43-02-03-25 (Deviation Tests and Directional Surveys) states in part (that) the survey contractor shall file a certified copy of all surveys with the director free of charge within thirty days of completion. Surveys must be submitted as one electronic copy, or in a form approved by the director. However, the director may require the directional survey to be filed immediately after completion if the survey is needed to conduct the operation of the director's office in a timely manner. Certified surveys must be submitted via email in one adobe document, with a certification cover page to certsurvey@nd.gov.

Survey points shall be of such frequency to accurately determine the entire location of the well bore.

Specifically, the Horizontal and Directional well survey frequency is 100 feet in the vertical, 30 feet in the curve (or when sliding) and 90 feet in the lateral.

Surface casing cement

Tail cement utilized on surface casing must have a minimum compressive strength of 500 psi within 12 hours, and tail cement utilized on production casing must have a minimum compressive strength of 500 psi before drilling the plug or initiating tests.

Logs

NDAC Section 43-02-03-31 requires the running of (1) a suite of open hole logs from which formation tops and porosity zones can be determined, (2) a Gamma Ray Log run from total depth to ground level elevation of the well bore, and (3) a log from which the presence and quality of cement can be determined (Standard CBL or Ultrasonic cement evaluation log) in every well in which production or intermediate casing has been set, this log must be run prior to completing the well. All logs run must be submitted free of charge, as one digital TIFF (tagged image file format) copy and one digital LAS (log ASCII) formatted copy. Digital logs may be submitted on a standard CD, DVD, or attached to an email sent to digitallogs@nd.gov

Thank you for your cooperation.

Sincerely,

David Burns
Engineering Technician



APPLICATION FOR PERMIT TO DRILL HORIZONTAL WELL - FORM 1H

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 54269 (08-2005)

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

Type of Work New Location	Type of Well Oil & Gas	Approximate Date Work Will Start 11 / 1 / 2014	Confidential Status No
Operator OASIS PETROLEUM NORTH AMERICA LLC			Telephone Number 281-404-9563
Address 1001 Fannin Suite 1500		City Houston	State TX Zip Code 77002

☒ Notice has been provided to the owner of any permanently occupied dwelling within 1,320 feet.

☒ This well is not located within five hundred feet of an occupied dwelling.

WELL INFORMATION (If more than one lateral proposed, enter data for additional laterals on page 2)

Well Name LEWIS FEDERAL				Well Number 5300 11-31 3B			
Surface Footages 1005 F N L 233 F W L		Qtr-Qtr LOT1	Section 31	Township 153 N	Range 100 W	County McKenzie	
Longstring Casing Point Footages 997 F N L 749 F W L		Qtr-Qtr LOT1	Section 31	Township 153 N	Range 100 W	County McKenzie	
Longstring Casing Point Coordinates From Well Head 8 N From WH 516 E From WH		Azimuth 91.98 °	Longstring Total Depth 11046 Feet MD 10760 Feet TVD				
Bottom Hole Footages From Nearest Section Line 998 F N L 210 F E L		Qtr-Qtr LOT4	Section 32	Township 153 N	Range 100 W	County Williams	
Bottom Hole Coordinates From Well Head 7 N From WH 10079 E From WH		KOP Lateral 1 10283 Feet MD	Azimuth Lateral 1 90 °		Estimated Total Depth Lateral 1 20610 Feet MD 10844 Feet TVD		
Latitude of Well Head 48 ° 02 ' 09.74 "		Longitude of Well Head -103 ° 36 ' 11.59 "		NAD Reference NAD83	Description of (Subject to NDIC Approval) Spacing Unit: Sections 31 & 32 T153N R100W		
Ground Elevation 2104 Feet Above S.L.	Acres in Spacing/Drilling Unit 1280	Spacing/Drilling Unit Setback Requirement 500 Feet N/S 200 Feet E/W			Industrial Commission Order 23752		
North Line of Spacing/Drilling Unit 10522 Feet	South Line of Spacing/Drilling Unit 10535 Feet	East Line of Spacing/Drilling Unit 5280 Feet			West Line of Spacing/Drilling Unit 5248 Feet		
Objective Horizons Bakken					Pierre Shale Top 1984		
Proposed Surface Casing	Size 9 - 5/8 "	Weight 36 Lb./Ft.	Depth 2100 Feet	Cement Volume 1184 Sacks	NOTE: Surface hole must be drilled with fresh water and surface casing must be cemented back to surface.		
Proposed Longstring Casing	Size 7 - "	Weight(s) 29/32 Lb./Ft.	Longstring Total Depth 11046 Feet MD 10760 Feet TVD		Cement Volume 794 Sacks	Cement Top 4885 Feet	Top Dakota Sand 5385 Feet
Base Last Charles Salt (If Applicable) 9230 Feet		NOTE: Intermediate or longstring casing string must be cemented above the top Dakota Group Sand.					
Proposed Logs Triple Combo: KOP to Kibby GR/Res to BSC GR to surf CND through the Dakota							
Drilling Mud Type (Vertical Hole - Below Surface Casing) Invert				Drilling Mud Type (Lateral) Salt Water Gel			
Survey Type in Vertical Portion of Well MWD Every 100 Feet		Survey Frequency: Build Section 30 Feet		Survey Frequency: Lateral 90 Feet		Survey Contractor Ryan	

NOTE: A Gamma Ray log must be run to ground surface and a CBL must be run on intermediate or longstring casing string if set.

Surveys are required at least every 30 feet in the build section and every 90 feet in the lateral section of a horizontal well. Measurement inaccuracies are not considered when determining compliance with the spacing/drilling unit boundary setback requirement except in the following scenarios: 1) When the angle between the well bore and the respective boundary is 10 degrees or less; or 2) If Industry standard methods and equipment are not utilized. Consult the applicable field order for exceptions.

If measurement inaccuracies are required to be considered, a 2° MWD measurement inaccuracy will be applied to the horizontal portion of the well bore. This measurement inaccuracy is applied to the well bore from KOP to TD.

REQUIRED ATTACHMENTS: Certified surveyor's plat, horizontal section plat, estimated geological tops, proposed mud/cementing plan, directional plot/plan, \$100 fee.
See Page 2 for Comments section and signature block.

COMMENTS, ADDITIONAL INFORMATION, AND/OR LIST OF ATTACHMENTS

Documents forwarded by email: Drill plan with drilling fluids, Well Summary with casing/cement plans, Directional Plan & Plot, Plats

Lateral 2

KOP Lateral 2	Azimuth Lateral 2	Estimated Total Depth Lateral 2		KOP Coordinates From Well Head		
Feet MD	°	Feet MD	Feet TVD	From WH	From WH	
Formation Entry Point Coordinates From Well Head		Bottom Hole Coordinates From Well Head				
From WH	From WH	From WH	From WH			
KOP Footages From Nearest Section Line		Qtr-Qtr	Section	Township	Range	County
F	L	F	L	N	W	
Bottom Hole Footages From Nearest Section Line		Qtr-Qtr	Section	Township	Range	County
F	L	F	L	N	W	

Lateral 3

KOP Lateral 3 Feet MD	Azimuth Lateral 3 °	Estimated Total Depth Lateral 3 Feet MD Feet TVD		KOP Coordinates From Well Head From WH From WH	
Formation Entry Point Coordinates From Well Head From WH From WH		Bottom Hole Coordinates From Well Head From WH From WH			
KOP Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County
Bottom Hole Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County

Lateral 4

KOP Lateral 4 Feet MD	Azimuth Lateral 4 °	Estimated Total Depth Lateral 4 Feet MD Feet TVD		KOP Coordinates From Well Head From WH From WH	
Formation Entry Point Coordinates From Well Head From WH From WH		Bottom Hole Coordinates From Well Head From WH From WH			
KOP Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County
Bottom Hole Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County

Lateral 5

KOP Lateral 5 Feet MD	Azimuth Lateral 5 °	Estimated Total Depth Lateral 5 Feet MD Feet TVD		KOP Coordinates From Well Head From WH From WH		
Formation Entry Point Coordinates From Well Head From WH From WH		Bottom Hole Coordinates From Well Head From WH From WH				
KOP Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W	County
Bottom Hole Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W	County

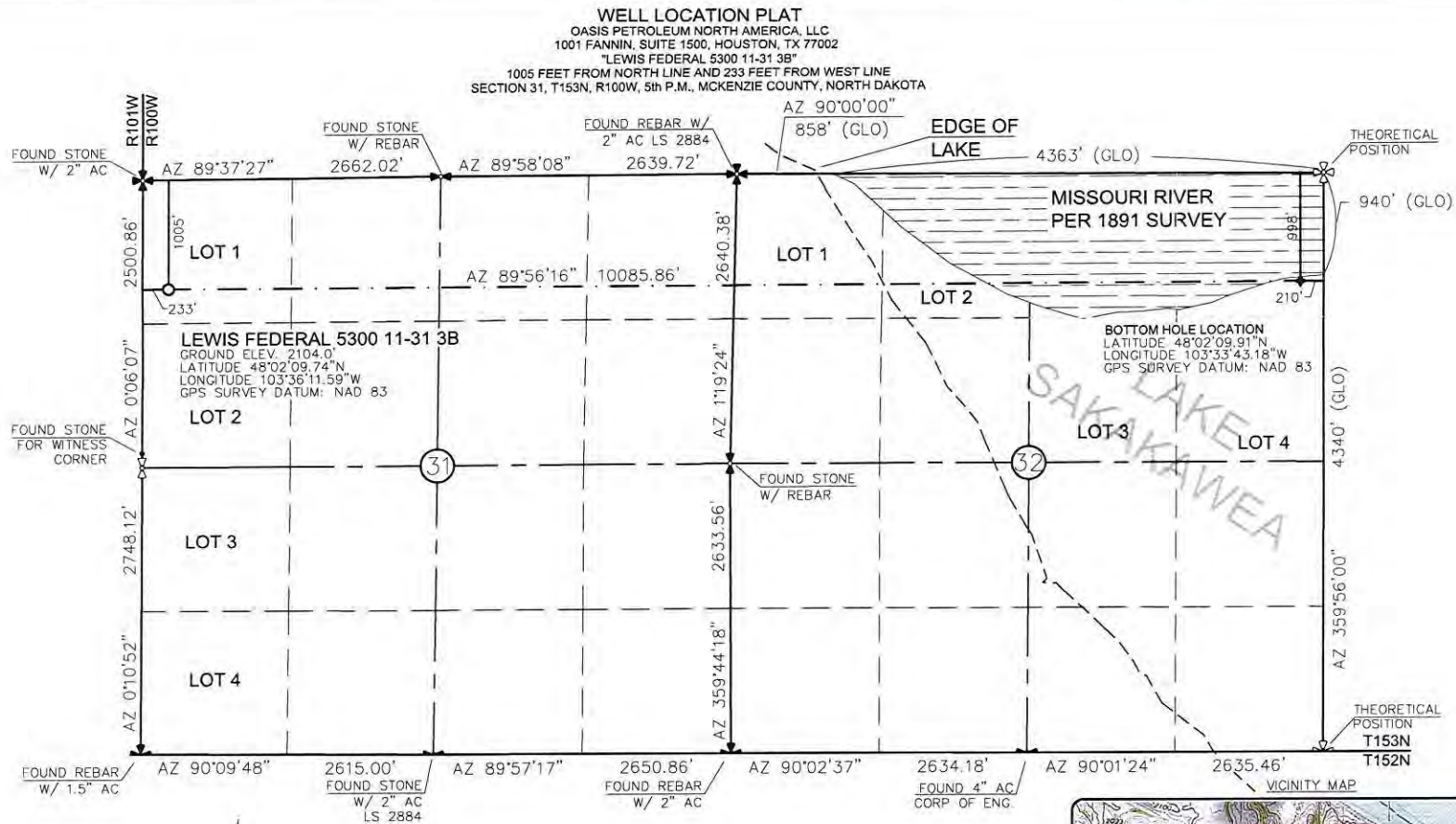
I hereby swear or affirm the information provided is true, complete and correct as determined from all available records.		Date 3 / 25 / 2014
ePermit	Printed Name Heather McCowan	Title Regulatory Assistant

FOR STATE USE ONLY

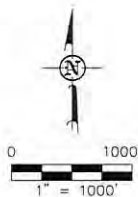
Permit and File Number 30188	API Number 33 - 053 - 06548
Field BAKER	
Pool BAKKEN	Permit Type DEVELOPMENT

FOR STATE USE ONLY

Date Approved 12 / 11 / 2014
By David Burns
Title Engineering Technician



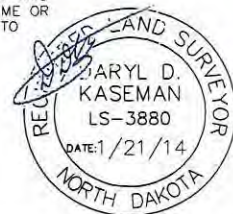
THIS DOCUMENT WAS ORIGINALLY ISSUED AND SEALED BY DARYL D. KASEMAN, PLS, REGISTRATION NUMBER 3880 ON 1/21/14, AND THE ORIGINAL DOCUMENTS ARE STORED AT THE OFFICES OF INTERSTATE ENGINEERING, INC.



- ⊕ = MONUMENT - RECOVERED
- ⊗ = MONUMENT - NOT RECOVERED

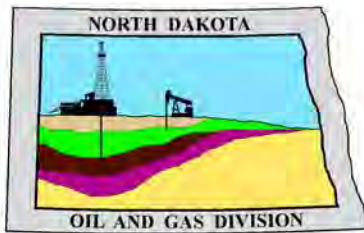
STAKED ON 1/14/14
 VERTICAL CONTROL DATUM WAS BASED UPON CONTROL POINT 705 WITH AN ELEVATION OF 2158.3'
 THIS SURVEY AND PLAT IS BEING PROVIDED AT THE REQUEST OF ERIC BAYES OF OASIS PETROLEUM. I CERTIFY THAT THIS PLAT CORRECTLY REPRESENTS WORK PERFORMED BY ME OR UNDER MY SUPERVISION AND IS TRUE AND CORRECT TO THE BEST OF MY KNOWLEDGE AND BELIEF.

[Signature]
 DARYL D. KASEMAN LS-3880



© 2014, INTERSTATE ENGINEERING, INC.

OASIS PETROLEUM NORTH AMERICA, LLC WELL LOCATION PLAT SECTION 31, T153N, R100W MCKENZIE COUNTY, NORTH DAKOTA	
Interstate Engineering, Inc. P.O. Box 648 335 East Main Street Salem, MA 01970 Ph: (406) 433-5617 Fax: (406) 433-5618 www.interstateeng.com	Project No.: 813-2013-002 Drawn By: J.B.K. Checked By: J.B.K. Date: JAN 2014
1/8 SHEET NO.	



Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.oilgas.nd.gov

April 9, 2014

**RE: Filter Socks and Other Filter Media
Leakproof Container Required
Oil and Gas Wells**

Dear Operator,

North Dakota Administrative Code Section 43-02-03-19.2 states in part that all waste material associated with exploration or production of oil and gas must be properly disposed of in an authorized facility in accord with all applicable local, state, and federal laws and regulations.

Filtration systems are commonly used during oil and gas operations in North Dakota. The Commission is very concerned about the proper disposal of used filters (including filter socks) used by the oil and gas industry.

Effective June 1, 2014, a container must be maintained on each well drilled in North Dakota beginning when the well is spud and must remain on-site during clean-out, completion, and flow-back whenever filtration operations are conducted. The on-site container must be used to store filters until they can be properly disposed of in an authorized facility. Such containers must be:

- leakproof to prevent any fluids from escaping the container
- covered to prevent precipitation from entering the container
- placard to indicate only filters are to be placed in the container

If the operator will not utilize a filtration system, a waiver to the container requirement will be considered, but only upon the operator submitting a Sundry Notice (Form 4) justifying their request.

As previously stated in our March 13, 2014 letter, North Dakota Administrative Code Section 33-20-02.1-01 states in part that every person who transports solid waste (which includes oil and gas exploration and production wastes) is required to have a valid permit issued by the North Dakota Department of Health, Division of Waste Management. Please contact the Division of Waste Management at (701) 328-5166 with any questions on the solid waste program. Note oil and gas exploration and production wastes include produced water, drilling mud, invert mud, tank bottom sediment, pipe scale, filters, and fly ash.

Thank you for your cooperation.

Sincerely,

Bruce E. Hicks

Assistant Director

DRILLING PLAN									
OPERATOR		Oasis Petroleum			COUNTY/STATE		McKenzie Co., ND		
WELL NAME		Lewis Federal 5300 11-31 3B			RIG		N/A		
WELL TYPE		Horizontal Middle Bakken							
LOCATION		NW NW 31-153N-100W			Surface Location (survey plat):		1005' FNL 233' FWL		
EST. T.D.		20,610'			GROUND ELEV:		2,112'		Sub Height: 22'
TOTAL LATERAL:		9,566'			KB ELEV:		2,134'		
MARKER	TVD	Subsea TVD	LOGS: Type Interval						
Pierre	NDIC MAP	1,984	150	OH Logs: Request Log waiver based on the Lewis Federal 5300 31-31H 2,850' S of surface location					
Greenhorn		4,584	-2,450	CBL/GR: Above top of cement/GR to base of casing					
Mowry		4,997	-2,863	MWD GR: KOP to lateral TD					
Dakota		5,385	-3,251	DEVIATION: Surf: 3 deg. max., 1 deg / 100'; srvy every 500'					
Rierdon		6,347	-4,213	Prod: 5 deg. max., 1 deg / 100'; srvy every 100'					
Dunham Salt		6,877	-4,743	DST'S: None planned					
Dunham Salt Base		6,947	-4,813	CORES: None planned					
Pine Salt		7,249	-5,115	MUDLOGGING: Two-Man: Begin 200' above Kibbey					
Pine Salt Base		7,314	-5,180	30' samples in curve and lateral					
Opeche Salt		7,380	-5,246	BOP: 11" 5000 psi blind, pipe & annular					
Opeche Salt Base		7,471	-5,337						
Amsden		7,693	-5,559						
Tyler		7,841	-5,707						
Otter/Base Minnelusa		8,065	-5,931						
Kibbey Lime		8,408	-6,274						
Charles Salt		8,561	-6,427						
Base Last Salt		9,230	-7,096						
Mission Canyon		9,446	-7,312						
Lodgepole		9,987	-7,853						
False Bakken		10,716	-8,582						
Upper Bakken Shale		10,728	-8,594						
Middle Bakken (Top of Target)		10,755	-8,621						
Middle Bakken (Base of target)		10,765	-8,631						
Lower Bakken Shale		10,780	-8,646						
Threeforks		10,810	-8,676						
Est. Dip Rate:		-0.50			Surface Formation: Glacial till				
Max. Anticipated BHP:		4671							
MUD:	Interval	Type	WT	Vis	WL	Remarks			
Surface:	0' -	2,100'	FW/Gel - Lime Sweeps	8.4-9.0	28-32	NC	Circ Mud Tanks		
Intermediate:	2,100' -	11,044'	Invert	9.5-10.4	40-50	30+HHp	Circ Mud Tanks		
Lateral:	11,044' -	20,610'	Salt Water	9.8-10.2	28-32	NC	Circ Mud Tanks		
CASING:	Size	Wt ppf	Hole	Depth	Cement	WOC	Remarks		
Surface:	9-5/8"	36#	13-1/2"	2,100'	To Surface	12	100' into Pierre		
Intermediate:	7"	29/32#	8-3/4"	11,044'	4885	24	500' above Dakota		
Production Liner:	4.5"	11.6#	6"	20,610'	TOL @ 10,233'		50' above KOP		
PROBABLE PLUGS, IF REQ'D:									
OTHER:	MD	TVD	FNL/FSL	FEL/FWL	S-T-R	AZI			
Surface:	2,100	2,100	1005' FNL	233' FWL	31-T153N-R100W	Survey Company:			
KOP:	10,283'	10,283'	980' FNL	258' FWL	31-T153N-R100W	Build Rate:	12 deg /100'		
EOC:	11,029'	10,760'	996' FNL	731' FWL	31-T153N-R100W				
Casing Point:	11,044'	10,760'	997' FNL	746' FWL	31-T153N-R100W				
Middle Bakken Lateral TD:	20,610'	10,844'	998' FNL	210' FEL	32-T153N-R100W				
Comments:									
Request Log waiver based on the Lewis Federal 5300 31-31H 2,850' S of surface location									
No Frac String									
35 packers & 15 Sleeves									
Oasis Petroleum does not use diesel fuel, as defined by the US EPA in the list below, in our hydraulic fracture operations.									
		68334-30-5	Fuels, diesel						
		68476-34-6	Fuels, diesel, No. 2						
		68476-34-6	Fuel oil No. 2						
		68476-31-3	Fuel Oil No. 4						
		8008-20-6	Kerosene						
Geology: NAG		1/22/2014		Engineering: A. Soto 3/13/14					

**Oasis Petroleum
Well Summary
Lewis Federal 5300 11-31 3B
Sec. 31 T153N R100W
McKenzie County, North Dakota**

SURFACE CASING AND CEMENT DESIGN

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Make-up Torque (ft-lbs)		
							Minimum	Optimum	Max
13-3/8"	0' to 2100	54.5	J-55	STC	12.615"	12.459"	4,100	5,470	6,840

Interval	Description	Collapse	Burst	Tension
		(psi) a	(psi) b	(1000 lbs) c
0' to 2100	13-3/8", 54.5#, J-55, STC, 8rd	1400 / 2.04	2730 / 2.76	689 / 3.45

API Rating & Safety Factor

- a) Based on full casing evacuation with 9 ppg fluid on backside (2100' setting depth).
- b) Burst pressure based on 9 ppg fluid with no fluid on backside (2100' setting depth).
- c) Based on string weight in 9 ppg fluid at 2100' TVD plus 100k# overpull. (Buoyed weight equals 99k lbs.)

Cement volumes are based on 9-5/8" casing set in 13-1/2" hole with 40% excess to circulate cement back to surface.
Mix and pump the following slurry.

Pre-flush (Spacer): **10 bbls** fresh water

Lead Slurry: **884 sks** (313 bbls) Conventional system with 75 lb/sk cement, 2% extender, 10% expanding agent, 2% CaCl₂ and 0.5 lb/sk lost circulation control agent

Tail Slurry: **300 sks** (62 bbls) Conventional system with 94 lb/sk cement, 0.2% CaCl₂, and .3 lb/sk lost circulation control agent

**Oasis Petroleum
Well Summary
Lewis Federal 5300 11-31 3B
Sec. 31 T153N R100W
McKenzie County, North Dakota**

CONTINGENCY INTERMEDIATE CASING AND CEMENT DESIGN

Intermediate Casing Design

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Make-up Torque (ft-lbs)		
							Minimum	Optimum	Max
9-5/8"	0' - 6400'	40	L-80	LTC	8.835"	8.75"***	5,450	7,270	9,090

**Special Drift

Interval	Description	Collapse	Burst	Tension
		(psi) a	(psi) b	(1000 lbs) c
0' - 6400'	9-5/8", 40#, HCP-110, LTC, 8rd	3090 / 3.71*	5750 / 1.24	837 / 3.86

API Rating & Safety Factor

- a) Burst pressure calculated from a gas kick coming from the production zone (Bakken Pool) at 9,000psi and a subsequent breakdown at the 9-5/8" shoe, based on a 13.5#/ft fracture gradient. Backup of 9 ppg fluid.
- b) Collapse pressure based on 11.5ppg fluid on backside and 9ppg fluid inside of casing.
- c) Yield based on string weight in 10 ppg fluid, (217k lbs buoyed weight) plus 100k lbs overpull.

Cement volumes are estimates based on 7" casing set in an 8-3/4" hole with 30% excess.

Pre-flush (Spacer): 20 bbls Chem wash

Lead Slurry: 592 sks (210 bbls) Conventional system with 75 lb/sk cement, 0.5lb/sk lost circulation, 10% expanding agent, 2% extender, 2% CaCl₂, 0.2% anti foam, and 0.4% fluid loss

Tail Slurry: 521 sks (108 bbls) Conventional system with 94 lb/sk cement, 0.3% anti-settling agent, 0.3% fluid loss agent, 0.3 lb/sk lost circulation control agent, 0.2% anti foam, and 0.1% retarder

**Oasis Petroleum
Well Summary
Lewis Federal 5300 11-31 3B
Sec. 31 T153N R100W
McKenzie County, North Dakota**

INTERMEDIATE CASING AND CEMENT DESIGN

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Make-up Torque (ft-lbs)		
							Minimum	Optimum	Max
7"	0' - 6677'	29	P-110	LTC	6.184"	6.059"	5,980	7,970	9,960
7"	6677' - 10283'	32	HCP-110	LTC	6.094"	6.000"	6,730	8,970	11210
7"	10283' - 11044'	29	P-110	LTC	6.184"	6.059"	5,980	7,970	9,960

****Special Drift**

Interval	Description	Collapse	Burst	Tension
		(psi) a	(psi) b	(1000 lbs) c
0' - 6677'	7", 29#, HCP-110, LTC, 8rd	8530 / 2.43*	11220 / 1.19	797 / 2.09
6677' - 10283'	7", 32#, HCP-110, LTC, 8rd	11820 / 2.20*	12460 / 1.29	
6677' - 10283'	7", 32#, HCP-110, LTC, 8rd	11820 / 1.05**	12460 / 1.29	
10283' - 11044'	7", 32#, HCP-110, LTC, 8rd	8530 / 1.52*	11220 / 1.15	

API Rating & Safety Factor

- a. *Assume full casing evacuation with 10 ppg fluid on backside. **Assume full casing evacuation with 1.2 psi/ft equivalent fluid gradient across salt intervals.
- b. Burst pressure based on 9000 psig max press for stimulation plus 10.2 ppg fluid in casing and 9 ppg fluid on backside-to 10760' TVD.
- c. Based on string weight in 10 ppg fluid, (280k lbs buoyed weight) plus 100k lbs overpull.

Cement volumes are estimates based on 7" casing set in an 8-3/4" hole with 30% excess.

Pre-flush (Spacer): **20 bbls** Chem wash
70 bbls 10.6# Scavenger

Lead Slurry: **187 sks** (85 bbls) Conventional system with 24 lb/sk cement, 54lb/sk extender, 3% KCl, 0.5% viscosifier, 0.2% anti foam, 0.5lb/sk lost circulation

Tail Slurry: **607 sks** (168 bbls) Conventional system with 94 lb/sk cement, 3% KCl, 35% Silica, 0.2% fluid loss agent, 0.5 lb/sk lost circulation control agent and 0.4% retarder

**Oasis Petroleum
Well Summary
Lewis Federal 5300 11-31 3B
Sec. 31 T153N R100W
McKenzie County, North Dakota**

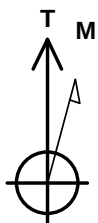
PRODUCTION LINER

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Estimated Torque
4-1/2"	10233' - 20610'	11.6	P-110	BTC	4.000"	3.875"	4,500psi

Interval	Description	Collapse	Burst	Tension	Condition
		(psi) a	(psi) b	(1000 lbs) c	
10233' - 20610'	4-1/2", 11.6 lb, P-110, BTC, 8rd	7560 / 1.40	10690 / 1.10	385 / 1.88	New

API Rating & Safety Factor

- a) Based on full casing evacuation with 9.5 ppg fluid on backside @ 10844' TVD.
- b) Burst pressure based on 9000 psi treating pressure with 10.2 ppg internal fluid gradient and 9 ppg external fluid gradient @ 10844' TVD.
- c) Based on string weight in 9.5 ppg fluid (Buoyed weight: 107k lbs.) plus 100k lbs overpull.



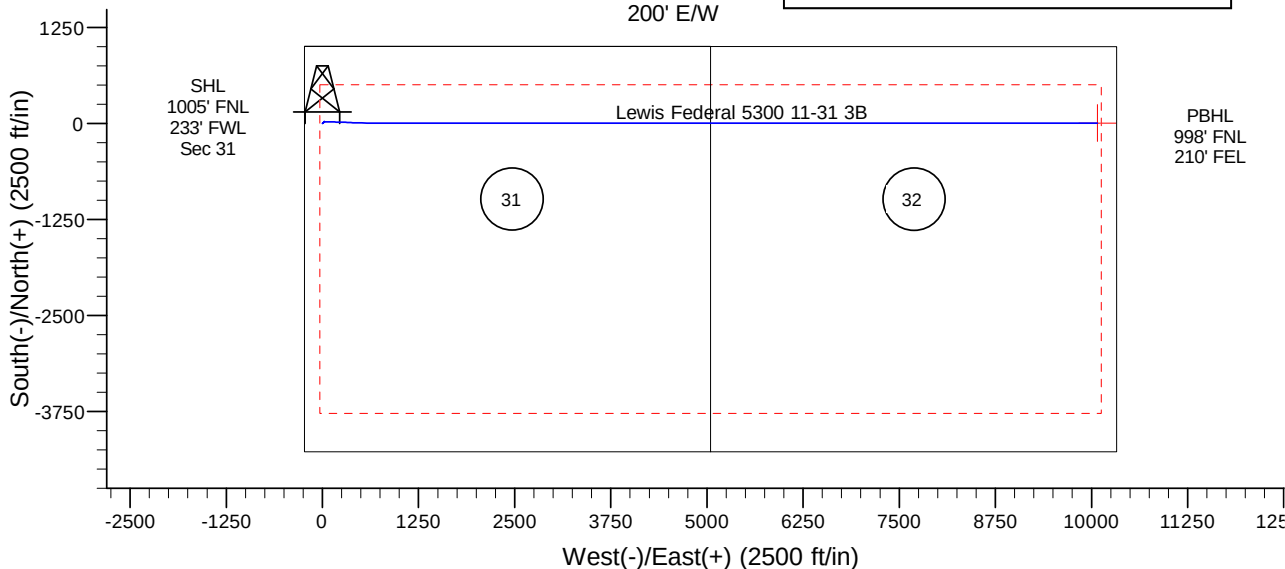
Azimuths to True North
Magnetic North: 8.16°

Magnetic Field
Strength: 56474.1snT
Dip Angle: 72.94°
Date: 3/11/2014
Model: IGRF200510



Project: Indian Hills
Site: 153N-100W-31/32
Well: Lewis Federal 5300 11-31 3B
Wellbore: Lewis Federal 5300 11-31 3B
Design: Plan #1

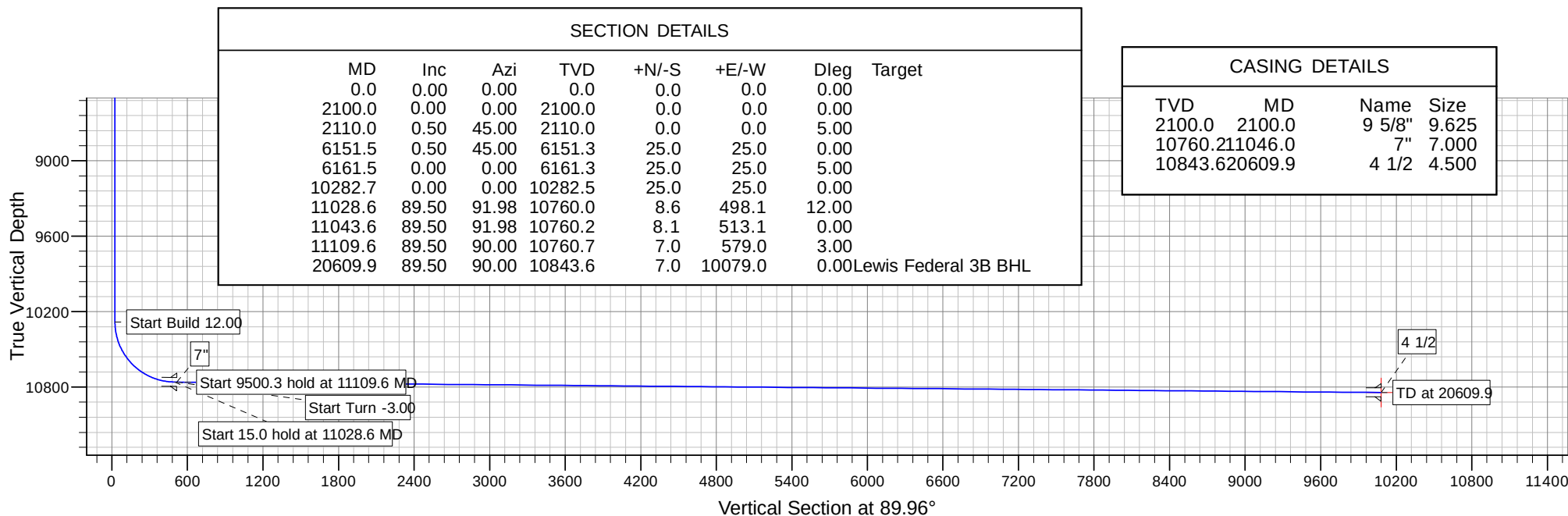
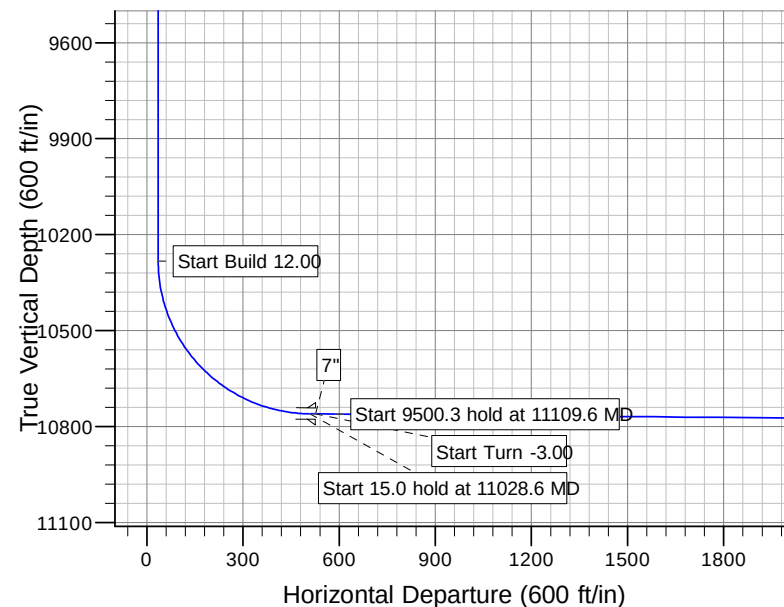
Setbacks
500' N/S
200' E/W



SITE DETAILS: 153N-100W-31/32

Site Centre Latitude: 48° 1' 42.010 N
Longitude: 103° 36' 10.620 W

Positional Uncertainty: 0.0
Convergence: -2.31
Local North: True



Oasis

Indian Hills

153N-100W-31/32

Lewis Federal 5300 11-31 3B

Lewis Federal 5300 11-31 3B

Plan: Plan #1

Standard Planning Report

14 April, 2014

Oasis Petroleum

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well Lewis Federal 5300 11-31 3B
Company:	Oasis	TVD Reference:	WELL @ 2134.0ft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2134.0ft (Original Well Elev)
Site:	153N-100W-31/32	North Reference:	True
Well:	Lewis Federal 5300 11-31 3B	Survey Calculation Method:	Minimum Curvature
Wellbore:	Lewis Federal 5300 11-31 3B		
Design:	Plan #1		

Project	Indian Hills		
Map System:	US State Plane 1983	System Datum:	Mean Sea Level
Geo Datum:	North American Datum 1983		
Map Zone:	North Dakota Northern Zone		

Site		153N-100W-31/32			
Site Position:		Northing:	390,399.32 ft	Latitude:	48° 1' 42.010 N
From:	Lat/Long	Easting:	1,209,468.83 ft	Longitude:	103° 36' 10.620 W
Position Uncertainty:	0.0 ft	Slot Radius:	13.200 in	Grid Convergence:	-2.31 °

Well	Lewis Federal 5300 11-31 3B					
Well Position	+N/-S	2,809.8 ft	Northing:	393,209.49 ft	Latitude:	48° 2' 9.740 N
	+E/-W	-65.9 ft	Easting:	1,209,516.17 ft	Longitude:	103° 36' 11.590 W
Position Uncertainty		0.0 ft	Wellhead Elevation:		Ground Level:	2,112.0 ft

Wellbore	Lewis Federal 5300 11-31 3B				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	IGRF200510	3/11/2014	8.16	72.94	56,474

Design	Plan #1				
Audit Notes:					
Version:	Phase:	PROTOTYPE	Tie On Depth:	0.0	
Vertical Section:	Depth From (TVD) (ft)	+N/-S (ft)	+E/-W (ft)	Direction (°)	
	0.0	0.0	0.0	89.96	

Plan Sections										
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)	TFO (°)	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00	0.00	0.00	0.00	
2,100.0	0.00	0.00	2,100.0	0.0	0.0	0.00	0.00	0.00	0.00	
2,110.0	0.50	45.00	2,110.0	0.0	0.0	5.00	5.00	0.00	45.00	
6,151.5	0.50	45.00	6,151.3	25.0	25.0	0.00	0.00	0.00	0.00	
6,161.5	0.00	0.00	6,161.3	25.0	25.0	5.00	-5.00	0.00	180.00	
10,282.7	0.00	0.00	10,282.5	25.0	25.0	0.00	0.00	0.00	0.00	
11,028.6	89.50	91.98	10,760.0	8.6	498.1	12.00	12.00	0.00	91.98	
11,043.6	89.50	91.98	10,760.2	8.1	513.1	0.00	0.00	0.00	0.00	
11,109.6	89.50	90.00	10,760.7	7.0	579.0	3.00	0.00	-3.00	-90.00	
20,609.9	89.50	90.00	10,843.6	7.0	10,079.0	0.00	0.00	0.00	0.00	Lewis Federal 3B BHI

Oasis Petroleum

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well Lewis Federal 5300 11-31 3B
Company:	Oasis	TVD Reference:	WELL @ 2134.0ft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2134.0ft (Original Well Elev)
Site:	153N-100W-31/32	North Reference:	True
Well:	Lewis Federal 5300 11-31 3B	Survey Calculation Method:	Minimum Curvature
Wellbore:	Lewis Federal 5300 11-31 3B		
Design:	Plan #1		

Planned Survey									
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
0.0	0.00	0.00	0.0	0.0	0.0	0.0	0.00	0.00	0.00
100.0	0.00	0.00	100.0	0.0	0.0	0.0	0.00	0.00	0.00
200.0	0.00	0.00	200.0	0.0	0.0	0.0	0.00	0.00	0.00
300.0	0.00	0.00	300.0	0.0	0.0	0.0	0.00	0.00	0.00
400.0	0.00	0.00	400.0	0.0	0.0	0.0	0.00	0.00	0.00
500.0	0.00	0.00	500.0	0.0	0.0	0.0	0.00	0.00	0.00
600.0	0.00	0.00	600.0	0.0	0.0	0.0	0.00	0.00	0.00
700.0	0.00	0.00	700.0	0.0	0.0	0.0	0.00	0.00	0.00
800.0	0.00	0.00	800.0	0.0	0.0	0.0	0.00	0.00	0.00
900.0	0.00	0.00	900.0	0.0	0.0	0.0	0.00	0.00	0.00
1,000.0	0.00	0.00	1,000.0	0.0	0.0	0.0	0.00	0.00	0.00
1,100.0	0.00	0.00	1,100.0	0.0	0.0	0.0	0.00	0.00	0.00
1,200.0	0.00	0.00	1,200.0	0.0	0.0	0.0	0.00	0.00	0.00
1,300.0	0.00	0.00	1,300.0	0.0	0.0	0.0	0.00	0.00	0.00
1,400.0	0.00	0.00	1,400.0	0.0	0.0	0.0	0.00	0.00	0.00
1,500.0	0.00	0.00	1,500.0	0.0	0.0	0.0	0.00	0.00	0.00
1,600.0	0.00	0.00	1,600.0	0.0	0.0	0.0	0.00	0.00	0.00
1,700.0	0.00	0.00	1,700.0	0.0	0.0	0.0	0.00	0.00	0.00
1,800.0	0.00	0.00	1,800.0	0.0	0.0	0.0	0.00	0.00	0.00
1,900.0	0.00	0.00	1,900.0	0.0	0.0	0.0	0.00	0.00	0.00
1,984.0	0.00	0.00	1,984.0	0.0	0.0	0.0	0.00	0.00	0.00
Pierre									
2,000.0	0.00	0.00	2,000.0	0.0	0.0	0.0	0.00	0.00	0.00
2,100.0	0.00	0.00	2,100.0	0.0	0.0	0.0	0.00	0.00	0.00
9 5/8"									
2,110.0	0.50	45.00	2,110.0	0.0	0.0	0.0	5.00	5.00	0.00
2,200.0	0.50	45.00	2,200.0	0.6	0.6	0.6	0.00	0.00	0.00
2,300.0	0.50	45.00	2,300.0	1.2	1.2	1.2	0.00	0.00	0.00
2,400.0	0.50	45.00	2,400.0	1.8	1.8	1.8	0.00	0.00	0.00
2,500.0	0.50	45.00	2,500.0	2.4	2.4	2.4	0.00	0.00	0.00
2,600.0	0.50	45.00	2,600.0	3.1	3.1	3.1	0.00	0.00	0.00
2,700.0	0.50	45.00	2,700.0	3.7	3.7	3.7	0.00	0.00	0.00
2,800.0	0.50	45.00	2,800.0	4.3	4.3	4.3	0.00	0.00	0.00
2,900.0	0.50	45.00	2,900.0	4.9	4.9	4.9	0.00	0.00	0.00
3,000.0	0.50	45.00	3,000.0	5.5	5.5	5.5	0.00	0.00	0.00
3,100.0	0.50	45.00	3,100.0	6.1	6.1	6.1	0.00	0.00	0.00
3,200.0	0.50	45.00	3,200.0	6.8	6.8	6.8	0.00	0.00	0.00
3,300.0	0.50	45.00	3,300.0	7.4	7.4	7.4	0.00	0.00	0.00
3,400.0	0.50	45.00	3,400.0	8.0	8.0	8.0	0.00	0.00	0.00
3,500.0	0.50	45.00	3,499.9	8.6	8.6	8.6	0.00	0.00	0.00
3,600.0	0.50	45.00	3,599.9	9.2	9.2	9.2	0.00	0.00	0.00
3,700.0	0.50	45.00	3,699.9	9.8	9.8	9.8	0.00	0.00	0.00
3,800.0	0.50	45.00	3,799.9	10.5	10.5	10.5	0.00	0.00	0.00
3,900.0	0.50	45.00	3,899.9	11.1	11.1	11.1	0.00	0.00	0.00
4,000.0	0.50	45.00	3,999.9	11.7	11.7	11.7	0.00	0.00	0.00
4,100.0	0.50	45.00	4,099.9	12.3	12.3	12.3	0.00	0.00	0.00
4,200.0	0.50	45.00	4,199.9	12.9	12.9	12.9	0.00	0.00	0.00
4,300.0	0.50	45.00	4,299.9	13.5	13.5	13.6	0.00	0.00	0.00
4,400.0	0.50	45.00	4,399.9	14.2	14.2	14.2	0.00	0.00	0.00
4,500.0	0.50	45.00	4,499.9	14.8	14.8	14.8	0.00	0.00	0.00
4,584.1	0.50	45.00	4,584.0	15.3	15.3	15.3	0.00	0.00	0.00
Greenhorn									
4,600.0	0.50	45.00	4,599.9	15.4	15.4	15.4	0.00	0.00	0.00
4,700.0	0.50	45.00	4,699.9	16.0	16.0	16.0	0.00	0.00	0.00
4,800.0	0.50	45.00	4,799.9	16.6	16.6	16.6	0.00	0.00	0.00

Oasis Petroleum

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well Lewis Federal 5300 11-31 3B
Company:	Oasis	TVD Reference:	WELL @ 2134.0ft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2134.0ft (Original Well Elev)
Site:	153N-100W-31/32	North Reference:	True
Well:	Lewis Federal 5300 11-31 3B	Survey Calculation Method:	Minimum Curvature
Wellbore:	Lewis Federal 5300 11-31 3B		
Design:	Plan #1		

Planned Survey									
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
4,900.0	0.50	45.00	4,899.9	17.2	17.2	17.3	0.00	0.00	0.00
4,997.1	0.50	45.00	4,997.0	17.8	17.8	17.9	0.00	0.00	0.00
Mowry									
5,000.0	0.50	45.00	4,999.9	17.9	17.9	17.9	0.00	0.00	0.00
5,100.0	0.50	45.00	5,099.9	18.5	18.5	18.5	0.00	0.00	0.00
5,200.0	0.50	45.00	5,199.9	19.1	19.1	19.1	0.00	0.00	0.00
5,300.0	0.50	45.00	5,299.9	19.7	19.7	19.7	0.00	0.00	0.00
5,385.1	0.50	45.00	5,385.0	20.2	20.2	20.3	0.00	0.00	0.00
Dakota									
5,400.0	0.50	45.00	5,399.9	20.3	20.3	20.3	0.00	0.00	0.00
5,500.0	0.50	45.00	5,499.9	20.9	20.9	21.0	0.00	0.00	0.00
5,600.0	0.50	45.00	5,599.9	21.6	21.6	21.6	0.00	0.00	0.00
5,700.0	0.50	45.00	5,699.9	22.2	22.2	22.2	0.00	0.00	0.00
5,800.0	0.50	45.00	5,799.9	22.8	22.8	22.8	0.00	0.00	0.00
5,900.0	0.50	45.00	5,899.9	23.4	23.4	23.4	0.00	0.00	0.00
6,000.0	0.50	45.00	5,999.9	24.0	24.0	24.1	0.00	0.00	0.00
6,100.0	0.50	45.00	6,099.8	24.7	24.7	24.7	0.00	0.00	0.00
6,151.5	0.50	45.00	6,151.3	25.0	25.0	25.0	0.00	0.00	0.00
6,161.5	0.00	0.00	6,161.3	25.0	25.0	25.0	5.00	-5.00	0.00
6,200.0	0.00	0.00	6,199.8	25.0	25.0	25.0	0.00	0.00	0.00
6,300.0	0.00	0.00	6,299.8	25.0	25.0	25.0	0.00	0.00	0.00
6,347.2	0.00	0.00	6,347.0	25.0	25.0	25.0	0.00	0.00	0.00
Rierdon									
6,400.0	0.00	0.00	6,399.8	25.0	25.0	25.0	0.00	0.00	0.00
6,500.0	0.00	0.00	6,499.8	25.0	25.0	25.0	0.00	0.00	0.00
6,600.0	0.00	0.00	6,599.8	25.0	25.0	25.0	0.00	0.00	0.00
6,700.0	0.00	0.00	6,699.8	25.0	25.0	25.0	0.00	0.00	0.00
6,800.0	0.00	0.00	6,799.8	25.0	25.0	25.0	0.00	0.00	0.00
6,877.2	0.00	0.00	6,877.0	25.0	25.0	25.0	0.00	0.00	0.00
Dunham Salt									
6,900.0	0.00	0.00	6,899.8	25.0	25.0	25.0	0.00	0.00	0.00
6,947.2	0.00	0.00	6,947.0	25.0	25.0	25.0	0.00	0.00	0.00
Dunham Salt Base									
7,000.0	0.00	0.00	6,999.8	25.0	25.0	25.0	0.00	0.00	0.00
7,100.0	0.00	0.00	7,099.8	25.0	25.0	25.0	0.00	0.00	0.00
7,200.0	0.00	0.00	7,199.8	25.0	25.0	25.0	0.00	0.00	0.00
7,249.2	0.00	0.00	7,249.0	25.0	25.0	25.0	0.00	0.00	0.00
Pine Salt									
7,300.0	0.00	0.00	7,299.8	25.0	25.0	25.0	0.00	0.00	0.00
7,314.2	0.00	0.00	7,314.0	25.0	25.0	25.0	0.00	0.00	0.00
Pine Salt Base									
7,380.2	0.00	0.00	7,380.0	25.0	25.0	25.0	0.00	0.00	0.00
Opeche Salt									
7,400.0	0.00	0.00	7,399.8	25.0	25.0	25.0	0.00	0.00	0.00
7,471.2	0.00	0.00	7,471.0	25.0	25.0	25.0	0.00	0.00	0.00
Opeche Salt Base									
7,500.0	0.00	0.00	7,499.8	25.0	25.0	25.0	0.00	0.00	0.00
7,600.0	0.00	0.00	7,599.8	25.0	25.0	25.0	0.00	0.00	0.00
7,693.2	0.00	0.00	7,693.0	25.0	25.0	25.0	0.00	0.00	0.00
Amsden									
7,700.0	0.00	0.00	7,699.8	25.0	25.0	25.0	0.00	0.00	0.00
7,800.0	0.00	0.00	7,799.8	25.0	25.0	25.0	0.00	0.00	0.00
7,841.2	0.00	0.00	7,841.0	25.0	25.0	25.0	0.00	0.00	0.00
Tyler									

Oasis Petroleum

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well Lewis Federal 5300 11-31 3B
Company:	Oasis	TVD Reference:	WELL @ 2134.0ft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2134.0ft (Original Well Elev)
Site:	153N-100W-31/32	North Reference:	True
Well:	Lewis Federal 5300 11-31 3B	Survey Calculation Method:	Minimum Curvature
Wellbore:	Lewis Federal 5300 11-31 3B		
Design:	Plan #1		

Planned Survey									
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
7,900.0	0.00	0.00	7,899.8	25.0	25.0	25.0	0.00	0.00	0.00
8,000.0	0.00	0.00	7,999.8	25.0	25.0	25.0	0.00	0.00	0.00
8,065.2	0.00	0.00	8,065.0	25.0	25.0	25.0	0.00	0.00	0.00
Otter/Base Minnelusa									
8,100.0	0.00	0.00	8,099.8	25.0	25.0	25.0	0.00	0.00	0.00
8,200.0	0.00	0.00	8,199.8	25.0	25.0	25.0	0.00	0.00	0.00
8,300.0	0.00	0.00	8,299.8	25.0	25.0	25.0	0.00	0.00	0.00
8,400.0	0.00	0.00	8,399.8	25.0	25.0	25.0	0.00	0.00	0.00
8,408.2	0.00	0.00	8,408.0	25.0	25.0	25.0	0.00	0.00	0.00
Kibbey Lime									
8,500.0	0.00	0.00	8,499.8	25.0	25.0	25.0	0.00	0.00	0.00
8,561.2	0.00	0.00	8,561.0	25.0	25.0	25.0	0.00	0.00	0.00
Charles Salt									
8,600.0	0.00	0.00	8,599.8	25.0	25.0	25.0	0.00	0.00	0.00
8,700.0	0.00	0.00	8,699.8	25.0	25.0	25.0	0.00	0.00	0.00
8,800.0	0.00	0.00	8,799.8	25.0	25.0	25.0	0.00	0.00	0.00
8,900.0	0.00	0.00	8,899.8	25.0	25.0	25.0	0.00	0.00	0.00
9,000.0	0.00	0.00	8,999.8	25.0	25.0	25.0	0.00	0.00	0.00
9,100.0	0.00	0.00	9,099.8	25.0	25.0	25.0	0.00	0.00	0.00
9,200.0	0.00	0.00	9,199.8	25.0	25.0	25.0	0.00	0.00	0.00
9,230.2	0.00	0.00	9,230.0	25.0	25.0	25.0	0.00	0.00	0.00
Base Last Salt									
9,300.0	0.00	0.00	9,299.8	25.0	25.0	25.0	0.00	0.00	0.00
9,400.0	0.00	0.00	9,399.8	25.0	25.0	25.0	0.00	0.00	0.00
9,446.2	0.00	0.00	9,446.0	25.0	25.0	25.0	0.00	0.00	0.00
Mission Canyon									
9,500.0	0.00	0.00	9,499.8	25.0	25.0	25.0	0.00	0.00	0.00
9,600.0	0.00	0.00	9,599.8	25.0	25.0	25.0	0.00	0.00	0.00
9,700.0	0.00	0.00	9,699.8	25.0	25.0	25.0	0.00	0.00	0.00
9,800.0	0.00	0.00	9,799.8	25.0	25.0	25.0	0.00	0.00	0.00
9,900.0	0.00	0.00	9,899.8	25.0	25.0	25.0	0.00	0.00	0.00
9,987.2	0.00	0.00	9,987.0	25.0	25.0	25.0	0.00	0.00	0.00
Lodgepole									
10,000.0	0.00	0.00	9,999.8	25.0	25.0	25.0	0.00	0.00	0.00
10,100.0	0.00	0.00	10,099.8	25.0	25.0	25.0	0.00	0.00	0.00
10,200.0	0.00	0.00	10,199.8	25.0	25.0	25.0	0.00	0.00	0.00
10,282.7	0.00	0.00	10,282.5	25.0	25.0	25.0	0.00	0.00	0.00
10,300.0	2.08	91.98	10,299.8	25.0	25.3	25.3	12.00	12.00	0.00
10,325.0	5.08	91.98	10,324.8	24.9	26.9	26.9	12.00	12.00	0.00
10,350.0	8.08	91.98	10,349.6	24.8	29.7	29.8	12.00	12.00	0.00
10,375.0	11.08	91.98	10,374.3	24.7	33.9	33.9	12.00	12.00	0.00
10,400.0	14.08	91.98	10,398.7	24.5	39.3	39.3	12.00	12.00	0.00
10,425.0	17.08	91.98	10,422.7	24.3	46.0	46.1	12.00	12.00	0.00
10,450.0	20.08	91.98	10,446.4	24.0	54.0	54.0	12.00	12.00	0.00
10,475.0	23.08	91.98	10,469.7	23.7	63.2	63.2	12.00	12.00	0.00
10,500.0	26.08	91.98	10,492.4	23.3	73.6	73.6	12.00	12.00	0.00
10,525.0	29.08	91.98	10,514.6	22.9	85.1	85.2	12.00	12.00	0.00
10,550.0	32.08	91.98	10,536.1	22.5	97.8	97.9	12.00	12.00	0.00
10,575.0	35.08	91.98	10,556.9	22.0	111.7	111.7	12.00	12.00	0.00
10,600.0	38.07	91.98	10,577.0	21.5	126.6	126.6	12.00	12.00	0.00
10,625.0	41.07	91.98	10,596.3	20.9	142.5	142.5	12.00	12.00	0.00
10,650.0	44.07	91.98	10,614.7	20.4	159.4	159.4	12.00	12.00	0.00
10,675.0	47.07	91.98	10,632.2	19.7	177.2	177.2	12.00	12.00	0.00
10,700.0	50.07	91.98	10,648.7	19.1	195.9	196.0	12.00	12.00	0.00
10,725.0	53.07	91.98	10,664.2	18.4	215.5	215.5	12.00	12.00	0.00

Oasis Petroleum

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well Lewis Federal 5300 11-31 3B
Company:	Oasis	TVD Reference:	WELL @ 2134.0ft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2134.0ft (Original Well Elev)
Site:	153N-100W-31/32	North Reference:	True
Well:	Lewis Federal 5300 11-31 3B	Survey Calculation Method:	Minimum Curvature
Wellbore:	Lewis Federal 5300 11-31 3B		
Design:	Plan #1		

Planned Survey									
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
10,750.0	56.07	91.98	10,678.7	17.7	235.9	235.9	12.00	12.00	0.00
10,775.0	59.07	91.98	10,692.1	17.0	257.0	257.0	12.00	12.00	0.00
10,800.0	62.07	91.98	10,704.4	16.2	278.7	278.7	12.00	12.00	0.00
10,825.0	65.07	91.98	10,715.6	15.5	301.1	301.1	12.00	12.00	0.00
10,826.1	65.21	91.98	10,716.0	15.4	302.1	302.1	12.00	12.00	0.00
False Bakken									
10,850.0	68.07	91.98	10,725.5	14.7	324.0	324.0	12.00	12.00	0.00
10,856.9	68.90	91.98	10,728.0	14.4	330.4	330.4	12.00	12.00	0.00
Upper Bakken Shale									
10,875.0	71.07	91.98	10,734.2	13.9	347.4	347.4	12.00	12.00	0.00
10,900.0	74.07	91.98	10,741.7	13.0	371.3	371.3	12.00	12.00	0.00
10,925.0	77.07	91.98	10,747.9	12.2	395.5	395.5	12.00	12.00	0.00
10,950.0	80.07	91.98	10,752.9	11.3	419.9	419.9	12.00	12.00	0.00
10,963.5	81.69	91.98	10,755.0	10.9	433.2	433.2	12.00	12.00	0.00
Middle Bakken (Top of Target)									
10,975.0	83.07	91.98	10,756.6	10.5	444.6	444.7	12.00	12.00	0.00
11,000.0	86.07	91.98	10,758.9	9.6	469.5	469.5	12.00	12.00	0.00
11,025.0	89.07	91.98	10,760.0	8.8	494.5	494.5	12.00	12.00	0.00
11,028.6	89.50	91.98	10,760.0	8.6	498.1	498.1	12.00	12.00	0.00
11,043.6	89.50	91.98	10,760.2	8.1	513.1	513.1	0.00	0.00	0.00
11,046.0	89.50	91.91	10,760.2	8.0	515.5	515.5	3.00	0.00	-3.00
7"									
11,109.6	89.50	90.00	10,760.7	7.0	579.0	579.0	3.00	0.00	-3.00
11,200.0	89.50	90.00	10,761.5	7.0	669.4	669.5	0.00	0.00	0.00
11,300.0	89.50	90.00	10,762.4	7.0	769.4	769.4	0.00	0.00	0.00
11,400.0	89.50	90.00	10,763.3	7.0	869.4	869.4	0.00	0.00	0.00
11,500.0	89.50	90.00	10,764.1	7.0	969.4	969.4	0.00	0.00	0.00
11,600.0	89.50	90.00	10,765.0	7.0	1,069.4	1,069.4	0.00	0.00	0.00
11,700.0	89.50	90.00	10,765.9	7.0	1,169.4	1,169.4	0.00	0.00	0.00
11,800.0	89.50	90.00	10,766.8	7.0	1,269.4	1,269.4	0.00	0.00	0.00
11,900.0	89.50	90.00	10,767.6	7.0	1,369.4	1,369.4	0.00	0.00	0.00
12,000.0	89.50	90.00	10,768.5	7.0	1,469.4	1,469.4	0.00	0.00	0.00
12,100.0	89.50	90.00	10,769.4	7.0	1,569.4	1,569.4	0.00	0.00	0.00
12,200.0	89.50	90.00	10,770.2	7.0	1,669.4	1,669.4	0.00	0.00	0.00
12,300.0	89.50	90.00	10,771.1	7.0	1,769.4	1,769.4	0.00	0.00	0.00
12,400.0	89.50	90.00	10,772.0	7.0	1,869.4	1,869.4	0.00	0.00	0.00
12,500.0	89.50	90.00	10,772.9	7.0	1,969.4	1,969.4	0.00	0.00	0.00
12,600.0	89.50	90.00	10,773.7	7.0	2,069.4	2,069.4	0.00	0.00	0.00
12,700.0	89.50	90.00	10,774.6	7.0	2,169.4	2,169.4	0.00	0.00	0.00
12,800.0	89.50	90.00	10,775.5	7.0	2,269.4	2,269.4	0.00	0.00	0.00
12,900.0	89.50	90.00	10,776.4	7.0	2,369.4	2,369.4	0.00	0.00	0.00
13,000.0	89.50	90.00	10,777.2	7.0	2,469.4	2,469.4	0.00	0.00	0.00
13,100.0	89.50	90.00	10,778.1	7.0	2,569.4	2,569.4	0.00	0.00	0.00
13,200.0	89.50	90.00	10,779.0	7.0	2,669.4	2,669.4	0.00	0.00	0.00
13,300.0	89.50	90.00	10,779.8	7.0	2,769.4	2,769.4	0.00	0.00	0.00
13,400.0	89.50	90.00	10,780.7	7.0	2,869.4	2,869.4	0.00	0.00	0.00
13,500.0	89.50	90.00	10,781.6	7.0	2,969.4	2,969.4	0.00	0.00	0.00
13,600.0	89.50	90.00	10,782.5	7.0	3,069.4	3,069.4	0.00	0.00	0.00
13,700.0	89.50	90.00	10,783.3	7.0	3,169.4	3,169.4	0.00	0.00	0.00
13,800.0	89.50	90.00	10,784.2	7.0	3,269.3	3,269.4	0.00	0.00	0.00
13,900.0	89.50	90.00	10,785.1	7.0	3,369.3	3,369.3	0.00	0.00	0.00
14,000.0	89.50	90.00	10,786.0	7.0	3,469.3	3,469.3	0.00	0.00	0.00
14,100.0	89.50	90.00	10,786.8	7.0	3,569.3	3,569.3	0.00	0.00	0.00
14,200.0	89.50	90.00	10,787.7	7.0	3,669.3	3,669.3	0.00	0.00	0.00
14,300.0	89.50	90.00	10,788.6	7.0	3,769.3	3,769.3	0.00	0.00	0.00

Oasis Petroleum

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well Lewis Federal 5300 11-31 3B
Company:	Oasis	TVD Reference:	WELL @ 2134.0ft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2134.0ft (Original Well Elev)
Site:	153N-100W-31/32	North Reference:	True
Well:	Lewis Federal 5300 11-31 3B	Survey Calculation Method:	Minimum Curvature
Wellbore:	Lewis Federal 5300 11-31 3B		
Design:	Plan #1		

Planned Survey									
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
14,400.0	89.50	90.00	10,789.4	7.0	3,869.3	3,869.3	0.00	0.00	0.00
14,500.0	89.50	90.00	10,790.3	7.0	3,969.3	3,969.3	0.00	0.00	0.00
14,600.0	89.50	90.00	10,791.2	7.0	4,069.3	4,069.3	0.00	0.00	0.00
14,700.0	89.50	90.00	10,792.1	7.0	4,169.3	4,169.3	0.00	0.00	0.00
14,800.0	89.50	90.00	10,792.9	7.0	4,269.3	4,269.3	0.00	0.00	0.00
14,900.0	89.50	90.00	10,793.8	7.0	4,369.3	4,369.3	0.00	0.00	0.00
15,000.0	89.50	90.00	10,794.7	7.0	4,469.3	4,469.3	0.00	0.00	0.00
15,100.0	89.50	90.00	10,795.6	7.0	4,569.3	4,569.3	0.00	0.00	0.00
15,200.0	89.50	90.00	10,796.4	7.0	4,669.3	4,669.3	0.00	0.00	0.00
15,300.0	89.50	90.00	10,797.3	7.0	4,769.3	4,769.3	0.00	0.00	0.00
15,400.0	89.50	90.00	10,798.2	7.0	4,869.3	4,869.3	0.00	0.00	0.00
15,500.0	89.50	90.00	10,799.0	7.0	4,969.3	4,969.3	0.00	0.00	0.00
15,600.0	89.50	90.00	10,799.9	7.0	5,069.3	5,069.3	0.00	0.00	0.00
15,700.0	89.50	90.00	10,800.8	7.0	5,169.3	5,169.3	0.00	0.00	0.00
15,800.0	89.50	90.00	10,801.7	7.0	5,269.3	5,269.3	0.00	0.00	0.00
15,900.0	89.50	90.00	10,802.5	7.0	5,369.3	5,369.3	0.00	0.00	0.00
16,000.0	89.50	90.00	10,803.4	7.0	5,469.3	5,469.3	0.00	0.00	0.00
16,100.0	89.50	90.00	10,804.3	7.0	5,569.3	5,569.3	0.00	0.00	0.00
16,200.0	89.50	90.00	10,805.1	7.0	5,669.3	5,669.3	0.00	0.00	0.00
16,300.0	89.50	90.00	10,806.0	7.0	5,769.3	5,769.3	0.00	0.00	0.00
16,400.0	89.50	90.00	10,806.9	7.0	5,869.2	5,869.3	0.00	0.00	0.00
16,500.0	89.50	90.00	10,807.8	7.0	5,969.2	5,969.2	0.00	0.00	0.00
16,600.0	89.50	90.00	10,808.6	7.0	6,069.2	6,069.2	0.00	0.00	0.00
16,700.0	89.50	90.00	10,809.5	7.0	6,169.2	6,169.2	0.00	0.00	0.00
16,800.0	89.50	90.00	10,810.4	7.0	6,269.2	6,269.2	0.00	0.00	0.00
16,900.0	89.50	90.00	10,811.3	7.0	6,369.2	6,369.2	0.00	0.00	0.00
17,000.0	89.50	90.00	10,812.1	7.0	6,469.2	6,469.2	0.00	0.00	0.00
17,100.0	89.50	90.00	10,813.0	7.0	6,569.2	6,569.2	0.00	0.00	0.00
17,200.0	89.50	90.00	10,813.9	7.0	6,669.2	6,669.2	0.00	0.00	0.00
17,300.0	89.50	90.00	10,814.7	7.0	6,769.2	6,769.2	0.00	0.00	0.00
17,400.0	89.50	90.00	10,815.6	7.0	6,869.2	6,869.2	0.00	0.00	0.00
17,500.0	89.50	90.00	10,816.5	7.0	6,969.2	6,969.2	0.00	0.00	0.00
17,600.0	89.50	90.00	10,817.4	7.0	7,069.2	7,069.2	0.00	0.00	0.00
17,700.0	89.50	90.00	10,818.2	7.0	7,169.2	7,169.2	0.00	0.00	0.00
17,800.0	89.50	90.00	10,819.1	7.0	7,269.2	7,269.2	0.00	0.00	0.00
17,900.0	89.50	90.00	10,820.0	7.0	7,369.2	7,369.2	0.00	0.00	0.00
18,000.0	89.50	90.00	10,820.9	7.0	7,469.2	7,469.2	0.00	0.00	0.00
18,100.0	89.50	90.00	10,821.7	7.0	7,569.2	7,569.2	0.00	0.00	0.00
18,200.0	89.50	90.00	10,822.6	7.0	7,669.2	7,669.2	0.00	0.00	0.00
18,300.0	89.50	90.00	10,823.5	7.0	7,769.2	7,769.2	0.00	0.00	0.00
18,400.0	89.50	90.00	10,824.3	7.0	7,869.2	7,869.2	0.00	0.00	0.00
18,500.0	89.50	90.00	10,825.2	7.0	7,969.2	7,969.2	0.00	0.00	0.00
18,600.0	89.50	90.00	10,826.1	7.0	8,069.2	8,069.2	0.00	0.00	0.00
18,700.0	89.50	90.00	10,827.0	7.0	8,169.2	8,169.2	0.00	0.00	0.00
18,800.0	89.50	90.00	10,827.8	7.0	8,269.2	8,269.2	0.00	0.00	0.00
18,900.0	89.50	90.00	10,828.7	7.0	8,369.2	8,369.2	0.00	0.00	0.00
19,000.0	89.50	90.00	10,829.6	7.0	8,469.2	8,469.2	0.00	0.00	0.00
19,100.0	89.50	90.00	10,830.5	7.0	8,569.1	8,569.1	0.00	0.00	0.00
19,200.0	89.50	90.00	10,831.3	7.0	8,669.1	8,669.1	0.00	0.00	0.00
19,300.0	89.50	90.00	10,832.2	7.0	8,769.1	8,769.1	0.00	0.00	0.00
19,400.0	89.50	90.00	10,833.1	7.0	8,869.1	8,869.1	0.00	0.00	0.00
19,500.0	89.50	90.00	10,833.9	7.0	8,969.1	8,969.1	0.00	0.00	0.00
19,600.0	89.50	90.00	10,834.8	7.0	9,069.1	9,069.1	0.00	0.00	0.00
19,700.0	89.50	90.00	10,835.7	7.0	9,169.1	9,169.1	0.00	0.00	0.00
19,800.0	89.50	90.00	10,836.6	7.0	9,269.1	9,269.1	0.00	0.00	0.00

Oasis Petroleum

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well Lewis Federal 5300 11-31 3B
Company:	Oasis	TVD Reference:	WELL @ 2134.0ft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2134.0ft (Original Well Elev)
Site:	153N-100W-31/32	North Reference:	True
Well:	Lewis Federal 5300 11-31 3B	Survey Calculation Method:	Minimum Curvature
Wellbore:	Lewis Federal 5300 11-31 3B		
Design:	Plan #1		

Planned Survey									
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
19,900.0	89.50	90.00	10,837.4	7.0	9,369.1	9,369.1	0.00	0.00	0.00
20,000.0	89.50	90.00	10,838.3	7.0	9,469.1	9,469.1	0.00	0.00	0.00
20,100.0	89.50	90.00	10,839.2	7.0	9,569.1	9,569.1	0.00	0.00	0.00
20,200.0	89.50	90.00	10,840.1	7.0	9,669.1	9,669.1	0.00	0.00	0.00
20,300.0	89.50	90.00	10,840.9	7.0	9,769.1	9,769.1	0.00	0.00	0.00
20,400.0	89.50	90.00	10,841.8	7.0	9,869.1	9,869.1	0.00	0.00	0.00
20,500.0	89.50	90.00	10,842.7	7.0	9,969.1	9,969.1	0.00	0.00	0.00
20,609.9	89.50	90.00	10,843.6	7.0	10,079.0	10,079.0	0.00	0.00	0.00
4 1/2 - Lewis Federal 3B BHL									

Design Targets									
Target Name	Dip Angle (°)	Dip Dir. (°)	TVD (ft)	+N/-S (ft)	+E/-W (ft)	Northing (ft)	Easting (ft)	Latitude	Longitude
- hit/miss target									
- Shape									
Lewis Federal 3B BHL	0.00	0.00	10,843.9	7.0	10,079.0	392,810.37	1,219,587.29	48° 2' 9.782 N	103° 33' 43.277 W
- plan misses target center by 0.3ft at 20609.9ft MD (10843.6 TVD, 7.0 N, 10079.0 E)									
- Point									

Casing Points				
Measured Depth (ft)	Vertical Depth (ft)	Name	Casing Diameter (in)	Hole Diameter (in)
2,100.0	2,100.0	9 5/8"	9.625	13.500
11,046.0	10,760.2	7"	7.000	8.750
20,609.9	10,843.6	4 1/2	4.500	6.000

Oasis Petroleum

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well Lewis Federal 5300 11-31 3B
Company:	Oasis	TVD Reference:	WELL @ 2134.0ft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2134.0ft (Original Well Elev)
Site:	153N-100W-31/32	North Reference:	True
Well:	Lewis Federal 5300 11-31 3B	Survey Calculation Method:	Minimum Curvature
Wellbore:	Lewis Federal 5300 11-31 3B		
Design:	Plan #1		

Formations						
Measured Depth (ft)	Vertical Depth (ft)	Name	Lithology	Dip (°)	Dip Direction (°)	
1,984.0	1,984.0	Pierre				
4,584.1	4,584.0	Greenhorn				
4,997.1	4,997.0	Mowry				
5,385.1	5,385.0	Dakota				
6,347.2	6,347.0	Rierdon				
6,877.2	6,877.0	Dunham Salt				
6,947.2	6,947.0	Dunham Salt Base				
7,249.2	7,249.0	Pine Salt				
7,314.2	7,314.0	Pine Salt Base				
7,380.2	7,380.0	Opeche Salt				
7,471.2	7,471.0	Opeche Salt Base				
7,693.2	7,693.0	Amsden				
7,841.2	7,841.0	Tyler				
8,065.2	8,065.0	Otter/Base Minnelusa				
8,408.2	8,408.0	Kibbey Lime				
8,561.2	8,561.0	Charles Salt				
9,230.2	9,230.0	Base Last Salt				
9,446.2	9,446.0	Mission Canyon				
9,987.2	9,987.0	Lodgepole				
10,826.1	10,716.0	False Bakken				
10,856.9	10,728.0	Upper Bakken Shale				
10,963.5	10,755.0	Middle Bakken (Top of Target)				

STATEMENT

This statement is being sent in order to comply with NDAC 43-02-03-16 (Application for permit to drill and recomplete) which states (in part that) "confirmation that a legal street address has been requested for the well site, and well facility if separate from the well site, and the proposed road access to the nearest existing public road". On the date noted below a legal street address was requested from the appropriate county office.

McKenzie County

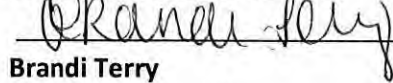
Aaron Chisolm – McKenzie County Dept.

Lewis Federal 5300 11-31 2T

Lewis Federal 5300 11-31 3B

Lewis Federal 5300 11-31 4T2

Lewis Federal 5300 11-31 5T



Brandi Terry

Regulatory Specialist

Oasis Petroleum North America, LLC

Hello Dave,

Below is the comments from construction regarding the wells on fill as requested. Page 4 of the plats shows the production layout. We only have pumping units planned for onsite with production going to an offsite CTB. Please let me know if you need further information on the CTB.

Once permits have been received, I will work with Cody Vanderbusch to get approval on the offsite pit. He has requested well file numbers be issued prior to submitting a sundry. With the sundry, we will include a landowner agreement and plat showing the offsite cutting pit. If for any reason the pit is not approved, we will haul cuttings to an appropriate disposal or work with the NDIC for another solution.

Sincerely,

Victoria Siemieniewski
Regulatory Specialist
Direct: 281-404-9652



From: Damon Jorgensen
Sent: Monday, November 24, 2014 4:44 PM
To: Victoria Siemieniewski
Cc: James Demorrett; Kristy Aasheim; Josh.Schmierer@interstateeng.com; Thomas Osen; Karyme Martin
Subject: RE: Lewis Federal 5300 11-31 2T, 3B, 4T2, 5T Well Pad

Victoria,

Regarding the question of geotech: While there are places in the Indian Hills field where glacial till overburden is present, this area is comprised primarily of bedrock clays, sandy clays and layers of sandstone. During construction this material is placed, compacted in lifts, and tested to ensure proper compaction has been achieved. Any material (coal, topsoil, large rocks/sandstone, etc) encountered during construction is removed and kept out of the compacted fill sections of the pad. Having successfully constructed numerous pads with extensive fill sections in this area, some with fills over 45', we feel confident in our supervision and seasoned contractors that the additional time and expense associated with formal geotechnical review of each pad is not warranted at this time.

Josh, Remove the pit and have the plats updated. produce an offsite pit plat. I'm assuming you have already gotten with facilities and have the production facility info Victoria needs (see comments below)....?

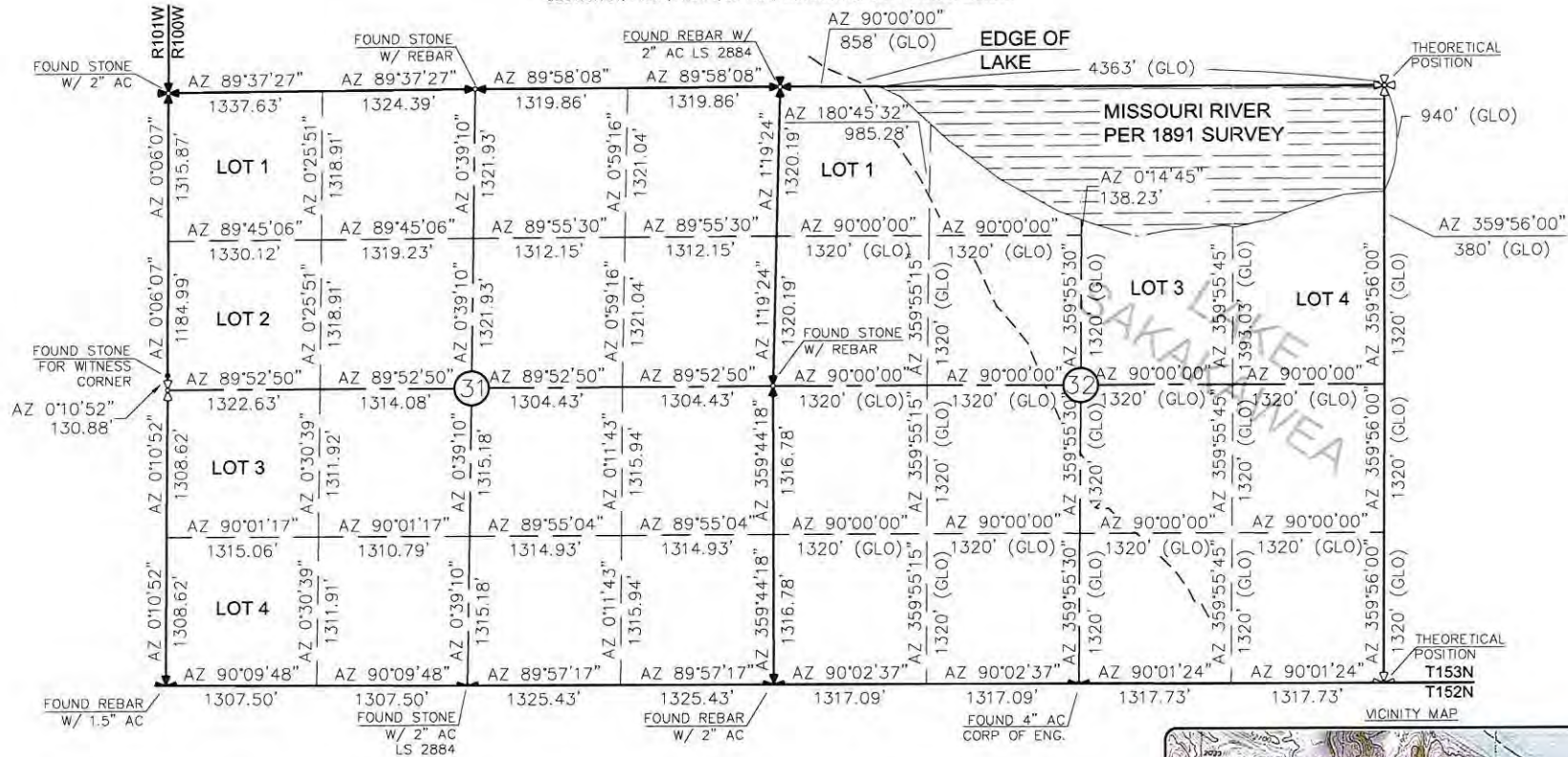
If there is anything else, please let me know.

Thanks

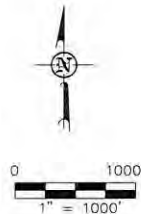


Damon Jorgensen
Construction Superintendent
701.577.1687 Office
701.400.1353 Mobile

SECTION BREAKDOWN
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "LEWIS FEDERAL 5300 11-31 3B"
 1005 FEET FROM NORTH LINE AND 233 FEET FROM WEST LINE
 SECTION 31, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



THIS DOCUMENT WAS ORIGINALLY
 ISSUED AND SEALED BY DARYL D.
 KASEMAN, PLS, REGISTRATION NUMBER
 3880 ON 1/21/14 AND THE
 ORIGINAL DOCUMENTS ARE STORED AT
 THE OFFICES OF INTERSTATE
 ENGINEERING, INC.



ALL AZIMUTHS ARE BASED ON G.P.S.
 OBSERVATIONS. THE ORIGINAL SURVEY OF THIS
 AREA FOR THE GENERAL LAND OFFICE (G.L.O.)
 WAS 1897. THE CORNERS FOUND ARE AS
 INDICATED AND ALL OTHERS ARE COMPUTED FROM
 THOSE CORNERS FOUND AND BASED ON G.L.O.
 DATA. THE MAPPING ANGLE FOR THIS AREA IS
 APPROXIMATELY 0°03'.



© 2014, INTERSTATE ENGINEERING, INC.

Revised	By	Date	Description

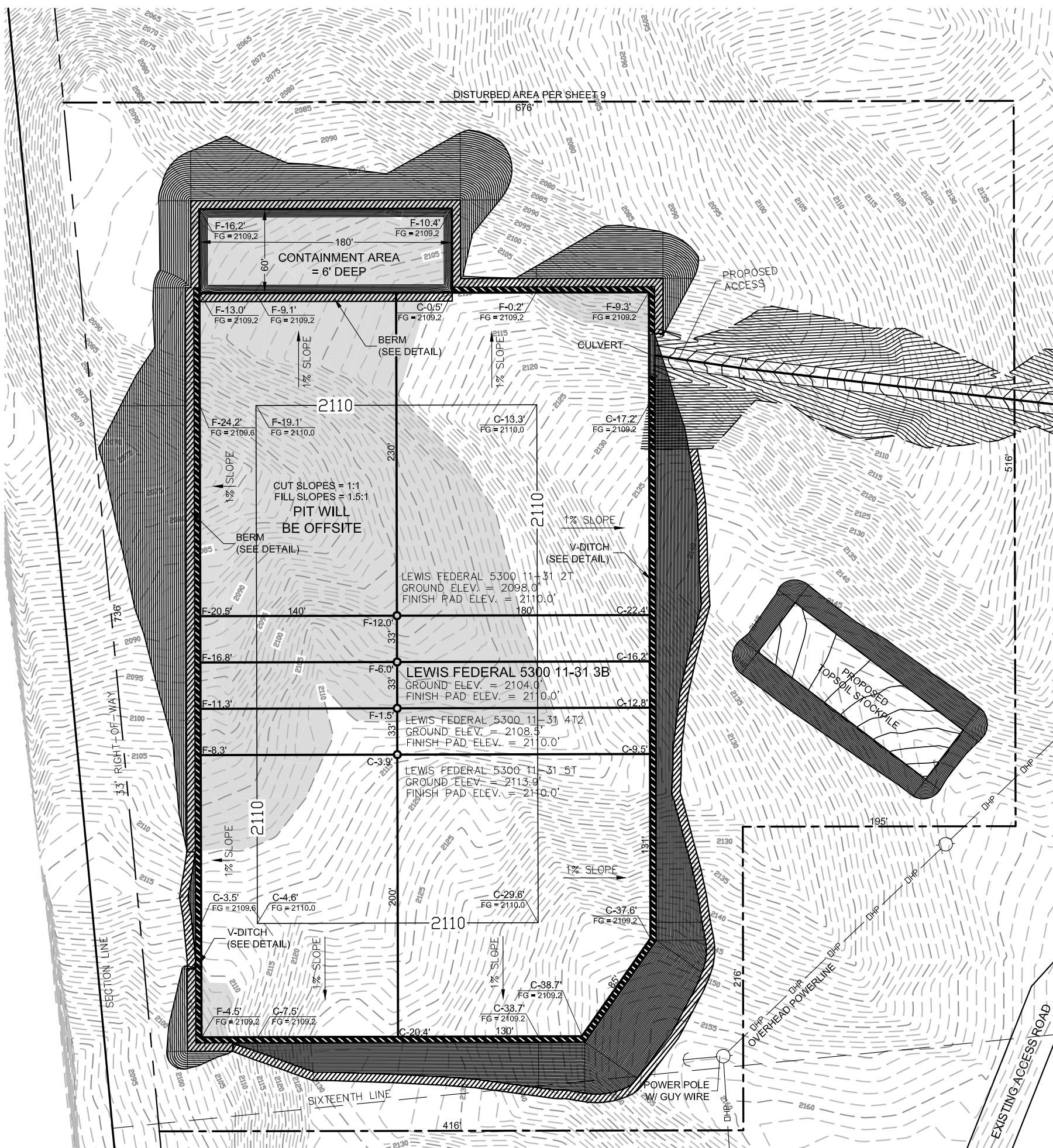
OASIS PETROLEUM NORTH AMERICA, LLC	
SECTION BREAKDOWN	
SECTIONS 31 & 32, T153N, R100W	
MCKENZIE COUNTY, NORTH DAKOTA	
Drawn By: J.D.K.	Project No.: 131202378.02
Checked By: J.D.K.	Date: JAN 2014

Interstate Engineering, Inc.
 P.O. Box 648
 435 S. Broadway
 Sierra, Montana 59270
 Ph: (406) 433-5617
 Fax: (406) 433-5618
 www.interstateeng.com
Other than in Montana, this document is not valid.



2/8
 SHEET NO.

PAD LAYOUT
OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"LEWIS FEDERAL 5300 11-31 3B"
1005 FEET FROM NORTH LINE AND 233 FEET FROM WEST LINE
SECTION 31, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

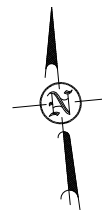
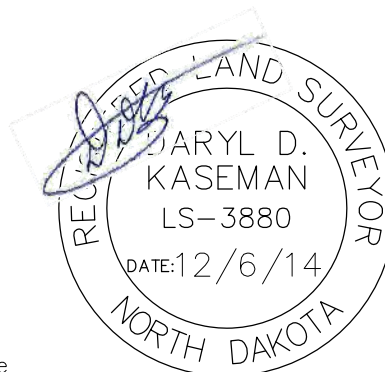


-  — BERM
 — DITCH
 Proposed Contours
 Original Contours

NOTE: Pad dimensions shown are to usable area, the v-ditch and berm areas shall be built to the outside of the pad dimensions.

NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.

© 2014, INTERSTATE ENGINEERING, INC.



3/9

SHEET NO.



Professionals you need, people you trust

Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617
Fax (406) 433-5618
www.interstateeng.com

Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
PAD LAYOUT
SECTION 31, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

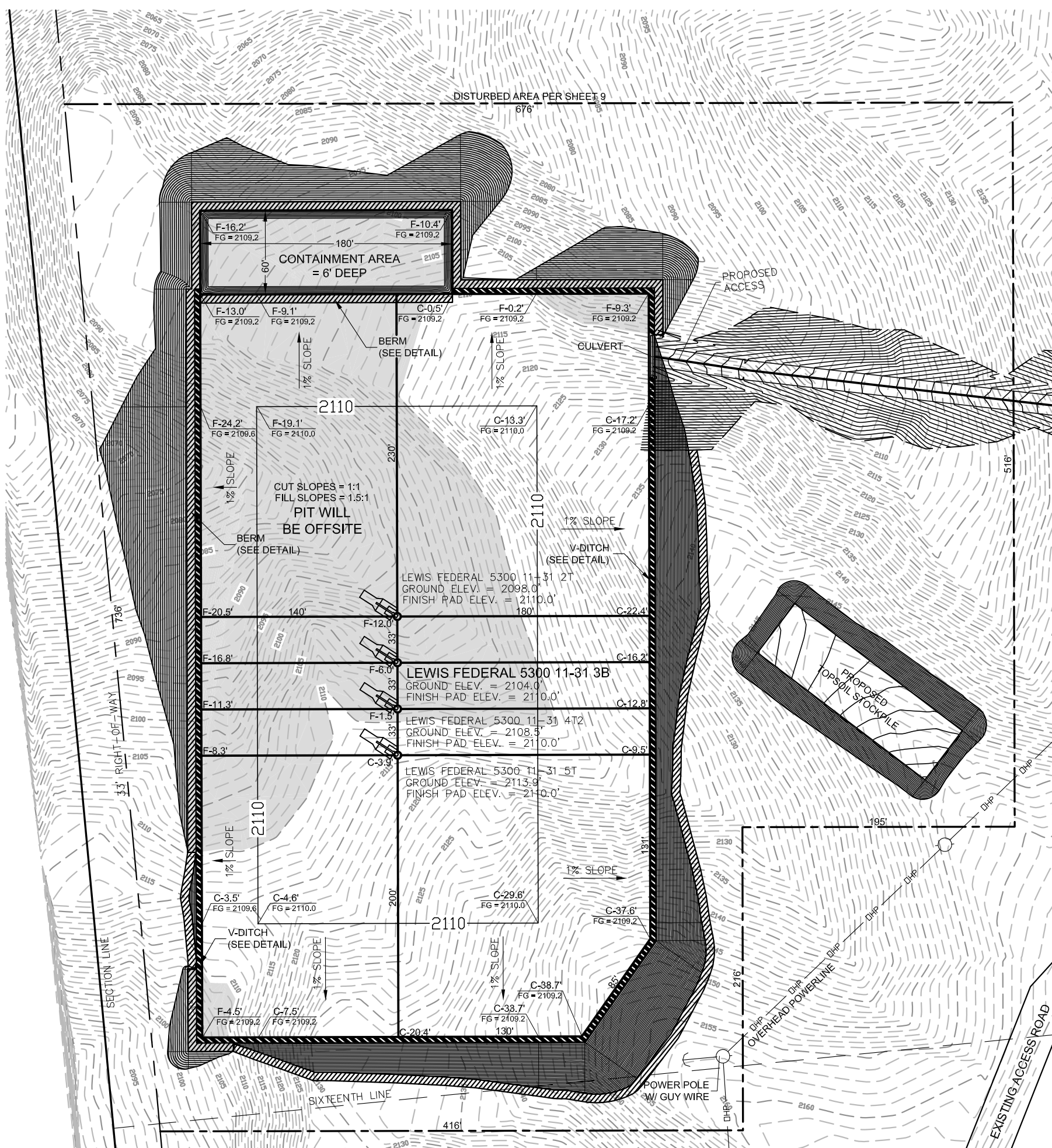
Drawn By: <u>B.H.H.</u>	Project No.: <u>S13-09-378.02</u>
Checked By: <u>D.D.K.</u>	Date: <u>JAN, 2014</u>

Revision No.	Date	By	Description
REV 1	12/3/14	BHH	REMOVED PIT FROM PAD

Quad Pad\CAD\Lewis Federal 5300 11-31_3B Revised 12-3-14.dwg - 12/6/2014 9:16

Quad Pad\CAD\Lewis Federal 5300 11-31 3B Revised 12-3-14.dwg - 12/6/2014 9:16

PRODUCTION LAYOUT
OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"LEWIS FEDERAL 5300 11-31 3B"
1005 FEET FROM NORTH LINE AND 233 FEET FROM WEST LINE
SECTION 31, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

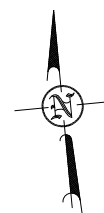
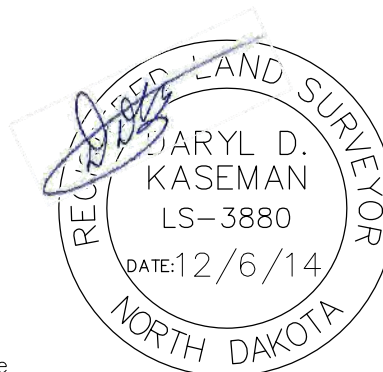


- 
-  — BERM
 — DITCH
 — Proposed Contours
 — Original Contours

NOTE: Pad dimensions shown are to usable area, the v-ditch and berm areas shall be built to the outside of the pad dimensions.

NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.

© 2014, INTERSTATE ENGINEERING, INC.



4/9

SHEET NO.



Professionals you need, people you trust

Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617
Fax (406) 433-5618
www.interstateeng.com

Other offices in Minnesota, North Dakota and South Dakota

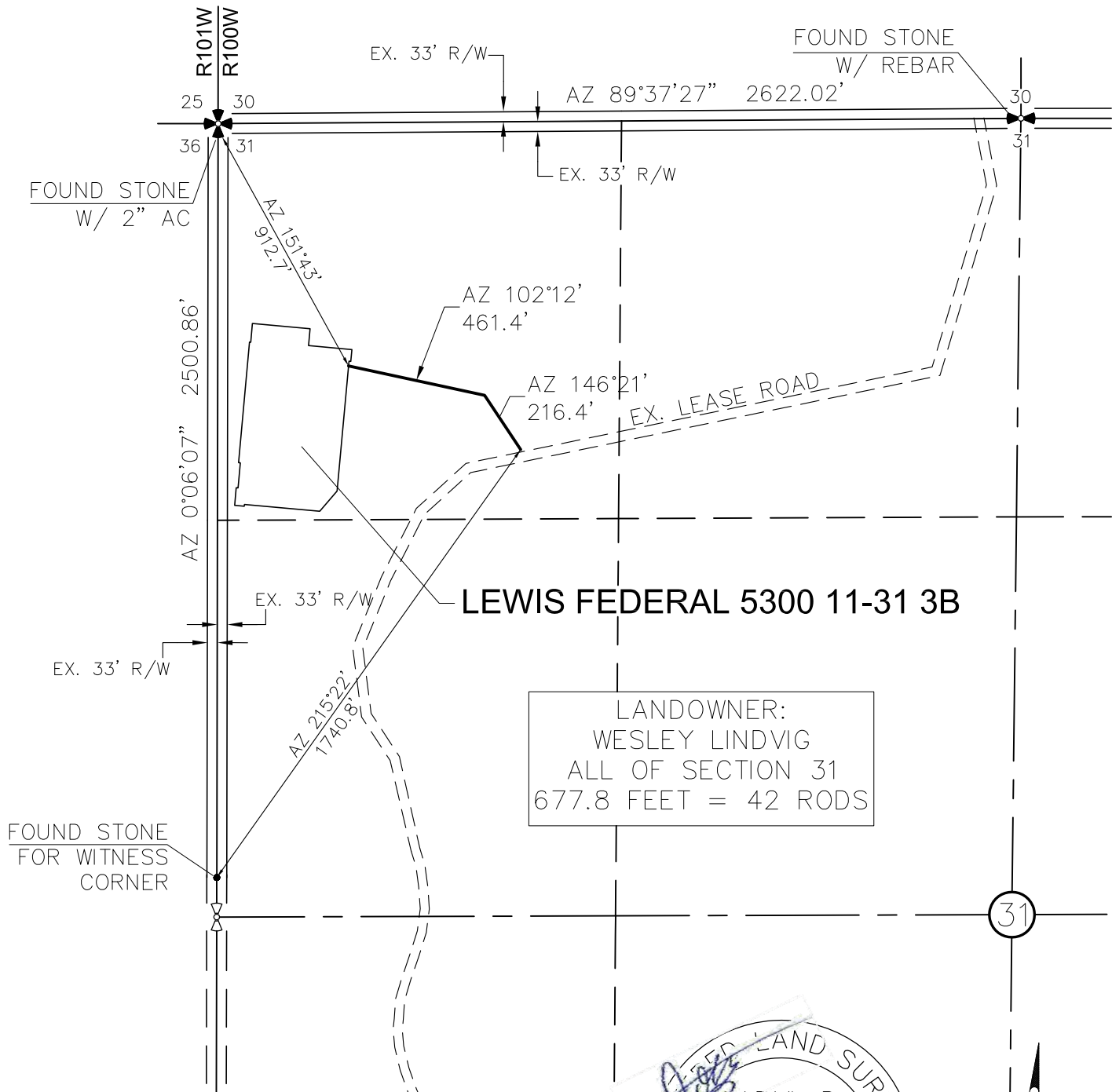
OASIS PETROLEUM NORTH AMERICA, LLC
PRODUCTION LAYOUT
SECTION 31, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: <u>B.H.H.</u>	Project No.: <u>S13-09-378.02</u>
Checked By: <u>D.D.K.</u>	Date: <u>JAN, 2014</u>

Revision No.	Date	By	Description
REV 1	12/3/14	BHH	REMOVED PIT FROM PAD

Quad Pad\CAD\Lewis Federal 5300 11-31_3B Revised 12-3-14.dwg - 12/6/2014 9:16

ACCESS APPROACH
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "LEWIS FEDERAL 5300 11-31 3B"
 1005 FEET FROM NORTH LINE AND 233 FEET FROM WEST LINE
 SECTION 31, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



THIS DOCUMENT WAS ORIGINALLY ISSUED AND SEALED BY DARYL D. KASEMAN, PLS, REGISTRATION NUMBER 3880 ON 12/6/14 AND THE ORIGINAL DOCUMENTS ARE STORED AT THE OFFICES OF INTERSTATE ENGINEERING, INC.

NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.

© 2014, INTERSTATE ENGINEERING, INC.

5/9

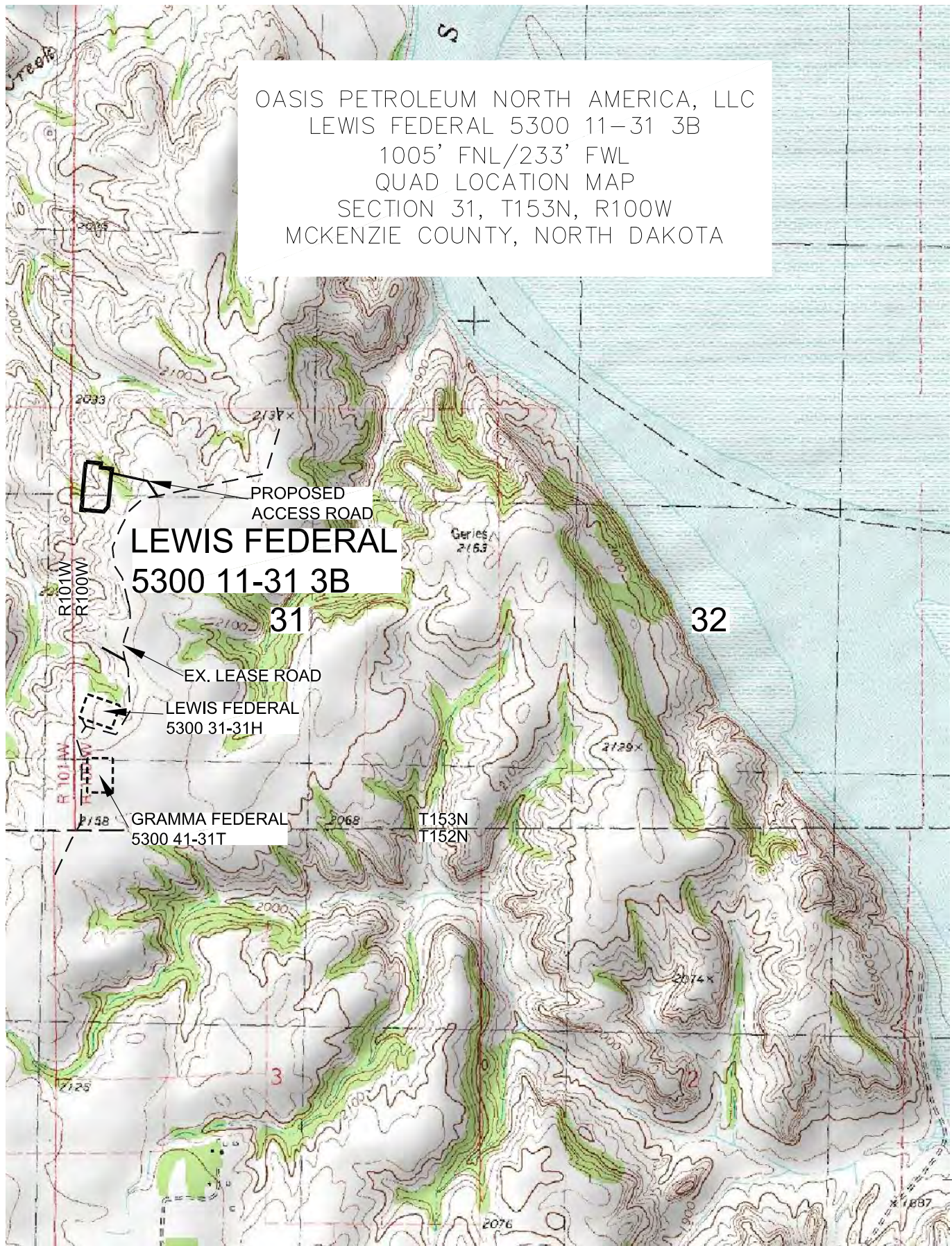


Interstate Engineering, Inc.
 P.O. Box 648
 425 East Main Street
 Sidney, Montana 59270
 Ph (406) 433-5617
 Fax (406) 433-5618
 www.interstateeng.com
 Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
 ACCESS APPROACH
 SECTION 31, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA
 Drawn By: B.H.H. Project No.: S13-09-378.02
 Checked By: D.D.K. Date: JAN, 2014

Revision No.	Date	By	Description
REV 1	12/3/14	BHH	REMOVED PIT FROM PAD

OASIS PETROLEUM NORTH AMERICA, LLC
 LEWIS FEDERAL 5300 11-31 3B
 1005' FNL/233' FWL
 QUAD LOCATION MAP
 SECTION 31, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA



© 2014, INTERSTATE ENGINEERING, INC.

6/9

SHEET NO.



Professionals you need, people you trust

Interstate Engineering, Inc.
 P.O. Box 648
 425 East Main Street
 Sidney, Montana 59270
 Ph (406) 433-5617
 Fax (406) 433-5618
 www.interstateeng.com

Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
 QUAD LOCATION MAP
 SECTION 31, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S13-09-378.02
 Checked By: D.D.K. Date: JAN, 2014

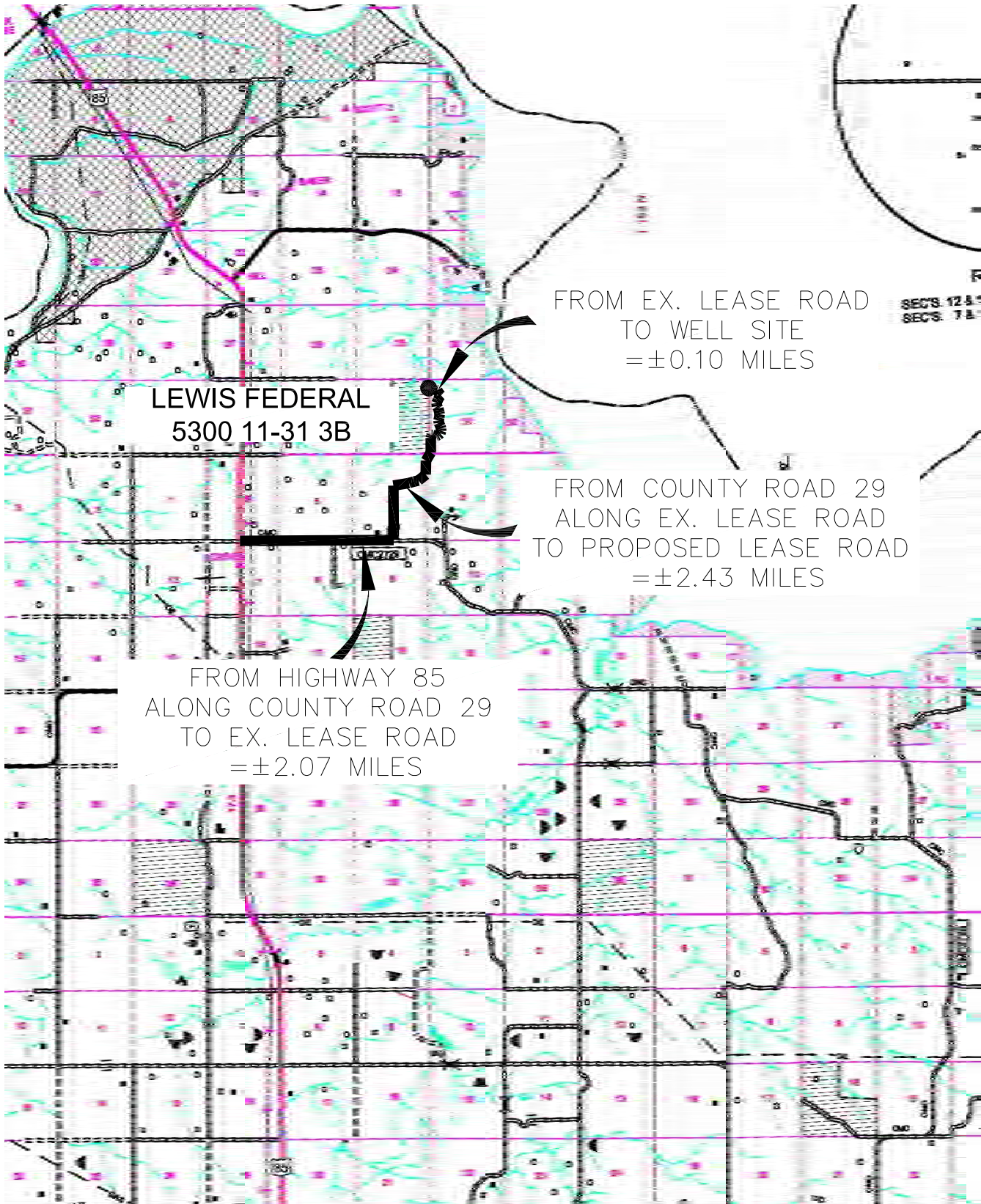
Revision No.	Date	By	Description
REV 1	12/3/14	BHH	REMOVED PIT FROM PAD

COUNTY ROAD MAP

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002

"LEWIS FEDERAL 5300 11-31 3B"

1005 FEET FROM NORTH LINE AND 233 FEET FROM WEST LINE
SECTION 31, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



© 2014, INTERSTATE ENGINEERING, INC.

SCALE: 1" = 2 MILE

7/9

SHEET NO.



Professionals you need, people you trust

Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617
Fax (406) 433-5618
www.interstateeng.com

Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
COUNTY ROAD MAP
SECTION 31, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.J.H. Project No.: S13-09-378.02
Checked By: D.D.K. Date: JAN, 2014

Revision No.	Date	By	Description
REV 1	12/3/14	BHH	REMOVED PIT FROM PAD

Oasis

Indian Hills

153N-100W-31/32

Lewis Federal 5300 11-31 3B

Lewis Federal 5300 11-31 3B

Plan #1

Anticollision Report

11 March, 2014

Oasis Petroleum

Anticollision Report

Company:	Oasis	Local Co-ordinate Reference:	Well Lewis Federal 5300 11-31 3B
Project:	Indian Hills	TVD Reference:	WELL @ 2134.0usft (Original Well Elev)
Reference Site:	153N-100W-31/32	MD Reference:	WELL @ 2134.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Lewis Federal 5300 11-31 3B	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Lewis Federal 5300 11-31 3B	Database:	OpenWellsCompass - EDM Prod
Reference Design:	Plan #1	Offset TVD Reference:	Offset Datum

Reference	Plan #1		
Filter type:	NO GLOBAL FILTER: Using user defined selection & filtering criteria		
Interpolation Method:	Stations	Error Model:	ISCWSA
Depth Range:	Unlimited	Scan Method:	Closest Approach 3D
Results Limited by:	Maximum center-center distance of 10,000.0 usft	Error Surface:	Elliptical Conic
Warning Levels Evaluated at:	2.00 Sigma	Casing Method:	Not applied

Survey Tool Program		Date	3/11/2014		
From (usft)	To (usft)	Survey (Wellbore)	Tool Name	Description	
0.0	20,640.4	Plan #1 (Lewis Federal 5300 11-31 3B)	MWD	MWD - Standard	

Summary							
Site Name	Reference Measured Depth (usft)	Offset Measured Depth (usft)	Distance		Separation Factor	Warning	
			Between Centres (usft)	Between Ellipses (usft)			
Offset Well - Wellbore - Design							
153N-100W-31/32							
Lewis Federal 5300 11-31 2T - Lewis Federal 5300 11-31	2,100.0	2,100.0	32.5	23.4	3.550	CC	
Lewis Federal 5300 11-31 2T - Lewis Federal 5300 11-31	20,640.9	20,803.1	454.6	-123.9	0.786	Level 1, ES, SF	
Lewis Federal 5300 11-31 4T2 - Lewis Federal 5300 11-3	2,100.0	2,100.0	33.6	24.4	3.667	CC	
Lewis Federal 5300 11-31 4T2 - Lewis Federal 5300 11-3	20,640.9	20,756.3	118.5	-31.6	0.789	Level 1, ES, SF	
Lewis Federal 5300 11-31 5T - Lewis Federal 5300 11-31	2,100.0	2,100.0	66.1	57.0	7.216	CC	
Lewis Federal 5300 11-31 5T - Lewis Federal 5300 11-31	20,640.9	20,812.3	502.4	-80.6	0.862	Level 1, ES, SF	

Offset Design		153N-100W-31/32 - Lewis Federal 5300 11-31 2T - Lewis Federal 5300 11-31 2T - Plan #1											Offset Site Error:	0.0 usft
Survey Program:		0-MWD											Offset Well Error:	0.0 usft
Reference		Offset		Semi Major Axis			Distance							
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning	
0.0	0.0	0.0	0.0	0.0	0.0	4.79	32.4	2.7	32.5					
100.0	100.0	100.0	100.0	0.1	0.1	4.79	32.4	2.7	32.5	32.4	0.18	185.596		
200.0	200.0	200.0	200.0	0.3	0.3	4.79	32.4	2.7	32.5	31.9	0.62	52.074		
300.0	300.0	300.0	300.0	0.5	0.5	4.79	32.4	2.7	32.5	31.5	1.07	30.286		
400.0	400.0	400.0	400.0	0.8	0.8	4.79	32.4	2.7	32.5	31.0	1.52	21.352		
500.0	500.0	500.0	500.0	1.0	1.0	4.79	32.4	2.7	32.5	30.6	1.97	16.488		
600.0	600.0	600.0	600.0	1.2	1.2	4.79	32.4	2.7	32.5	30.1	2.42	13.429		
700.0	700.0	700.0	700.0	1.4	1.4	4.79	32.4	2.7	32.5	29.7	2.87	11.327		
800.0	800.0	800.0	800.0	1.7	1.7	4.79	32.4	2.7	32.5	29.2	3.32	9.795		
900.0	900.0	900.0	900.0	1.9	1.9	4.79	32.4	2.7	32.5	28.8	3.77	8.627		
1,000.0	1,000.0	1,000.0	1,000.0	2.1	2.1	4.79	32.4	2.7	32.5	28.3	4.22	7.708		
1,100.0	1,100.0	1,100.0	1,100.0	2.3	2.3	4.79	32.4	2.7	32.5	27.9	4.67	6.967		
1,200.0	1,200.0	1,200.0	1,200.0	2.6	2.6	4.79	32.4	2.7	32.5	27.4	5.12	6.355		
1,300.0	1,300.0	1,300.0	1,300.0	2.8	2.8	4.79	32.4	2.7	32.5	27.0	5.57	5.842		
1,400.0	1,400.0	1,400.0	1,400.0	3.0	3.0	4.79	32.4	2.7	32.5	26.5	6.02	5.406		
1,500.0	1,500.0	1,500.0	1,500.0	3.2	3.2	4.79	32.4	2.7	32.5	26.1	6.47	5.030		
1,600.0	1,600.0	1,600.0	1,600.0	3.5	3.5	4.79	32.4	2.7	32.5	25.6	6.92	4.703		
1,700.0	1,700.0	1,700.0	1,700.0	3.7	3.7	4.79	32.4	2.7	32.5	25.2	7.37	4.416		
1,800.0	1,800.0	1,800.0	1,800.0	3.9	3.9	4.79	32.4	2.7	32.5	24.7	7.82	4.162		
1,900.0	1,900.0	1,900.0	1,900.0	4.1	4.1	4.79	32.4	2.7	32.5	24.3	8.27	3.936		
2,000.0	2,000.0	2,000.0	2,000.0	4.4	4.4	4.79	32.4	2.7	32.5	23.8	8.72	3.733		
2,100.0	2,100.0	2,100.0	2,100.0	4.6	4.6	4.79	32.4	2.7	32.5	23.4	9.17	3.550	CC	

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Oasis Petroleum

Anticollision Report

Company:	Oasis	Local Co-ordinate Reference:	Well Lewis Federal 5300 11-31 3B
Project:	Indian Hills	TVD Reference:	WELL @ 2134.0usft (Original Well Elev)
Reference Site:	153N-100W-31/32	MD Reference:	WELL @ 2134.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Lewis Federal 5300 11-31 3B	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Lewis Federal 5300 11-31 3B	Database:	OpenWellsCompass - EDM Prod
Reference Design:	Plan #1	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-31/32 - Lewis Federal 5300 11-31 2T - Lewis Federal 5300 11-31 2T - Plan #1													Offset Site Error:	0.0 usft
Survey Program: 0-MWD													Offset Well Error:	0.0 usft
Reference		Offset		Semi Major Axis		Distance								Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor		
2,110.0	2,110.0	2,109.7	2,109.7	4.6	4.6	-40.26	32.5	2.7	32.5	23.3	9.21	3.534		
2,200.0	2,200.0	2,199.7	2,199.7	4.8	4.8	-41.26	33.3	2.7	32.7	23.1	9.61	3.407		
2,300.0	2,300.0	2,299.7	2,299.7	5.0	5.0	-42.36	34.1	2.7	33.0	22.9	10.05	3.279		
2,400.0	2,400.0	2,399.7	2,399.7	5.2	5.3	-43.45	35.0	2.7	33.2	22.7	10.49	3.163		
2,500.0	2,500.0	2,499.7	2,499.7	5.5	5.5	-44.52	35.9	2.7	33.4	22.5	10.94	3.057		
2,600.0	2,600.0	2,599.7	2,599.7	5.7	5.7	-45.57	36.7	2.7	33.7	22.3	11.38	2.960		
2,700.0	2,700.0	2,699.7	2,699.7	5.9	5.9	-46.61	37.6	2.7	34.0	22.1	11.82	2.872		
2,800.0	2,800.0	2,799.7	2,799.7	6.1	6.2	-47.63	38.5	2.7	34.2	22.0	12.27	2.790		
2,900.0	2,900.0	2,899.7	2,899.7	6.3	6.4	-48.63	39.4	2.7	34.5	21.8	12.71	2.715		
3,000.0	3,000.0	2,999.7	2,999.7	6.6	6.6	-49.62	40.2	2.7	34.8	21.7	13.16	2.646		
3,100.0	3,100.0	3,099.7	3,099.7	6.8	6.8	-50.59	41.1	2.7	35.1	21.5	13.61	2.582		
3,200.0	3,200.0	3,199.7	3,199.7	7.0	7.1	-51.54	42.0	2.7	35.5	21.4	14.05	2.523		
3,300.0	3,300.0	3,299.7	3,299.6	7.2	7.3	-52.47	42.9	2.7	35.8	21.3	14.50	2.468		
3,400.0	3,400.0	3,399.7	3,399.6	7.4	7.5	-53.39	43.7	2.7	36.1	21.2	14.94	2.417		
3,500.0	3,499.9	3,499.7	3,499.6	7.7	7.7	-54.29	44.6	2.7	36.5	21.1	15.39	2.369		
3,600.0	3,599.9	3,599.7	3,599.6	7.9	8.0	-55.18	45.5	2.7	36.8	21.0	15.84	2.325		
3,700.0	3,699.9	3,699.7	3,699.6	8.1	8.2	-56.04	46.3	2.7	37.2	20.9	16.28	2.284		
3,800.0	3,799.9	3,799.7	3,799.6	8.3	8.4	-56.89	47.2	2.7	37.6	20.8	16.73	2.245		
3,900.0	3,899.9	3,899.7	3,899.6	8.6	8.6	-57.72	48.1	2.7	37.9	20.8	17.18	2.209		
4,000.0	3,999.9	3,999.7	3,999.6	8.8	8.9	-58.54	49.0	2.7	38.3	20.7	17.63	2.175		
4,100.0	4,099.9	4,099.7	4,099.6	9.0	9.1	-59.34	49.8	2.7	38.7	20.7	18.07	2.143		
4,200.0	4,199.9	4,199.7	4,199.6	9.2	9.3	-60.12	50.7	2.7	39.1	20.6	18.52	2.113		
4,300.0	4,299.9	4,299.7	4,299.6	9.4	9.5	-60.89	51.6	2.7	39.5	20.6	18.97	2.085		
4,400.0	4,399.9	4,399.7	4,399.6	9.7	9.8	-61.64	52.4	2.7	40.0	20.5	19.42	2.058		
4,500.0	4,499.9	4,499.7	4,499.6	9.9	10.0	-62.37	53.3	2.7	40.4	20.5	19.86	2.033		
4,600.0	4,599.9	4,599.7	4,599.6	10.1	10.2	-63.09	54.2	2.7	40.8	20.5	20.31	2.010		
4,700.0	4,699.9	4,699.7	4,699.6	10.3	10.4	-63.80	55.1	2.7	41.3	20.5	20.76	1.987		
4,800.0	4,799.9	4,799.7	4,799.6	10.6	10.7	-64.49	55.9	2.7	41.7	20.5	21.21	1.966		
4,900.0	4,899.9	4,899.7	4,899.5	10.8	10.9	-65.16	56.8	2.7	42.1	20.5	21.65	1.946		
5,000.0	4,999.9	4,999.7	4,999.5	11.0	11.1	-65.82	57.7	2.7	42.6	20.5	22.10	1.928		
5,100.0	5,099.9	5,099.7	5,099.5	11.2	11.3	-66.47	58.6	2.7	43.1	20.5	22.55	1.910		
5,200.0	5,199.9	5,199.6	5,199.5	11.5	11.6	-67.10	59.4	2.7	43.5	20.5	23.00	1.893		
5,300.0	5,299.9	5,299.6	5,299.5	11.7	11.8	-67.72	60.3	2.7	44.0	20.6	23.45	1.877		
5,400.0	5,399.9	5,399.6	5,399.5	11.9	12.0	-68.33	61.2	2.7	44.5	20.6	23.89	1.862		
5,500.0	5,499.9	5,499.6	5,499.5	12.1	12.2	-68.92	62.0	2.7	45.0	20.6	24.34	1.847		
5,600.0	5,599.9	5,599.6	5,599.5	12.3	12.5	-69.50	62.9	2.7	45.4	20.7	24.79	1.833		
5,700.0	5,699.9	5,699.6	5,699.5	12.6	12.7	-70.07	63.8	2.7	45.9	20.7	25.24	1.820		
5,800.0	5,799.9	5,799.6	5,799.5	12.8	12.9	-70.62	64.7	2.7	46.4	20.7	25.69	1.808		
5,900.0	5,899.9	5,899.6	5,899.5	13.0	13.1	-71.17	65.5	2.7	46.9	20.8	26.14	1.796		
6,000.0	5,999.9	5,999.6	5,999.5	13.2	13.4	-71.70	66.4	2.7	47.4	20.9	26.58	1.784		
6,100.0	6,099.8	6,099.6	6,099.5	13.5	13.6	-72.22	67.3	2.7	47.9	20.9	27.03	1.774		
6,151.5	6,151.3	6,151.1	6,150.9	13.6	13.7	-72.49	67.7	2.7	48.2	20.9	27.26	1.768		
6,161.5	6,161.3	6,161.1	6,160.9	13.6	13.7	-72.49	67.8	2.7	48.3	21.0	27.29	1.769		
6,200.0	6,199.8	6,199.6	6,199.5	13.7	13.8	-72.31	68.2	2.7	48.6	21.1	27.45	1.769		
6,300.0	6,299.8	6,299.6	6,299.5	13.9	14.0	-72.84	69.0	2.7	49.3	21.5	27.90	1.769		
6,400.0	6,399.8	6,399.6	6,399.5	14.1	14.2	-73.39	69.9	2.7	50.1	21.8	28.34	1.769		
6,500.0	6,499.8	6,499.6	6,499.4	14.3	14.5	-73.96	70.8	2.7	50.9	22.1	28.79	1.769		
6,600.0	6,599.8	6,599.6	6,599.4	14.6	14.7	-74.53	71.6	2.7	51.7	22.5	29.23	1.768		
6,700.0	6,699.8	6,699.6	6,699.4	14.8	14.9	-75.12	72.5	2.7	52.5	22.8	29.68	1.768		
6,800.0	6,799.8	6,799.6	6,799.4	15.0	15.1	-75.72	73.4	2.7	53.3	23.2	30.12	1.769		
6,900.0	6,899.8	6,899.6	6,899.4	15.2	15.4	-76.34	74.3	2.7	54.1	23.5	30.57	1.769		
7,000.0	6,999.8	6,999.6	6,999.4	15.5	15.6	-76.96	75.1	2.7	54.9	23.9	31.02	1.769		
7,100.0	7,099.8	7,099.6	7,099.4	15.7	15.8	-77.60	76.0	2.7	55.7	24.2	31.46	1.769		

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Oasis Petroleum

Anticollision Report

Company:	Oasis	Local Co-ordinate Reference:	Well Lewis Federal 5300 11-31 3B
Project:	Indian Hills	TVD Reference:	WELL @ 2134.0usft (Original Well Elev)
Reference Site:	153N-100W-31/32	MD Reference:	WELL @ 2134.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Lewis Federal 5300 11-31 3B	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Lewis Federal 5300 11-31 3B	Database:	OpenWellsCompass - EDM Prod
Reference Design:	Plan #1	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-31/32 - Lewis Federal 5300 11-31 2T - Lewis Federal 5300 11-31 2T - Plan #1													Offset Site Error:	0.0 usft
Survey Program: 0-MWD													Offset Well Error:	0.0 usft
Reference		Offset		Semi Major Axis		Distance								Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor		
7,200.0	7,199.8	7,199.6	7,199.4	15.9	16.0	-23.24	76.9	2.7	56.5	24.6	31.91	1.770		
7,300.0	7,299.8	7,299.6	7,299.4	16.1	16.3	-22.90	77.8	2.7	57.3	24.9	32.35	1.770		
7,400.0	7,399.8	7,399.6	7,399.4	16.3	16.5	-22.56	78.6	2.7	58.1	25.3	32.80	1.770		
7,500.0	7,499.8	7,499.6	7,499.4	16.6	16.7	-22.24	79.5	2.7	58.9	25.6	33.25	1.771		
7,600.0	7,599.8	7,599.6	7,599.4	16.8	16.9	-21.92	80.4	2.7	59.7	26.0	33.69	1.772		
7,700.0	7,699.8	7,699.6	7,699.4	17.0	17.2	-21.61	81.2	2.7	60.5	26.4	34.14	1.772		
7,800.0	7,799.8	7,799.6	7,799.3	17.2	17.4	-21.31	82.1	2.7	61.3	26.7	34.59	1.773		
7,900.0	7,899.8	7,900.1	7,899.8	17.5	17.6	-21.21	82.4	2.7	61.6	26.6	35.02	1.759		
8,000.0	7,999.8	8,000.1	7,999.8	17.7	17.8	-21.21	82.4	2.7	61.6	26.1	35.46	1.737		
8,100.0	8,099.8	8,100.1	8,099.8	17.9	18.1	-21.21	82.4	2.7	61.6	25.7	35.91	1.715		
8,200.0	8,199.8	8,200.1	8,199.8	18.1	18.3	-21.21	82.4	2.7	61.6	25.2	36.36	1.694		
8,300.0	8,299.8	8,300.1	8,299.8	18.3	18.5	-21.21	82.4	2.7	61.6	24.8	36.80	1.674		
8,400.0	8,399.8	8,400.1	8,399.8	18.6	18.7	-21.21	82.4	2.7	61.6	24.3	37.25	1.654		
8,500.0	8,499.8	8,500.1	8,499.8	18.8	19.0	-21.21	82.4	2.7	61.6	23.9	37.70	1.634		
8,600.0	8,599.8	8,600.1	8,599.8	19.0	19.2	-21.21	82.4	2.7	61.6	23.5	38.14	1.615		
8,700.0	8,699.8	8,700.1	8,699.8	19.2	19.4	-21.21	82.4	2.7	61.6	23.0	38.59	1.596		
8,800.0	8,799.8	8,800.1	8,799.8	19.5	19.6	-21.21	82.4	2.7	61.6	22.6	39.04	1.578		
8,900.0	8,899.8	8,900.1	8,899.8	19.7	19.8	-21.21	82.4	2.7	61.6	22.1	39.49	1.560		
9,000.0	8,999.8	9,000.1	8,999.8	19.9	20.1	-21.21	82.4	2.7	61.6	21.7	39.93	1.542		
9,100.0	9,099.8	9,100.1	9,099.8	20.1	20.3	-21.21	82.4	2.7	61.6	21.2	40.38	1.525		
9,200.0	9,199.8	9,200.1	9,199.8	20.3	20.5	-21.21	82.4	2.7	61.6	20.8	40.83	1.509		
9,300.0	9,299.8	9,300.1	9,299.8	20.6	20.7	-21.21	82.4	2.7	61.6	20.3	41.28	1.492 Level 3	Level 3	
9,400.0	9,399.8	9,400.1	9,399.8	20.8	21.0	-21.21	82.4	2.7	61.6	19.9	41.72	1.476 Level 3	Level 3	
9,500.0	9,499.8	9,500.1	9,499.8	21.0	21.2	-21.21	82.4	2.7	61.6	19.4	42.17	1.461 Level 3	Level 3	
9,600.0	9,599.8	9,600.1	9,599.8	21.2	21.4	-21.21	82.4	2.7	61.6	19.0	42.62	1.445 Level 3	Level 3	
9,700.0	9,699.8	9,700.1	9,699.8	21.5	21.6	-21.21	82.4	2.7	61.6	18.5	43.07	1.430 Level 3	Level 3	
9,800.0	9,799.8	9,800.1	9,799.8	21.7	21.9	-21.21	82.4	2.7	61.6	18.1	43.51	1.416 Level 3	Level 3	
9,900.0	9,899.8	9,900.1	9,899.8	21.9	22.1	-21.21	82.4	2.7	61.6	17.6	43.96	1.401 Level 3	Level 3	
10,000.0	9,999.8	10,000.1	9,999.8	22.1	22.3	-21.21	82.4	2.7	61.6	17.2	44.41	1.387 Level 3	Level 3	
10,100.0	10,099.8	10,100.1	10,099.8	22.4	22.5	-21.21	82.4	2.7	61.6	16.7	44.86	1.373 Level 3	Level 3	
10,200.0	10,199.8	10,200.1	10,199.8	22.6	22.8	-21.21	82.4	2.7	61.6	16.3	45.31	1.360 Level 3	Level 3	
10,282.7	10,282.5	10,282.7	10,282.5	22.8	23.0	-21.21	82.4	2.7	61.6	15.9	45.68	1.349 Level 3	Level 3	
10,300.0	10,299.8	10,300.1	10,299.8	22.8	23.0	-113.44	82.4	2.7	61.7	15.9	45.78	1.348 Level 3	Level 3	
10,325.0	10,324.8	10,325.0	10,324.8	22.9	23.1	-114.69	82.4	2.7	62.4	16.5	45.87	1.359 Level 3	Level 3	
10,350.0	10,349.6	10,349.8	10,349.6	22.9	23.1	-116.88	82.4	2.7	63.6	17.7	45.94	1.385 Level 3	Level 3	
10,375.0	10,374.3	10,374.7	10,374.4	23.0	23.2	-119.58	82.6	3.1	65.6	19.6	45.98	1.426 Level 3	Level 3	
10,400.0	10,398.7	10,399.6	10,399.4	23.0	23.2	-121.94	83.3	4.6	68.3	22.3	45.99	1.485 Level 3	Level 3	
10,425.0	10,422.7	10,424.7	10,424.3	23.1	23.3	-123.88	84.5	7.3	71.6	25.7	45.97	1.559		
10,450.0	10,446.4	10,449.9	10,449.1	23.1	23.3	-125.40	86.3	11.2	75.6	29.7	45.91	1.647		
10,475.0	10,469.7	10,475.2	10,473.8	23.2	23.4	-126.51	88.6	16.3	80.2	34.4	45.83	1.750		
10,500.0	10,492.4	10,500.6	10,498.2	23.3	23.4	-127.25	91.4	22.6	85.3	39.6	45.73	1.865		
10,525.0	10,514.6	10,526.0	10,522.2	23.3	23.5	-127.65	94.8	30.1	90.9	45.3	45.62	1.992		
10,550.0	10,536.1	10,551.5	10,545.8	23.4	23.6	-127.75	98.8	38.7	97.0	51.5	45.52	2.131		
10,575.0	10,556.9	10,576.9	10,568.9	23.5	23.6	-127.59	103.2	48.6	103.5	58.1	45.42	2.280		
10,600.0	10,577.0	10,602.4	10,591.4	23.6	23.7	-127.21	108.2	59.5	110.5	65.2	45.35	2.438		
10,625.0	10,596.3	10,627.9	10,613.2	23.7	23.8	-126.63	113.6	71.5	118.0	72.7	45.31	2.604		
10,650.0	10,614.7	10,653.4	10,634.3	23.8	23.8	-125.89	119.5	84.5	125.8	80.5	45.30	2.777		
10,675.0	10,632.2	10,678.8	10,654.5	23.9	23.9	-125.02	125.9	98.5	134.0	88.7	45.34	2.955		
10,700.0	10,648.7	10,704.3	10,673.9	24.1	24.0	-124.03	132.7	113.5	142.6	97.1	45.44	3.138		
10,725.0	10,664.2	10,729.7	10,692.4	24.2	24.1	-122.95	139.9	129.4	151.5	105.9	45.59	3.324		
10,750.0	10,678.7	10,755.1	10,709.9	24.4	24.2	-121.78	147.5	146.1	160.8	115.0	45.80	3.511		
10,775.0	10,692.1	10,780.4	10,726.4	24.5	24.4	-120.55	155.4	163.7	170.4	124.3	46.07	3.699		
10,800.0	10,704.4	10,805.8	10,741.8	24.7	24.5	-119.26	163.7	182.0	180.3	133.9	46.41	3.885		

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Oasis Petroleum

Anticollision Report

Company:	Oasis	Local Co-ordinate Reference:	Well Lewis Federal 5300 11-31 3B
Project:	Indian Hills	TVD Reference:	WELL @ 2134.0usft (Original Well Elev)
Reference Site:	153N-100W-31/32	MD Reference:	WELL @ 2134.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Lewis Federal 5300 11-31 3B	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Lewis Federal 5300 11-31 3B	Database:	OpenWellsCompass - EDM Prod
Reference Design:	Plan #1	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-31/32 - Lewis Federal 5300 11-31 2T - Lewis Federal 5300 11-31 2T - Plan #1												Offset Site Error:	0.0 usft
Survey Program: 0-MWD												Offset Well Error:	0.0 usft
Reference		Offset		Semi Major Axis			Distance						Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	Offset Wellbore Centre +E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	
10,825.0	10,715.5	10,831.1	10,756.2	24.9	24.6	-117.93	172.3	201.0	190.5	143.7	46.81	4.069	
10,850.0	10,725.5	10,856.5	10,769.4	25.2	24.8	-116.56	181.3	220.7	200.9	153.6	47.27	4.250	
10,875.0	10,734.2	10,881.9	10,781.5	25.4	25.0	-115.15	190.5	241.1	211.5	163.8	47.78	4.428	
10,900.0	10,741.7	10,907.3	10,792.3	25.7	25.2	-113.73	200.0	262.0	222.4	174.1	48.34	4.601	
10,925.0	10,747.9	10,932.7	10,802.0	25.9	25.4	-112.28	209.7	283.4	233.4	184.5	48.95	4.769	
10,950.0	10,752.9	10,958.2	10,810.4	26.2	25.6	-110.82	219.6	305.3	244.6	195.0	49.59	4.932	
10,975.0	10,756.5	10,983.8	10,817.5	26.5	25.8	-109.35	229.8	327.7	255.9	205.6	50.28	5.090	
11,000.0	10,758.9	11,009.6	10,823.4	26.8	26.1	-107.88	240.1	350.6	267.3	216.3	50.98	5.243	
11,025.0	10,760.0	11,035.4	10,827.8	27.2	26.4	-106.41	250.7	373.8	278.7	227.0	51.71	5.390	
11,028.6	10,760.0	11,039.1	10,828.4	27.2	26.4	-106.20	252.2	377.1	280.4	228.6	51.82	5.411	
11,043.6	10,760.1	11,054.8	10,830.3	27.4	26.6	-106.16	258.6	391.2	287.2	235.1	52.15	5.507	
11,109.6	10,760.7	11,122.3	10,833.2	28.4	27.4	-104.75	286.4	452.7	315.1	261.2	53.98	5.838	
11,200.0	10,761.5	11,222.8	10,834.1	29.8	28.7	-103.05	325.2	545.3	349.2	292.4	56.79	6.149	
11,300.0	10,762.4	11,338.9	10,835.2	31.6	30.6	-101.69	363.8	654.8	381.9	321.5	60.34	6.328	
11,400.0	10,763.2	11,459.6	10,836.4	33.6	32.7	-100.74	396.9	770.9	408.8	344.5	64.34	6.354	
11,500.0	10,764.1	11,584.3	10,837.6	35.7	35.2	-100.09	423.3	892.7	429.7	360.9	68.76	6.249	
11,600.0	10,765.0	11,712.1	10,838.8	37.9	37.8	-99.68	442.0	1,019.1	444.1	370.6	73.55	6.038	
11,700.0	10,765.9	11,841.8	10,840.0	40.2	40.7	-99.47	452.4	1,148.3	452.0	373.3	78.67	5.746	
11,800.0	10,766.7	11,962.2	10,841.1	42.6	43.4	-99.44	454.4	1,268.7	453.5	369.8	83.75	5.415	
11,900.0	10,767.6	12,062.2	10,842.1	45.0	45.7	-99.45	454.4	1,368.7	453.6	365.0	88.51	5.124	
12,000.0	10,768.5	12,162.2	10,843.0	47.5	48.1	-99.46	454.4	1,468.7	453.6	360.2	93.38	4.857	
12,100.0	10,769.3	12,262.2	10,843.9	50.1	50.5	-99.47	454.4	1,568.7	453.6	355.2	98.35	4.612	
12,200.0	10,770.2	12,362.2	10,844.9	52.6	53.0	-99.47	454.4	1,668.7	453.6	350.2	103.40	4.387	
12,300.0	10,771.1	12,462.2	10,845.8	55.3	55.6	-99.48	454.4	1,768.7	453.6	345.1	108.53	4.180	
12,400.0	10,772.0	12,562.2	10,846.8	57.9	58.2	-99.49	454.4	1,868.7	453.6	339.9	113.72	3.989	
12,500.0	10,772.8	12,662.2	10,847.7	60.6	60.8	-99.50	454.4	1,968.7	453.6	334.7	118.96	3.813	
12,600.0	10,773.7	12,762.2	10,848.7	63.3	63.4	-99.51	454.4	2,068.7	453.6	329.4	124.25	3.651	
12,700.0	10,774.6	12,862.2	10,849.6	66.0	66.1	-99.52	454.4	2,168.7	453.6	324.1	129.58	3.501	
12,800.0	10,775.5	12,962.2	10,850.5	68.7	68.7	-99.53	454.4	2,268.7	453.7	318.7	134.94	3.362	
12,900.0	10,776.3	13,062.2	10,851.5	71.5	71.4	-99.54	454.4	2,368.7	453.7	313.3	140.34	3.233	
13,000.0	10,777.2	13,162.2	10,852.4	74.2	74.2	-99.54	454.4	2,468.7	453.7	307.9	145.77	3.112	
13,100.0	10,778.1	13,262.2	10,853.4	77.0	76.9	-99.55	454.4	2,568.7	453.7	302.5	151.23	3.000	
13,200.0	10,778.9	13,362.2	10,854.3	79.8	79.6	-99.56	454.4	2,668.7	453.7	297.0	156.70	2.895	
13,300.0	10,779.8	13,462.2	10,855.3	82.6	82.4	-99.57	454.4	2,768.7	453.7	291.5	162.20	2.797	
13,400.0	10,780.7	13,562.2	10,856.2	85.4	85.2	-99.58	454.4	2,868.7	453.7	286.0	167.72	2.705	
13,500.0	10,781.6	13,662.2	10,857.1	88.2	87.9	-99.59	454.4	2,968.7	453.7	280.5	173.25	2.619	
13,600.0	10,782.4	13,762.2	10,858.1	91.0	90.7	-99.60	454.4	3,068.7	453.7	275.0	178.80	2.538	
13,700.0	10,783.3	13,862.2	10,859.0	93.9	93.5	-99.60	454.4	3,168.7	453.8	269.4	184.36	2.461	
13,800.0	10,784.2	13,962.2	10,860.0	96.7	96.3	-99.61	454.4	3,268.7	453.8	263.8	189.93	2.389	
13,900.0	10,785.1	14,062.2	10,860.9	99.5	99.2	-99.62	454.4	3,368.7	453.8	258.3	195.52	2.321	
14,000.0	10,785.9	14,162.2	10,861.8	102.4	102.0	-99.63	454.4	3,468.6	453.8	252.7	201.11	2.256	
14,100.0	10,786.8	14,262.2	10,862.8	105.2	104.8	-99.64	454.4	3,568.6	453.8	247.1	206.72	2.195	
14,200.0	10,787.7	14,362.2	10,863.7	108.1	107.6	-99.65	454.4	3,668.6	453.8	241.5	212.33	2.137	
14,300.0	10,788.5	14,462.2	10,864.7	110.9	110.5	-99.66	454.4	3,768.6	453.8	235.9	217.95	2.082	
14,400.0	10,789.4	14,562.2	10,865.6	113.8	113.3	-99.67	454.4	3,868.6	453.8	230.3	223.58	2.030	
14,500.0	10,790.3	14,662.2	10,866.6	116.7	116.2	-99.67	454.4	3,968.6	453.9	224.6	229.22	1.980	
14,600.0	10,791.2	14,762.2	10,867.5	119.5	119.0	-99.68	454.4	4,068.6	453.9	219.0	234.86	1.932	
14,700.0	10,792.0	14,862.2	10,868.4	122.4	121.9	-99.69	454.4	4,168.6	453.9	213.4	240.51	1.887	
14,800.0	10,792.9	14,962.2	10,869.4	125.3	124.7	-99.70	454.4	4,268.6	453.9	207.7	246.17	1.844	
14,900.0	10,793.8	15,062.2	10,870.3	128.2	127.6	-99.71	454.4	4,368.6	453.9	202.1	251.82	1.802	
15,000.0	10,794.7	15,162.2	10,871.3	131.0	130.5	-99.72	454.4	4,468.6	453.9	196.4	257.49	1.763	
15,100.0	10,795.5	15,262.2	10,872.2	133.9	133.3	-99.73	454.4	4,568.6	453.9	190.8	263.16	1.725	
15,200.0	10,796.4	15,362.2	10,873.2	136.8	136.2	-99.74	454.4	4,668.6	453.9	185.1	268.83	1.689	

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Oasis Petroleum

Anticollision Report

Company:	Oasis	Local Co-ordinate Reference:	Well Lewis Federal 5300 11-31 3B
Project:	Indian Hills	TVD Reference:	WELL @ 2134.0usft (Original Well Elev)
Reference Site:	153N-100W-31/32	MD Reference:	WELL @ 2134.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Lewis Federal 5300 11-31 3B	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Lewis Federal 5300 11-31 3B	Database:	OpenWellsCompass - EDM Prod
Reference Design:	Plan #1	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-31/32 - Lewis Federal 5300 11-31 2T - Lewis Federal 5300 11-31 2T - Plan #1													Offset Site Error:	0.0 usft
Survey Program: 0-MWD													Offset Well Error:	0.0 usft
Reference		Offset		Semi Major Axis			Distance							Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor		
15,300.0	10,797.3	15,462.2	10,874.1	139.7	139.1	-99.74	454.4	4,768.6	453.9	179.4	274.51	1.654		
15,400.0	10,798.1	15,562.2	10,875.0	142.6	141.9	-99.75	454.4	4,868.6	454.0	173.8	280.19	1.620		
15,500.0	10,799.0	15,662.2	10,876.0	145.5	144.8	-99.76	454.4	4,968.6	454.0	168.1	285.87	1.588		
15,600.0	10,799.9	15,762.2	10,876.9	148.3	147.7	-99.77	454.4	5,068.6	454.0	162.4	291.55	1.557		
15,700.0	10,800.8	15,862.2	10,877.9	151.2	150.6	-99.78	454.4	5,168.6	454.0	156.8	297.24	1.527		
15,800.0	10,801.6	15,962.2	10,878.8	154.1	153.5	-99.79	454.4	5,268.6	454.0	151.1	302.94	1.499 Level 3	Level 3	
15,900.0	10,802.5	16,062.2	10,879.8	157.0	156.3	-99.80	454.4	5,368.6	454.0	145.4	308.63	1.471 Level 3	Level 3	
16,000.0	10,803.4	16,162.2	10,880.7	159.9	159.2	-99.80	454.4	5,468.6	454.0	139.7	314.33	1.444 Level 3	Level 3	
16,100.0	10,804.3	16,262.2	10,881.6	162.8	162.1	-99.81	454.4	5,568.6	454.0	134.0	320.03	1.419 Level 3	Level 3	
16,200.0	10,805.1	16,362.2	10,882.6	165.7	165.0	-99.82	454.4	5,668.5	454.1	128.3	325.73	1.394 Level 3	Level 3	
16,300.0	10,806.0	16,462.2	10,883.5	168.6	167.9	-99.83	454.4	5,768.5	454.1	122.6	331.43	1.370 Level 3	Level 3	
16,400.0	10,806.9	16,562.2	10,884.5	171.5	170.8	-99.84	454.4	5,868.5	454.1	116.9	337.14	1.347 Level 3	Level 3	
16,500.0	10,807.7	16,662.2	10,885.4	174.4	173.7	-99.85	454.4	5,968.5	454.1	111.2	342.84	1.324 Level 3	Level 3	
16,600.0	10,808.6	16,762.2	10,886.4	177.3	176.6	-99.86	454.4	6,068.5	454.1	105.6	348.55	1.303 Level 3	Level 3	
16,700.0	10,809.5	16,862.2	10,887.3	180.2	179.5	-99.87	454.4	6,168.5	454.1	99.9	354.26	1.282 Level 3	Level 3	
16,800.0	10,810.4	16,962.2	10,888.2	183.1	182.4	-99.87	454.4	6,268.5	454.1	94.2	359.97	1.262 Level 3	Level 3	
16,900.0	10,811.2	17,062.2	10,889.2	186.0	185.3	-99.88	454.4	6,368.5	454.1	88.5	365.69	1.242 Level 2		
17,000.0	10,812.1	17,162.2	10,890.1	188.9	188.2	-99.89	454.4	6,468.5	454.2	82.8	371.40	1.223 Level 2		
17,100.0	10,813.0	17,262.2	10,891.1	191.8	191.1	-99.90	454.4	6,568.5	454.2	77.0	377.12	1.204 Level 2		
17,200.0	10,813.9	17,362.2	10,892.0	194.7	194.0	-99.91	454.4	6,668.5	454.2	71.3	382.83	1.186 Level 2		
17,300.0	10,814.7	17,462.2	10,892.9	197.6	196.9	-99.92	454.4	6,768.5	454.2	65.6	388.55	1.169 Level 2		
17,400.0	10,815.6	17,562.2	10,893.9	200.5	199.8	-99.93	454.4	6,868.5	454.2	59.9	394.27	1.152 Level 2		
17,500.0	10,816.5	17,662.2	10,894.8	203.5	202.7	-99.93	454.4	6,968.5	454.2	54.2	399.99	1.136 Level 2		
17,600.0	10,817.3	17,762.2	10,895.8	206.4	205.6	-99.94	454.4	7,068.5	454.2	48.5	405.71	1.120 Level 2		
17,700.0	10,818.2	17,862.2	10,896.7	209.3	208.5	-99.95	454.4	7,168.5	454.2	42.8	411.43	1.104 Level 2		
17,800.0	10,819.1	17,962.2	10,897.7	212.2	211.4	-99.96	454.4	7,268.5	454.2	37.1	417.16	1.089 Level 2		
17,900.0	10,820.0	18,062.2	10,898.6	215.1	214.3	-99.97	454.4	7,368.5	454.3	31.4	422.88	1.074 Level 2		
18,000.0	10,820.8	18,162.2	10,899.5	218.0	217.2	-99.98	454.4	7,468.5	454.3	25.7	428.61	1.060 Level 2		
18,100.0	10,821.7	18,262.2	10,900.5	220.9	220.1	-99.99	454.4	7,568.5	454.3	20.0	434.33	1.046 Level 2		
18,200.0	10,822.6	18,362.2	10,901.4	223.8	223.0	-100.00	454.4	7,668.5	454.3	14.2	440.06	1.032 Level 2		
18,300.0	10,823.5	18,462.2	10,902.4	226.7	225.9	-100.00	454.4	7,768.5	454.3	8.5	445.78	1.019 Level 2		
18,400.0	10,824.3	18,562.2	10,903.3	229.7	228.8	-100.01	454.4	7,868.5	454.3	2.8	451.51	1.006 Level 2		
18,500.0	10,825.2	18,662.2	10,904.3	232.6	231.7	-100.02	454.4	7,968.4	454.3	-2.9	457.24	0.994 Level 1		
18,600.0	10,826.1	18,762.2	10,905.2	235.5	234.6	-100.03	454.4	8,068.4	454.3	-8.6	462.97	0.981 Level 1		
18,700.0	10,826.9	18,862.2	10,906.1	238.4	237.5	-100.04	454.4	8,168.4	454.4	-14.3	468.69	0.969 Level 1		
18,800.0	10,827.8	18,962.2	10,907.1	241.3	240.5	-100.05	454.4	8,268.4	454.4	-20.1	474.42	0.958 Level 1		
18,900.0	10,828.7	19,062.2	10,908.0	244.2	243.4	-100.06	454.4	8,368.4	454.4	-25.8	480.15	0.946 Level 1		
19,000.0	10,829.6	19,162.2	10,909.0	247.1	246.3	-100.06	454.4	8,468.4	454.4	-31.5	485.88	0.935 Level 1		
19,100.0	10,830.4	19,262.2	10,909.9	250.1	249.2	-100.07	454.4	8,568.4	454.4	-37.2	491.61	0.924 Level 1		
19,200.0	10,831.3	19,362.2	10,910.9	253.0	252.1	-100.08	454.4	8,668.4	454.4	-42.9	497.34	0.914 Level 1		
19,300.0	10,832.2	19,462.2	10,911.8	255.9	255.0	-100.09	454.4	8,768.4	454.4	-48.6	503.07	0.903 Level 1		
19,400.0	10,833.1	19,562.2	10,912.7	258.8	257.9	-100.10	454.4	8,868.4	454.4	-54.4	508.81	0.893 Level 1		
19,500.0	10,833.9	19,662.2	10,913.7	261.7	260.8	-100.11	454.4	8,968.4	454.5	-60.1	514.54	0.883 Level 1		
19,600.0	10,834.8	19,762.2	10,914.6	264.6	263.8	-100.12	454.4	9,068.4	454.5	-65.8	520.27	0.874 Level 1		
19,700.0	10,835.7	19,862.2	10,915.6	267.6	266.7	-100.13	454.4	9,168.4	454.5	-71.5	526.00	0.864 Level 1		
19,800.0	10,836.5	19,962.2	10,916.5	270.5	269.6	-100.13	454.4	9,268.4	454.5	-77.2	531.73	0.855 Level 1		
19,900.0	10,837.4	20,062.2	10,917.5	273.4	272.5	-100.14	454.4	9,368.4	454.5	-83.0	537.47	0.846 Level 1		
20,000.0	10,838.3	20,162.2	10,918.4	276.3	275.4	-100.15	454.4	9,468.4	454.5	-88.7	543.20	0.837 Level 1		
20,100.0	10,839.2	20,262.2	10,919.3	279.2	278.3	-100.16	454.4	9,568.4	454.5	-94.4	548.93	0.828 Level 1		
20,200.0	10,840.0	20,362.2	10,920.3	282.1	281.2	-100.17	454.4	9,668.4	454.5	-100.1	554.67	0.819 Level 1		
20,300.0	10,840.9	20,462.2	10,921.2	285.1	284.2	-100.18	454.4	9,768.4	454.6	-105.8	560.40	0.811 Level 1		
20,400.0	10,841.8	20,562.2	10,922.2	288.0	287.1	-100.19	454.4	9,868.4	454.6	-111.6	566.13	0.803 Level 1		
20,500.0	10,842.7	20,662.2	10,923.1	290.9	290.0	-100.19	454.4	9,968.4	454.6	-117.3	571.87	0.795 Level 1		

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Oasis Petroleum

Anticollision Report

Company:	Oasis	Local Co-ordinate Reference:	Well Lewis Federal 5300 11-31 3B
Project:	Indian Hills	TVD Reference:	WELL @ 2134.0usft (Original Well Elev)
Reference Site:	153N-100W-31/32	MD Reference:	WELL @ 2134.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Lewis Federal 5300 11-31 3B	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Lewis Federal 5300 11-31 3B	Database:	OpenWellsCompass - EDM Prod
Reference Design:	Plan #1	Offset TVD Reference:	Offset Datum

Offset Design	153N-100W-31/32 - Lewis Federal 5300 11-31 2T - Lewis Federal 5300 11-31 2T - Plan #1											Offset Site Error:	0.0 usft
Survey Program:	0-MWD											Offset Well Error:	0.0 usft
Reference		Offset		Semi Major Axis		Distance							
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
20,600.0	10,843.5	20,762.2	10,924.1	293.8	292.3	-100.20	454.4	10,068.4	454.6	-122.4	576.99	0.788	Level 1
20,640.9	10,843.9	20,803.1	10,924.4	294.6	293.1	-100.21	454.4	10,109.2	454.6	-123.9	578.53	0.786	Level 1, ES, SF

Oasis Petroleum

Anticollision Report

Company:	Oasis	Local Co-ordinate Reference:	Well Lewis Federal 5300 11-31 3B
Project:	Indian Hills	TVD Reference:	WELL @ 2134.0usft (Original Well Elev)
Reference Site:	153N-100W-31/32	MD Reference:	WELL @ 2134.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Lewis Federal 5300 11-31 3B	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Lewis Federal 5300 11-31 3B	Database:	OpenWellsCompass - EDM Prod
Reference Design:	Plan #1	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-31/32 - Lewis Federal 5300 11-31 4T2 - Lewis Federal 5300 11-31 4T2 - Plan #1												Offset Site Error:	0.0 usft
Survey Program: 0-MWD												Offset Well Error:	0.0 usft
Reference		Offset		Semi Major Axis		Distance							Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	
0.0	0.0	0.0	0.0	0.0	0.0	-174.20	-33.4	-3.4	33.6				
100.0	100.0	100.0	100.0	0.1	0.1	-174.20	-33.4	-3.4	33.6	33.4	0.18	191.709	
200.0	200.0	200.0	200.0	0.3	0.3	-174.20	-33.4	-3.4	33.6	33.0	0.62	53.789	
300.0	300.0	300.0	300.0	0.5	0.5	-174.20	-33.4	-3.4	33.6	32.5	1.07	31.283	
400.0	400.0	400.0	400.0	0.8	0.8	-174.20	-33.4	-3.4	33.6	32.1	1.52	22.055	
500.0	500.0	500.0	500.0	1.0	1.0	-174.20	-33.4	-3.4	33.6	31.6	1.97	17.031	
600.0	600.0	600.0	600.0	1.2	1.2	-174.20	-33.4	-3.4	33.6	31.2	2.42	13.871	
700.0	700.0	700.0	700.0	1.4	1.4	-174.20	-33.4	-3.4	33.6	30.7	2.87	11.701	
800.0	800.0	800.0	800.0	1.7	1.7	-174.20	-33.4	-3.4	33.6	30.3	3.32	10.117	
900.0	900.0	900.0	900.0	1.9	1.9	-174.20	-33.4	-3.4	33.6	29.8	3.77	8.911	
1,000.0	1,000.0	1,000.0	1,000.0	2.1	2.1	-174.20	-33.4	-3.4	33.6	29.4	4.22	7.962	
1,100.0	1,100.0	1,100.0	1,100.0	2.3	2.3	-174.20	-33.4	-3.4	33.6	28.9	4.67	7.196	
1,200.0	1,200.0	1,200.0	1,200.0	2.6	2.6	-174.20	-33.4	-3.4	33.6	28.5	5.12	6.564	
1,300.0	1,300.0	1,300.0	1,300.0	2.8	2.8	-174.20	-33.4	-3.4	33.6	28.0	5.57	6.034	
1,400.0	1,400.0	1,400.0	1,400.0	3.0	3.0	-174.20	-33.4	-3.4	33.6	27.6	6.02	5.584	
1,500.0	1,500.0	1,500.0	1,500.0	3.2	3.2	-174.20	-33.4	-3.4	33.6	27.1	6.47	5.196	
1,600.0	1,600.0	1,600.0	1,600.0	3.5	3.5	-174.20	-33.4	-3.4	33.6	26.7	6.92	4.858	
1,700.0	1,700.0	1,700.0	1,700.0	3.7	3.7	-174.20	-33.4	-3.4	33.6	26.2	7.37	4.562	
1,800.0	1,800.0	1,800.0	1,800.0	3.9	3.9	-174.20	-33.4	-3.4	33.6	25.8	7.82	4.299	
1,900.0	1,900.0	1,900.0	1,900.0	4.1	4.1	-174.20	-33.4	-3.4	33.6	25.3	8.27	4.066	
2,000.0	2,000.0	2,000.0	2,000.0	4.4	4.4	-174.20	-33.4	-3.4	33.6	24.9	8.72	3.856	
2,100.0	2,100.0	2,100.0	2,100.0	4.6	4.6	-174.20	-33.4	-3.4	33.6	24.4	9.17	3.667 CC	
2,110.0	2,110.0	2,109.8	2,109.8	4.6	4.6	140.79	-33.5	-3.4	33.7	24.5	9.21	3.657	
2,200.0	2,200.0	2,199.8	2,199.8	4.8	4.8	140.61	-34.0	-2.8	34.8	25.2	9.56	3.637	
2,300.0	2,300.0	2,299.8	2,299.8	5.0	4.9	140.41	-34.6	-2.2	36.0	26.0	9.96	3.615	
2,400.0	2,400.0	2,399.8	2,399.8	5.2	5.1	140.24	-35.3	-1.6	37.2	26.9	10.36	3.593	
2,500.0	2,500.0	2,499.8	2,499.8	5.5	5.3	140.07	-35.9	-1.0	38.5	27.7	10.77	3.572	
2,600.0	2,600.0	2,599.8	2,599.8	5.7	5.5	139.91	-36.5	-0.3	39.7	28.5	11.18	3.551	
2,700.0	2,700.0	2,699.8	2,699.7	5.9	5.7	139.76	-37.1	0.3	40.9	29.3	11.59	3.531	
2,800.0	2,800.0	2,799.8	2,799.7	6.1	5.9	139.62	-37.7	0.9	42.2	30.1	12.00	3.511	
2,900.0	2,900.0	2,899.8	2,899.7	6.3	6.1	139.49	-38.3	1.5	43.4	31.0	12.42	3.492	
3,000.0	3,000.0	2,999.7	2,999.7	6.6	6.3	139.37	-39.0	2.1	44.6	31.8	12.84	3.474	
3,100.0	3,100.0	3,099.7	3,099.7	6.8	6.5	139.25	-39.6	2.7	45.8	32.6	13.26	3.456	
3,200.0	3,200.0	3,199.7	3,199.7	7.0	6.7	139.14	-40.2	3.4	47.1	33.4	13.69	3.439	
3,300.0	3,300.0	3,299.7	3,299.7	7.2	6.9	139.03	-40.8	4.0	48.3	34.2	14.11	3.423	
3,400.0	3,400.0	3,399.7	3,399.7	7.4	7.1	138.93	-41.4	4.6	49.5	35.0	14.54	3.407	
3,500.0	3,499.9	3,499.7	3,499.7	7.7	7.3	138.84	-42.0	5.2	50.8	35.8	14.97	3.392	
3,600.0	3,599.9	3,599.7	3,599.6	7.9	7.5	138.75	-42.7	5.8	52.0	36.6	15.40	3.377	
3,700.0	3,699.9	3,699.7	3,699.6	8.1	7.7	138.66	-43.3	6.4	53.2	37.4	15.83	3.363	
3,800.0	3,799.9	3,799.7	3,799.6	8.3	7.9	138.58	-43.9	7.1	54.5	38.2	16.26	3.350	
3,900.0	3,899.9	3,899.7	3,899.6	8.6	8.1	138.50	-44.5	7.7	55.7	39.0	16.69	3.337	
4,000.0	3,999.9	3,999.7	3,999.6	8.8	8.4	138.42	-45.1	8.3	56.9	39.8	17.12	3.324	
4,100.0	4,099.9	4,099.7	4,099.6	9.0	8.6	138.35	-45.7	8.9	58.2	40.6	17.56	3.312	
4,200.0	4,199.9	4,199.7	4,199.6	9.2	8.8	138.28	-46.4	9.5	59.4	41.4	17.99	3.301	
4,300.0	4,299.9	4,299.6	4,299.6	9.4	9.0	138.21	-47.0	10.1	60.6	42.2	18.43	3.289	
4,400.0	4,399.9	4,399.6	4,399.6	9.7	9.2	138.15	-47.6	10.8	61.9	43.0	18.87	3.279	
4,500.0	4,499.9	4,499.6	4,499.5	9.9	9.4	138.09	-48.2	11.4	63.1	43.8	19.30	3.268	
4,600.0	4,599.9	4,599.6	4,599.5	10.1	9.6	138.03	-48.8	12.0	64.3	44.6	19.74	3.258	
4,700.0	4,699.9	4,699.6	4,699.5	10.3	9.9	137.97	-49.4	12.6	65.6	45.4	20.18	3.249	
4,800.0	4,799.9	4,799.6	4,799.5	10.6	10.1	137.92	-50.1	13.2	66.8	46.2	20.62	3.239	
4,900.0	4,899.9	4,899.6	4,899.5	10.8	10.3	137.86	-50.7	13.8	68.0	47.0	21.06	3.230	
5,000.0	4,999.9	4,999.6	4,999.5	11.0	10.5	137.81	-51.3	14.5	69.2	47.8	21.50	3.221	
5,100.0	5,099.9	5,099.6	5,099.5	11.2	10.7	137.76	-51.9	15.1	70.5	48.5	21.94	3.213	

CC - Min centre to center distance or covergent point, SF - min separation factor, ES - min ellipse separation

Oasis Petroleum

Anticollision Report

Company:	Oasis	Local Co-ordinate Reference:	Well Lewis Federal 5300 11-31 3B
Project:	Indian Hills	TVD Reference:	WELL @ 2134.0usft (Original Well Elev)
Reference Site:	153N-100W-31/32	MD Reference:	WELL @ 2134.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Lewis Federal 5300 11-31 3B	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Lewis Federal 5300 11-31 3B	Database:	OpenWellsCompass - EDM Prod
Reference Design:	Plan #1	Offset TVD Reference:	Offset Datum

Offset Design												Offset Site Error:	0.0 usft
Survey Program: 0-MWD												Offset Well Error:	0.0 usft
Reference		Offset		Semi Major Axis		Distance							
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
5,200.0	5,199.9	5,199.6	5,199.5	11.5	10.9	137.71	-52.5	15.7	71.7	49.3	22.38	3.205	
5,300.0	5,299.9	5,299.6	5,299.4	11.7	11.2	137.67	-53.2	16.3	72.9	50.1	22.82	3.197	
5,400.0	5,399.9	5,399.6	5,399.4	11.9	11.4	137.62	-53.8	16.9	74.2	50.9	23.26	3.189	
5,500.0	5,499.9	5,499.6	5,499.4	12.1	11.6	137.58	-54.4	17.5	75.4	51.7	23.70	3.182	
5,600.0	5,599.9	5,599.5	5,599.4	12.3	11.8	137.54	-55.0	18.2	76.6	52.5	24.14	3.175	
5,700.0	5,699.9	5,699.5	5,699.4	12.6	12.0	137.50	-55.6	18.8	77.9	53.3	24.58	3.168	
5,800.0	5,799.9	5,799.5	5,799.4	12.8	12.2	137.46	-56.2	19.4	79.1	54.1	25.03	3.161	
5,900.0	5,899.9	5,899.5	5,899.4	13.0	12.5	137.42	-56.9	20.0	80.3	54.9	25.47	3.154	
6,000.0	5,999.9	5,999.5	5,999.4	13.2	12.7	137.39	-57.5	20.6	81.6	55.7	25.91	3.148	
6,100.0	6,099.8	6,099.5	6,099.4	13.5	12.9	137.35	-58.1	21.3	82.8	56.5	26.36	3.142	
6,151.5	6,151.3	6,151.0	6,150.8	13.6	13.0	137.33	-58.4	21.6	83.4	56.9	26.58	3.139	
6,161.5	6,161.3	6,161.5	6,161.3	13.6	13.0	-177.67	-58.4	21.6	83.5	56.9	26.61	3.138	
6,200.0	6,199.8	6,200.0	6,199.8	13.7	13.1	-177.67	-58.4	21.6	83.5	56.7	26.76	3.120	
6,300.0	6,299.8	6,300.0	6,299.8	13.9	13.3	-177.67	-58.4	21.6	83.5	56.3	27.19	3.072	
6,400.0	6,399.8	6,400.0	6,399.8	14.1	13.5	-177.67	-58.4	21.6	83.5	55.9	27.61	3.025	
6,500.0	6,499.8	6,500.0	6,499.8	14.3	13.7	-177.67	-58.4	21.6	83.5	55.5	28.03	2.979	
6,600.0	6,599.8	6,600.0	6,599.8	14.6	13.9	-177.67	-58.4	21.6	83.5	55.1	28.46	2.935	
6,700.0	6,699.8	6,700.0	6,699.8	14.8	14.1	-177.67	-58.4	21.6	83.5	54.6	28.88	2.891	
6,800.0	6,799.8	6,800.0	6,799.8	15.0	14.3	-177.67	-58.4	21.6	83.5	54.2	29.31	2.849	
6,900.0	6,899.8	6,900.0	6,899.8	15.2	14.5	-177.67	-58.4	21.6	83.5	53.8	29.74	2.808	
7,000.0	6,999.8	7,000.0	6,999.8	15.5	14.8	-177.67	-58.4	21.6	83.5	53.3	30.16	2.769	
7,100.0	7,099.8	7,100.0	7,099.8	15.7	15.0	-177.67	-58.4	21.6	83.5	52.9	30.59	2.730	
7,200.0	7,199.8	7,200.0	7,199.8	15.9	15.2	-177.67	-58.4	21.6	83.5	52.5	31.02	2.692	
7,300.0	7,299.8	7,300.0	7,299.8	16.1	15.4	-177.67	-58.4	21.6	83.5	52.1	31.45	2.655	
7,400.0	7,399.8	7,400.0	7,399.8	16.3	15.6	-177.67	-58.4	21.6	83.5	51.6	31.88	2.619	
7,500.0	7,499.8	7,500.0	7,499.8	16.6	15.8	-177.67	-58.4	21.6	83.5	51.2	32.31	2.585	
7,600.0	7,599.8	7,600.0	7,599.8	16.8	16.0	-177.67	-58.4	21.6	83.5	50.8	32.74	2.551	
7,700.0	7,699.8	7,700.0	7,699.8	17.0	16.2	-177.67	-58.4	21.6	83.5	50.3	33.17	2.517	
7,800.0	7,799.8	7,800.0	7,799.8	17.2	16.4	-177.67	-58.4	21.6	83.5	49.9	33.60	2.485	
7,900.0	7,899.8	7,900.0	7,899.8	17.5	16.6	-177.67	-58.4	21.6	83.5	49.5	34.04	2.453	
8,000.0	7,999.8	8,000.0	7,999.8	17.7	16.8	-177.67	-58.4	21.6	83.5	49.0	34.47	2.423	
8,100.0	8,099.8	8,100.0	8,099.8	17.9	17.0	-177.67	-58.4	21.6	83.5	48.6	34.90	2.393	
8,200.0	8,199.8	8,200.0	8,199.8	18.1	17.3	-177.67	-58.4	21.6	83.5	48.2	35.34	2.363	
8,300.0	8,299.8	8,300.0	8,299.8	18.3	17.5	-177.67	-58.4	21.6	83.5	47.7	35.77	2.335	
8,400.0	8,399.8	8,400.0	8,399.8	18.6	17.7	-177.67	-58.4	21.6	83.5	47.3	36.20	2.307	
8,500.0	8,499.8	8,500.0	8,499.8	18.8	17.9	-177.67	-58.4	21.6	83.5	46.9	36.64	2.279	
8,600.0	8,599.8	8,600.0	8,599.8	19.0	18.1	-177.67	-58.4	21.6	83.5	46.4	37.07	2.252	
8,700.0	8,699.8	8,700.0	8,699.8	19.2	18.3	-177.67	-58.4	21.6	83.5	46.0	37.51	2.226	
8,800.0	8,799.8	8,800.0	8,799.8	19.5	18.5	-177.67	-58.4	21.6	83.5	45.6	37.95	2.201	
8,900.0	8,899.8	8,900.0	8,899.8	19.7	18.7	-177.67	-58.4	21.6	83.5	45.1	38.38	2.176	
9,000.0	8,999.8	9,000.0	8,999.8	19.9	19.0	-177.67	-58.4	21.6	83.5	44.7	38.82	2.151	
9,100.0	9,099.8	9,100.0	9,099.8	20.1	19.2	-177.67	-58.4	21.6	83.5	44.3	39.26	2.127	
9,200.0	9,199.8	9,200.0	9,199.8	20.3	19.4	-177.67	-58.4	21.6	83.5	43.8	39.69	2.104	
9,300.0	9,299.8	9,300.0	9,299.8	20.6	19.6	-177.67	-58.4	21.6	83.5	43.4	40.13	2.081	
9,400.0	9,399.8	9,400.0	9,399.8	20.8	19.8	-177.67	-58.4	21.6	83.5	42.9	40.57	2.058	
9,500.0	9,499.8	9,500.0	9,499.8	21.0	20.0	-177.67	-58.4	21.6	83.5	42.5	41.00	2.037	
9,600.0	9,599.8	9,600.0	9,599.8	21.2	20.2	-177.67	-58.4	21.6	83.5	42.1	41.44	2.015	
9,700.0	9,699.8	9,700.0	9,699.8	21.5	20.5	-177.67	-58.4	21.6	83.5	41.6	41.88	1.994	
9,800.0	9,799.8	9,800.0	9,799.8	21.7	20.7	-177.67	-58.4	21.6	83.5	41.2	42.32	1.973	
9,900.0	9,899.8	9,900.0	9,899.8	21.9	20.9	-177.67	-58.4	21.6	83.5	40.7	42.76	1.953	
10,000.0	9,999.8	10,000.0	9,999.8	22.1	21.1	-177.67	-58.4	21.6	83.5	40.3	43.20	1.933	
10,100.0	10,099.8	10,100.0	10,099.8	22.4	21.3	-177.67	-58.4	21.6	83.5	39.9	43.64	1.914	
10,200.0	10,199.8	10,200.0	10,199.8	22.6	21.5	-177.67	-58.4	21.6	83.5	39.4	44.08	1.895	

CC - Min centre to center distance or covergent point, SF - min separation factor, ES - min ellipse separation

Oasis Petroleum

Anticollision Report

Company:	Oasis	Local Co-ordinate Reference:	Well Lewis Federal 5300 11-31 3B
Project:	Indian Hills	TVD Reference:	WELL @ 2134.0usft (Original Well Elev)
Reference Site:	153N-100W-31/32	MD Reference:	WELL @ 2134.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Lewis Federal 5300 11-31 3B	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Lewis Federal 5300 11-31 3B	Database:	OpenWellsCompass - EDM Prod
Reference Design:	Plan #1	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-31/32 - Lewis Federal 5300 11-31 4T2 - Lewis Federal 5300 11-31 4T2 - Plan #1													Offset Site Error:	0.0 usft
Survey Program: 0-MWD													Offset Well Error:	0.0 usft
Reference		Offset		Semi Major Axis		Distance								Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor		
10,282.7	10,282.5	10,282.7	10,282.5	22.8	21.7	-177.67	-58.4	21.6	83.5	39.1	44.44	1.879		
10,300.0	10,299.8	10,300.0	10,299.8	22.8	21.8	90.57	-58.4	21.6	83.5	39.0	44.55	1.874		
10,325.0	10,324.8	10,324.9	10,324.8	22.9	21.8	91.63	-58.4	21.6	83.5	38.9	44.66	1.870		
10,350.0	10,349.6	10,349.8	10,349.6	22.9	21.9	93.56	-58.4	21.6	83.7	38.9	44.77	1.869		
10,375.0	10,374.3	10,374.4	10,374.3	23.0	21.9	96.31	-58.4	21.6	84.0	39.2	44.88	1.873		
10,400.0	10,398.7	10,399.2	10,399.0	23.0	22.0	99.82	-58.4	21.6	84.8	39.8	44.97	1.886		
10,425.0	10,422.7	10,425.1	10,424.9	23.1	22.0	103.64	-58.3	22.7	85.9	40.8	45.05	1.906		
10,450.0	10,446.4	10,451.4	10,451.1	23.1	22.1	107.39	-58.0	25.2	87.1	42.0	45.11	1.931		
10,475.0	10,469.7	10,478.0	10,477.4	23.2	22.2	111.06	-57.6	29.2	88.5	43.3	45.12	1.961		
10,500.0	10,492.4	10,505.0	10,503.8	23.3	22.2	114.64	-57.0	34.7	90.0	44.9	45.08	1.996		
10,525.0	10,514.6	10,532.3	10,530.2	23.3	22.3	118.14	-56.2	41.8	91.6	46.6	44.98	2.036		
10,550.0	10,536.1	10,560.0	10,556.4	23.4	22.3	121.53	-55.3	50.4	93.3	48.5	44.81	2.082		
10,575.0	10,556.9	10,588.0	10,582.5	23.5	22.4	124.83	-54.1	60.8	95.1	50.5	44.57	2.133		
10,600.0	10,577.0	10,616.4	10,608.2	23.6	22.5	128.03	-52.8	72.8	96.9	52.6	44.26	2.189		
10,625.0	10,596.3	10,645.2	10,633.4	23.7	22.6	131.14	-51.3	86.4	98.7	54.8	43.87	2.250		
10,650.0	10,614.7	10,674.3	10,658.1	23.8	22.7	134.16	-49.6	101.8	100.5	57.1	43.40	2.315		
10,675.0	10,632.2	10,703.8	10,682.1	23.9	22.8	137.09	-47.7	118.8	102.2	59.4	42.87	2.385		
10,700.0	10,648.7	10,733.6	10,705.2	24.1	22.9	139.94	-45.7	137.5	103.9	61.7	42.26	2.459		
10,725.0	10,664.2	10,763.7	10,727.3	24.2	23.0	142.72	-43.4	157.8	105.5	63.9	41.60	2.536		
10,750.0	10,678.7	10,794.1	10,748.3	24.4	23.1	145.42	-41.0	179.7	107.0	66.1	40.90	2.616		
10,775.0	10,692.1	10,824.9	10,768.1	24.5	23.3	148.06	-38.4	203.1	108.3	68.2	40.15	2.698		
10,800.0	10,704.4	10,855.9	10,786.4	24.7	23.5	150.64	-35.7	227.9	109.5	70.2	39.38	2.782		
10,825.0	10,715.5	10,887.1	10,803.2	24.9	23.7	153.17	-32.8	254.1	110.6	72.0	38.59	2.865		
10,850.0	10,725.5	10,918.6	10,818.4	25.2	23.9	155.66	-29.8	281.5	111.4	73.6	37.80	2.947		
10,875.0	10,734.2	10,950.2	10,831.7	25.4	24.2	158.11	-26.6	310.1	112.0	75.0	37.03	3.026		
10,900.0	10,741.7	10,982.1	10,843.3	25.7	24.5	160.52	-23.4	339.6	112.5	76.2	36.29	3.099		
10,925.0	10,747.9	11,014.1	10,852.8	25.9	24.8	162.91	-20.0	369.9	112.7	77.1	35.61	3.164		
10,950.0	10,752.9	11,046.1	10,860.3	26.2	25.2	165.28	-16.6	400.9	112.6	77.6	34.99	3.219		
10,975.0	10,756.5	11,078.3	10,865.7	26.5	25.6	167.63	-13.2	432.4	112.4	77.9	34.47	3.260		
11,000.0	10,758.9	11,110.4	10,868.9	26.8	26.0	169.98	-9.7	464.2	111.8	77.8	34.05	3.285		
11,025.0	10,760.0	11,141.6	10,870.0	27.2	26.4	172.24	-6.2	495.1	111.1	77.3	33.76	3.291		
11,028.6	10,760.0	11,145.1	10,870.1	27.2	26.4	172.51	-5.9	498.6	111.0	77.3	33.74	3.290		
11,043.6	10,760.1	11,159.9	10,870.2	27.4	26.6	173.59	-4.3	513.3	110.8	77.1	33.66	3.291		
11,109.6	10,760.7	11,225.2	10,870.8	28.4	27.6	177.18	1.6	578.4	110.3	76.7	33.57	3.285		
11,164.9	10,761.2	11,280.4	10,871.4	29.3	28.5	178.83	4.7	633.4	110.2	76.5	33.68	3.272		
11,200.0	10,761.5	11,315.4	10,871.7	29.8	29.1	179.45	5.9	668.4	110.2	76.4	33.78	3.263		
11,300.0	10,762.4	11,415.4	10,872.7	31.6	31.0	179.76	6.5	768.4	110.3	76.1	34.16	3.229		
11,400.0	10,763.2	11,515.4	10,873.6	33.6	33.0	179.76	6.5	868.4	110.4	75.8	34.59	3.191		
11,500.0	10,764.1	11,615.4	10,874.6	35.7	35.1	179.76	6.5	968.4	110.5	75.4	35.09	3.149		
11,600.0	10,765.0	11,715.4	10,875.5	37.9	37.3	179.76	6.5	1,068.4	110.6	74.9	35.63	3.103		
11,700.0	10,765.9	11,815.4	10,876.5	40.2	39.6	179.76	6.5	1,168.4	110.6	74.4	36.23	3.054		
11,800.0	10,766.7	11,915.4	10,877.5	42.6	42.0	179.76	6.5	1,268.4	110.7	73.9	36.88	3.003		
11,900.0	10,767.6	12,015.4	10,878.4	45.0	44.5	179.76	6.5	1,368.4	110.8	73.3	37.57	2.950		
12,000.0	10,768.5	12,115.4	10,879.4	47.5	47.0	179.76	6.5	1,468.4	110.9	72.6	38.31	2.895		
12,100.0	10,769.3	12,215.4	10,880.3	50.1	49.6	179.76	6.5	1,568.4	111.0	71.9	39.08	2.840		
12,200.0	10,770.2	12,315.4	10,881.3	52.6	52.2	179.76	6.5	1,668.4	111.1	71.2	39.90	2.784		
12,300.0	10,771.1	12,415.4	10,882.3	55.3	54.8	179.76	6.5	1,768.4	111.2	70.4	40.75	2.728		
12,400.0	10,772.0	12,515.4	10,883.2	57.9	57.5	179.76	6.5	1,868.4	111.3	69.6	41.64	2.672		
12,500.0	10,772.8	12,615.4	10,884.2	60.6	60.2	179.76	6.5	1,968.4	111.3	68.8	42.56	2.616		
12,600.0	10,773.7	12,715.4	10,885.1	63.3	62.9	179.76	6.5	2,068.3	111.4	67.9	43.51	2.561		
12,700.0	10,774.6	12,815.4	10,886.1	66.0	65.6	179.76	6.5	2,168.3	111.5	67.0	44.49	2.507		
12,800.0	10,775.5	12,915.4	10,887.1	68.7	68.3	179.76	6.5	2,268.3	111.6	66.1	45.49	2.453		
12,900.0	10,776.3	13,015.4	10,888.0	71.5	71.1	179.76	6.5	2,368.3	111.7	65.2	46.52	2.401		

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Oasis Petroleum

Anticollision Report

Company:	Oasis	Local Co-ordinate Reference:	Well Lewis Federal 5300 11-31 3B
Project:	Indian Hills	TVD Reference:	WELL @ 2134.0usft (Original Well Elev)
Reference Site:	153N-100W-31/32	MD Reference:	WELL @ 2134.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Lewis Federal 5300 11-31 3B	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Lewis Federal 5300 11-31 3B	Database:	OpenWellsCompass - EDM Prod
Reference Design:	Plan #1	Offset TVD Reference:	Offset Datum

Offset Design												Offset Site Error:	0.0 usft
Survey Program: 0-MWD												Offset Well Error:	0.0 usft
Reference		Offset		Semi Major Axis		Distance							
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
13,000.0	10,777.2	13,115.4	10,889.0	74.2	73.9	179.76	6.5	2,468.3	111.8	64.2	47.57	2.350	
13,100.0	10,778.1	13,215.4	10,889.9	77.0	76.7	179.76	6.5	2,568.3	111.9	63.2	48.65	2.300	
13,200.0	10,778.9	13,315.4	10,890.9	79.8	79.5	179.76	6.5	2,668.3	112.0	62.2	49.74	2.251	
13,300.0	10,779.8	13,415.4	10,891.9	82.6	82.3	179.76	6.5	2,768.3	112.0	61.2	50.85	2.203	
13,400.0	10,780.7	13,515.4	10,892.8	85.4	85.1	179.76	6.5	2,868.3	112.1	60.2	51.98	2.157	
13,500.0	10,781.6	13,615.4	10,893.8	88.2	87.9	179.76	6.5	2,968.3	112.2	59.1	53.13	2.112	
13,600.0	10,782.4	13,715.4	10,894.7	91.0	90.7	179.76	6.5	3,068.3	112.3	58.0	54.29	2.069	
13,700.0	10,783.3	13,815.4	10,895.7	93.9	93.6	179.76	6.5	3,168.3	112.4	56.9	55.46	2.026	
13,800.0	10,784.2	13,915.4	10,896.7	96.7	96.4	179.76	6.5	3,268.3	112.5	55.8	56.65	1.985	
13,900.0	10,785.1	14,015.4	10,897.6	99.5	99.2	179.76	6.5	3,368.3	112.6	54.7	57.86	1.946	
14,000.0	10,785.9	14,115.4	10,898.6	102.4	102.1	179.76	6.5	3,468.3	112.7	53.6	59.07	1.907	
14,100.0	10,786.8	14,215.4	10,899.5	105.2	104.9	179.76	6.5	3,568.3	112.7	52.4	60.30	1.870	
14,200.0	10,787.7	14,315.4	10,900.5	108.1	107.8	179.76	6.5	3,668.3	112.8	51.3	61.53	1.834	
14,300.0	10,788.5	14,415.4	10,901.5	110.9	110.7	179.76	6.5	3,768.3	112.9	50.1	62.78	1.799	
14,400.0	10,789.4	14,515.4	10,902.4	113.8	113.5	179.76	6.5	3,868.3	113.0	49.0	64.03	1.765	
14,500.0	10,790.3	14,615.4	10,903.4	116.7	116.4	179.76	6.5	3,968.3	113.1	47.8	65.30	1.732	
14,600.0	10,791.2	14,715.4	10,904.3	119.5	119.3	179.76	6.5	4,068.3	113.2	46.6	66.57	1.700	
14,700.0	10,792.0	14,815.4	10,905.3	122.4	122.1	179.76	6.5	4,168.2	113.3	45.4	67.85	1.669	
14,800.0	10,792.9	14,915.4	10,906.3	125.3	125.0	179.76	6.5	4,268.2	113.4	44.2	69.14	1.639	
14,900.0	10,793.8	15,015.4	10,907.2	128.2	127.9	179.76	6.5	4,368.2	113.4	43.0	70.44	1.611	
15,000.0	10,794.7	15,115.4	10,908.2	131.0	130.8	179.76	6.5	4,468.2	113.5	41.8	71.74	1.583	
15,100.0	10,795.5	15,215.4	10,909.1	133.9	133.7	179.76	6.5	4,568.2	113.6	40.6	73.05	1.555	
15,200.0	10,796.4	15,315.4	10,910.1	136.8	136.5	179.76	6.5	4,668.2	113.7	39.3	74.36	1.529	
15,300.0	10,797.3	15,415.4	10,911.1	139.7	139.4	179.76	6.5	4,768.2	113.8	38.1	75.68	1.504	
15,400.0	10,798.1	15,515.4	10,912.0	142.6	142.3	179.76	6.5	4,868.2	113.9	36.9	77.00	1.479 Level 3	Level 3
15,500.0	10,799.0	15,615.4	10,913.0	145.5	145.2	179.76	6.5	4,968.2	114.0	35.6	78.33	1.455 Level 3	Level 3
15,600.0	10,799.9	15,715.4	10,913.9	148.3	148.1	179.76	6.5	5,068.2	114.1	34.4	79.67	1.432 Level 3	Level 3
15,700.0	10,800.8	15,815.4	10,914.9	151.2	151.0	179.76	6.5	5,168.2	114.1	33.1	81.01	1.409 Level 3	Level 3
15,800.0	10,801.6	15,915.4	10,915.9	154.1	153.9	179.76	6.5	5,268.2	114.2	31.9	82.35	1.387 Level 3	Level 3
15,900.0	10,802.5	16,015.4	10,916.8	157.0	156.8	179.76	6.5	5,368.2	114.3	30.6	83.70	1.366 Level 3	Level 3
16,000.0	10,803.4	16,115.4	10,917.8	159.9	159.7	179.76	6.5	5,468.2	114.4	29.3	85.05	1.345 Level 3	Level 3
16,100.0	10,804.3	16,215.4	10,918.7	162.8	162.6	179.76	6.5	5,568.2	114.5	28.1	86.41	1.325 Level 3	Level 3
16,200.0	10,805.1	16,315.4	10,919.7	165.7	165.5	179.76	6.5	5,668.2	114.6	26.8	87.76	1.305 Level 3	Level 3
16,300.0	10,806.0	16,415.4	10,920.7	168.6	168.4	179.77	6.5	5,768.2	114.7	25.5	89.13	1.286 Level 3	Level 3
16,400.0	10,806.9	16,515.4	10,921.6	171.5	171.3	179.77	6.5	5,868.2	114.7	24.3	90.49	1.268 Level 3	Level 3
16,500.0	10,807.7	16,615.4	10,922.6	174.4	174.2	179.77	6.5	5,968.2	114.8	23.0	91.86	1.250 Level 3	Level 3
16,600.0	10,808.6	16,715.4	10,923.5	177.3	177.1	179.77	6.5	6,068.2	114.9	21.7	93.23	1.233 Level 2	
16,700.0	10,809.5	16,815.4	10,924.5	180.2	180.0	179.77	6.5	6,168.2	115.0	20.4	94.61	1.216 Level 2	
16,800.0	10,810.4	16,915.4	10,925.5	183.1	182.9	179.77	6.5	6,268.2	115.1	19.1	95.99	1.199 Level 2	
16,900.0	10,811.2	17,015.4	10,926.4	186.0	185.8	179.77	6.5	6,368.1	115.2	17.8	97.37	1.183 Level 2	
17,000.0	10,812.1	17,115.4	10,927.4	188.9	188.7	179.77	6.5	6,468.1	115.3	16.5	98.75	1.167 Level 2	
17,100.0	10,813.0	17,215.4	10,928.3	191.8	191.6	179.77	6.5	6,568.1	115.4	15.2	100.13	1.152 Level 2	
17,200.0	10,813.9	17,315.4	10,929.3	194.7	194.5	179.77	6.5	6,668.1	115.4	13.9	101.52	1.137 Level 2	
17,300.0	10,814.7	17,415.4	10,930.3	197.6	197.4	179.77	6.5	6,768.1	115.5	12.6	102.91	1.123 Level 2	
17,400.0	10,815.6	17,515.4	10,931.2	200.5	200.3	179.77	6.5	6,868.1	115.6	11.3	104.30	1.109 Level 2	
17,500.0	10,816.5	17,615.4	10,932.2	203.5	203.2	179.77	6.5	6,968.1	115.7	10.0	105.70	1.095 Level 2	
17,600.0	10,817.3	17,715.4	10,933.1	206.4	206.2	179.77	6.5	7,068.1	115.8	8.7	107.09	1.081 Level 2	
17,700.0	10,818.2	17,815.4	10,934.1	209.3	209.1	179.77	6.5	7,168.1	115.9	7.4	108.49	1.068 Level 2	
17,800.0	10,819.1	17,915.4	10,935.1	212.2	212.0	179.77	6.5	7,268.1	116.0	6.1	109.89	1.055 Level 2	
17,900.0	10,820.0	18,015.4	10,936.0	215.1	214.9	179.77	6.5	7,368.1	116.1	4.8	111.29	1.043 Level 2	
18,000.0	10,820.8	18,115.4	10,937.0	218.0	217.8	179.77	6.5	7,468.1	116.1	3.5	112.69	1.031 Level 2	
18,100.0	10,821.7	18,215.4	10,937.9	220.9	220.7	179.77	6.5	7,568.1	116.2	2.1	114.10	1.019 Level 2	
18,200.0	10,822.6	18,315.4	10,938.9	223.8	223.6	179.77	6.5	7,668.1	116.3	0.8	115.50	1.007 Level 2	

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Oasis Petroleum

Anticollision Report

Company:	Oasis	Local Co-ordinate Reference:	Well Lewis Federal 5300 11-31 3B
Project:	Indian Hills	TVD Reference:	WELL @ 2134.0usft (Original Well Elev)
Reference Site:	153N-100W-31/32	MD Reference:	WELL @ 2134.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Lewis Federal 5300 11-31 3B	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Lewis Federal 5300 11-31 3B	Database:	OpenWellsCompass - EDM Prod
Reference Design:	Plan #1	Offset TVD Reference:	Offset Datum

Offset Design												Offset Site Error:	0.0 usft
Survey Program: 0-MWD												Offset Well Error:	0.0 usft
Reference		Offset		Semi Major Axis			Distance						
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
18,300.0	10,823.5	18,415.4	10,939.9	226.7	226.5	179.77	6.5	7,768.1	116.4	-0.5	116.91	0.996	Level 1
18,400.0	10,824.3	18,515.4	10,940.8	229.7	229.5	179.77	6.5	7,868.1	116.5	-1.8	118.32	0.985	Level 1
18,500.0	10,825.2	18,615.4	10,941.8	232.6	232.4	179.77	6.5	7,968.1	116.6	-3.1	119.73	0.974	Level 1
18,600.0	10,826.1	18,715.4	10,942.7	235.5	235.3	179.77	6.5	8,068.1	116.7	-4.5	121.14	0.963	Level 1
18,700.0	10,826.9	18,815.4	10,943.7	238.4	238.2	179.77	6.5	8,168.1	116.8	-5.8	122.56	0.953	Level 1
18,800.0	10,827.8	18,915.4	10,944.7	241.3	241.1	179.77	6.5	8,268.1	116.8	-7.1	123.97	0.943	Level 1
18,900.0	10,828.7	19,015.4	10,945.6	244.2	244.0	179.77	6.5	8,368.1	116.9	-8.5	125.38	0.933	Level 1
19,000.0	10,829.6	19,115.4	10,946.6	247.1	246.9	179.77	6.5	8,468.0	117.0	-9.8	126.80	0.923	Level 1
19,100.0	10,830.4	19,215.4	10,947.5	250.1	249.9	179.77	6.5	8,568.0	117.1	-11.1	128.22	0.913	Level 1
19,200.0	10,831.3	19,315.4	10,948.5	253.0	252.8	179.77	6.5	8,668.0	117.2	-12.4	129.64	0.904	Level 1
19,300.0	10,832.2	19,415.4	10,949.5	255.9	255.7	179.77	6.5	8,768.0	117.3	-13.8	131.06	0.895	Level 1
19,400.0	10,833.1	19,515.4	10,950.4	258.8	258.6	179.77	6.5	8,868.0	117.4	-15.1	132.48	0.886	Level 1
19,500.0	10,833.9	19,615.4	10,951.4	261.7	261.5	179.77	6.5	8,968.0	117.5	-16.4	133.90	0.877	Level 1
19,600.0	10,834.8	19,715.4	10,952.3	264.6	264.4	179.77	6.5	9,068.0	117.5	-17.8	135.32	0.869	Level 1
19,700.0	10,835.7	19,815.4	10,953.3	267.6	267.4	179.77	6.5	9,168.0	117.6	-19.1	136.75	0.860	Level 1
19,800.0	10,836.5	19,915.4	10,954.3	270.5	270.3	179.77	6.5	9,268.0	117.7	-20.5	138.17	0.852	Level 1
19,900.0	10,837.4	20,015.4	10,955.2	273.4	273.2	179.77	6.5	9,368.0	117.8	-21.8	139.60	0.844	Level 1
20,000.0	10,838.3	20,115.4	10,956.2	276.3	276.1	179.77	6.5	9,468.0	117.9	-23.1	141.02	0.836	Level 1
20,100.0	10,839.2	20,215.4	10,957.1	279.2	279.0	179.77	6.5	9,568.0	118.0	-24.5	142.45	0.828	Level 1
20,200.0	10,840.0	20,315.4	10,958.1	282.1	282.0	179.77	6.5	9,668.0	118.1	-25.8	143.88	0.821	Level 1
20,300.0	10,840.9	20,415.4	10,959.1	285.1	284.9	179.77	6.5	9,768.0	118.2	-27.2	145.31	0.813	Level 1
20,400.0	10,841.8	20,515.4	10,960.0	288.0	287.8	179.77	6.5	9,868.0	118.2	-28.5	146.74	0.806	Level 1
20,500.0	10,842.7	20,615.4	10,961.0	290.9	290.7	179.77	6.5	9,968.0	118.3	-29.8	148.17	0.799	Level 1
20,600.0	10,843.5	20,715.4	10,961.9	293.8	293.6	179.77	6.5	10,068.0	118.4	-31.2	149.60	0.792	Level 1
20,640.9	10,843.9	20,756.3	10,962.3	294.6	294.8	179.77	6.5	10,108.9	118.5	-31.6	150.04	0.789	Level 1, ES, SF

Oasis Petroleum

Anticollision Report

Company:	Oasis	Local Co-ordinate Reference:	Well Lewis Federal 5300 11-31 3B
Project:	Indian Hills	TVD Reference:	WELL @ 2134.0usft (Original Well Elev)
Reference Site:	153N-100W-31/32	MD Reference:	WELL @ 2134.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Lewis Federal 5300 11-31 3B	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Lewis Federal 5300 11-31 3B	Database:	OpenWellsCompass - EDM Prod
Reference Design:	Plan #1	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-31/32 - Lewis Federal 5300 11-31 5T - Lewis Federal 5300 11-31 5T - Plan #1												Offset Site Error:	0.0 usft
Survey Program: 0-MWD												Offset Well Error:	0.0 usft
Reference		Offset		Semi Major Axis		Distance							Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	
0.0	0.0	0.0	0.0	0.0	0.0	-174.69	-65.9	-6.1	66.1				
100.0	100.0	100.0	100.0	0.1	0.1	-174.69	-65.9	-6.1	66.1	66.0	0.18	377.290	
200.0	200.0	200.0	200.0	0.3	0.3	-174.69	-65.9	-6.1	66.1	65.5	0.62	105.859	
300.0	300.0	300.0	300.0	0.5	0.5	-174.69	-65.9	-6.1	66.1	65.1	1.07	61.566	
400.0	400.0	400.0	400.0	0.8	0.8	-174.69	-65.9	-6.1	66.1	64.6	1.52	43.405	
500.0	500.0	500.0	500.0	1.0	1.0	-174.69	-65.9	-6.1	66.1	64.2	1.97	33.518	
600.0	600.0	600.0	600.0	1.2	1.2	-174.69	-65.9	-6.1	66.1	63.7	2.42	27.299	
700.0	700.0	700.0	700.0	1.4	1.4	-174.69	-65.9	-6.1	66.1	63.3	2.87	23.027	
800.0	800.0	800.0	800.0	1.7	1.7	-174.69	-65.9	-6.1	66.1	62.8	3.32	19.911	
900.0	900.0	900.0	900.0	1.9	1.9	-174.69	-65.9	-6.1	66.1	62.4	3.77	17.538	
1,000.0	1,000.0	1,000.0	1,000.0	2.1	2.1	-174.69	-65.9	-6.1	66.1	61.9	4.22	15.670	
1,100.0	1,100.0	1,100.0	1,100.0	2.3	2.3	-174.69	-65.9	-6.1	66.1	61.5	4.67	14.162	
1,200.0	1,200.0	1,200.0	1,200.0	2.6	2.6	-174.69	-65.9	-6.1	66.1	61.0	5.12	12.919	
1,300.0	1,300.0	1,300.0	1,300.0	2.8	2.8	-174.69	-65.9	-6.1	66.1	60.6	5.57	11.876	
1,400.0	1,400.0	1,400.0	1,400.0	3.0	3.0	-174.69	-65.9	-6.1	66.1	60.1	6.02	10.989	
1,500.0	1,500.0	1,500.0	1,500.0	3.2	3.2	-174.69	-65.9	-6.1	66.1	59.7	6.47	10.225	
1,600.0	1,600.0	1,600.0	1,600.0	3.5	3.5	-174.69	-65.9	-6.1	66.1	59.2	6.92	9.561	
1,700.0	1,700.0	1,700.0	1,700.0	3.7	3.7	-174.69	-65.9	-6.1	66.1	58.8	7.37	8.978	
1,800.0	1,800.0	1,800.0	1,800.0	3.9	3.9	-174.69	-65.9	-6.1	66.1	58.3	7.82	8.461	
1,900.0	1,900.0	1,900.0	1,900.0	4.1	4.1	-174.69	-65.9	-6.1	66.1	57.9	8.27	8.001	
2,000.0	2,000.0	2,000.0	2,000.0	4.4	4.4	-174.69	-65.9	-6.1	66.1	57.4	8.72	7.589	
2,100.0	2,100.0	2,100.0	2,100.0	4.6	4.6	-174.69	-65.9	-6.1	66.1	57.0	9.17	7.216 CC	
2,110.0	2,110.0	2,109.5	2,109.5	4.6	4.6	140.32	-65.9	-6.1	66.2	57.0	9.21	7.193	
2,200.0	2,200.0	2,199.4	2,199.4	4.8	4.8	140.69	-66.7	-6.1	67.6	58.1	9.55	7.077	
2,300.0	2,300.0	2,299.4	2,299.4	5.0	4.9	141.07	-67.6	-6.1	69.2	59.2	9.94	6.955	
2,400.0	2,400.0	2,399.4	2,399.4	5.2	5.1	141.44	-68.4	-6.1	70.7	60.4	10.34	6.840	
2,500.0	2,500.0	2,499.4	2,499.4	5.5	5.3	141.80	-69.3	-6.1	72.3	61.5	10.74	6.730	
2,600.0	2,600.0	2,599.4	2,599.3	5.7	5.5	142.13	-70.2	-6.1	73.8	62.7	11.14	6.626	
2,700.0	2,700.0	2,699.3	2,699.3	5.9	5.7	142.46	-71.0	-6.1	75.4	63.8	11.54	6.528	
2,800.0	2,800.0	2,799.3	2,799.3	6.1	5.8	142.77	-71.9	-6.1	76.9	65.0	11.95	6.435	
2,900.0	2,900.0	2,899.3	2,899.3	6.3	6.0	143.07	-72.8	-6.1	78.5	66.1	12.37	6.347	
3,000.0	3,000.0	2,999.3	2,999.3	6.6	6.2	143.36	-73.7	-6.1	80.0	67.3	12.78	6.263	
3,100.0	3,100.0	3,099.3	3,099.3	6.8	6.4	143.63	-74.5	-6.1	81.6	68.4	13.20	6.184	
3,200.0	3,200.0	3,199.3	3,199.2	7.0	6.6	143.90	-75.4	-6.1	83.2	69.6	13.62	6.109	
3,300.0	3,300.0	3,299.3	3,299.2	7.2	6.8	144.16	-76.3	-6.1	84.7	70.7	14.04	6.037	
3,400.0	3,400.0	3,399.3	3,399.2	7.4	7.0	144.40	-77.2	-6.1	86.3	71.9	14.46	5.970	
3,500.0	3,499.9	3,499.2	3,499.2	7.7	7.2	144.64	-78.0	-6.1	87.9	73.0	14.88	5.905	
3,600.0	3,599.9	3,599.2	3,599.2	7.9	7.4	144.87	-78.9	-6.1	89.5	74.1	15.31	5.844	
3,700.0	3,699.9	3,699.2	3,699.2	8.1	7.6	145.09	-79.8	-6.1	91.0	75.3	15.74	5.785	
3,800.0	3,799.9	3,799.2	3,799.1	8.3	7.8	145.31	-80.6	-6.1	92.6	76.4	16.16	5.729	
3,900.0	3,899.9	3,899.2	3,899.1	8.6	8.0	145.52	-81.5	-6.1	94.2	77.6	16.59	5.676	
4,000.0	3,999.9	3,999.2	3,999.1	8.8	8.3	145.72	-82.4	-6.1	95.8	78.7	17.02	5.625	
4,100.0	4,099.9	4,099.2	4,099.1	9.0	8.5	145.91	-83.3	-6.1	97.3	79.9	17.46	5.577	
4,200.0	4,199.9	4,199.2	4,199.1	9.2	8.7	146.10	-84.1	-6.1	98.9	81.0	17.89	5.530	
4,300.0	4,299.9	4,299.1	4,299.1	9.4	8.9	146.28	-85.0	-6.1	100.5	82.2	18.32	5.486	
4,400.0	4,399.9	4,399.1	4,399.0	9.7	9.1	146.45	-85.9	-6.1	102.1	83.3	18.75	5.443	
4,500.0	4,499.9	4,499.1	4,499.0	9.9	9.3	146.63	-86.8	-6.1	103.7	84.5	19.19	5.402	
4,600.0	4,599.9	4,599.1	4,599.0	10.1	9.5	146.79	-87.6	-6.1	105.2	85.6	19.62	5.363	
4,700.0	4,699.9	4,699.1	4,699.0	10.3	9.7	146.95	-88.5	-6.1	106.8	86.8	20.06	5.325	
4,800.0	4,799.9	4,799.1	4,799.0	10.6	10.0	147.11	-89.4	-6.1	108.4	87.9	20.50	5.289	
4,900.0	4,899.9	4,899.1	4,899.0	10.8	10.2	147.26	-90.2	-6.1	110.0	89.1	20.94	5.255	
5,000.0	4,999.9	4,999.0	4,998.9	11.0	10.4	147.41	-91.1	-6.1	111.6	90.2	21.37	5.221	
5,100.0	5,099.9	5,099.0	5,098.9	11.2	10.6	147.55	-92.0	-6.1	113.2	91.4	21.81	5.189	

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Oasis Petroleum

Anticollision Report

Company:	Oasis	Local Co-ordinate Reference:	Well Lewis Federal 5300 11-31 3B
Project:	Indian Hills	TVD Reference:	WELL @ 2134.0usft (Original Well Elev)
Reference Site:	153N-100W-31/32	MD Reference:	WELL @ 2134.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Lewis Federal 5300 11-31 3B	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Lewis Federal 5300 11-31 3B	Database:	OpenWellsCompass - EDM Prod
Reference Design:	Plan #1	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-31/32 - Lewis Federal 5300 11-31 5T - Lewis Federal 5300 11-31 5T - Plan #1													Offset Site Error:	0.0 usft
Survey Program: 0-MWD													Offset Well Error:	0.0 usft
Reference		Offset		Semi Major Axis		Distance							Warning	
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor		
5,200.0	5,199.9	5,199.0	5,198.9	11.5	10.8	147.69	-92.9	-6.1	114.8	92.5	22.25	5.158		
5,300.0	5,299.9	5,299.0	5,298.9	11.7	11.0	147.82	-93.7	-6.1	116.4	93.7	22.69	5.128		
5,400.0	5,399.9	5,399.0	5,398.9	11.9	11.2	147.96	-94.6	-6.1	117.9	94.8	23.13	5.100		
5,500.0	5,499.9	5,499.0	5,498.9	12.1	11.5	148.08	-95.5	-6.1	119.5	96.0	23.57	5.072		
5,600.0	5,599.9	5,599.0	5,598.8	12.3	11.7	148.21	-96.4	-6.1	121.1	97.1	24.01	5.045		
5,700.0	5,699.9	5,699.0	5,698.8	12.6	11.9	148.33	-97.2	-6.1	122.7	98.3	24.45	5.019		
5,800.0	5,799.9	5,798.9	5,798.8	12.8	12.1	148.45	-98.1	-6.1	124.3	99.4	24.89	4.994		
5,900.0	5,899.9	5,898.9	5,898.8	13.0	12.3	148.56	-99.0	-6.1	125.9	100.6	25.33	4.970		
6,000.0	5,999.9	5,998.9	5,998.8	13.2	12.6	148.68	-99.8	-6.1	127.5	101.7	25.77	4.947		
6,100.0	6,099.8	6,098.9	6,098.8	13.5	12.8	148.79	-100.7	-6.1	129.1	102.9	26.22	4.924		
6,151.5	6,151.3	6,150.4	6,150.2	13.6	12.9	148.84	-101.2	-6.1	129.9	103.5	26.44	4.913		
6,161.5	6,161.3	6,160.4	6,160.2	13.6	12.9	-166.15	-101.3	-6.1	130.0	103.5	26.49	4.909		
6,200.0	6,199.8	6,198.9	6,198.7	13.7	13.0	-166.19	-101.6	-6.1	130.4	103.7	26.65	4.891		
6,300.0	6,299.8	6,298.9	6,298.7	13.9	13.2	-166.28	-102.5	-6.1	131.2	104.1	27.09	4.843		
6,400.0	6,399.8	6,398.9	6,398.7	14.1	13.4	-166.37	-103.3	-6.1	132.1	104.5	27.53	4.796		
6,500.0	6,499.8	6,498.9	6,498.7	14.3	13.7	-166.46	-104.2	-6.1	132.9	104.9	27.97	4.751		
6,600.0	6,599.8	6,598.9	6,598.7	14.6	13.9	-166.55	-105.1	-6.1	133.8	105.3	28.41	4.707		
6,700.0	6,699.8	6,698.9	6,698.7	14.8	14.1	-166.63	-106.0	-6.1	134.6	105.7	28.86	4.664		
6,800.0	6,799.8	6,798.9	6,798.7	15.0	14.3	-166.72	-106.8	-6.1	135.5	106.2	29.30	4.623		
6,900.0	6,899.8	6,898.9	6,898.7	15.2	14.5	-166.80	-107.7	-6.1	136.3	106.6	29.74	4.583		
7,000.0	6,999.8	6,998.9	6,998.7	15.5	14.8	-166.89	-108.6	-6.1	137.2	107.0	30.18	4.544		
7,100.0	7,099.8	7,098.9	7,098.7	15.7	15.0	-166.97	-109.4	-6.1	138.0	107.4	30.63	4.506		
7,200.0	7,199.8	7,198.9	7,198.7	15.9	15.2	-167.05	-110.3	-6.1	138.9	107.8	31.07	4.469		
7,300.0	7,299.8	7,298.9	7,298.7	16.1	15.4	-167.13	-111.2	-6.1	139.7	108.2	31.51	4.433		
7,400.0	7,399.8	7,398.9	7,398.6	16.3	15.6	-167.21	-112.1	-6.1	140.6	108.6	31.96	4.398		
7,500.0	7,499.8	7,498.8	7,498.6	16.6	15.9	-167.29	-112.9	-6.1	141.4	109.0	32.40	4.364		
7,600.0	7,599.8	7,598.8	7,598.6	16.8	16.1	-167.36	-113.8	-6.1	142.3	109.4	32.84	4.331		
7,700.0	7,699.8	7,698.8	7,698.6	17.0	16.3	-167.44	-114.7	-6.1	143.1	109.8	33.29	4.299		
7,800.0	7,799.8	7,798.8	7,798.6	17.2	16.5	-167.52	-115.5	-6.1	144.0	110.2	33.73	4.268		
7,900.0	7,899.8	7,900.1	7,899.8	17.5	16.7	-167.54	-115.9	-6.1	144.3	110.1	34.17	4.222		
8,000.0	7,999.8	8,000.1	7,999.8	17.7	16.9	-167.54	-115.9	-6.1	144.3	109.7	34.58	4.171		
8,100.0	8,099.8	8,100.1	8,099.8	17.9	17.1	-167.54	-115.9	-6.1	144.3	109.3	35.00	4.122		
8,200.0	8,199.8	8,200.1	8,199.8	18.1	17.3	-167.54	-115.9	-6.1	144.3	108.8	35.41	4.074		
8,300.0	8,299.8	8,300.1	8,299.8	18.3	17.5	-167.54	-115.9	-6.1	144.3	108.4	35.83	4.027		
8,400.0	8,399.8	8,400.1	8,399.8	18.6	17.7	-167.54	-115.9	-6.1	144.3	108.0	36.24	3.980		
8,500.0	8,499.8	8,500.1	8,499.8	18.8	17.9	-167.54	-115.9	-6.1	144.3	107.6	36.66	3.935		
8,600.0	8,599.8	8,600.1	8,599.8	19.0	18.1	-167.54	-115.9	-6.1	144.3	107.2	37.08	3.891		
8,700.0	8,699.8	8,700.1	8,699.8	19.2	18.3	-167.54	-115.9	-6.1	144.3	106.8	37.50	3.847		
8,800.0	8,799.8	8,800.1	8,799.8	19.5	18.5	-167.54	-115.9	-6.1	144.3	106.3	37.91	3.805		
8,900.0	8,899.8	8,900.1	8,899.8	19.7	18.7	-167.54	-115.9	-6.1	144.3	105.9	38.33	3.763		
9,000.0	8,999.8	9,000.1	8,999.8	19.9	18.9	-167.54	-115.9	-6.1	144.3	105.5	38.75	3.722		
9,100.0	9,099.8	9,100.1	9,099.8	20.1	19.1	-167.54	-115.9	-6.1	144.3	105.1	39.18	3.682		
9,200.0	9,199.8	9,200.1	9,199.8	20.3	19.3	-167.54	-115.9	-6.1	144.3	104.7	39.60	3.643		
9,300.0	9,299.8	9,300.1	9,299.8	20.6	19.5	-167.54	-115.9	-6.1	144.3	104.2	40.02	3.605		
9,400.0	9,399.8	9,400.1	9,399.8	20.8	19.7	-167.54	-115.9	-6.1	144.3	103.8	40.44	3.567		
9,500.0	9,499.8	9,500.1	9,499.8	21.0	19.9	-167.54	-115.9	-6.1	144.3	103.4	40.87	3.530		
9,600.0	9,599.8	9,600.1	9,599.8	21.2	20.1	-167.54	-115.9	-6.1	144.3	103.0	41.29	3.494		
9,700.0	9,699.8	9,700.1	9,699.8	21.5	20.3	-167.54	-115.9	-6.1	144.3	102.5	41.72	3.458		
9,800.0	9,799.8	9,800.1	9,799.8	21.7	20.5	-167.54	-115.9	-6.1	144.3	102.1	42.14	3.423		
9,900.0	9,899.8	9,900.1	9,899.8	21.9	20.7	-167.54	-115.9	-6.1	144.3	101.7	42.57	3.389		
10,000.0	9,999.8	10,000.1	9,999.8	22.1	20.9	-167.54	-115.9	-6.1	144.3	101.3	42.99	3.355		
10,100.0	10,099.8	10,100.1	10,099.8	22.4	21.1	-167.54	-115.9	-6.1	144.3	100.8	43.42	3.322		
10,200.0	10,199.8	10,200.1	10,199.8	22.6	21.3	-167.54	-115.9	-6.1	144.3	100.4	43.85	3.290		

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Oasis Petroleum

Anticollision Report

Company:	Oasis	Local Co-ordinate Reference:	Well Lewis Federal 5300 11-31 3B
Project:	Indian Hills	TVD Reference:	WELL @ 2134.0usft (Original Well Elev)
Reference Site:	153N-100W-31/32	MD Reference:	WELL @ 2134.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Lewis Federal 5300 11-31 3B	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Lewis Federal 5300 11-31 3B	Database:	OpenWellsCompass - EDM Prod
Reference Design:	Plan #1	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-31/32 - Lewis Federal 5300 11-31 5T - Lewis Federal 5300 11-31 5T - Plan #1												Offset Site Error:	0.0 usft
Survey Program: 0-MWD												Offset Well Error:	0.0 usft
Reference		Offset		Semi Major Axis		Distance							Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	
10,282.7	10,282.5	10,282.7	10,282.5	22.8	21.5	-167.54	-115.9	-6.1	144.3	100.1	44.20	3.264	
10,300.0	10,299.8	10,300.1	10,299.8	22.8	21.5	100.59	-115.9	-6.1	144.3	100.0	44.27	3.260	
10,325.0	10,324.8	10,325.0	10,324.8	22.9	21.6	101.16	-115.9	-6.1	144.6	100.2	44.37	3.259	
10,350.0	10,349.6	10,349.8	10,349.6	22.9	21.6	102.20	-115.9	-6.1	145.2	100.7	44.47	3.265	
10,375.0	10,374.3	10,373.7	10,373.4	23.0	21.7	103.47	-116.0	-5.8	146.2	101.6	44.56	3.280	
10,400.0	10,398.7	10,397.3	10,397.1	23.0	21.7	104.68	-116.6	-4.5	147.8	103.1	44.65	3.309	
10,425.0	10,422.7	10,421.1	10,420.7	23.1	21.8	105.81	-117.7	-2.0	149.9	105.2	44.73	3.352	
10,450.0	10,446.4	10,445.0	10,444.3	23.1	21.8	106.85	-119.3	1.5	152.6	107.8	44.80	3.407	
10,475.0	10,469.7	10,469.0	10,467.7	23.2	21.9	107.79	-121.4	6.1	155.9	111.1	44.86	3.476	
10,500.0	10,492.4	10,493.1	10,491.0	23.3	21.9	108.62	-124.0	11.8	159.8	114.8	44.92	3.557	
10,525.0	10,514.6	10,517.4	10,514.1	23.3	22.0	109.33	-127.1	18.6	164.1	119.1	44.97	3.649	
10,550.0	10,536.1	10,541.7	10,536.8	23.4	22.0	109.93	-130.7	26.4	169.0	124.0	45.02	3.753	
10,575.0	10,556.9	10,566.1	10,559.1	23.5	22.1	110.40	-134.7	35.4	174.3	129.3	45.07	3.868	
10,600.0	10,577.0	10,590.5	10,581.0	23.6	22.1	110.75	-139.3	45.4	180.1	135.0	45.13	3.992	
10,625.0	10,596.3	10,615.1	10,602.3	23.7	22.2	110.98	-144.3	56.4	186.4	141.2	45.19	4.124	
10,650.0	10,614.7	10,639.7	10,623.0	23.8	22.3	111.10	-149.8	68.5	193.1	147.8	45.28	4.264	
10,675.0	10,632.2	10,664.4	10,643.1	23.9	22.4	111.11	-155.7	81.6	200.1	154.7	45.38	4.410	
10,700.0	10,648.7	10,689.1	10,662.5	24.1	22.4	111.02	-162.1	95.6	207.6	162.1	45.51	4.561	
10,725.0	10,664.2	10,713.9	10,681.1	24.2	22.5	110.82	-168.9	110.6	215.4	169.7	45.68	4.715	
10,750.0	10,678.7	10,738.8	10,698.8	24.4	22.6	110.53	-176.1	126.5	223.5	177.6	45.87	4.872	
10,775.0	10,692.1	10,763.8	10,715.7	24.5	22.8	110.16	-183.7	143.2	231.9	185.8	46.11	5.029	
10,800.0	10,704.4	10,788.9	10,731.7	24.7	22.9	109.71	-191.7	160.8	240.6	194.2	46.39	5.186	
10,825.0	10,715.5	10,814.1	10,746.6	24.9	23.0	109.18	-200.1	179.2	249.6	202.8	46.72	5.342	
10,850.0	10,725.5	10,839.3	10,760.6	25.2	23.2	108.58	-208.8	198.4	258.8	211.7	47.08	5.495	
10,875.0	10,734.2	10,864.7	10,773.4	25.4	23.4	107.93	-217.9	218.3	268.1	220.6	47.50	5.645	
10,900.0	10,741.7	10,890.2	10,785.2	25.7	23.6	107.22	-227.2	238.9	277.7	229.8	47.96	5.791	
10,925.0	10,747.9	10,915.8	10,795.7	25.9	23.8	106.46	-236.9	260.2	287.5	239.0	48.46	5.931	
10,950.0	10,752.9	10,941.6	10,805.1	26.2	24.0	105.66	-246.8	282.1	297.3	248.3	49.01	6.067	
10,975.0	10,756.5	10,967.6	10,813.2	26.5	24.3	104.81	-257.1	304.6	307.3	257.7	49.59	6.196	
11,000.0	10,758.9	10,993.8	10,819.9	26.8	24.6	103.94	-267.5	327.6	317.3	267.1	50.21	6.320	
11,025.0	10,760.0	11,020.2	10,825.4	27.2	24.9	103.04	-278.2	351.1	327.4	276.5	50.86	6.437	
11,028.6	10,760.0	11,023.9	10,826.0	27.2	24.9	102.91	-279.7	354.5	328.9	277.9	50.96	6.453	
11,043.6	10,760.1	11,039.9	10,828.5	27.4	25.1	103.08	-286.3	368.8	334.9	283.6	51.30	6.528	
11,109.6	10,760.7	11,109.5	10,833.1	28.4	26.0	102.89	-315.0	432.0	361.3	308.4	52.91	6.828	
11,200.0	10,761.5	11,211.2	10,834.0	29.8	27.6	101.53	-354.9	525.5	396.1	340.4	55.77	7.103	
11,300.0	10,762.4	11,330.8	10,835.2	31.6	29.6	100.41	-395.4	638.0	429.5	370.0	59.48	7.221	
11,400.0	10,763.2	11,455.3	10,836.3	33.6	32.1	99.62	-430.0	757.6	456.9	393.3	63.69	7.174	
11,500.0	10,764.1	11,584.2	10,837.6	35.7	34.7	99.07	-457.5	883.5	478.1	409.7	68.36	6.993	
11,600.0	10,765.0	11,716.3	10,838.8	37.9	37.7	98.73	-476.9	1,014.1	492.6	419.1	73.44	6.707	
11,700.0	10,765.9	11,850.5	10,840.1	40.2	40.8	98.56	-487.2	1,147.9	500.2	421.4	78.82	6.346	
11,800.0	10,766.7	11,971.4	10,841.2	42.6	43.6	98.54	-488.9	1,268.7	501.4	417.4	84.06	5.965	
11,900.0	10,767.6	12,071.4	10,842.1	45.0	46.0	98.55	-488.9	1,368.7	501.5	412.5	88.93	5.639	
12,000.0	10,768.5	12,171.4	10,843.1	47.5	48.5	98.56	-488.9	1,468.7	501.5	407.6	93.90	5.340	
12,100.0	10,769.3	12,271.4	10,844.0	50.1	51.0	98.56	-488.9	1,568.7	501.5	402.5	98.97	5.067	
12,200.0	10,770.2	12,371.4	10,845.0	52.6	53.5	98.57	-488.9	1,668.7	501.5	397.4	104.11	4.817	
12,300.0	10,771.1	12,471.4	10,845.9	55.3	56.1	98.58	-488.9	1,768.7	501.5	392.2	109.32	4.588	
12,400.0	10,772.0	12,571.4	10,846.9	57.9	58.8	98.59	-488.9	1,868.7	501.5	386.9	114.58	4.377	
12,500.0	10,772.8	12,671.4	10,847.8	60.6	61.4	98.60	-488.9	1,968.7	501.5	381.6	119.89	4.183	
12,600.0	10,773.7	12,771.4	10,848.7	63.3	64.1	98.60	-488.9	2,068.7	501.5	376.3	125.25	4.004	
12,700.0	10,774.6	12,871.4	10,849.7	66.0	66.8	98.61	-488.9	2,168.7	501.5	370.9	130.64	3.839	
12,800.0	10,775.5	12,971.4	10,850.6	68.7	69.5	98.62	-488.9	2,268.7	501.6	365.5	136.06	3.686	
12,900.0	10,776.3	13,071.4	10,851.6	71.5	72.2	98.63	-488.9	2,368.7	501.6	360.0	141.52	3.544	
13,000.0	10,777.2	13,171.4	10,852.5	74.2	75.0	98.64	-488.9	2,468.7	501.6	354.6	147.00	3.412	

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Oasis Petroleum

Anticollision Report

Company:	Oasis	Local Co-ordinate Reference:	Well Lewis Federal 5300 11-31 3B
Project:	Indian Hills	TVD Reference:	WELL @ 2134.0usft (Original Well Elev)
Reference Site:	153N-100W-31/32	MD Reference:	WELL @ 2134.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Lewis Federal 5300 11-31 3B	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Lewis Federal 5300 11-31 3B	Database:	OpenWellsCompass - EDM Prod
Reference Design:	Plan #1	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-31/32 - Lewis Federal 5300 11-31 5T - Lewis Federal 5300 11-31 5T - Plan #1													Offset Site Error:	0.0 usft
Survey Program: 0-MWD													Offset Well Error:	0.0 usft
Reference		Offset		Semi Major Axis		Distance								Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	Offset Wellbore Centre +E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor		
13,100.0	10,778.1	13,271.4	10,853.5	77.0	77.8	98.64	-488.9	2,568.7	501.6	349.1	152.50	3.289		
13,200.0	10,778.9	13,371.4	10,854.4	79.8	80.5	98.65	-488.9	2,668.7	501.6	343.6	158.03	3.174		
13,300.0	10,779.8	13,471.4	10,855.3	82.6	83.3	98.66	-488.9	2,768.7	501.6	338.0	163.57	3.067		
13,400.0	10,780.7	13,571.4	10,856.3	85.4	86.1	98.67	-488.9	2,868.7	501.6	332.5	169.13	2.966		
13,500.0	10,781.6	13,671.4	10,857.2	88.2	88.9	98.67	-488.9	2,968.7	501.6	326.9	174.70	2.871		
13,600.0	10,782.4	13,771.4	10,858.2	91.0	91.7	98.68	-488.9	3,068.7	501.6	321.3	180.29	2.782		
13,700.0	10,783.3	13,871.4	10,859.1	93.9	94.5	98.69	-488.9	3,168.7	501.6	315.8	185.89	2.699		
13,800.0	10,784.2	13,971.4	10,860.0	96.7	97.4	98.70	-488.9	3,268.7	501.7	310.2	191.51	2.620		
13,900.0	10,785.1	14,071.4	10,861.0	99.5	100.2	98.71	-488.9	3,368.7	501.7	304.5	197.13	2.545		
14,000.0	10,785.9	14,171.4	10,861.9	102.4	103.0	98.71	-488.9	3,468.6	501.7	298.9	202.76	2.474		
14,100.0	10,786.8	14,271.4	10,862.9	105.2	105.9	98.72	-488.9	3,568.6	501.7	293.3	208.40	2.407		
14,200.0	10,787.7	14,371.4	10,863.8	108.1	108.7	98.73	-488.9	3,668.6	501.7	287.7	214.05	2.344		
14,300.0	10,788.5	14,471.4	10,864.8	110.9	111.6	98.74	-488.9	3,768.6	501.7	282.0	219.71	2.284		
14,400.0	10,789.4	14,571.4	10,865.7	113.8	114.4	98.75	-488.9	3,868.6	501.7	276.4	225.37	2.226		
14,500.0	10,790.3	14,671.4	10,866.6	116.7	117.3	98.75	-488.9	3,968.6	501.7	270.7	231.04	2.172		
14,600.0	10,791.2	14,771.4	10,867.6	119.5	120.2	98.76	-488.9	4,068.6	501.7	265.0	236.71	2.120		
14,700.0	10,792.0	14,871.4	10,868.5	122.4	123.0	98.77	-488.9	4,168.6	501.8	259.4	242.39	2.070		
14,800.0	10,792.9	14,971.4	10,869.5	125.3	125.9	98.78	-488.9	4,268.6	501.8	253.7	248.08	2.023		
14,900.0	10,793.8	15,071.4	10,870.4	128.2	128.8	98.79	-488.9	4,368.6	501.8	248.0	253.77	1.977		
15,000.0	10,794.7	15,171.4	10,871.4	131.0	131.6	98.79	-488.9	4,468.6	501.8	242.3	259.46	1.934		
15,100.0	10,795.5	15,271.4	10,872.3	133.9	134.5	98.80	-488.9	4,568.6	501.8	236.6	265.16	1.892		
15,200.0	10,796.4	15,371.4	10,873.2	136.8	137.4	98.81	-488.9	4,668.6	501.8	230.9	270.86	1.853		
15,300.0	10,797.3	15,471.4	10,874.2	139.7	140.3	98.82	-488.9	4,768.6	501.8	225.3	276.57	1.814		
15,400.0	10,798.1	15,571.4	10,875.1	142.6	143.1	98.82	-488.9	4,868.6	501.8	219.6	282.27	1.778		
15,500.0	10,799.0	15,671.4	10,876.1	145.5	146.0	98.83	-488.9	4,968.6	501.8	213.9	287.99	1.743		
15,600.0	10,799.9	15,771.4	10,877.0	148.3	148.9	98.84	-488.9	5,068.6	501.9	208.2	293.70	1.709		
15,700.0	10,800.8	15,871.4	10,878.0	151.2	151.8	98.85	-488.9	5,168.6	501.9	202.4	299.42	1.676		
15,800.0	10,801.6	15,971.4	10,878.9	154.1	154.7	98.86	-488.9	5,268.6	501.9	196.7	305.14	1.645		
15,900.0	10,802.5	16,071.4	10,879.8	157.0	157.6	98.86	-488.9	5,368.6	501.9	191.0	310.86	1.615		
16,000.0	10,803.4	16,171.4	10,880.8	159.9	160.5	98.87	-488.9	5,468.6	501.9	185.3	316.58	1.585		
16,100.0	10,804.3	16,271.4	10,881.7	162.8	163.4	98.88	-488.9	5,568.6	501.9	179.6	322.31	1.557		
16,200.0	10,805.1	16,371.4	10,882.7	165.7	166.3	98.89	-488.9	5,668.5	501.9	173.9	328.04	1.530		
16,300.0	10,806.0	16,471.4	10,883.6	168.6	169.2	98.90	-488.9	5,768.5	501.9	168.2	333.77	1.504		
16,400.0	10,806.9	16,571.4	10,884.6	171.5	172.1	98.90	-488.9	5,868.5	501.9	162.4	339.50	1.478 Level 3	Level 3	
16,500.0	10,807.7	16,671.4	10,885.5	174.4	175.0	98.91	-488.9	5,968.5	501.9	156.7	345.23	1.454 Level 3	Level 3	
16,600.0	10,808.6	16,771.4	10,886.4	177.3	177.9	98.92	-488.9	6,068.5	502.0	151.0	350.97	1.430 Level 3	Level 3	
16,700.0	10,809.5	16,871.4	10,887.4	180.2	180.8	98.93	-488.9	6,168.5	502.0	145.3	356.70	1.407 Level 3	Level 3	
16,800.0	10,810.4	16,971.4	10,888.3	183.1	183.7	98.93	-488.9	6,268.5	502.0	139.5	362.44	1.385 Level 3	Level 3	
16,900.0	10,811.2	17,071.4	10,889.3	186.0	186.6	98.94	-488.9	6,368.5	502.0	133.8	368.18	1.363 Level 3	Level 3	
17,000.0	10,812.1	17,171.4	10,890.2	188.9	189.5	98.95	-488.9	6,468.5	502.0	128.1	373.92	1.343 Level 3	Level 3	
17,100.0	10,813.0	17,271.4	10,891.2	191.8	192.4	98.96	-488.9	6,568.5	502.0	122.4	379.66	1.322 Level 3	Level 3	
17,200.0	10,813.9	17,371.4	10,892.1	194.7	195.3	98.97	-488.9	6,668.5	502.0	116.6	385.41	1.303 Level 3	Level 3	
17,300.0	10,814.7	17,471.4	10,893.0	197.6	198.2	98.97	-488.9	6,768.5	502.0	110.9	391.15	1.283 Level 3	Level 3	
17,400.0	10,815.6	17,571.4	10,894.0	200.5	201.1	98.98	-488.9	6,868.5	502.0	105.2	396.89	1.265 Level 3	Level 3	
17,500.0	10,816.5	17,671.4	10,894.9	203.5	204.0	98.99	-488.9	6,968.5	502.1	99.4	402.64	1.247 Level 2		
17,600.0	10,817.3	17,771.4	10,895.9	206.4	206.9	99.00	-488.9	7,068.5	502.1	93.7	408.39	1.229 Level 2		
17,700.0	10,818.2	17,871.4	10,896.8	209.3	209.8	99.01	-488.9	7,168.5	502.1	87.9	414.13	1.212 Level 2		
17,800.0	10,819.1	17,971.4	10,897.7	212.2	212.7	99.01	-488.9	7,268.5	502.1	82.2	419.88	1.196 Level 2		
17,900.0	10,820.0	18,071.4	10,898.7	215.1	215.6	99.02	-488.9	7,368.5	502.1	76.5	425.63	1.180 Level 2		
18,000.0	10,820.8	18,171.4	10,899.6	218.0	218.5	99.03	-488.9	7,468.5	502.1	70.7	431.38	1.164 Level 2		
18,100.0	10,821.7	18,271.4	10,900.6	220.9	221.4	99.04	-488.9	7,568.5	502.1	65.0	437.13	1.149 Level 2		
18,200.0	10,822.6	18,371.4	10,901.5	223.8	224.3	99.04	-488.9	7,668.5	502.1	59.3	442.88	1.134 Level 2		
18,300.0	10,823.5	18,471.4	10,902.5	226.7	227.3	99.05	-488.9	7,768.5	502.1	53.5	448.63	1.119 Level 2		

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Oasis Petroleum

Anticollision Report

Company:	Oasis	Local Co-ordinate Reference:	Well Lewis Federal 5300 11-31 3B
Project:	Indian Hills	TVD Reference:	WELL @ 2134.0usft (Original Well Elev)
Reference Site:	153N-100W-31/32	MD Reference:	WELL @ 2134.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Lewis Federal 5300 11-31 3B	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Lewis Federal 5300 11-31 3B	Database:	OpenWellsCompass - EDM Prod
Reference Design:	Plan #1	Offset TVD Reference:	Offset Datum

Offset Design												Offset Site Error:	0.0 usft
Survey Program: 0-MWD												Offset Well Error:	0.0 usft
Reference		Offset		Semi Major Axis		Highside Toolface (°)	Distance		Minimum Separation (usft)	Separation Factor	Warning		
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)		Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)					
18,400.0	10,824.3	18,571.4	10,903.4	229.7	230.2	99.06	-488.9	7,868.4	502.2	47.8	454.39	1.105	Level 2
18,500.0	10,825.2	18,671.4	10,904.3	232.6	233.1	99.07	-488.9	7,968.4	502.2	42.0	460.14	1.091	Level 2
18,600.0	10,826.1	18,771.4	10,905.3	235.5	236.0	99.08	-488.9	8,068.4	502.2	36.3	465.89	1.078	Level 2
18,700.0	10,826.9	18,871.4	10,906.2	238.4	238.9	99.08	-488.9	8,168.4	502.2	30.5	471.65	1.065	Level 2
18,800.0	10,827.8	18,971.4	10,907.2	241.3	241.8	99.09	-488.9	8,268.4	502.2	24.8	477.40	1.052	Level 2
18,900.0	10,828.7	19,071.4	10,908.1	244.2	244.7	99.10	-488.9	8,368.4	502.2	19.1	483.16	1.039	Level 2
19,000.0	10,829.6	19,171.4	10,909.1	247.1	247.6	99.11	-488.9	8,468.4	502.2	13.3	488.91	1.027	Level 2
19,100.0	10,830.4	19,271.4	10,910.0	250.1	250.6	99.12	-488.9	8,568.4	502.2	7.6	494.67	1.015	Level 2
19,200.0	10,831.3	19,371.4	10,910.9	253.0	253.5	99.12	-488.9	8,668.4	502.2	1.8	500.42	1.004	Level 2
19,300.0	10,832.2	19,471.4	10,911.9	255.9	256.4	99.13	-488.9	8,768.4	502.3	-3.9	506.18	0.992	Level 1
19,400.0	10,833.1	19,571.4	10,912.8	258.8	259.3	99.14	-488.9	8,868.4	502.3	-9.7	511.94	0.981	Level 1
19,500.0	10,833.9	19,671.4	10,913.8	261.7	262.2	99.15	-488.9	8,968.4	502.3	-15.4	517.69	0.970	Level 1
19,600.0	10,834.8	19,771.4	10,914.7	264.6	265.1	99.15	-488.9	9,068.4	502.3	-21.2	523.45	0.960	Level 1
19,700.0	10,835.7	19,871.4	10,915.7	267.6	268.1	99.16	-488.9	9,168.4	502.3	-26.9	529.21	0.949	Level 1
19,800.0	10,836.5	19,971.4	10,916.6	270.5	271.0	99.17	-488.9	9,268.4	502.3	-32.7	534.96	0.939	Level 1
19,900.0	10,837.4	20,071.4	10,917.5	273.4	273.9	99.18	-488.9	9,368.4	502.3	-38.4	540.72	0.929	Level 1
20,000.0	10,838.3	20,171.4	10,918.5	276.3	276.8	99.19	-488.9	9,468.4	502.3	-44.1	546.48	0.919	Level 1
20,100.0	10,839.2	20,271.4	10,919.4	279.2	279.7	99.19	-488.9	9,568.4	502.3	-49.9	552.24	0.910	Level 1
20,200.0	10,840.0	20,371.4	10,920.4	282.1	282.6	99.20	-488.9	9,668.4	502.4	-55.6	558.00	0.900	Level 1
20,300.0	10,840.9	20,471.4	10,921.3	285.1	285.6	99.21	-488.9	9,768.4	502.4	-61.4	563.76	0.891	Level 1
20,400.0	10,841.8	20,571.4	10,922.3	288.0	288.5	99.22	-488.9	9,868.4	502.4	-67.1	569.51	0.882	Level 1
20,500.0	10,842.7	20,671.4	10,923.2	290.9	291.4	99.23	-488.9	9,968.4	502.4	-72.9	575.27	0.873	Level 1
20,600.0	10,843.5	20,771.4	10,924.1	293.8	294.3	99.23	-488.9	10,068.4	502.4	-78.6	581.03	0.865	Level 1
20,640.9	10,843.9	20,812.3	10,924.5	294.6	295.5	99.24	-488.9	10,109.2	502.4	-80.6	582.98	0.862	Level 1, ES, SF

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Oasis Petroleum

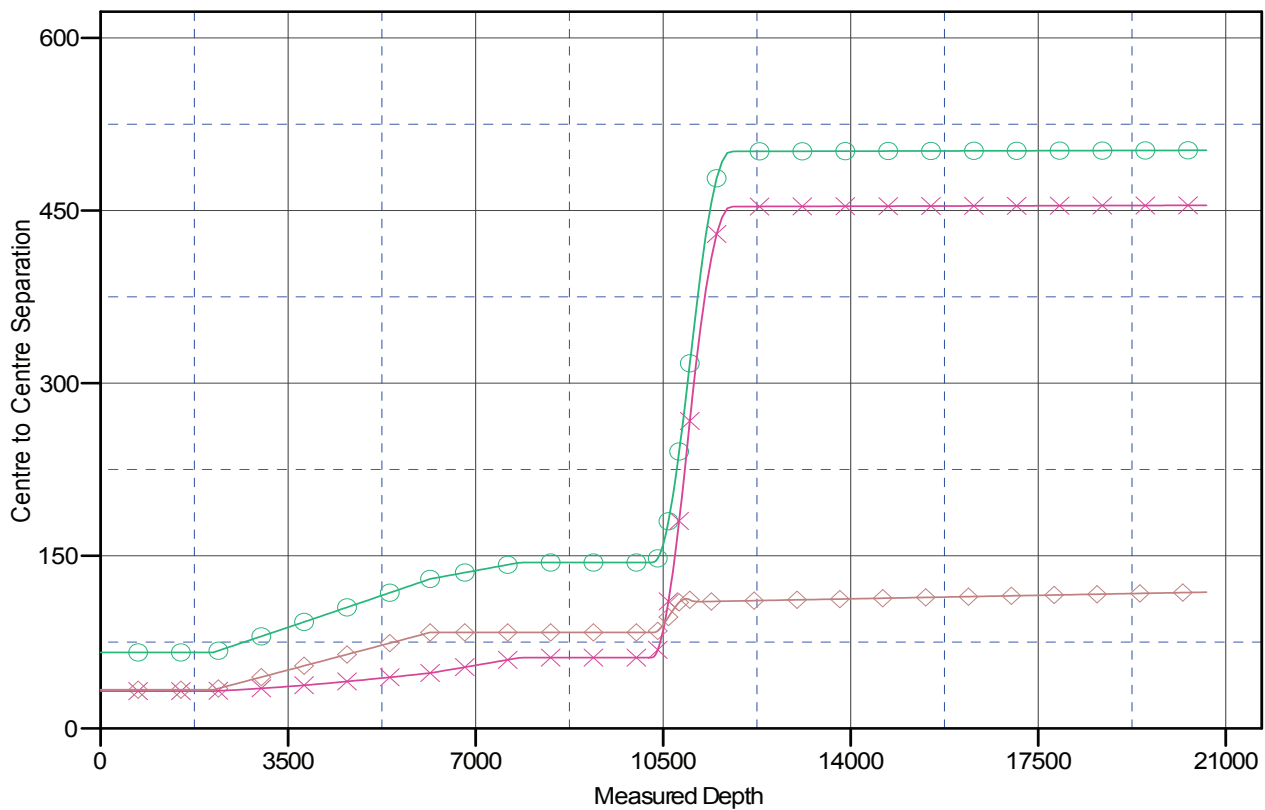
Anticollision Report

Company:	Oasis	Local Co-ordinate Reference:	Well Lewis Federal 5300 11-31 3B
Project:	Indian Hills	TVD Reference:	WELL @ 2134.0usft (Original Well Elev)
Reference Site:	153N-100W-31/32	MD Reference:	WELL @ 2134.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Lewis Federal 5300 11-31 3B	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Lewis Federal 5300 11-31 3B	Database:	OpenWellsCompass - EDM Prod
Reference Design:	Plan #1	Offset TVD Reference:	Offset Datum

Reference Depths are relative to WELL @ 2134.0usft (Original Well Ele
Offset Depths are relative to Offset Datum
Central Meridian is 100° 30' 0.000 W

Coordinates are relative to: Lewis Federal 5300 11-31 3B
Coordinate System is US State Plane 1983, North Dakota Northern Zone
Grid Convergence at Surface is: -2.31°

Ladder Plot



LEGEND

Lewis Federal 5300 11-31 1T, Plan #1 V0 Lewis Federal 5300 11-31 4T2, Lewis Federal 5300 11-31 4T2, Plan #1 V0 Lewis Federal 5300 11-31 5T, Lewis F

Oasis Petroleum

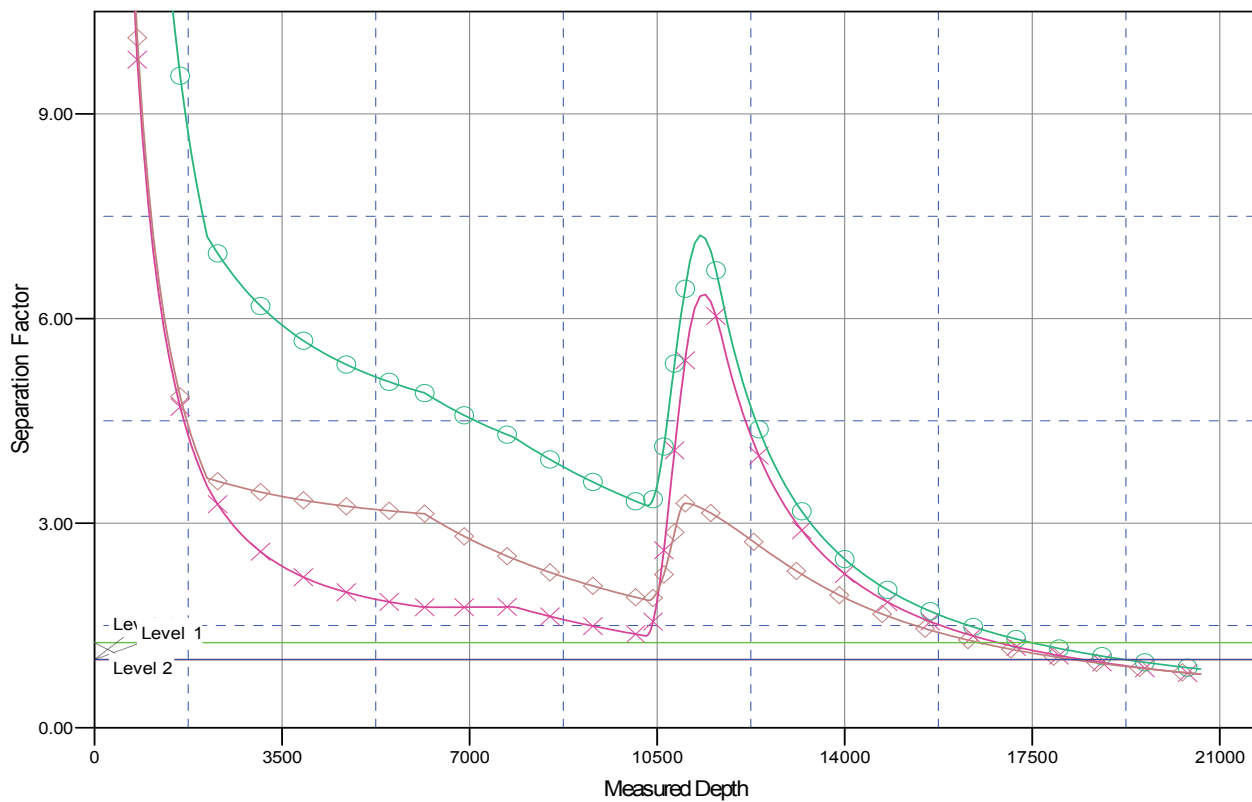
Anticollision Report

Company:	Oasis	Local Co-ordinate Reference:	Well Lewis Federal 5300 11-31 3B
Project:	Indian Hills	TVD Reference:	WELL @ 2134.0usft (Original Well Elev)
Reference Site:	153N-100W-31/32	MD Reference:	WELL @ 2134.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Lewis Federal 5300 11-31 3B	Survey Calculation Method:	Minimum Curvature
Well Error:	0.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Lewis Federal 5300 11-31 3B	Database:	OpenWellsCompass - EDM Prod
Reference Design:	Plan #1	Offset TVD Reference:	Offset Datum

Reference Depths are relative to WELL @ 2134.0usft (Original Well Ele
Offset Depths are relative to Offset Datum
Central Meridian is 100° 30' 0.000 W

Coordinates are relative to: Lewis Federal 5300 11-31 3B
Coordinate System is US State Plane 1983, North Dakota Northern Zone
Grid Convergence at Surface is: -2.31°

Separation Factor Plot



LEGEND

—●— Lewis Federal 5300 11-31 2T, Plan #1 V0
 —◆— Lewis Federal 5300 11-31 4T2, Lewis Federal 5300 11-31 4T2, Plan #1 V0
 —×— Lewis Federal 5300 11-31 5T, Lewis Federal 5300 11-31 5T, Plan #1 V0